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DAVID SKLANSKY

GETTING THE BEST OF IT!

MATHEMATICS OF GAMBLING

GENERAL GAMBLING CONCEPTS

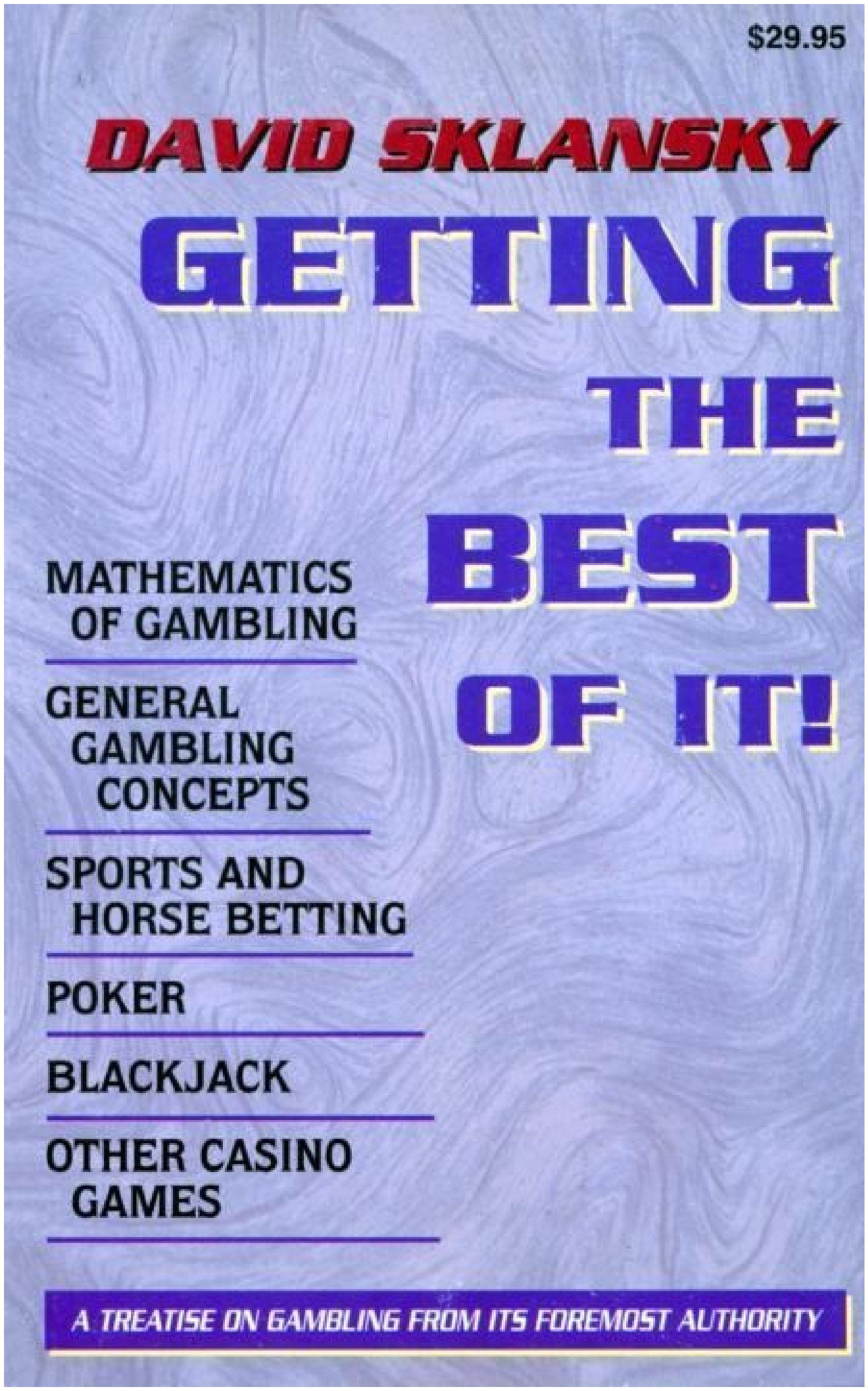
SPORTS AND HORSE BETTING

POKER

BLACKJACK

OTHER CASINO GAMES

A TREATISE ON GAMBLING FROM ITS FOREMOST AUTHORITY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Other Books From Two Plus Two Publishing

Hold 'em Poker For Advanced Players by David Sklansky and Mason Malmuth is the definitive work on this very complicated game. In fact, it may be the most difficult of poker games to play at an expert level. Even though much has appeared in print on how to play hold 'em, very little of it is accurate. There are several reasons for this. First, is that the structure of the game as played in most major card rooms has changed from what it was when hold 'em was first introduced. Second, the players have gotten much better as the years have gone by. And third, most authors only have a very limited understanding of true expert play.

This book should correct all of this. Anyone who studies this text, is well disciplined, and gets the proper experience should become a significant winner at the game. Some of the ideas discussed include play on the first two cards, semi-bluffing, the free card, inducing bluffs, being beaten on the river, staying with a draw, playing when a pair flops, playing trash hands, fourth street play, playing in loose games, playing in short handed games, and much more. A great deal of this material has never appeared in print before.

Seven-Card Stud For Advanced Players by David Sklansky, Mason Malmuth, and Ray Zee is the definitive work on this extremely complex game. There always seems to be exceptions to how a hand should be played. Deciding exactly what the right strategy should be in any particular situation can be very difficult. Perhaps this is why very few authors have attempted to analyze this game even though it is widely played.

With the writing of this book, a major gap in the poker literature is closed. Anyone who studies this text, is well disciplined, and gets the proper experience should become a significant winner. Some of the ideas discussed include the cards that are out, ante stealing, playing big pairs, reraising the possible bigger pair, playing little and medium pairs, playing three flushes, playing three straights, playing weak hands, fourth street, pairing your door card on fourth street, fifth street, sixth street, seventh street, defending against the possible ante steal, playing against a paired door card, continuing with a draw, scare card strategy, buying the free card on fourth street, playing in tightly structured games, playing in loose games, playing in short handed games, and much more. A great deal of this material has never appeared in print before.

High-Low-Split Poker, Seven-Card Stud and Omaha Eight-or-Better For Advanced Players by Ray Zee is the third book in the "For Advanced Players" series. In reality, it is really books 3 and 4 in the progression for two reasons. First, many of the concepts are similar for both games. And second, players mastering one game can easily make the transition to the other.

Some of the ideas discussed in the seven-card stud eight-or-better section include starting hands, disguising your hand on third street, when an ace raises, fourth street, fifth street, sixth street, seventh street, position, bluffing, staying to the end, scare cards, and much more. Some of the ideas discussed in the Omaha eight-or-better section include general concepts, position, low hands, high hands, your starting hand, how to play your hand, play on the flop, multi-way versus short handed play, scare cards, getting counterfeited, your playing style, and much more. A great deal of this material has never appeared in print before.

Gambling For a Living by David Sklansky and Mason Malmuth is the ultimate book for anyone interested in making a living at the gaming tables. The book includes chapters on horse racing, slot and poker machines, blackjack, poker, sports betting, and casino tournaments. They also identify the games that are not beatable. It is designed to show anyone new to this field how to be successful and what it takes to become a professional gambler. As with the other books by these authors much of this material has never appeared in print before.

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About David Sklansky

David Sklansky is generally considered the number one authority on gambling in the world today. Besides his eight books on the subject, David also has produced two videos and numerous writings for various gaming publications. His occasional poker seminars are always given an enthusiastic reception including those given at the Taj Mahal in Atlantic City and the World Series of Poker in Las Vegas.

More recently David has been doing consulting work for casinos, Internet gaming sites, and gaming device companies.

David attributes his standing in the gambling community to three things:

1. The fact that he presents his ideas as simply as possible (sometimes with Mason Malmuth) even though these ideas frequently involve concepts that are deep, subtle, and not to be found elsewhere.

2. The fact that the things he says and writes can be counted on to be accurate.

3. The fact that to this day a large proportion of his income is still derived from gambling (usually poker but also occasionally blackjack, sports betting, horses, video games, casino promotions, or casino tournaments).

Thus, those who depend on David's advice know that he still depends on it himself.

Other Books by David Sklansky

Hold 'em Poker
The Theory of Poker
Sklansky on Poker
Poker, Gaming, and Life

Other Books With David Sklansky

Hold 'em Poker For Advanced Players by David Sklansky and Mason Malmuth

Seven Card Stud for Advanced Players by David Sklansky, Mason Malmuth, and Ray Zee

Gambling For a Living by David Sklansky and Mason Malmuth

Introduction

People gamble for many reasons. I can think of three good ones.

1. To have fun

2. To make a "score"

3. To win money consistently

Any of these three reasons are perfectly acceptable to me. (The fourth common reason, namely a psychological craving for action is obviously unacceptable. This is the motive behind a compulsive gambler.) Those of you who have read my writings in the past may be surprised to learn that I have no objection to gambling just for the fun of it as long as you know that is what you are doing. I see nothing wrong with a couple coming out to Las Vegas to enjoy the shows and to try their luck at the tables. As long as they can easily afford the loss and they are having a good time, there is nothing wrong with some occasional frivolous gambling with the percentages against them. They might even win!

Likewise it is fine to make an occasional more serious effort to overcome slight unfavorable odds if your goal is to win a lot of money with a small initial investment. I even know some professional gamblers who every once in a while take a hundred dollar bill over to the craps table, expecting to lose it, but planning to bet aggressively in order to win at least five grand if they get lucky. This is fine, as long as they don't make it a habit and they stick to the best bets available, i.e., those with the smallest percentage disadvantage. However, this book was written mainly for those people whose main reason for gambling is reason No. 3.

As I said, there is nothing innately wrong in making bad bets as long as you know that you are doing it and you have a good reason for it. However, the fact is that most serious gamblers (and even the not-so-serious ones) are not gambling for the fun of it or to make a big score but rather to try and consistently win more times and more money than they lose. Unfortunately, this is easier said than done. Most gamblers want to win, but don't. The reason for this stems from the irrefutable fact that the only way you can show a profit over the long run when you are gambling is to be getting the best of it.

There are only two ways you can have the best of it. One way is to cheat. The other way is to have the overall odds or the skill factor in your favor. I leave the discussion of cheating to others. This book will simply show you how to become a consistent, honest, winning player.

Former Gambling Times readers will notice that many of the chapters in this book are basically reprints of my past columns in that publication. While a compilation of my past columns has been in great demand for quite a while, (for convenience sake if nothing else) this is more than that. Many changes from the original columns have been made. Slight errors have been corrected. Certain difficult sections were made more easily understood. Articles were written especially for this book. Some inappropriate columns were deleted. I also have written a short introduction to most chapters. More important, however, is the fact that the articles have been arranged into sections, each dealing with a different aspect of gambling. This is extremely important. Now the varied material in my past writings has been put in a form that will be most effective to the serious and aspiring gamblers and students of gambling.

♣ ♦ ♥ ♠

This book is separated into six sections (with a newly written introduction to each section). The first section is on probability. Readers may find the section tough going, but I urge that you at least get through the first three chapters. You might want to skip the other chapters in this section and come back to them later.

The second section is on my bread and butter game: poker. I can never write enough about this immensely complicated game. I think you will find that some of the chapters in this section explore some concepts that have never been analyzed by any other writer.

Blackjack gets its own section as it is the one other casino game (besides poker) that can be consistently beaten. Once again the five chapters in this section approach the subject in a unique way.

The fourth section is on Other Casino Games. The percentages usually are against you in these other games. However, as befits this book, the articles in this section are intended to help you find those rare times when even these games can offer you the best of it.

Section five is about Sports and Horse Betting. First, I explain how to make a sports bet and where the bookie's edge comes from. Then I show you how to overcome this edge by betting on one of those occasional games where the bookies line is wrong. After that, I discuss hedging and middling and then get into some important horse betting concepts.

Finally, the last section is on General Gambling Concepts. Many of the points in these chapters can be applied to a variety of games. The chapters in this section are all must-reading for the serious gambler.

Part One
Probability: The Mathematics of Gambling

Probability: The Mathematics of Gambling

Introduction

One of the major decisions I had to make when putting this book together was whether to make this section first or last. There were some good reasons to put it at the end, almost like an appendix. After all, most readers will buy this book to read about gambling, not mathematics. In fact, some people may even elect not to buy this book after only leafing through this first section and deciding that it is too difficult and not very entertaining.

Since all the other sections are very interesting as well as informative, I definitely had good reason to relegate the probability section to the back of the book. Not only might I cost myself some sales, I also might keep some readers from reading the whole book under the mistaken impression that the whole book is mathematically technical. (Hopefully I have altered that impression in this introduction.)

In spite of the preceding arguments I have to, in good conscience, put the Probability section first. I have two reasons:

1. By putting it first I stress the importance of at least a cursory knowledge of probability for the aspiring serious or professional gambler.

2. Many of the chapters in the later sections assume some knowledge of probability (especially the first three chapters on expectation, odds and probabilities, and parlays).

There are pro gamblers who have almost no knowledge of probability. However, they are freaks. They have a natural gift much like the rare athlete who doesn't have to train. It is safe to assume that you don't fall into this category. Therefore, you must understand the fundamentals before becoming a champion. In gambling the fundamentals are probability.

I have endeavored to make the chapters in this section as simple as possible. The subjects are approached much differently than they are in the typical probability textbook. Most readers should be able to understand them if they read slowly and carefully. Reading these chapters is work but it is well worth it. It will mean thousands and thousands of dollars in your gambling career. It is especially important that you understand the first three chapters in this section as clearly as possible.

If you have a particularly strong aversion to mathematics, it may be better to skip most of this section and come back to it later. The rest of the book can still be of much value. However, once again, I would urge you to try and get through the first three chapters as they play such a large part in all forms of gambling.

Expectation and Percentage Edge

The following chapter is the first article I ever wrote for Gambling Times. It is also the most important one. If you want to be a consistent winner, you must have a positive expectation as often as possible. When faced with a decision that yields two or more positive expectations, you must normally choose the one with the highest expectation.

Likewise when you are forced into a situation with a negative expectation (as when you are dealt a 16 in blackjack) you must know how to reduce this negative expectation as much as possible. That is what "getting the best of it" is all about.

Only by thinking in terms of expectation can you get the most out of your gambling. The concept.comes up in all games and in fact in all financial decisions. However, before you can use this concept you must first understand it. I therefore urge you to read this chapter (and the next one) even if you decide to skip the other chapters in the Mathematics of Gambling section.

You've probably heard that the casino has a 1.4 percent advantage in craps, a 5.3 percent advantage in roulette, and so forth. Yet few gamblers really understand what those figures mean even though they are part of the most important concept in gambling - what a bet is worth.

Mathematicians call what the bet is worth the "mathematical expectation" or simply "the expectation" of players.

Here's how to arrive at it.

In any fair bet situation - say, flipping an honest coin at even money - both participants can expect to break even in the very long run, so the expectation is zero.

But in most betting situations one player has an advantage - gamblers say this player "has the best of it. "

One way you might have the best of it is by betting even money at any game that you can win more than half the time. Suppose you're playing a 100 point game of gin (without doubles for blitzes) for $10, and you think your chances to win are 60 percent.

After 100 games you figure to have won 60 and lost 40. You will have collected $600 and paid out $400, for a profit of $200. You have averaged $2 profit per game, even though you will win or lose $10 on each individual game. $2 is your average profit per game.

Thus, $2 is your mathematical expectation. It is what each bet is worth to you. Every time you play the game you can figure you earned $2 whether you actually won or lost the $10.

Since $2 is 20 percent of the $10 bet, you have a 20 percent edge, while your opponent has a 20 percent disadvantage.

For a more complicated example let's say you were allowing double pay for blitzes, and could somehow predict that your opponent would win 52 percent of the games, 3 percent of which were blitzes, while you would blitz him 10 percent of the time and win a simple game 38 percent of the time.

In 100 games you would pay $10 forty-nine times, pay $20 three times, collect $20 ten times and collect $10 thirty-eight times. After 100 games you would be winning $30 for a 30 cent per game positive expectation and a three percent edge-despite the fact your opponent would have won more games than you.

A similar situation exists with a game such as blackjack. Even professional card counters do not win quite half their bets, but they win well over half their splits and double downs and get 3-to-2 on blackjack; they also bet more when their edge is greater, according to the cards which have already been played. They thus have an overall edge of I to 2 percent, depending upon the quality of their play and the rules of the game they are playing.

Figuring expectation is only slightly more difficult when bets are not for even money. You have the best or worst of it depending on your chances in comparison to the odds you are getting or laying. Suppose someone is laying you 4-to-1 on a proposition when you are only a 3-to-1 underdog. Obviously you have the best of it; let's see by how much.

Say he's betting $4 to your $1 on something that you know will on average win once in every four times you play. After four times you will win $4 once and lose $1 three times for a net profit of $1. This is an expectation of $1 per four bets, or 25 cents per $1 bet. The percentage edge is 25 percent. It's that simple!

But watch this: Even though your opponent figures to average losing 25 cents per shot, his percentage disadvantage is only 6'/4 percent. Because he only loses 25 cents each time he bets $4. It often works out that the percentage advantage of the favorite isn't the same as the percentage disadvantage for the underdog.

Of course it's possible to have a positive expectation laying the odds just as long as the odds you're laving aren't as high as they should be. The crap table lays you 9-to-1 (they call it 10-for-1) on the hard eight. The right price is 10-to-1.

Here's how to figure that out: You're betting that double fours will be rolled before the seven or any other eight. There are six ways to make a seven (1-6, 2-5, 3-4, 4-3, 5-2, and 6-1) and four ways to make an easy eight (2-6, 3-5, 5-3, 6-2) for a total of 10, against only one way to make a hard eight (4-4). All the other possible rolls do not affect this bet.

Now suppose you're betting $1 a roll. In 11 rolls the house will win 10 times and collect a total of $10 from you. You will win once and they will pay you $9, so you will be down $1.

Now, the house has bet $110 to make that $1, so they could tell you quite honestly that their advantage here is only a little less than one percent. That doesn't sound like much .

But your disadvantage is percent, because your loss is $1, and that's about percent of the $11 you bet.

It's important not to confuse percentage edge with expectation. One is a percentage, the other is an actual dollar amount. If you know your percentage advantage or disadvantage, you can get your positive or negative expectation by multiplying the amount of your bet(s) by the percentage. This is what you figure to win or lose in the long run. If $1 million is bet on the pass line over a period of time the casino should win about $14,100 of it, since the player has a 1.41 percent disadvantage. This is what those percentages really mean.

These are concepts that are important for the serious player to understand. And incidentally, if you are gambling for money, you should certainly be a serious player. Here are some practical applications that you can easily follow.

In blackjack we are constantly faced with strategic decisions. Thankfully, computers already have formulated the right decisions for these situations. In all cases the right play is the one with the highest positive expectation or the lowest negative one.

This is not always the play with the highest percentage edge. Remember that percentage and expectation are different concepts.

Suppose you are dealt a six-five against a dealer's ten. If you didn't double down you would win about 56 percent of the time, which is a 12 percent edge. If you do double down you will win only 54 percent of the time for an 8 percent edge. But you will be betting twice as much and 8 percent of two units is greater than 12 percent of one unit, thus giving you a greater expectation.

This example shows that the most important aspect of any bet is its money expectation rather than percentage edge. Say we have our choice of taking $20-to-$10 or $70-to-$50 on an even money proposition. We should take the second proposition since it gives us an expectation of $5 more per bet than the first proposition. The fact there is a greater percentage edge in the first bet shouldn't enter our considerations in most cases. The one exception is when you don't have a big enough bankroll to comfortably cover the large bet.

It isn't necessary in most cases to bet just because you have a little bit the best of it, but you must if you're playing poker.

If the pot offers 5-to-1 odds and you are only 4-to-1 underdog you must call because you are giving up too much by folding. You have to ante every hand and you can win only if you take advantage of almost every favorable situation (where you have a positive expectation) just as your opponents will. Likewise if you are 6-to-1 underdog in this spot you must fold. This is also true for blackjack; you must double down just about every time the computer indicates to do so, or you won't win.

When betting sports, however, there is nothing wrong with avoiding bets that are only marginally good because nobody makes you bet on games where you have the worst of it.

We said this before, but it bears repeating. These concepts are the most important for you to understand if you want to be a successful gambler. Realize that each bet has earned you something when the odds or percentage is in your favor.

It isn't necessary to have repeated trials to accept a bet. If you get $6-to-$5 on an even money bet you have earned 50 cents regardless of the outcome. Keep on "getting the best of it," and you will find your winnings have totaled the sum of your expectations.

Odds and Probabilities

When I first came to Vegas and met other professional gamblers I was struck by the fact that they always put the chances of any gambling propositions in the terms of "odds."

Everything was a "2-to-I dog" or a "7-to-5 favorite." While this might be a convenient way to think of the chances of something happening as far as money payoffs are concerned it is very inconvenient if you need to calculate the chances of a proposition involving more than one event.

This is why so many of the old-time pros are so weak in probability problems. The first step is to think of things in terms of fractions or percentages rather than odds. Thus, you must know how to change some particular odds to its equivalent fraction. This chapter will show you how.

When you speak of the 'odds against" some event or the probability of this event occurring, you actually are speaking of the same thing. However the specific numbers would be different. Thus, a football team, for example, that is a 3-to-1 underdog to win has a probability of/4.

It is important to understand the difference in expressing the likelihood of an event in terms of odds as opposed to probability. You also should know the specific relationship between these two numbers. This is especially important for gamblers since odds are used in their day-to-day "work." In calculating the odds of combined events such as parlays, it is necessary to use the probabilities (usually expressed as a fraction) of these events to perform the calculations. I will have more on these methods of calculation in future chapters. However, the first step is to learn how to change the odds to their equivalent probabilities and vice versa.

Before doing this, let us clarify the meaning of these two terms. If an attempt at a particular task will be successful one time for every four times that it will be unsuccessful, the odds against its success are said to be 4-to-1 (read: four-to-one). It might be called a 4-to-1 "shot" or a 4-to-1 underdog. (It is better to use the term "shot" for reasons I will explain shortly.)

Similarly if something should lose eight times for every five times it wins it is an 8-to-5 shot. In other words the odds against some event's occurrence (e.g. winning) is simply the ratio of non occurrences to occurrences (e.g. losses-to-wins). If you win twice for every one time you lose you are a 1-to-2 shot. The first number is always the number of losses or failures. (Gamblers occasionally state the odds on big favorites backwards. Thus they might call someone who will win three times for every two losses as "3-to-2 favorite" when, in fact, they mean a "2-to-3 shot." This is very confusing since a "3-to-2 favorite" may also mean a 3-to-2 shot (40 percent) that has the best chance among many competitors (e.g. horses) and, thus, is the favorite. It is because of these ambiguities that I prefer the word "shot" after the odds rather than "underdog" or "favorite.")

Since odds are simply ratios, you always can reduce them to lowest terms if you desire simply by dividing both numbers by the same number. Thus 25-to-10 is the same as 5-to-2 (dividing by 5), 9-to-6 also is 3-to-2, and so on.

Another way to measure the chances of an event occurring is by speaking of its probability. If an attempt will be successful two times out of five trials, the probability of its success is said to be 2/5 (or 40 percent or .4). In general, the probability of any particular event's occurrence is simply the number of successes over the total number of trials. Notice that in the two-fifth's example just given, we would be unsuccessful three times out of five and the odds would therefore be 3-to-2.

The thoughtful reader should now pause to see if he can derive the general technique that will change any odds to a fraction or any fraction to odds. He will find it much easier to remember if he derives it himself. If you are too lazy to even try to do this, I have grave doubts about your success as a gambler.

Let's see if you came up with the right answer. To try a specific example first, we will change 9-to-5 odds to its equivalent probability. 9-to-5 tells us that we will win five times for every nine times we lose. Our total number of attempts would be fourteen (5 wins + 9 losses). Thus, we should win 5 times for every 14 trials (on the average, of course). Therefore, the equivalent probability is 5/14.

$$\frac{5}{14} = \frac{5}{9+5}$$

It should now be apparent that the general rule is this: To change odds to their equivalent fraction, first add both numbers in the odds. This becomes the denominator (bottom) of the fraction. The numerator (top) of the fraction is simply the second number of the odds.

Examples: 5-to-3 is ⅜, 3-to-1 is ¼, 1-to-2 is ⅔. In algebra, a-to-b is b/(a + b).

Whereas it is necessary to use probabilities in order to do sophisticated calculations, this means that the answer will be expressed as a probability. However the result is often more convenient if it is reverted to odds.

To change a fraction to its equivalent odds, keep in mind that the numerator of the fraction is the number of successes or wins and the denominator is the total number of trials. The number of failures is simply the total number of trials minus the number of successes. Since odds are always in the form of failures-to-successes, we now have our rule: *To change a probability fraction to its equivalent odds, first subtract the numerator from the denominator.* This becomes the first number of the odds. The second number of the odds is simply the numerator of the fraction. *Examples:* ⅜ is 5-to-3, ⅓ is 2-to-1, ¾ is 1-to-3. In algebra, a/b is (b-a)-to-a.

The ability to change odds to fractions and vice versa almost instantaneously in your head is indispensable for the gambler who wants to win.

Mathematics of Parlays

This chapter explains one of the simpler types of probability problems. While simple, it is also extremely important to the serious gambler. If you know the chances of each of a variety of events taking place, what are the chances that they all take place? The answer depends on whether the events are related to one another.

I show you how to answer the question whether or not they are related. I also show you an analogous technique to calculate the probability that none of a variety of events takes place.

Finally, if you know the chances that none of the events takes place it is a simple matter to calculate the chances that at least one (or more) can take place. It is all explained in the following chapter.

One of the most common and important types of probability problems is what gamblers call "parlays." When you bet a parlay, you are betting on the outcome of two or more events with the stipulation that all of your selections must be right in order for you to win. Of course, betting this way greatly reduces your chances of winning, but you are paid off at correspondingly higher odds when you do win. The ability to figure the exact probability against any particular parlay is an important weapon in the gambler's arsenal.

Most gamblers think solely of sporting events when they think of parlays. Actually, any time you figure the probability that all of a number of events (each with its own separate probability) will occur you are figuring a parlay. For example, the probability of being dealt a pat heart flush in draw poker is simply a parlay problem since you are really calculating the chances that all five cards are hearts.

If the events in question are not related to one another - that is, if the outcome of one event has no effect on the outcome of the other event - there is then a very simple procedure for obtaining the parlay probability. Simply multiply the probability of each event. Notice I said the probability, not the odds. Many gamblers mistakenly multiply the odds. If the original event's chances were expressed in terms of odds, they must first be changed to probabilities (fractions or percentages) and then multiplied. This result can now be changed back into odds. (See previous chapter.)

Let us try a few examples. Suppose you want to make a bet on the Yankees and Dodgers. The Yankees are 6-to-5 underdogs and the Dodgers are 3-to-2 underdogs. The Yankees' probability of winning is 5/11. The Dodgers' is 2/5. The probability that they both will win is thus 10/55, which reduces to 2/11.

$$\left(\frac{5}{11}\right)\left(\frac{2}{5}\right) = \frac{2}{11} = \frac{10}{55}$$

The odds against it are therefore 9-to-2. Simple.

What are the chances of red coming up on the roulette wheel three times in a row?

$$\left(\frac{18}{38}\right)\left(\frac{18}{38}\right)\left(\frac{18}{38}\right) = \frac{739}{6{,}859} \ (\textit{after reducing})$$

This figure is less than 1/9, which shows how your disadvantage increases as you make more than one bet. If red were even money, three consecutive reds would have a probability of 1/8.

$$\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{8}$$

What is the probability of throwing two dice five times without throwing a seven? Since the probability of throwing a seven in one roll is 1/6, the probability of not throwing a seven is 5/6. The probability of throwing five consecutive non-sevens is

$$\left(\frac{5}{6}\right)\left(\frac{5}{6}\right)\left(\frac{5}{6}\right)\left(\frac{5}{6}\right)\left(\frac{5}{6}\right) = \frac{3{,}125}{7{,}776}$$

or about 40 percent. In other words, you will roll a seven at least once in five rolls about 60 percent of the time. Now you see why hot rolls on the dice table are so infrequent. This example also illustrates how even heavy favorites are unlikely to repeat several times without a loss.

In football it is generally felt that a six-point favorite is a I -to-2 shot to win the game. (Gamblers incorrectly call them 2-to-I favorites.) In other words, they should win two out of three. Suppose someone offers to bet you even money that at least one of two six-point favorites that you choose will lose the game outright. (Neglect ties.) Who has the best of it? Well, the chances that you will win both games are

$$\left(\frac{2}{3}\right)\left(\frac{2}{3}\right) = \frac{4}{9}$$

You are a 5-to-4 underdog! Those of you who bet six-point "teasers" might learn something from this example.

Up to this point we have been talking about events which have no effect on one another. They are called independent events. Very often, however, events are dependent. Hence, the more general rule for figuring parlays is to multiply the probability of the first event by the probability of the second event given that the first event did, in fact, occur. If you are talking about three or more events, the third probability is based on the assumption that the first two events occurred, and so on. Keeping in mind that we are once again talking about probability, not odds, what we are doing is adjusting the second and successive probabilities, assuming that the first and successive events occurred.

Example: What is the probability of being dealt two aces "wired" in five-card stud? This is simply a parlay problem. In other words, what is the probability that your first card will be an ace and your second card will be an ace? The probability that the first card will be an ace is 4/52 = 1/13. The figure for the second card must now be changed to the probability of its being an ace if the first card was, in fact, an ace. With three aces left in a 51 card deck this figure is thus 3/51 = 1/17. The parlay is therefore

$$\left(\frac{1}{13}\right)\left(\frac{1}{17}\right) = \frac{1}{221}$$

This is the probability of being dealt two aces wired. Three aces wired in seven-card stud is

$$\left(\frac{4}{52}\right)\left(\frac{3}{51}\right)\left(\frac{2}{50}\right) = \frac{1}{5{,}525}$$

Any three-of-a-kind is simply

$$(13)\left(\frac{1}{5{,}525}\right) = \frac{1}{425}$$

or 424-to-1.

What is the probability of being dealt a pat flush in draw poker? The simplest way to answer this question is to figure the probability for a specific suit, say spades, and then multiply by four. The probability that the first card is a spade is 13/52 or '/4. If it is, the probability that the second card is a spade is 12/51. If the first two cards are spades the third probability becomes 11/50. Working down, your equation becomes

$$\left(\frac{13}{52}\right)\left(\frac{12}{51}\right)\left(\frac{11}{50}\right)\left(\frac{10}{49}\right)\left(\frac{9}{48}\right) = \frac{33}{66{,}640}$$

(after reducing). Any flush, then, is

$$(4)\left(\frac{33}{66{,}640}\right) = \frac{132}{66{,}640}$$

which is about 500-to-1. (These figures includes the slight possibility of a straight flush.)

What is the probability that your first three cards in seven-card stud contain no pair'? The best way to figure this problem is to neglect the first card, find the probability that the second card doesn't match it (48/51), and the probability that the third card doesn't match the first two (assuming that these first two didn't match). With six "bad" cards left in the deck, this second probability is 44/50. Our answer is therefore

$$\frac{352}{425} = \left(\frac{48}{51}\right)\left(\frac{44}{50}\right)$$

The odds against a pair are therefore 352-to-73 or a little less than 5-to-1.

Whenever you have a problem involving the chances of at least one event happening you can use the tools we have just learned. On the surface it may seem hard to calculate the chances of "one or more" successes. However, if we realize that "one or more" is simply the opposite of "none," we have our method. Simply figure the probability that all events are unsuccessful. Subtract this result from one and you have the probability of at least one successful event.

A common hustle is to bet that one of your first three cards in seven-card stud will be an ace, deuce, or trey. In other words, the bet is that at least one of these cards is an ace, deuce, or trey. To figure the price, simply ask yourself, what are the chances that none of the cards are one of these three denominations. The answer to this second question is

$$\left(\frac{40}{52}\right)\left(\frac{39}{51}\right)\left(\frac{38}{50}\right) = \frac{59{,}280}{132{,}600} = \frac{38}{85}$$

Therefore, 38 out of 85 times no ace, deuce, or trey will appear. One of these cards will appear 47 out of 85 times.

With practice any parlay problem should become easy. Just remember to work with fractions, not odds, and to change the probabilities when the events are related.

Combinations

This chapter and the next one start to get a little heavy. It is not necessary to fully understand them in order to become a successful gambler. However, having a basic understanding of them certainly can be helpful even if you don't master them completely. You might want to look them over quickly and then come back to them after reading the rest of the book. Of course, if you are mathematically inclined you can go ahead and read them thoroughly now. You will never see these concepts explained any simpler any place else.

The mathematical subject of permutations and combinations is simply short cut methods of counting. What you are counting is the number of distinct ways of picking a few objects out of a collection of many. If the order in which we pick them makes a difference, it is called permutation. Otherwise it is called a combination. It just so happens that the ability to quickly count the number of different combinations of a few things that can be picked out of a larger collection (such as picking five cards from a deck of 52) is very important when figuring out many gambling problems. That, of course, is why this chapter is included in this book.

One of the most useful tools a player has for figuring out probability problems (and therefore gambling problems) is the concept of combinations. When we speak of combinations, we mean the number of ways in which a group of elements can combine into smaller groups. To put it another way, combinations tell us how many different possible groups could be formed from this larger group.

As an example, suppose I have 3 extra season tickets to the Yankee home games. I can therefore take 3 guests to every one of them. I have 9 friends who I think will make good companions, and I need to pick 3 of them to invite. I do this every time the Yanks play at home. So as not to insult anybody and to add variety for myself, I decide to invite a different group of 3 (from these 9) every night. It is important to realize that we can form a different group by substituting just one person for another. Thus, Mark, Ted and George are counted as a different group from Mark, Ted and Peter.

The question is, how many games can I attend before I have to go with a group of friends that I have already gone with? In other words, how many different combinations of 3 friends out of 9 can be made? The answer is 84 (enough for the whole season), and I'll explain how I got this figure in a moment.

For the time being, it's important for you to understand that the foregoing problem is simply a special case of taking 9 things 3 at a time. Obviously, it is not necessary that we be speaking of people here. As a second example, some bookies offer what they call a Trinella bet. To win, you must pick the first 3 horses to finish a race (the order doesn't matter). In a 9 horse race, how many different Trinellas do you have to choose from`? Once again the answer is 84. It is simply another case of 9 things taken 3 at a time. You would have to bet 84 different Trinellas in a nine-horse race in order to assure yourself a winner.

The mathematical symbol for combinations is either a capital "C" or tall parenthesis O. Nine things taken three at a time is written.

$${}_{9}C_{3}$$

or, more frequently,

$$\binom{9}{3}$$

12 things taken 5 at a time would be written

$${}_{12}C_{5} \quad or \quad \binom{12}{5}$$

From here on I will use only the more commonly used parenthesis symbol. "N" things taken "r" at a time, which is the way mathematicians like to describe a general combination problem, is thus written as

$$\binom{n}{r}$$

To evaluate any individual combination, it is necessary to form a large fraction. To show you how to do this, let us take

$$\binom{10}{4}$$

as an example. The bottom of the fraction, or denominator, is simply the numbers 1 through 4 multiplied together

$$\frac{}{(1)(2)(3)(4)}$$

This is because 4 is the bottom number in the combination symbol. If it had been, for instance, a 6, the denominator of the fraction would be

$$(1)(2)(3)(4)(5)(6)$$

To obtain the numerator of the fraction, you start with the number on top of the combination symbol and start multiplying each successive lower number until there are as many numbers in the numerator as there are in the denominator. Thus,

$$\binom{10}{4}$$

would be

$$\frac{(10)(9)(8)(7)}{(1)(2)(3)(4)}$$

or

$$\frac{5{,}040}{24}$$

For those of you who may have forgotten that a fraction is really just a division sign,

$$\frac{5{,}040}{24}$$

simply means

$$24\overline{)5{,}040}$$

The answer in this case is 210.

To makes sure you understand the technique, I will list some more examples without bothering to calculate them out.

1. $\binom{13}{5} = \dfrac{(13)(12)(11)(10)(9)}{(1)(2)(3)(4)(5)}$

2. $\binom{9}{7} = \dfrac{(9)(8)(7)(6)(5)(4)(3)}{(1)(2)(3)(4)(5)(6)(7)}$

3. $\binom{52}{5} = \dfrac{(52)(51)(50)(49)(48)}{(1)(2)(3)(4)(5)}$

4. $\binom{52}{2} = \dfrac{(52)(51)}{(1)(2)}$

5. $\binom{9}{2} = \dfrac{(9)(8)}{(1)(2)}$

Once again, realize that you multiply "downward" on top as many numbers as you multiply "upward" on the bottom of the fraction. You will always have as many numbers in the numerator as in the denominator.

Before showing how combinations are used to solve gambling problems, let me give you some hints on how best to calculate out these rather cumbersome fractions. The trick is canceling out. It should be used even when you have a calculator available. You cancel out similar numbers from both the numerator and denominator of the fraction before you do any multiplying. To give a detailed example of this technique, let us use 10 things taken 6 at a time or

$$\binom{10}{6}$$

In expanded form we have

$$\frac{(10)(9)(8)(7)(6)(5)}{(1)(2)(3)(4)(5)(6)}$$

First notice that there is a 5 and 6 on both the top and the bottom. They can be eliminated:

$$\frac{(10)(9)(8)(7)(\cancel{6})(\cancel{5})}{(1)(2)(3)(4)(\cancel{5})(\cancel{6})} = \frac{(10)(9)(8)(7)}{(1)(2)(3)(4)}$$

Next we notice that the 2 and 4 on the bottom multiply out to be 8, and there is an 8 on the top:

$$\frac{(10)(9)(\cancel{8})(7)}{(1)(\cancel{2})(3)(\cancel{4})} = \frac{(10)(9)(7)}{(1)(3)}$$

Finally, we may eliminate the 3 on the bottom by dividing the 9 on top by that same 3:

$$\frac{(10)(\overset{3}{\cancel{9}})(8)}{(1)(\cancel{3})} = \frac{(10)(3)(7)}{1}$$

With hardly any multiplying at all, this canceling technique has allowed us to easily calculate out

$$\binom{10}{6}$$

to a figure of 210. There is obviously more than one way to cancel out numbers in this fashion, but as long as you always divide the top and the bottom by the same number, you will get the same result.

Some of you may notice that our result for

$$\binom{10}{6}$$

was the same as for

$$\binom{10}{4}$$

Was this coincidence? Suppose my girlfriend and I were entertaining 10 guests at home. I decided to take 6 of them with me to a poker game and leave the other 4 with Sherry. As we have seen, there are 210 different combinations of 6 out of 10 that I could pick. However, Sherry could have just as easily picked 4 people to stay with her, leaving me 6 people to take. From her point of view, she is picking 4 out of 10. However, for every possible 4 out of 10 she chooses, there is a corresponding combination of six out of 10 not chosen. Thus, it makes sense that it should be the same number. In mathematical terms

$$\binom{n}{r} = \binom{n}{n-r}$$

Other examples:

$$\binom{12}{8} = \binom{12}{4}$$

$$\binom{11}{8} = \binom{11}{3}$$

$$\binom{15}{10} = \binom{15}{5}$$

Those readers who might like to verify this by calculation will soon see the obvious pattern as they cancel out numbers. As an example:

$$\binom{8}{6} = \frac{(8)(7)(6)(5)(4)(3)}{(1)(2)(3)(4)(5)(6)}$$

$$= \frac{(8)(7)(\cancel{6})(\cancel{5})(\cancel{4})(\cancel{3})}{(1)(2)(\cancel{3})(\cancel{4})(\cancel{5})(\cancel{6})}$$

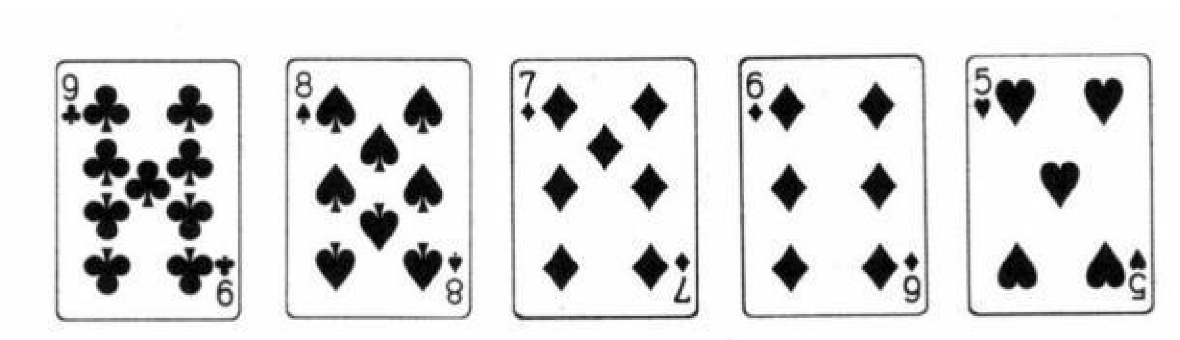
$$= \frac{(8)(7)}{(1)(2)} = \binom{8}{2}$$

In practice this means that whenever a problem asks you to evaluate a combination where you have to take more than half of the total number of elements, you save time by calculating its simpler counterpart.

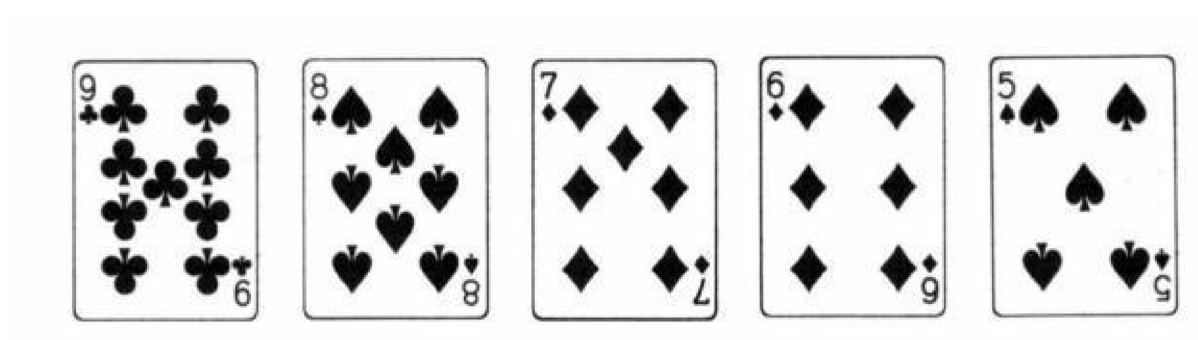
Let us now see how to use this new-found knowledge to figure out some gambling problems. Many people know that 2,598,960 is the total number of possible poker hands. This number is simply

$$\binom{52}{5}$$

It is the total number of ways of choosing 5 cards from a 52 card deck. It is important to realize that any change in 1 card or I suit signifies a different hand. Thus,



is a different hand from



even though both hands are called nine-high straights. In fact, there are 1,020 different nine-high straights (and 4 more nine-high straight flushes).

Now that we understand what is meant by 2,598,960 possible five-card poker hands, we can begin calculating poker odds. It is simply a matter of counting or computing the number of ways of being dealt any type of hand and comparing it with the total number of possible hands. For instance, there is only 1 hand that is called a royal flush in spades. Thus, your chance of being dealt this hand is

$$\frac{1}{2{,}598{,}960}$$

There are 4 hands altogether that are called royal flushes. The chances of being dealt one pat is therefore

$$\frac{4}{2{,}598{,}960}$$

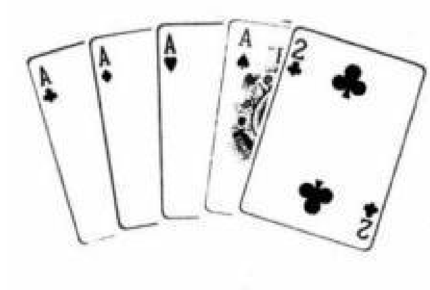
or about 650,000-to- 1. Since there are 40 hands altogether that are called straight flushes, including royal flushes, our chances of being dealt one of them is

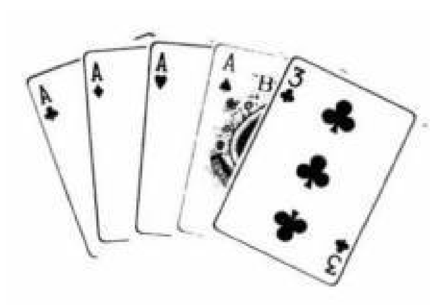
$$\frac{40}{2{,}598{,}960}$$

or about 65,000-to-1. A simple straight flush is

$$\frac{36}{2{,}598{,}960}$$

or about 72,000-to-1. How many hands are called four aces? Not 1, but 48. We could have





etc. The same applies to all other four-of-a-kind hands. Therefore, the total number of hands that are called four-of-a-kind is

$$(48)(13) = 624$$

The chances of being dealt four-of-a-kind are therefore

$$\frac{624}{2,598,960}$$

or 4,164-to-1.

How many spade flushes are there? This in itself is a combination problem. We need to choose five spades out of the thirteen available to use. There are

$$\binom{13}{5}$$

different ways to do this. This works out to 1,287. Since this goes for any suit, we have 5,148 different hands that are called flushes (1,287)(4) minus 40 of these hands which are called straight flushes. Our chances of being dealt a pat flush are

$$\frac{5,108}{2,598,960}$$

or about 508-to-1.

Before we can calculate the odds against other poker hands we must learn another mathematical principle. Take the following example: the Knicks travel to Boston to play basketball against the Celtics. The Knicks bring 10 players with them. The Celtics have eleven. Since basketball is played with five players, there are

$$\binom{10}{5} = 252$$

different five-man teams the Knicks can have on the floor at any one time, while the Celtics can form any one of

$$\binom{11}{5} = 462$$

teams. The question is, how many different possible configurations of 2 five-man teams, 1 team from each club, can be put on the floor. The answer is not

$$\binom{21}{10}$$

which is the number of ways 10 players can be picked from a total of 21. This is because

$$\binom{21}{10}$$

would include things like picking seven players from the Knicks and only three from the Celtics as well as any other combination of players that adds up to 10. But by the rules of basketball, we must have 5 from each team. The easiest way to see how to calculate the total numbers of ways these 2 teams can appear on the floor is by numbering the possible Knick teams 1,2,3,4,....252 and the possible Celtics teams 1,2,3,4,...462 It is now easy to see that any Knicks team that is out on the floor could be playing against any 1 of 462 Celtics teams. In other words, it could be Knicks team number I vs. Celtics team 1-462 or Knicks team number 2 vs. Celtics team 1-462 or Knicks team 3 vs. Celtics team 1-462. Since each one of the possible 252 Knicks teams could be playing any one of 462 different possible Cetics teams, the total number of different floor configurations is simply

$$(252)(462) = 116{,}424$$

This principle is called the multiplication principle. It simply means that if you are trying to reach a goal that requires more than one step, you merely calculate the number of ways you can obtain each individual step and multiply these numbers together to get the total number of ways of reaching your goal. Thus, if I have to fly to Moscow from Las Vegas, changing planes in New York and London, and if I have a choice of three airlines for the first leg, four airlines for the second leg and two airlines for the third leg, I can reach my destination in one of

$$(3)(4)(2) = 24$$

different ways.

How many poker hands are called "aces full of deuces?" We need 3 aces and

2 deuces. Since we have 4 each of aces and deuces available to us in the deck, we should be able to see the analogy between this problem and the basketball problem. The number of possible "teams" of 3 aces out of 4 to choose from is simply

$$\binom{4}{3} = 4$$

The number of possible pairs of deuces is

$$\binom{4}{2} = 6$$

We therefore have

$$(4)(6) = 24$$

24 hands, all of which are aces full of deuces. The total number of aces full is simply

$$(24)(12) = 288$$

since there are also aces full of treys, fours, fives, etc. There are also 288 hands called kings full, queens full, jacks full, etc. there are

$$(288)(13) = 3{,}744$$

full houses altogether. The chances of being dealt 1 pat are thus

$$\frac{3{,}744}{2{,}598{,}960}$$

or 693-to-1

Strangely enough, the weaker poker hands are more difficult to calculate than the stronger hands. So as not to push too much on my readers at one time, I will leave these more difficult problems for a future book, but I don't want to end this subject without talking a little about keno. Virtually no one, including keno bosses, understands the correct way to compute keno prices. Yet it can be done rather simply, once again, with the use of combinations. While it is true that the total number of ways that 20 numbers can be lit on the eighty-number board is

$$\binom{80}{20}$$

this gigantic number is not normally needed to calculate keno propositions.

As an example, let us calculate the probability of marking an eight-spot ticket and hitting exactly 6 numbers. What this means is that 6 of our numbers will be in the twenty-number lit section of the board, while two of our numbers are in the unlit section. The best way to do this problem is by imagining that we made out every possible eight-spot ticket. The number of different eight-spot tickets is

$$\binom{80}{8}$$

since we are simply choosing 8 numbers out of the 80 available numbers when we mark an eight spot. We now look up at the keno board after the 20 numbers have been chosen. Purely for the sake of argument, I will say that the numbers lit up on the board are the numbers 1 through 20. It is now time to look back at our enormous pile of marked keno tickets and find those where we correctly picked 6 out of 8 numbers. As an example, the ticket marked 2, 5, 8, 9,14, 20, 36, and 50 would qualify as a ticket with six hits. However, this is only one of many. In fact, there are many other tickets that hit specifically the numbers 2, 5, 8, 9,14 and 20, since the two other numbers that didn't hit can be any two of the sixty unlit numbers. There are

$$\binom{60}{2} = 1{,}770$$

different tickets, all of which hit specifically the six numbers 2, 5, 8, 9,14, 20 in combination with 2 losing numbers. Remember that this is only one of many combinations of 6 numbers out of the 20 numbers (1-20) lit up on the board. In fact, there are

$$\binom{20}{6} = 38{,}760$$

combinations of six winning numbers out of the 20 on the board, each of which has 1,770 variations based on the two losing numbers. Thus, the total number of tickets with 6 hits out of 8 is

$$\binom{20}{6}\binom{60}{2}$$

Since the total possible number of eight spots is

$$\binom{80}{8}$$

we have our probability of writing an eight spot that hit exactly 6 numbers:

$$\frac{\binom{20}{6}\binom{60}{2}}{\binom{80}{8}} = .002367, \textit{ or about } 422\text{-}\textit{to}\text{-}1$$

Those readers who have followed me so far should see that if we mark an x-spot ticket and want to find the probability of exactly y hits, we must first form the number

$$\binom{80}{x}$$

for the total number of possible x spots. We next form two "teams," one containing y numbers out of 20 lit numbers.

$$\binom{20}{y}$$

and one containing the remaining (x - y) numbers from the unlit 60

$$\binom{60}{x-y}$$

We thus have the general formula for keno. The chances of hitting exactly y numbers in keno after marking an x spot is simply

$$\frac{\binom{20}{y}\binom{60}{x-y}}{\binom{80}{x}}$$

Bayes’ Theorem

Suppose you were playing in a private craps game and witnessed a friend of yours making 24 passes in a row. Someone now accuses him of cheating. You immediately come to his defense. You have known George all your life and in your opinion it is a million-to-1 against him being a cheater. Unfortunately there is a flaw in the argument. You see, the odds against 24 straight passes (honestly thrown) are about 10 million-to-1. Therefore, once George does perform this amazing feat, it is about 10 times more likely that he did it by cheating than by luck.

Problems like this are handled by Bayes' Theorem. They come up quite frequently in gambling (especially poker). Whenever something has occurred that could have two or more different "causes" behind it, Bayes' Theorem shows us how to calculate the probability for each cause. For instance, if you know that a man will raise with any pat hand in draw poker, you can use Bayes's Theorem to calculate the chances that your queen high straight is good if he does raise. This chapter shows you the specific technique (in a much simpler way than most math books) to do problems of this sort.

Bayes' Theorem is one of the most important and misunderstood concepts in probability theory. Its applications are very important to the serious gambler. I would advise you to read what follows slowly and carefully, and more than once in order to get a full understanding of the subject.

On the surface, Bayes' Theorem is easy enough. To illustrate this, let us look at the following problem: Suppose in a ten-horse race there are three fillies. Assume further that we know that the true probabilities of each of the fillies winning the race are 5 percent for filly A, 10 percent for filly B and 15 percent for filly C. If we were asked the chances of a filly winning the race, we could immediately say 30 percent. This, however, is not a Bayes' Theorem problem.

On the other hand, suppose we left the track before the race was run and were later told that a filly had won. What is the probability, based on this new information, that filly A won the race?

Many readers will intuitively realize that the answer is 5 out of 30 or 1 out of 6 or 1/6 or 5-to-1 against. Those readers have grasped the principle of Bayes' Theorem without ever having learned it.

By the way, some mathematicians get very upset when someone speaks of the probability of an event that has already happened, such as this race. I'm not one of them. To avoid this objection, I could have stated the problem differently: Suppose I wanted to make a proposition bet with someone that filly A would win the race if in fact, some filly does win? (If a male horse wins there is no bet.) For me to have a good bet, I need better than 5-to-1 odds.

I will now define Bayes' Theorem as simply as I can: if something can happen by any of a variety of different causes or means, the probability that it will happen in any particular way is the probability of the event happening by that means, as compared to the total probability of the event occurring at all. In other words, the answer will be a fraction in this form:

$$\frac{\textit{Probability of event occurring by desired means}}{\textit{Total probability of event occuring at all}}$$

The best way to get this concept across is by way of many examples. Even if you're having trouble grasping the above definition, you'll catch on by following these examples carefully.

One of the most important applications of Bayes' Theorem is in draw poker (especially jacks or better to open). Suppose a loose player, who will open on any openers, opens the pot. What is the chance he has two pair or better? To answer this question, we must make use of the following chart:

Approximate Chances of Being Dealt

Full house or better	.2%
Flush	.2%
Straight	.4%
Three-of-a-kind	2.0%
Two pair	5.0%
Pair of aces	3.2%
Pair of kings	3.2%
Pair of queens	3.2%
Pair of jacks	3.2%
Other pairs (3.2% each)	28.8%
No pair	50.6%
Total	100.0%

We can see immediately that the chances of a player being dealt two pair or better is 7.8 percent.

$$7.8 = .2 + .2 + .4 + 2 + 5$$

However, now that he has opened, that probability has changed. We must compare the 7.8 percent with the total probability of him being dealt openers. Which is about 20.6 percent.

Thus, the chances of a player having two pair or better, if you know he has at least two jacks, is about

$$\frac{7.8\%}{20.6\%}$$

or about 38 percent. (If instead, we know the player to be someone who needed at least two kings to open, the probability that he has at least two pair becomes

$$\frac{7.8\%}{14.2\%}$$

or about 55 percent.)

This simple example dramatically proves why it is so important to know both your math and your player when playing poker. It is usually correct to fold threes and deuces against a man who you know needs at least kings to open; but a player, who will open with jacks, usually should be raised with this same two pair. Your math does you little good if you don't know your player.

Let's leave poker for a moment. Assume the correct probability of the following mythical football teams winning the Super Bowl is:

N.Y. Gulls	8%
Chicago Bees	15%
Miami Birds	10%
Atlanta Caps	2%
L.A. Flames	20%
N.Y. Guards	20%
Detroit Wings	7%
Cleveland Bucs	18%
Total	100%

What are the chances that the Gulls will win the Super Bowl if a New York team wins it? This is like the horse race problem. The number on the top (numerator) is simply 8 percent. The number on the bottom is the total chances that a New York team wins, or 28 percent. The answer is

$$\frac{8\%}{28\%} = \frac{2}{7}$$

Before going on to more difficult problems, I'd like to point something out to nonmathematically inclined readers. Thus far, I have kept all probabilities in the form of percentages. By doing this, I've kept you from manipulating fractions within fractions.

When a fraction is composed of one percentage over another percentage, you evaluate it just as if the percentage signs aren't even there since they cancel out: 10%/30% is the same as 10/30 or 1/3. A fraction such as 11/43 divided by 33/43 is also simple.

As long as the denominators of both the small fractions are the same, they cancel out and can be ignored. Therefore, 11/43 divided by 33/43 reduces to 11/33 or 1/3. If the fractions are not in this form, such as 3/a divided by 3/5, you may have to review your seventh grade math or use a calculator.

Now for a hard one. Suppose you have five quarters in a piggy bank. One of these coins is a double-headed quarter. Someone now proceeds to reach into the piggy bank, grabs a coin at random and flips it three times. He gets three heads. What are the chances he grabbed the double headed coin?

This is a classic Bayes' Theorem problem. We must compare the chances of the event occurring in the specified way to the total chances that it occurs at all. The difficult part is defining what event we are talking about. In this case, the event in question is picking a coin and then getting three heads in a row.

What are the chances of picking the double-headed coin and then getting three heads? It is simply 1/5. That's because there's one chance in five of grabbing the bad coin and, if grabbed, three heads will always follow.

What are the chances of picking a good coin and then getting three heads? It is

$$\left(\frac{4}{5}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{4}{40} = \frac{1}{10}$$

(There is a 4/5 chance of picking a good coin, but this must be multiplied by the chances of getting three straight heads with it.) Thus, the total chances of picking a coin and then getting three straight heads is

$$\frac{1}{5} + \frac{1}{10} = \frac{2}{10} + \frac{1}{10} = \frac{3}{10}$$

The chances that we have picked the double-headed coin after we tossed three heads is

$$\frac{\left(\frac{1}{5}\right)}{\left(\frac{3}{10}\right)} = \frac{2}{3}$$

In other words, if you perform this experiment many times and count only those times when you get three heads in a row, you'll find that you've picked the double headed coin in 2 out of 3 cases.

What has happened here is that we've compared two unlikely events. It is rare that we will pick the double-headed coin at random, but even more rare that we will pick a normal coin and toss three straight heads.

Now suppose we had 100 coins and only one was double headed. If we select a coin randomly, toss it three times and each time get heads, what are the chances that the bad coin has been selected? It is

$$\frac{\left(\frac{1}{100}\right)}{\frac{1}{100} + \left[\left(\frac{99}{100}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\right]}$$

or about 7'/2 percent.

The first 1/100 represents the probability of picking the double-headed coin and then throwing three heads. (It does not merely represent the probability of selecting the irregular coin, although that is also 1/100 in this case.)

Next, the first probability is divided by the total probability of the event, (choosing a coin and flipping three heads), occurring at all.

That total probability includes: (1) the chances that the coin was double-headed thereby accounting for the three heads (1/100); plus (2) the chances that the odd coin was not selected (99/100), multiplied by the chances that three heads were obtained ('/2)('h)('/2).

It remains unlikely that we've picked the double-headed coin since there were so many others.

What if we had gotten seven heads in a row? The chances that we picked the bad one has now become

$$\frac{\left(\frac{1}{100}\right)}{\frac{1}{100}+\left[\left(\frac{99}{100}\right)\left(\frac{1}{2}\right)^{7}\right]} = \frac{128}{227}$$

or about 56.4%. Now we're favored to have picked the irregular coin, since it's only 99-to-I against picking the odd coin, and approximately 130-to-1 against picking a good coin, and then getting seven heads in a row with it.

To give a more practical example of how Bayes' Theorem compares two different ways of getting some unlikely results, let us look at medical tests. Suppose you went to your doctor to have a routine test for some rare disease, say tuberculosis. Let's suppose that the chances of an American, who has no symptoms, having TB are 1/1,000. Let's say the TB test the doctor gives you isn't flawless.

The test will never tell you that you are okay when you do have TB, but is one in a hundred tests will indicate that a healthy person has it. (Doctors call this a "false positive.")

With this information, what are the chances that you have TB if the test indicates you do? The event in question is that the test says you have TB. The total probability that this will happen is 1/100 (false positive) + 1 /1,000 (you really have it) or 1.1 percent.

The probability that you had it before the event took place was 1/1,000 or .1 percent. The probability that you have TB after the test revealed you do is

$$\frac{.1\%}{1.1\%} = \frac{1}{11}$$

You are still a 10-to-1 underdog against having TB even after the test says you do. That's because there's a better chance that the test is wrong than you have it.

This situation actually comes up frequently in medicine, especially concerning rare diseases. This is why doctors usually insist on more than one test to diagnose them.

Here's a similar problem from the realm of gambling. Imagine you somehow knew that one in 500 blackjack dealers in Las Vegas cheated. Let us further assume that in spite of the fact that you are a good counter, one particular dealer beat you out of 200 bets over a period of months. What is the probability that you were cheated?

To answer this question, we must remember that Thorp and others tell us that there is about a 1 percent chance of losing 200 bets when playing a typical good strategy. Thus, supposing that a cheating dealer also figures to beat you out of 200 bets, the chances that you were cheated are simply the chances we picked the cheat (1/500) over the total chances we lost 200 bets (1/500 + 1/100)' which is

$$\frac{\left(\frac{1}{500}\right)}{\frac{1}{500}+\frac{5}{500}} = \frac{1}{6}$$

If the assumptions are correct, it is still 5-to-1 against you playing a cheating dealer.

By this time, sharp readers may have noticed that when an event can happen in only one of two ways, you can simply form a ratio of the two probabilities to get the odds, (not the probability), that it happened one way versus another. For instance, if a man will raise with either a three card flush or a pair on the first three cards in seven-card stud, what are the chances that he has a flush?

Since a three flush occurs about 5.2 percent of the time in seven card stud, and a pair occurs about 17 percent of the time, it is simply 17-to-5.2 or about 3'/4-to-I against his having a three flush when he does raise. This is a quicker way (but no more accurate),

$$\left(\frac{499}{500}\right)\left(\frac{1}{100}\right)$$

than saying the probability is

$$\frac{5.2}{5.2+17}$$

that he has a three flush.

An Interesting Dice Proposition

The other day I heard a proposition that I had first heard about 15 years ago back in New York. Upon hearing it again I realized that it could form the basis for a chapter in this book.

First of all, this wasn't a typical sucker bet. Many knowledgeable people would go for it. It is far from obvious that you have the worst of it. Secondly, it is a great problem to illustrate most of the precepts that are encountered in probability problems. (Thirdly, of course, knowing this proposition may make you money if you can trap someone else. At least now you won't be trapped yourself.)

What are the chances of throwing both a six and an eight before throwing two sevens (using two dice, or course)? Many dice players know that throwing a six before a seven is a 6-to-5 underdog. This is because there are six ways to roll a seven (6-1,1-6, 5-2, 2-5, 4-3, 3-4), and only five ways to roll a six (5-1,1-5, 4-2, 2-4, 3-3). The same goes for rolling an eight before a seven. It is also a 6-to-5 underdog. Thus, it stands to reason that you are an underdog to roll both a six and an eight before you roll two sevens. However, it's not true! You will roll at least one six and eight before two sevens about 54'/2 percent of the time. Now, let's see why.

The basic reason the six-eight is favored is because they are big favorites to jump off into the lead. They are 10-6 favorites to do this since there are 10 ways to roll either a six or an eight but still only six ways to roll a seven. This does not by itself prove that the sixeight is the favorite, however, as we shall see later. We must calculate the answer to know for sure.

First, realize that there are three different ways that the six-eight can win. One way is for both a six and eight to come up before even one seven. A second way is for the six-eight to hit one of its number, then hit a seven, and then come back with the other number before the second seven appears. Finally, the six-eight may come from behind with both numbers after one seven has appeared first.

In other words, calling a good number a success (S) and a bad number (namely a seven) a failure (7) the three ways six-eight can win can be represented by: SS, S7S, and 7SS.

What we must do now is calculate the chances for each of these three ways of winning and then add them all up.

First, let's calculate the chances of two consecutive wins (SS). (They don't really have to be consecutive in that other numbers besides seven can show in between; these other numbers can be disregarded.) Since there are 10 ways to jump off the lead with a six or eight and only six ways to roll a seven, the first S is a 10-to-6 favorite. This must now be changed to a fraction, which is 10/16. Do you see why? The six-eight jumps off into the lead 10 out of 16 times.

Now what are the chances for a second success before a seven? It is no longer 10/16 since this second winning number must now be specifically a six or an eight, whatever the case may be.

In other words, if you start off rolling a six you must now roll an eight before two sevens come up. Any further sixes rolled don't count. The chances for this second S, whether it be a six or eight, are therefore simply six-to-five against or 5/11.

To figure out the probability of winning without even one seven (SS) is obtained by multiplying these two probabilities:

$$\left(\frac{10}{16}\right)\left(\frac{5}{11}\right) = \frac{50}{176}$$

or 28.409 percent.

Now let's calculate "S7S." This means jumping off into the lead, then rolling a seven before rolling the second winning number and then rolling this second number before the second seven. (Remember, the other numbers on the dice are disregarded when they are thrown.) Once again, the probability for the first "S" is 10/16.

Now we must multiply by the chances that a seven comes up before the second number. The probability of this is 6/11 since it is a 6-to-5 favorite. Finally, the chances that the third relevant number is the needed winning number rather than a seven are once again 5/11. We thus have:

$$\left(\frac{10}{16}\right)\left(\frac{6}{11}\right)\left(\frac{5}{11}\right) = \frac{300}{1936}$$

or 15.496 percent.

Finally, what are the chances of 7SS? The first seven is a 10-to-6 underdog to the other two numbers. Its chances are 6/16. The first S can be either number and is therefore a favorite over the second seven. This is 10/16. Finally, the second S must be specifically an eight or a six, whichever is needed. The chances of this are 5/11. We therefore have:

$$\left(\frac{6}{16}\right)\left(\frac{10}{16}\right)\left(\frac{5}{11}\right) = \frac{300}{2,816}$$

or 10.653 percent.

To find out the probability of rolling a six and an eight before two sevens, we simply add up the chances for each of the three ways of doing it:

$$28.409\% + 15.496\% + 10.653\% = 54.558\%$$

This is our answer. Translated back to odds makes it almost exactly a 6-to-5 favorite, in spite of the fact that both numbers were originally 6-to-5 underdogs by themselves to a seven.

(Incidentally, this proposition is mentioned in Scarnes' Complete Guide to Gambling. However, he calculates it incorrectly. His method makes the six-eight a 50-to-36 favorite or 58.140 percent. You can trap anyone who uses his figure by asking for 13-to-10 odds with the two sevens. They are only about a 12-to-10 underdog, but Scarnes' figure makes them about a 14-to-10 underdog.)

Just to double check the method I use, let's figure the chances for the two sevens coming up first. The answer had better be about 45.44 percent since this is 100 percent - 54.56 percent. Let's try it the long way. Two sevens can also win three different ways, namely: 77, 7S7, and S77, where S once again stands for six or eight. Calculating each one separately, we get:

$$77: \left(\frac{6}{16}\right)\left(\frac{6}{16}\right) = \frac{36}{256} \text{ or } 14.063\%$$

$$7S7: \left(\frac{6}{16}\right)\left(\frac{10}{16}\right)\left(\frac{6}{11}\right) = \frac{360}{2{,}816} \text{ or } 12.784\%$$

$$S77: \left(\frac{10}{16}\right)\left(\frac{6}{11}\right)\left(\frac{6}{11}\right) = \frac{360}{1{,}936} \text{ or } 18.595\%$$

$$14.063\% + 12.784\% + 18.595\% = 45.442\%$$

Voila! It works.

To make sure you understand exactly how this method works, let's try a second problem. What are the chances of rolling both a five and a nine before two sevens?

We know that there are four ways to roll a five or nine. Therefore, the five-nine has eight ways to jump into the lead compared to six ways for the seven. Does this make the five-nine the favorite in this proposition? Let's see. 5-9 (S) can win three ways: SS, S7S, and 7SS. Calculating each one separately, we get:

$$\left(\frac{8}{14}\right)\left(\frac{4}{10}\right) = \frac{32}{140}$$

or 22.857 percent.

Do you see how I got these figures? The first success had eight ways to hit versus six ways for the seven. The second success had only four ways to hit, however, while the seven still had six ways. I then changed these odds to fractions (8-to-6 = 8/14, 4-to-6 = 4/10). Now let's calculate the other two ways to win with the fivenine:

$$\left(\frac{8}{14}\right)\left(\frac{6}{10}\right)\left(\frac{4}{10}\right) = \frac{192}{1{,}400}$$

or 13.714 percent. Finally

$$\left(\frac{6}{14}\right)\left(\frac{8}{14}\right)\left(\frac{4}{10}\right) = \frac{192}{1{,}960}$$

or 9.796 percent.

Adding up gives us 46.367 percent.

$$46.367\% = 22.857\% + 13.714\% + 9.796\%$$

Even though the five-nine is a favorite to jump off into the lead it is still the underdog in this proposition. Thus you cannot simply depend on "common sense" to get you the right answer all the time. You've got to know your math.

Part Two
Poker

Poker Introduction

The vast majority of professional gamblers are probably poker players. This is because poker is the easiest game to "get the best of it." The reason for this is that you are playing against other individuals rather than the house and there is little or no cut or vigorish against you. Thus, all that is necessary is that you play somewhat above average in order to show a long run profit in the typical poker game.

Of course, if you actually want to make a living from poker it is a good idea to get better than just "above average." This is especially true if you intend to play in bigger games where the competition is tougher. It is even more important if you intend to play in casino games where the competition is tougher still and you have to pay to play on top of that.

When I write about poker it is with the intention of bringing my readers up to expert or even world class level. I usually do not write for beginners. This is especially true of most of the chapters in this section. Since they were originally articles in Gambling Times the chapters are, in many cases, about isolated aspects of advanced poker play.

Those of you who are interested in a more general work on poker should read my book The Theory Poker. (You might also want to read my other books Hold 'em Poker, Sklansky on Poker, and Poker, Gaming, and Life. In addition, you might want to read Hold 'em Poker For Advanced Players, which I co-authored with Mason Malmuth, Seven-Card Stud For Advanced Players, which I co-authored with Mason Malmuth and Ray Zee, and Gambling For a Living which I co-authored with Mason Malmuth). However, even The Theory of'Poker is aimed toward the non-beginner. If you are just starting to play you might want to read the Fundamentals of Poker by Mason Malmuth and Lynne Loomis.

The following chapters (with the exception of the hold 'em introduction) are for experienced players only. If you fit into this category I think you will find many concepts in these chapters that you have never thought of before, concepts that, - once fully understand - will make you a lot of money.

The Eight Mistakes in Poker

Not too long ago. I was trying to decide on what my next poker article should be about. Normally I would pick some sort of bad play that a novice was likely to make. I would dissect it and show why it was wrong. Then I would give the correct play (in most circumstances). Since I have seen hundreds of bad plays in poker I had plenty to choose from.

Then something hit me. With all the arguments and analysis as to how to play a poker hand, there are really only eight things you can do wrong in a flat limit game. After all, there are really very few options in poker. When it is up to you to bet, you can bet or check. When there is a bet to you, you can call, raise or fold. Thus any "bad play" that you might make during a poker hand must fall into one of eight categories.

Upon realizing this I had the subject for this chapter: Define and analyze the eight general mistakes in poker and give some specific examples where they might come up.

I consider the following chapter one of my most important and useful to the serious poker player and have thus placed it first in this section.

What makes one person a winning poker player and a second person a losing one'? Assuming they both play in equally tough games, the obvious answer is that the winning player plays better.

What does it mean to play better'? This question can be answered in many ways, but it all boils down to one thing: the better player makes the correct play more often in a particular situation. In other words, he makes fewer major mistakes. What kind of mistakes is a bad player apt to make? On first glance, it may seem that there are many bad plays that one can make in poker, the fact of the matter is that they all fall into one of eight categories.

When someone is playing limit poker, there are only five possible options. If he is first to act, he can check or bet. If someone else has bet, he can call, raise or fold. (How much to bet or raise is an additional decision in no-limit poker. Thus, no-limit adds the possibility of betting or raising the wrong amount. This mistake will not be covered in this chapter.) All limit poker plays can be reduced to one of these five decisions. Making the wrong decision is the only error that can be made. Every poker mistake can thus be placed in one of the following eight categories:

1. Checking when you should bet.

2. Betting when you should check.

3. Calling when you should fold.

4. Calling when you should raise.

5. Folding when you should call.

6. Folding when you should raise.

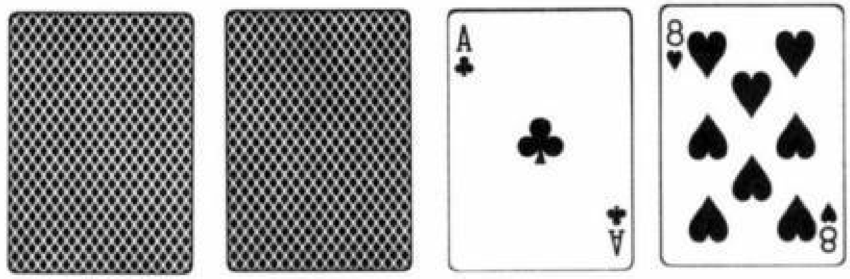
7. Raising when you should call.

8. Raising when you should fold.

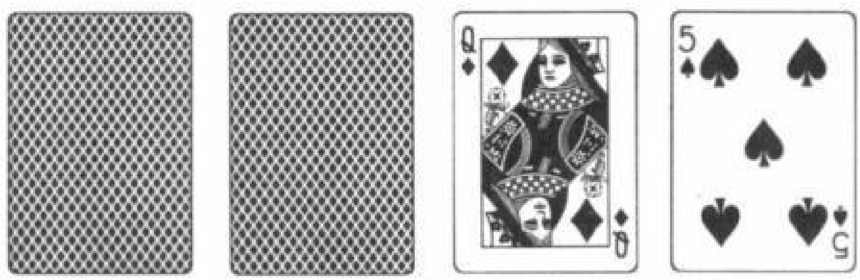
These mistakes are made by good and bad players. However, some of these errors are worse than others, and the bad player is more apt to make one of the critical mistakes. Let us examine each of the foregoing possible mistakes.

Mistake No. 1. Checking When You Should Bet. This is one of the most common and critical mistakes in poker. When there are more cards to come, it is usually correct to bet a mediocre hand if you are first to act. This is true even if your hand figures to be second best, and it is especially true if you have to call after you have checked and your opponent has bet. Betting accomplishes two things. Your opponent may fold if his hand isn't as good as you thought. Also, even if he doesn't fold, you have shown strength that may allow you to steal the pot on a later round.

A common example of this mistake is checking something like



into a



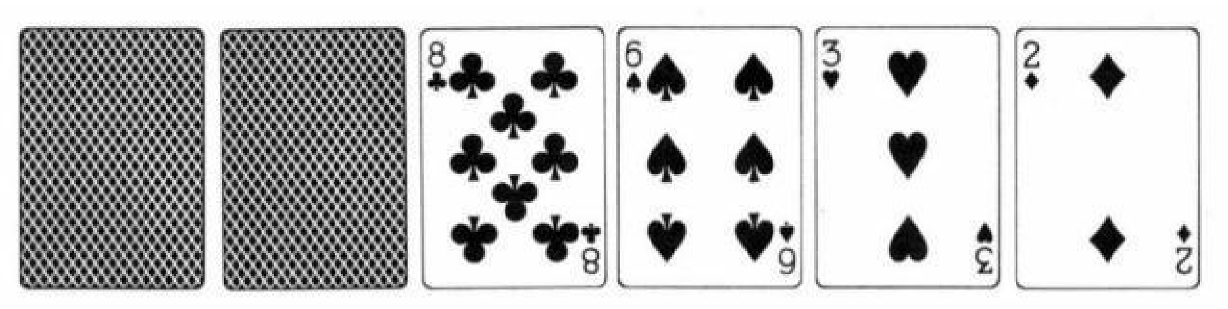
on fourth street in seven-card stud, even if you only have two eights and think he has two queens. Not betting is wrong.

Another situation in which players tend to check when they should bet is on the end with a hopeless hand. In other words, they don't try to bluff. It is not necessary that the bluff be favored to work in order for it to be worthwhile. You are usually getting odds of anywhere from 5-to-1 to 10-to-I on your bluff, so it only has to work occasionally to show a profit. You also gain advertising value when you are caught.

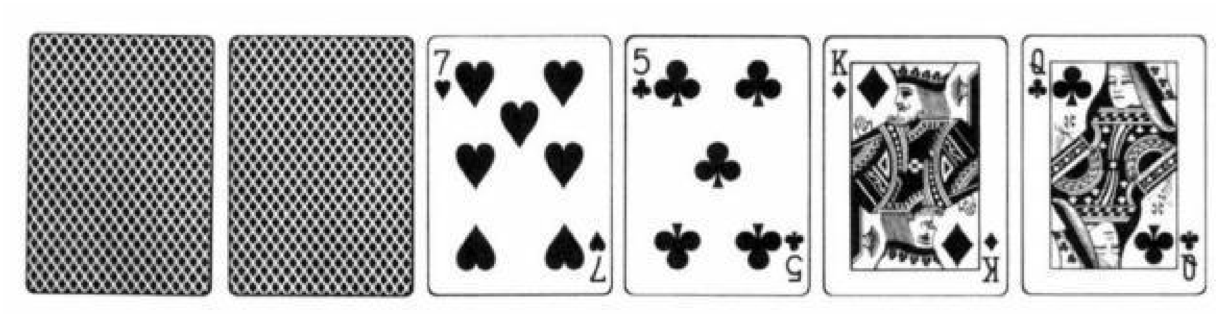
A third example of this error is checking on the end with a big hand in hopes of getting in a check raise. It is frequently better just to bet than to attempt to maneuver a check raise. This is even more true against tough players who are capable of folding when you check raise. Thus, for instance, you should usually bet a hidden full house into a possible flush in seven-card stud. (You may also win three bets by betting rather than checking.)

Mistake No. 2. Betting When You Should Check. This is not as common or critical an error as Mistake No. 1, but it can come up. Suckers frequently make this mistake when all the cards are out and they are betting a decent hand, but can only get called if they are beaten. Trying a hopeless bluff is another example of this mistake. Better players will sometimes make this error when they bet a fair hand on the end for value in situations when they are more likely to win an extra bet on the end by checking and calling than by betting (usually because checking gives them a chance to snap off a bluff).

Players will also occasionally bet instead of going for a check raise, the correct play in certain situations. For example, suppose you are playing seven-card lowball; you have an

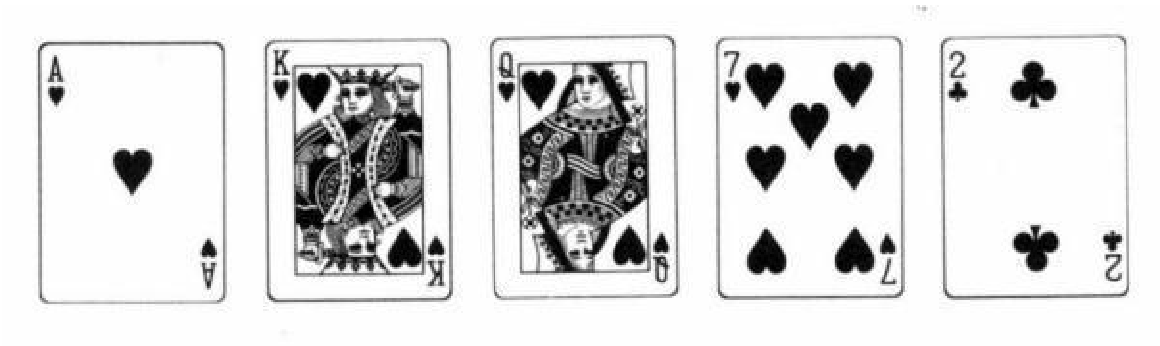


your opponent has



The right play is to check raise with a 6-low, but players don't. If you bet, your opponent will probably just call with a 7; if you check, he will probably bet.

Another instance when the opportunity to check raise is often missed is illustrated by the following example. Suppose you are playing draw poker, jacks or better to open. The player to your right opens, and you call with a



(flush draw). Two or more players come in after initially checking and draw one card. obviously to flushes or straights. The opener also takes one card and checks. You should check your flush if you make it because the worst that can happen is that you lose one bet from the opener in the event that everyone else misses his hand. If you check and someone else bets (either because he made his hand or because he is trying to steal the pot), you figure to win three bets instead of one. If you bet yourself, you figure to just get called by the come hand; the opener won't overcall. An added advantage to checking in this spot is that you save money if the opener fills up. If he check raises one of the other players, youu just fold.

Mistake No. 3. Calling When You Should Fold. This mistake is made by most players in the early rounds of betting because they don't understand the concept of effective odds as explained in my book The Theory ofPoker. Briefly stated, the concept suggests that it is not worth chasing with a hand even when it appears you are getting good odds because of the bets to follow.

Another situation in which players make this mistake is when they do not consider the fact that they may make the hand they are drawing to but still lose. However, this mistake is only critical in an early round of betting. With one card to come, mistakenly drawing to two pair against an obvious flush getting 7-to-1 odds really costs you a fraction of a bet and is no big deal. (Which is why the "percentage players" with no real feel for the game can't beat some of the better players who may make this error.)

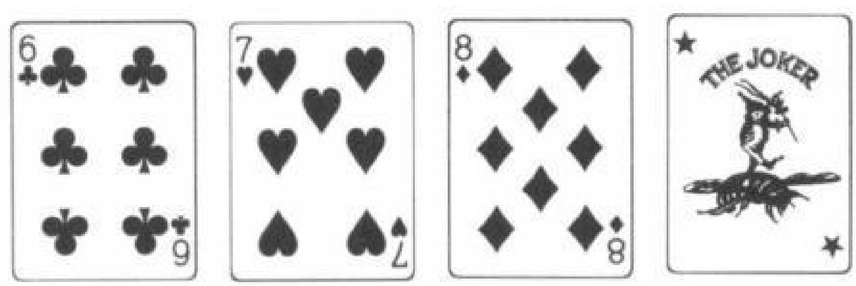
Another example of Mistake No. 3 is calling on the end "for the size of the pot' when it is impossible that you could have the best hand. I see this a lot.

Mistake No. 4. Calling When You Should Raise. This is probably the worst mistake and the most common. When there is a lot of money in the pot, you should usually try to narrow the field by raising when another player bets. Most players will do this if they figure to have a somewhat better hand than the bettor on their right. However, they will usually just call if they think their hand is much better in hopes of trapping players behind them. They are probably wrong. You really need a monster hand to slowplay in this situation. Even then there is no guarantee that the other players would not call a double bet.

In the situation where you probably have the second best hand, it is also worth raising in order to give yourself the best chance to win the pot. This play may also get you a free card on the next betting round, as you may now be checked to. Two obvious examples of this play are, in razz poker, to raise a possible four-card 7-low with a four-card 8-low or to raise an apparent two small pair with one pair of aces in seven-card stud when there are players behind you.

Calling instead of raising in head-up situations is a frequent mistake as well. Even when it appears you can't beat the bettor, a raise has three advantages: (1) Your opponent may fold instantly if he was bluffing or semi-bluffing; (2) Your raise might get you a free card on the next round, (3) Your raise might enable you to steal the pot on a subsequent round if you catch a good card.

There are so many examples of this mistake that I can't begin to enumerate them. Just to show, however, that it occurs in all games, I will give you an example from jacks or better draw poker, "Gardena style." Should a player open in late position, it is usually much better to raise than to call on his immediate left with a 16-way straight draw such as



By raising, you (a) may win it immediately. (b) may be able to steal it after the draw and (C) still may win even if he calls. You also figure to get a flush draw out behind you, which is obviously to your advantage.

All in all, remember that calling is frequently a bad play. Any time you start to call, consider raising instead. You will be surprised how often raising is the correct decision.

Mistake No. 5. Folding When You Should Call. This mistake is not too common, but it can be critical when it is made. It is more often made by good players than bad ones. When a good player makes it, it is usually when all the cards are out and he is trying to save the last bet, frequently seen in draw and lowball games. A player who will fold a pat nine-low 40 percent of the time when he is bet into may be costing himself a lot of money. Those players who play that way against me are going to have a lot of trouble. (See the chapter on game theory in The Theory Poker.) It is hard to save that last bet with a decent hand against someone who might take advantage of it. If you are wrong even once in 5 times, you can cost yourself a lot of money.

Mistake No. 6. Folding When You Should Raise. This doesn't come up too often. But one time it does arise is when you have the opportunity to try to bluff on the end against a tough player who may fold for one last bet. This is particularly effective if you check raise bluff a good player. However, your chances for success with this play must be higher than usual, as you are risking two bets to steal the pot.

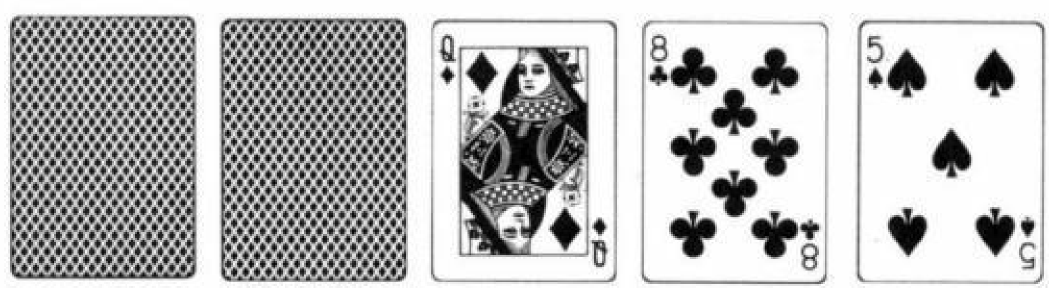
Another instance of this mistake can occur when you have a drawing hand with one card to come but aren't getting good enough odds to call. A semi-bluff raise is a better play than folding if there is some chance that you will win the pot right there. This play comes up most often in hold 'em when you have a straight draw or third pair with an ace kicker. In fact, even if you are getting good enough odds to call, it is usually better to raise.

Mistake No. 7. Raising When You Should Call. This is also a rare error. There is, however, one situation in which it is frequently made. It occurs in multi-way pots when all the cards are out. Let's say the player to your right bets. You are fairly sure you have him beaten and figure that he will call your raise. However, if he reraises. you are probably beaten. Meanwhile, there are one or more players to your left who will probably overcall with a worse hand than yours if you just call. Raising would normally be a bad play in this instance, as calling should win two bets with no risk.

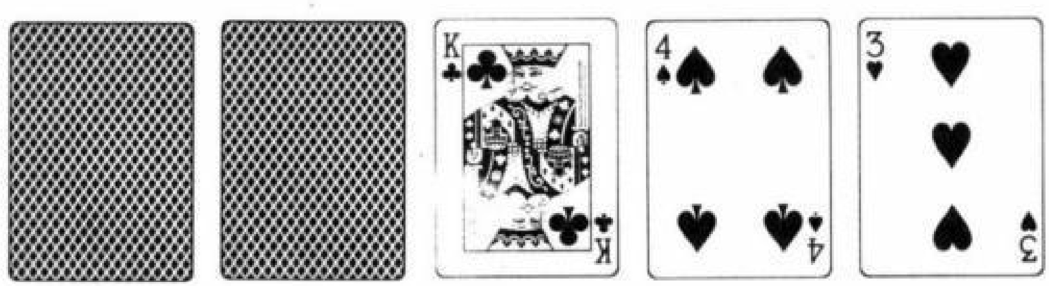
A less common example of this error occurs when you have a strong come hand in a multi-way pot. Most players will not raise a bettor to their right for fear of knocking other players out. With certain hands. it may not even be correct to raise a bettor to your left who gets a few callers. If he is a good player, he will reraise when you raise, which may be bad for you. However, this situation is not so clear cut. For instance, if you have a small straight-flush draw, you don't mind him knocking out other drawing hands.

Mistake No. 8. Raising When You Should Fold. This error is normally made only by expert players because they know there are many situations where the proper play is to raise or fold because calling is out of the question. However, sometimes fold, raise and call is a better ranking of potential plays than raise, fold, and call. In other words, even though raising is better than calling, folding is better still.

In a recent game, I raised when I should have folded. The game was seven-card stud. A player showing



bet and another with



raised. I re-raised with two small pair, but I should have folded, even though I thought I had the best hand. Raising was certainly better than calling, but folding was better still, as I had too many ways to lose. Many of the aggressive players in Vegas frequently make similar errors, but it is not usually that bad an error and can serve to enhance their "wild" image. It is seen most often on third street in seven-card stud. Many aggressive players will raise others behind them if no one has yet called and they have the high card showing. This is usually a bad play when they have two total blanks in the hole, but raising is certainly better than calling.

When determining how to play a hand, you can never be sure whether you may be making a mistake. When deciding whether to risk an error, you should realize that a mistake can do 1 of 2 things: cost you an extra bet or cost you the pot. Obviously, the second outcome is worse. That outcome can result from Mistakes No. 1. 4, 5, and 6. These are the costlier mistakes, the ones that the pros rarely make. When in doubt, it is much better to cost yourself a bet than-to cost yourself the pot. You cost yourself 1 or possibly 2 bets if you make Mistakes No. 2, 3. 7 or 8. (Notice, however, that Mistake No. 3 can cost you quite a few bets when committed early in a hand.) What this all comes down to is that the best way to play winning poker is to play relatively few hands, but play them aggressively.

Poker: Psychology or Percentages?

This chapter was originally written to clear up the argument over whether poker play requires mainly psychological or mathematical skills ... in fact it requires both simultaneously. One is of little use without the other. Furthermore. a third factor, logic, also must be used if one is to become a good poker player.

One of the most frequently discussed questions regarding expert poker play is whether the most important aspect is knowing how to play your cards" or knowing how to "play your players." Another way to put this is whether the most important factor is percentage or psychology.

Since I have often been accused of putting undue emphasis on the mathematics of poker while neglecting the "human" side of it. I have decided to set the record straight.

First of all. it is not correct to say that all poker decisions are based either on percentage or psychology. There is a third factor that is often times even more important than the other two. This factor is logic. How each of these factors is used in a poker game will be explained shortly.

The controversial question is what fraction of expert poker play can be attributed to each of these factors. In other words, is poker one-quarter percentages. one-half psychology and one quarter logic? If not, what is the breakdown?

Part of the answer depends on the game in question. Doyle Brunson has said poker is a "people game." It is important, however. to remember that Doyle's game is usually no-limit poker. Without question, no-limit is much more psychological than fixedlimit poker. This is because a player may very well fold a very good hand if the bet is high enough. Knowing whether you can bluff a certain player with a big bet is mainly a psychological question.

Similarly. when you do have a big hand, it takes psychological skill to know how much you can "sell it for" so that a lesser hand will call. This accounts for Doyle's feelings. But even no-imit play makes frequent use of both percentage and logic.

Before coming to any conclusion, let us take a closer look at these factors. Exactly how are each of these factors used in a poker game?

Percentages are most widely used when determining the probability that your hand will improve to the best hand. This probability is then compared to the pot odcls when deciding whether to call or fold with a hand that ought to improve to N\ in.

Like~\ise. you need to know percentages when you have a hand and your opponents are drawing to beat you. There are many situations in poker that are almost strictly percentage plays.

For instance, whether you should try for an inside straight (which you know will be good if you hit it) with one card to come in limit hold 'em poker, is almost purely mathematics. Even here, though, the play becomes a pure odds question only after you have psychologically evaluated the situation as one in which you could not get away with a bluff if you missed.

When playing seven-card stud, if you have, let's say, a pair of aces against what appears to be three different four flushes (in different suits) on fourth street, you are in trouble. You will get 3-to -1 odds on a bet. but you are more than a 3-to-1 underdog.

Knowing when to try to "knock players out" of a multi-way pot is another mathematical decision. Finally. your choice of starting hands in very easy, loose games, in which a bluff will rarely work, is mainly a card playing decision rather than a people playing decision.

Purely psychological decisions are most often encountered when you are considering a bluff, or a call with a hand that can only beat a bluff. In most cases, part of this decision is based on reading your opponent's hand and his actions in previous betting rounds. Even here logic as well as psychology play a part in your decision, since you must recreate the hand to make the best decision.

As already mentioned, psychological plays are at their purest in no-limit poker. There is no substitute for knowing your opponents in this game. The technical expert against a bunch of strangers will do far worse than a somewhat less knowledgeable player in this same game who knows all of the opponents.

Knowing your players is not just important for bluffing situations. For example, suppose a player, who you know to be tight, calls a raise with a deuce showing in seven-card stud.

Now. he pairs deuces in sight. Since you know he is a tight player, you can be fairly sure he does not have three deuces. Had he been a loose player, you couldn't be sure. This is just one of a multitude of poker situations in which your play is based mainly on psychology.

We now come to logic. It is this third rarely mentioned aspect of poker play that I think is the most important, especially in a limit game.

When I speak of logic, I am speaking of the formal type of reasoning that is characterized by frequent use of the words "if...then," "or," "and." etc.: "If I do that, then he will do this."

Good examples of almost purely logical plays occur in the last round of play. especially against a single opponent in a limit game. Suppose you are first to act: you have a good hand and suspect that your opponent has either a hand slightly worse than yours or that he has a complete bust. What should you do'?

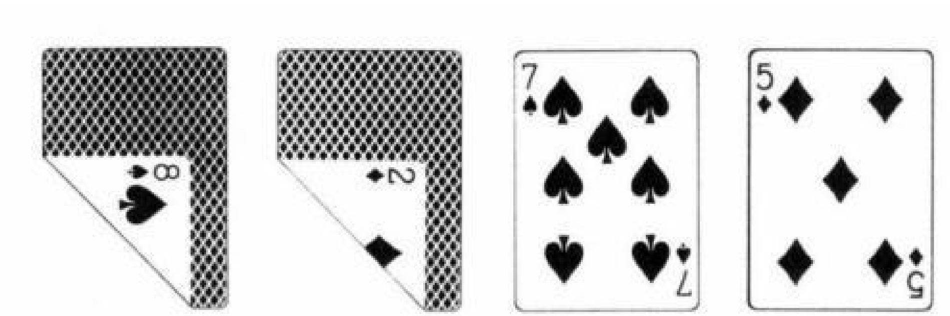
Logic says to check. If he has a decent hand, he will bet if you check. So, you don't lose any money by checking. You may, however, gain a bet by checking if this entices him to bluff with a busted hand that he would have folded had you bet.

This situation is mainly a logical one. But notice that there is also some psychology to it since you made a "read" on his possible hands as well as what he figures to do with them.

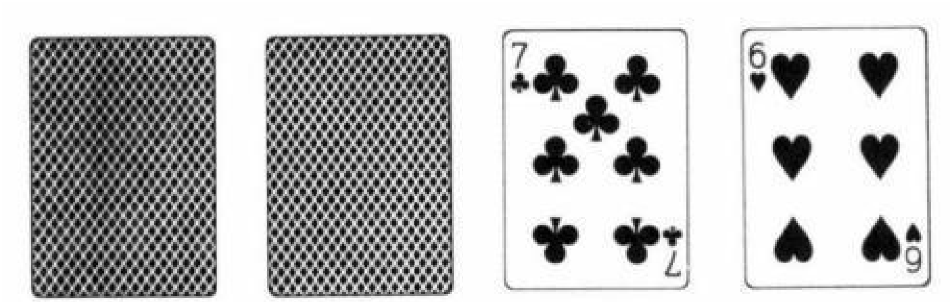
In this same instance, if it appears that his hand is quite a bit worse than yours, it may be right to bet, relinquishing your chances of snapping off a bluff. The reasoning is that he will probably call your bet but show his hand down if you check.

An example of a logical play in an earlier round is as follows: You are first to act in a head-up situation. You have what appears to be the slightly better hand. In reality, though, you suspect that your hand is somewhat worse. Still, if you check, you must call if he bets. Furthermore, if you check and he checks behind you, it probably means that your hand was in fact the best hand and now you have given the worse hand a free card! Therefore, you must bet in this situation especially since you needn't fear a raise.

A clear example of this case happens frequently in seven-card lowball. You have

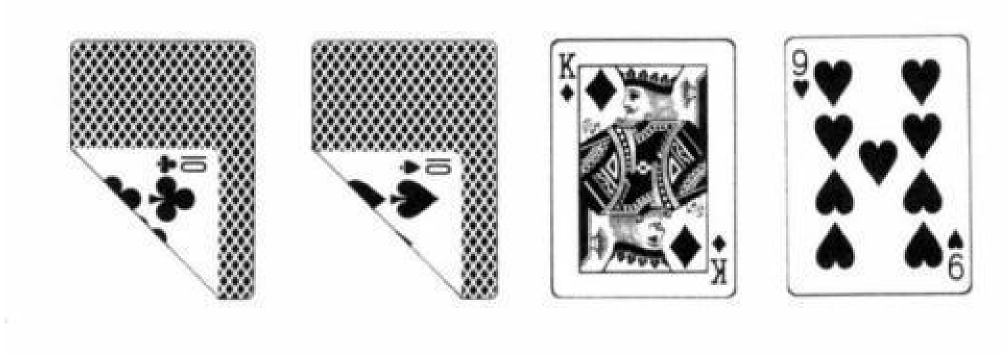


and your opponent has

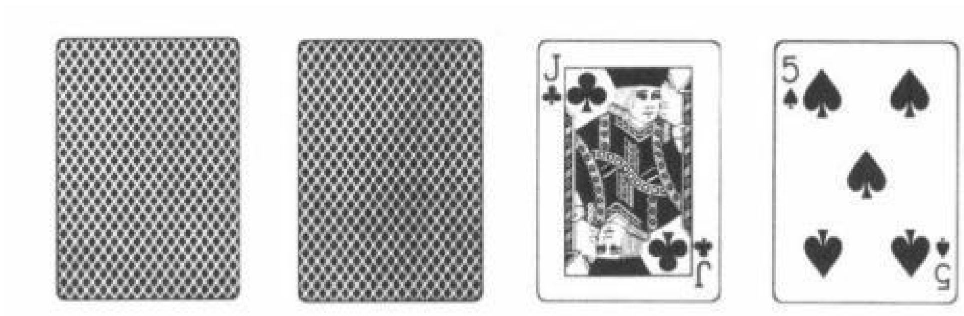


You must bet here.

An analogous situation in seven-card stud might be when you have



against an opponent's



Since your opponent is worried about two kings, you should bet even if you think there is a good chance he has two jacks.

Again, I could have given many more examples of poker plays based mainly on logic.

Lets get back to the original question. Is poker mainly a card game or a people game? Is it most important to know your psychology, your percentages, or your logic? Are the most frequent situations that arise in a game mathematical, logical or psychological?

The answer is none of them! Those situations that I mentioned earlier in this article are all quite rare. They are rare in that very few poker situations can be played well based solely on one of these three disciplines.

In fact, almost all poker situations require all three simultaneously. This is your answer. The true expert does not just play his player or play his cards; both factors are closely intertwined in nearly all poker situations.

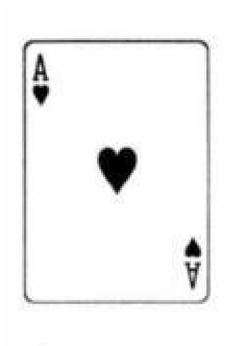
Even those situations mentioned earlier were not purely mathematical, psychological or logical. Typical poker situations are even less distinct in any one category. Let us look at a couple of examples.

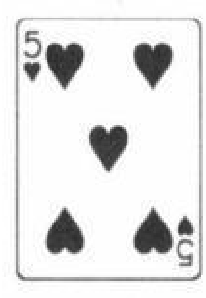
One of the clearest examples where you need to know both your math and your player, occurs in draw poker, jacks or better.

What do you do with two small pair if the man to your right, in fourth position, opens the pot'? You cannot answer this question accurately unless you know the minimum hand that he would open with in this position. You must know your player.

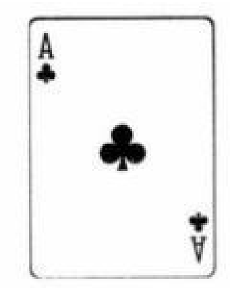
However, it does you no good to know his minimum opening hand if you cannot derive the mathematically correct play. Both psychology and mathematics alone are impotent in this situation. Only in combination are they most effective.

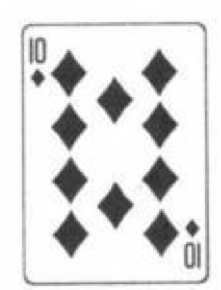
Here is a more complicated example from limit hold 'em. No one raised before the flop. The flop is





You have



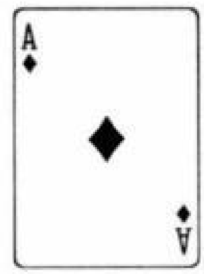


You bet. The last man raises and everyone else folds.

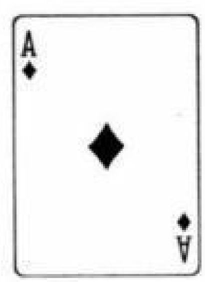
What to do from this point on is a delicate, finely tuned question of logic. mathematics and psychology. What you must do is decide the possible hands that this particular player would raise with and determine how he would play each of these possible hands on subsequent betting rounds.

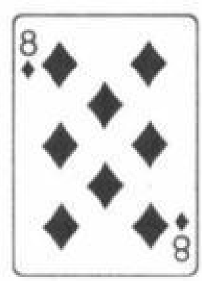
This involves knowing your player. Then, you must use both logic and percentages to ascertain your best course of action. For example, let's say that you think there is a good chance that he is raising on a four flush.

This would be a reasonable assumption against many players because you can almost rule out a hand like



since he would have raised before the flop. Likewise you can probably rule out a hand like three fives since he would probably just call on the flop to "suck other players in." Therefore, the only other reasonable hand, besides a- four flush, would be something like





Putting him on a four flush does not by itself automatically indicate the best line of play. Knowing how he intends to play this four flush on later rounds is very important, if you want to maximize your profits and minimize your losses.

For instance, if you think he will keep betting if you show no strength, the correct play is to check and call on every round. That is, as long as no more hearts fall. If, however, you expect him to check on the next round if he misses, you should reraise on the flop and bet on the next round if no hearts fall. (You would check on the end, of course, to try to snap off a bluff.) If you think he will bet on the fourth street, but give it up on the end if he misses, the proper play would be to call on the flop and check raise on fourth street if no heart comes. Once again, you would check on the end.

In this same situation, suppose the raiser was quite likely to have you beaten on the flop but also could possibly have a four flush. Your options would range from folding to calling on the flop and then betting out on fourth street. (You might fold because of the combined chances that are beaten plus the chances that he has a four flush and goes on to make it.) And there are other possibilities in between. As you can see, logic, math and psychology are all inextricably intertwined when making your decision.

I hope these examples have shed some light on the math vs. psychology question. The great poker players not only are very adept at figuring out their opponents' hands but they also know what to do with this information. Neither talent by itself well get you to the top.

How To Play in a Loose Game

Most of my books and articles on poker have been directed toward games with relatively tough opponents. Those readers who play in the expert manner that I advise in these chapters should show a profit in almost all games including the easier, looser games. However. strangely enough, the "expert" style of play will not show as much profit against bad players as a somewhat modified style will. This chapter explains how to play this modified style when the game dictates it.

Most of my writings on poker assume that the game in question is fairly tough and relatively tight. This is not usually the real world. Ninety percent of poker games are very loose and easy to beat.

Almost all private games are in this category, as are the small games in the casinos of Nevada, California and the Northwest. Unfortunately, those who play in these games and have been following the advice in my books, may not be extracting the maximum profit.

I have concentrated my writings on' the rare, tough games for three reasons: (1) if you can beat tough games, you are a favorite in easier games, even if you're not playing optimally; (2) tougher games are much more interesting to analyze and to write about; (3) I wanted to develop concepts that were unknown even to my fellow professional gamblers.

The basic characteristic of the typical private game or small casino game is that the players play much too loose. They play too many hands and go too far with them. Of the eight possible mistakes that a player can make in poker they basically make one. They call when they should fold. (See The Eight Mistakes in Poker" which appeared earlier in this section.)

If you play correctly in a game like this, you have the most profitable poker situation there is. The key is to play in a way that will take the greatest advantage of the above.

One obvious difference between playing in this game, as compared to one which is more tough, is that you should (almost) never bluff.' In The Theory of Poker I advocate bluffing as an important Nyeapon, but it isn't, in this kind of game. The general principle still holds that you should try a bluff on the end if you think that your chances of getting away with it are favorable compared to your pot odds. However, it is unlikely that a bluff would be indicated by this criteria. Your opponents don't fold.

The only possible exceptions would be where you think your opponent might also have a busted hand, or if others are starting to realize that you never bluff. Never even consider bluffing more than one or two players on the end. It is hopeless.

Semi-bluffs, with more cards to come are also much less useful in very loose games. Semi-bluffs are defined in all my books. They tend to span the hazy region between bluffs and value bets.

The most obvious case of semi-bluffing is ante-stealing with a mediocre hand that would be in trouble if called. Don't try it here. It is a very profitable play in a tight game where you may steal the ante, but still may win even if you are called. In a loose game, however, you don't have those two different ways of winning. If the hand isn't worth playing solely on its own merit, you can't add in the extra chance of stealing the ante since that chance is almost nonexistent.

Semi-bluffs later on in the hand are likewise not usually correct. This is especially true if you are last to act and your opponents have checked. Just give yourself a free card if you are fairly sure your hand is not legitimately worth a bet. You don't have that extra chance of winning by making them fold.

If you're first to act, semi-bluffs with fairly good hands may still be correct, because you actually may have the best hand. If you don't, you will have to call a bet after you check. Unfortunately, this play does not gain you the deception it would in a tougher game. Thus, a semi-bluff in early position is indicated only if your hand has quite a few "outs."

Keep in mind, however. that betting a drawing hand can be profitable in any position if you are in a multi-way pot (as you figure to be in loose games). Even though you don't have the best hand you are not semi-bluffing. You are betting to win money since you are getting 3-or 4-toI odds on your bet.

Straight and flush draws should bet and even raise in loose seven-card stud and hold 'em games if there is more than one card to come, unless a flush or straight is in danger of being beaten.

(Ironically, drawing hands also can be bet in tight games since there is a chance that the bet will win the pot right there. The drawing hand should not be bet in moderately tight games where it figures to get exactly one call.)

Not bluffing is a special case of the more general concept that it is incorrect to play deceptively or to disguise your hand in loose games. Bluffing is one form of deception that is not necessary. Slow-playing is the other form. It is almost never right to slow play a big hand in a loose game. Bet or raise whenever you get the chance.

The major exception to this rule would only occur if the man to your right bets and you have a monster hand. In this case, you can just call if there are many players behind you. Don't raise them out. Even bad players may very well fold if it is a "cold raise" to them.

However. you need a cinch hand to make it right to slowplay. A raise is never very bad, even in this situation. In fact, the only time to correctly slow-play in this game without a cinch is when you can set up a check-raise which will drive players out. This would occur when you have a good hand against many opponents.

Rather than simply betting, you should consider checking if you think one of the players to your right will bet. If he does, your check-raise will likely drive other players out. (There are, however, many private games where check-raising is frowned upon, although allowed. Too much check-raising, especially in a head-up situation, may make you unwelcome. In games like that, I would confine my check-raising to critical situations such as the one just described.) If you think a player on your left will bet if you check, you can consider checking an extremely strong hand in hopes of raising the bettor and his callers. If your hand is only good but not great, you're better off betting in hopes that the player on your left will raise you, thereby driving out the competition.

This brings us to another aspect of playing in loose games with multi-was pots. It is crucial to thin out competition whenever you have a decent hand.

Unless you are playing against absolute maniacs, a raise probably will make players fold if they have to call a double bet "cold." Thus, you frequently should raise a bettor on your right to knock out the other players even you are somewhat that you have the bettor beat. (I am assuming a limit game rather than no-limit or pot-limit. I also assume that there are more cards to come.)

While this play may cost you an extra bet or even two bets if the original bettor reraises. it is well worth the risk. Your raise easily may make the difference between winning or losing the pot.

As an example, suppose it is sixth street in seven-card stud. The player on your right bets with what you know is at least a pair of aces and could very well be aces-up. You have tens-up. The player to your left figures to have two jacks.

You must raise even though you are in jeopardy. Your extra bet has prevented the possibility that the jacks will call and make jacks-up while the other opponent only had two aces and failed to improve. Plays like this are extremely critical in multi-way pots.

They may even be right with "come" hands. For instance, there is nothing wrong with raising w ith a high four-flush with two cards to come in order to drive other players out. This play may allow you to win the pot by making a high pair. Had you let in all your opponents, this possibility is almost nil. Opportunities to make this play occur all the time in these types of games.

Remember, it isn't necessary that you be favored to have the original bettor beat, especially with more cards to come. (The reason you gain so much when you thin out the field is explained in more detail in The Theory of

This play may even come up when all the cards are out. If the bettor is prone to bluff, a raise may be indicated once again even if you are a small underdog to have him beat. This would occur if you are afraid an overcall will beat sou. Your risk of one extra bet may allow you to win a large pot with the second best hand if you make a better hand fold behind you.

Let us consider the play of starting hands. How tight should you play at the beginning of a hand in a loose easy game? Two other factors must be considered: (1) The size of the ante in comparison to the betting limits: (2) The site of the opening bet.

The larger the ante, the more starting hands you must play. This is just common sense and applies to any type of game. If you play very tight when the ante is high. anything you win when you play a pot will be lost back (plus more) in antes! But it is not enough to play loose in a large ante game with weak players. That is what they are doing. You must strive to thin out the field when you have a decent hand.

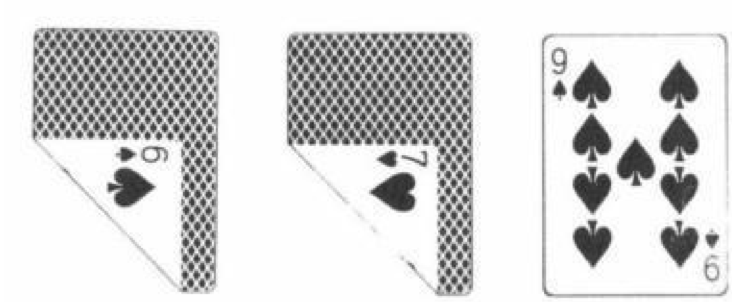
As mentioned before. you should almost never slow-play, especially at the beginning of a hand where you only have two or three cards. Since the ante forces you to play loose. you must also play some hands that are not that decent to start with. The key is which ones. It is important, particularly in games with multi-way pots, to pick hands that have chances of improving to a big hand.

Three card flushes to start with in seven-card stud is an obvious example. The higher the three-flush the better. Usually. you should take a second card with a good three-flush. even if you bust out on fourth street. especially in a good loose game.

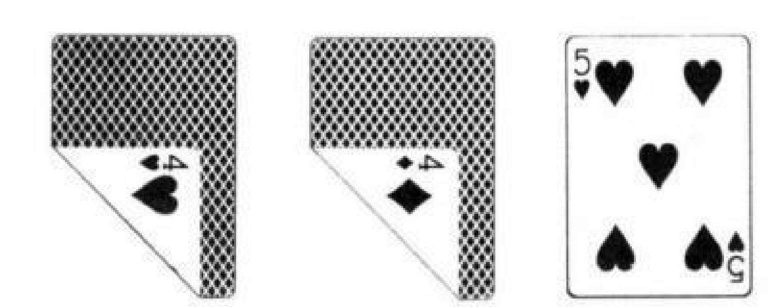
High three-card straights are also okay, though not as good as a three flush. Broken three-card straights have a lot more value if they Include a two-flush.

All these examples are from seven-card stud. The readers should think of analogous examples from other games.

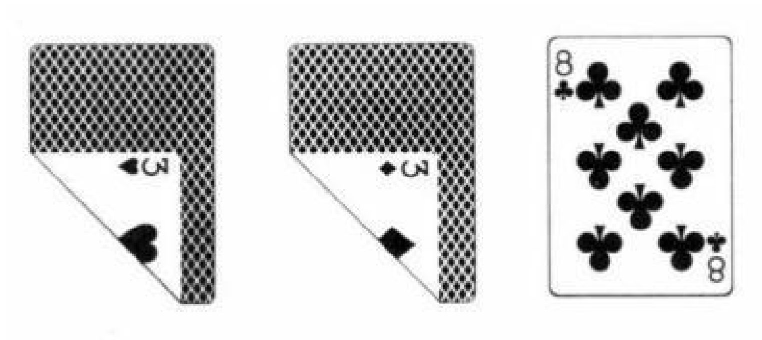
A hand like

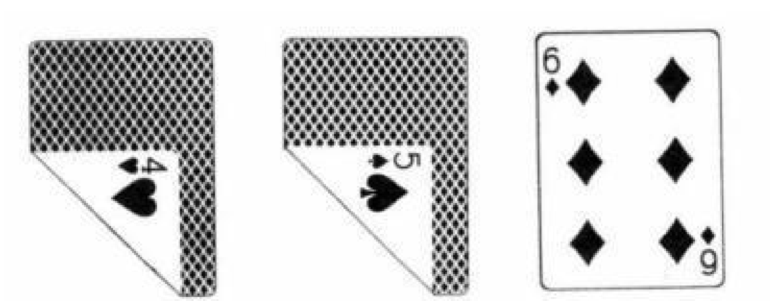


can definitely be played if your cards are live. The same is true for



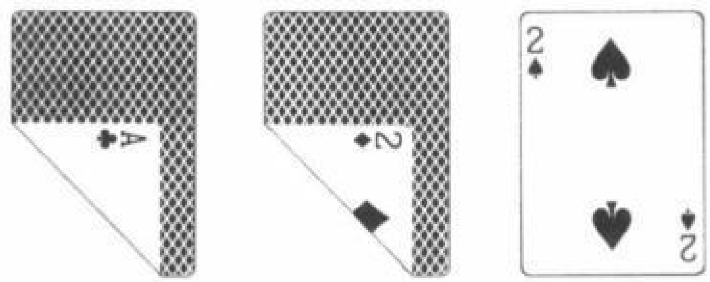
The kind of hands that should be thrown away are





or even

I would rather have



than any of these hands.

The problem is that these other hands not only need to improve in order to win they also have good chances of improving and still losing. Thus_ while a large ante forces you to take chances to make a big hand; it also forces you to throw away mediocre hands with little chance to improve to a hand that will beat all your opponents in a multi-way pot.

The initial bet is also a major consideration in determining how loose to start your play. Sometimes the opening bring-in bet is quite small as in the $ I -to-$3 game in Vegas where the first bet is 50 cents. You should play loose,fi this first bet only. even when the ante is relatively small. The kinds of hands you play in this situation would be much the same as the hands described in the previous paragraph.

Should you encounter a loose game where the ante is relatively small, and the first bet is relatively high, you should have the nuts! Just play tight. solid poker and watch the money roll in. It is interesting, however, that games like this seem to strip the expert poker player of many of his weapons.

Since psychological plays and the ability to read hands have very little value in these games, many superstars get disgusted leaving the easy pickings to the unimaginative. solid, percentage players. These percentage players are frequently bigger favorites in a very easy game than a world class player who can't "gear down." Keep this in mind when you play in such a game.

The one psychological strategy that can be used is noticing who the bluffers are and who the callers are. In a loose game, the bluffers are the suckers since they always get called.

It is important to change your strategy against bluffers. Let them bet your hand in spots where it is appropriate. Try to get it head-up with them if at all possible. You can even try to bluff them on the end when you miss your hand since they are likely to have very little. Against the callers. you play straightforwardly.

Finally. eye come to what may be the most important aspect of all: How to make sure you will be invited back. (Casino games do not have this problem associated with them. In casinos, the extra factor is the rake. If the rake is too high, you don't play unless the game is really great. You also wouldn't play marginal hands since your pot odds are reduced.)

Many private games do not like tight players. Thus, you play as loose asyou can get away with and avoid check-raising as much as possible.

Give the illusion of gambling. Don't be too serious. It may even be night to purposely lose back some of your winnings at the end of the night to leave a good taste in their mouths. If this is a steady game with the same players, try to see that no one gets hurt badly in any one session.

If you follow the advice in this chapter and stick to your steady, private games. you will never have to become a superstar. You will probably win more than they do while they buck their heads against each other.

Why You Lose in a Good Game

This essay covers a lot of the same material as the previous one but also mentions some new ideas that are very important.

There is a common misconception, even among professional poker players, that the smaller, wilder games are virtually unbeatable. "So many players stay in the pot," they say, "that your hand almost never holds up. Bluffing is, of course, unthinkable. It comes down to a card-holding contest."

I am constantly amazed when I hear these views expressed. After all, how is it possible that expert players would not be bigger favorites against rank amateurs than they are against their much tougher day-to-day opponents in bigger games? Nonetheless, this opinion has been expressed to me by many successful professional players.

The fact of the matter is that the detractors of super games are wrong. Expert players should do far better proportionally in a smaller, easier game (discounting the rake) than they would in a larger, tougher game. But the truth is that they don't.

Why is that? It turns out that the reasons are fairly complex. In this chapter I will explain those reasons as well as show a remedy.

There is no question that if a pro plays his normal strategy in a wild game he is costing himself money. This is obviously true if he stubbornly sticks to his normally profitable bluffing and semi-bluffing strategy which now has little hope of success. However, even if the pro eliminates all types of bluffs and sticks to the straightforward strategy that beats other games, he is still disappointed.

What is the problem'? The answer lies in the "Fundamental Theorem of Poker' (a term I coined and explain fully in my book The Theory of Poker). It has to do with the fact that if you have a good-but-not-great-hand, you might not want certain lesser hands in against you.

For instance, if you open with two small pair (say eights and fours) in draw poker (jacks-or-better to open), you don't want a pair of tens calling you before the draw if the ante is high. You will beat him about three out of four times, but this is not as good as a sure win of the ante. His call has cost you money. You do better in the long run if he folds.

This example illustrates the paradox inherent in good games. A tough player would throw away the two tens. The `'live one" doesn't and this costs you money. Does this mean that the live one is somehow playing correctly and the tough player isn't? Not at all. However, it is true for specifically that situation.

Good players know that you shouldn't call an opener who has at least jacks with a pair of tens, even with a high ante. What they might not know is that this strategy is clearly correct only because of the decent chance that the opener may already have jacks-up or better! It is only this possibility that makes the two tens a fold.

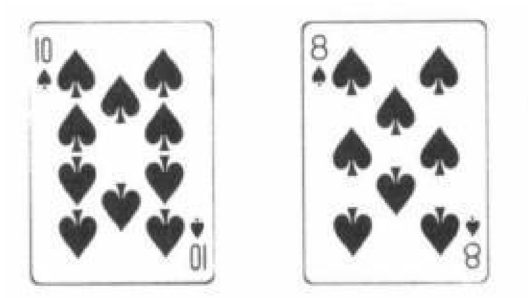
To put this another way. you. as a good player, are happy to be against players who will call you with two tens, but only because they will do it when you have jacks-up or better! In these cases their call makes you lots of money instead of costing you a little, but only in these cases.

This situation comes up all the time in lowball draw. You are playing against players who frequently call a bettor and then draw two cards. This is a bad play but only because the bettor might be pat. When the bettor takes one card, calling to take two is usually a slightly profitable play when you include the pot odds due to the antes. However, the play is disastrous when the bettor has a pat hand.

Thus you, as a good player. must understand that you will make far more money than you should on your pat hands (especially the good ones) in a wild game, but will do somewhat worse than normal with your routine one-card draws. In fact, your mediocre one-card draws should no longer be played in certain cases. More on this shortly.

The concept illustrated here is very important and doesn't apply only to wild games. It is the fact that many mediocre hands (in all forms of poker) should be thrown away (with more cards to come) only because of the possibility that your opponent may have a "monster.'"

Ifvou knew that his hand was only a little better than 'ours, you should usually play. In hold 'em games you throw away



for a cold raise because the raiser may have an over pair. Had you known that it was only two over cards you should probably have called. Conversely, if you are the raiser you only really gain from bad players when they call you and you have the over pair (before the flop).

Similar examples could be given for all forms of poker, but the general idea is this: Good players throw away weak hands against apparently good hands when they have more cards to come. Even good players. however, may not fully realize that their folds are only correct because of the possibility that their opponents hand is not just good but great. They would then be "drawing dead" or almost dead. If the possibility did not exist they would usually gain a little by calling.

Bad players do not understand nor practice this concept. How then should this affect your play against these players? First of all, it should change your psychological viewpoint. You must understand that your profits against these players come from far bigger than normal wins against them when you have them almost drawing dead. However, realize that your routine hands will do slightly worse. Your wins will not be as steady but will tend to come in bursts when you win inordinately big pots with your big hands.

Secondly, you must change your strategy a little. Opening two small pair in a position where it normally shows a slight profit will now become a loser in a game where players call with "shorts." Similarly, a fair one-card draw in lowball should no longer be opened in games where players will "incorrectly" play two-card draws behind you. (I knoxN this seems hard to believe, but it's true. You had a borderline hand and now some of your ante-stealing equity has been taken away.) High cards (unsuited) and small pairs in seven-card stud similarly go down in value. Ace jack becomes garbage in hold 'em.

On the other side of the coin, you yourself can loosen up on some hands - those that can win the highly probable big pot if you hit. Any ace suited is a good hand in a wild hold 'em game as is any pair in your hand. In lowball draw you should frequently play smooth two-card draws yourself, especially with the joker and especially with position. But you should stay away from eights.

Checking In the Dark

Frankly, this next chapter is not really too important as far as your poker income is concerned. Still, I wrote it just to show how nothing can be taken for granted in poker. (In fact, even a relatively minor point needs quite a few pages to analyze it properly.)

Most professional poker players believe it is never correct to check before looking at their last card. This chapter shows that situations may arise where this is not true. It also shows how to play against an opponent who "checks in the dark."

Most poker players have encountered an opponent who, before looking at his last card or cards, announced that he was checking "m the dark." If you're a serious player, this probably delighted you. By checking without looking, your opponent had provided you with extra information.

You knew that whatever he was checking was totally random, and you could choose your best strategy based on that knowledge. Most likely you were right to think that your opponent's play had increased your advantage.

There. however, are exceptions to this rule. Checking in the dark may be right in certain situations, especially against weaker players. In this chapter, I will show all the ramifications of this play when it is correct and when it is not. I'll also explain the best counterstrategy to employ when it's used against you.

Let me first define terms. When a player checks in the dark," he has not looked at his card or cards and has made this known to his opponents. If you check after pretending to look (because you know you're going to check no matter what you catch), you are checking in the dark but your opponents know it, so I don't consider that situation.

Likewise, those players who sneak a quick peek, then announce they are betting or checking in the dark, are making a play of questionable ethics, again not the subject of this chapter.

When is it right to check in the dark? Some experts say never. They reason that it never can be to your advantage to give an opponent this extra information - that your last card is completely random. Also, if there are cards you could catch which would make it correct to bet, checking in the dark erases that opportunity.

As logical as this sounds. these experts may be wrong. (They're right in suggesting that you should never check without looking if you are last to act. unless you know that you would check regardless of what card you caught.)

So, why isn't the "expert" advice to never check in the dark correct'? Well, it would be correct if you were pitied against perfectly programmed computers or errorless opponents. Fortunately. in the real world of poker, you're matched against human beings who may be logically or psychologically misled by your play.

Although checking in the dark will not cause a perfect opponent to err, it can frequently be used to confuse a human foe. Whenever your checking in the dark causes an opponent to make what figures to be the incorrect play, you'll eventually show a profit from it.

Opportunities to benefit by checking without looking arise mostly against weaker players. (although not against very bad players who pay no attention to what you do and act purely on the strength of their own hands. Against this latter type, you always should look, provided there are cards you could catch which would make a bet appropriate.)

However. there is one situation against a very tough player where you can check without looking. This occurs when it is completely obvious to both of you that you will check no matter what you catch.

Here's a clear-cut example that arises in draw poker. You open with a pair of aces and are raised. You call the raise and draw 3 cards. The raiser stands pat. If you are first to act, you will obviously check regardless of what you catch. (I'm assuming it's legal to check and then raise.)

In this situation, it will cost you nothing to check in the dark. Neither will it help you, except for one thing; you may have a tell. By not looking you can't possibly give away your hand. These automatic checking situations can occur in games such as sevencard stud, but there they are less clear cut.

The point is: If it is obvious that you're planning to check no matter what you catch, you can check in the dark against a tough opponent, especially if you fear you have a tell. Even in this case it may not be right. In my experience, I have noticed that even a tough player is psychologically more apt to call a check-raise when the raiser has looked at his last card before checking.

Let's look at some situations where checking in the dark is clearly the right play. This happens when your play will tend to make your opponent respond incorrectly. It will come up most often against a player to whom your checking in the dark will have a specific psychological impact.

Take this example from draw poker. You open with two pair. Someone calls who you are quite sure is "on the come" (drawing one card in the hope of making a straight or flush). Against many players, you should now check in the dark, because they will be afraid to bluff if they miss. They fear that if they do bet, there are two ways they could lose. You could have unknowingly made a full house and you could call even if you didn't. Had you looked and then checked, however, they would have occasionally bluffed - not so often that you could happily call them, but just often enough to force you to decide whether to make a "crying call" or to reluctantly fold what might be the better hand.

The only disadvantage to checking in the dark in this spot is that you may have made a full house and, at this point, your opponent will no longer bluff at it. This disadvantage doesn't come close to canceling out the edge you gain by stopping his bluffs when you don't fill up. Now you can simply fold if he bets without losing any more equity. (A fuller treatment of this subject can be found in The Theory of-Poker.)

The situation just described is the most common in which checking in the dark is correct. The usual purpose of this play is to cause an opponent to show down a hand that he might otherwise have bet. For instance, suppose you're playing seven-card stud. You have two red queens hidden in the hole with two fives, a seven, and a four, (including three spades) showing. Your opponent shows an ace, a jack, and a pair of threes. (See Figure A.)

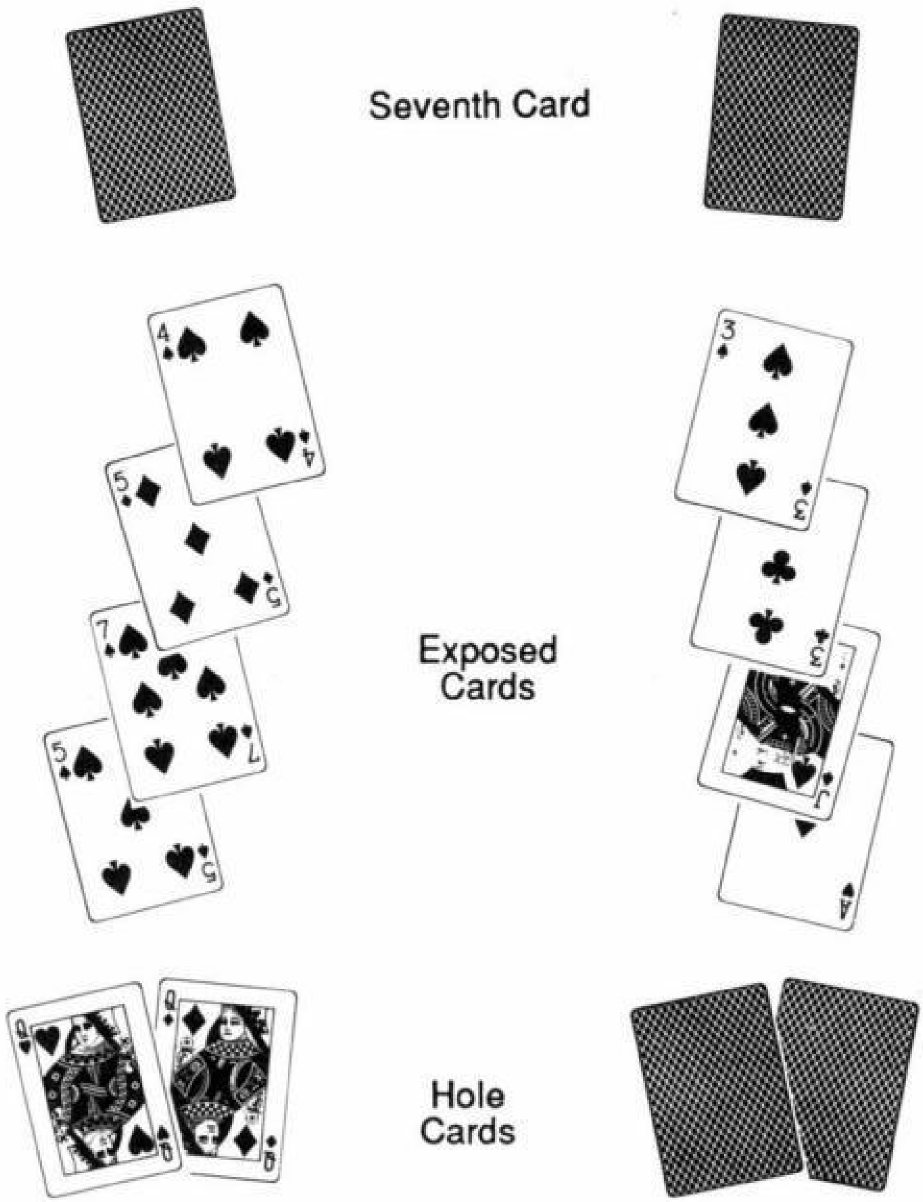


Figure A

You figure your opponent for aces-up or jacks-up, but you estimate that it is 75 percent that he has aces-up. You know that he will bet either hand if you look and then check, forcing you to call as a 3-to-I underdog when you don't improve. This call, on average, costs you half a bet.

On the other hand, if you check in the dark, you are sure he will only bet if he has queens-up beat. (He might even check aces-up, but this doesn't really help you since you're planning to fold if he bets. In fact, it hurts you since you now miss the opportunity to gain two bets on the check raise if you make a full house.) Thus, if your check in the dark causes your opponent to bet only if he has you beat, it saves you money in the long run.

Ifvou'd looked (in these previous two examples), you probably would have checked and called if you hadn't improved and check raised if you had. What's important is that you would have checked in either case. That's what made checking in the dark worth considering.

Against timid opponents. checking in the dark allows you to fold whenever they bet and you don't improve. Its important to understand that you'd have checked no matter what.

Can it ever be right to check in the dark when there are cards you can catch which will make it proper to come out betting? Let's examine this question.

In The 7heorv of Poker. I point out that many situations arise in which, as the first player to act. you should bet for value even though you figure to get called by hands better than yours which will usually. though not always, be shown down if you check.

Suppose you open with three fours in draw poker and are raised. You call and draw 2 cards. The raiser draws 1. You suspect disguised trips, but he may only have aces-up. You don't improve.

If your opponent is the type of player who, (1) will always call with aces-up or better if you bet; and (2) normally will bet medium trips or better (and only sometimes a worse hand than yours) if you check, then it is better to bet and make certain you win something when you have him beat - even though you are an underdog when he calls you.

It also is correct to come out betting when you improve and you are not sure he will call your check raise (or if he has only enough money left to call one bet).

In spite of all this, if you know that checking in the dark will scare him out of betting any hand that does beat you, do it! It'll save you a bet most of the time.

Let's look at an example from seven-card stud. (See Figure B.) After six cards, you have ace, nine, trey, deuce showing with 3 spades. You have another ace in the hole, but no flush draw. Your opponent shows king, queen, jack, ten. You put him on a straight or two pair going in, with a straight being slightly more likely. (The pot size forced you to call on sixth street.)

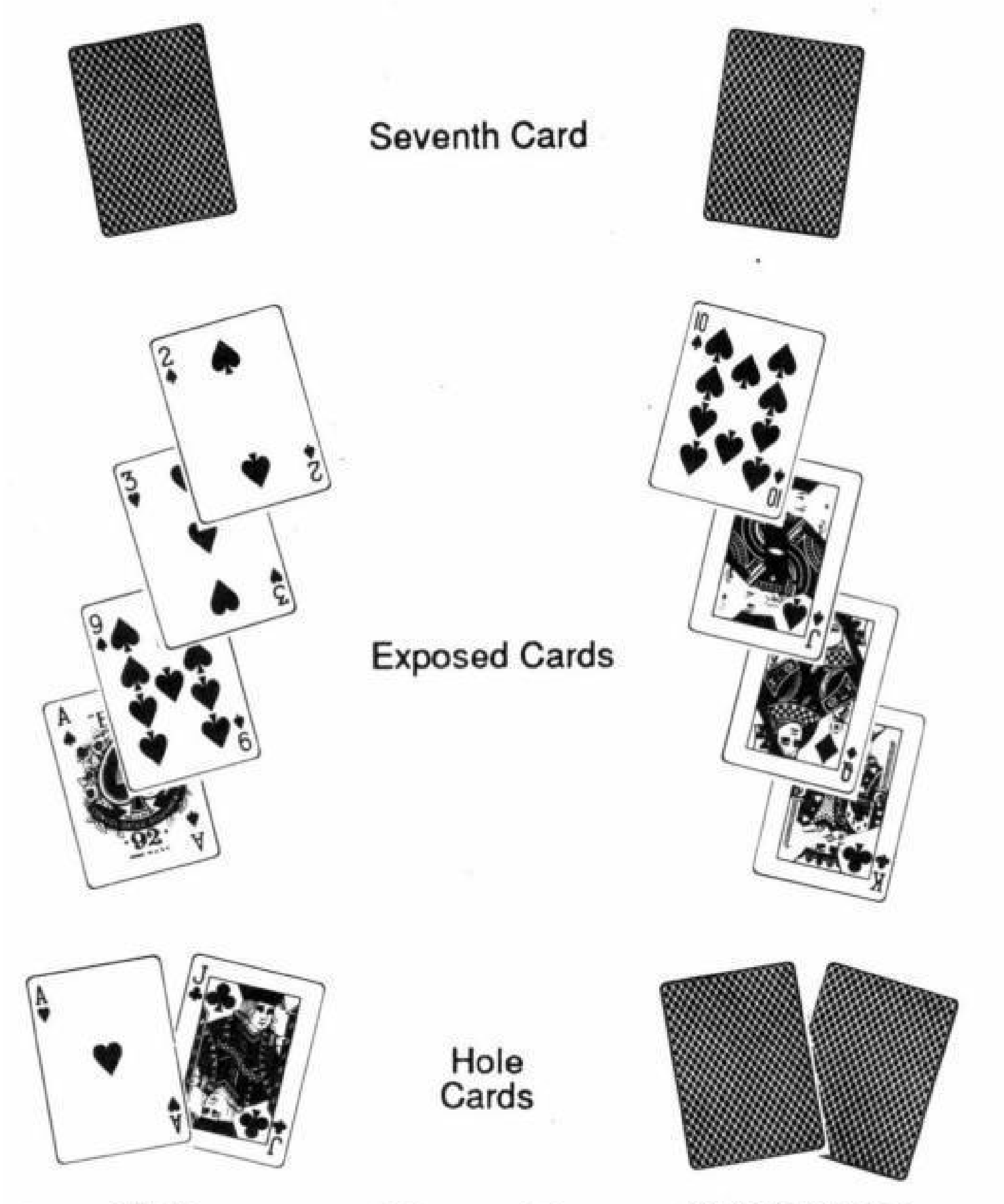


Figure B

If you make aces-up against good players, your correct strategy is to come out betting. That's because by betting you are representing a flush, and this will tend to keep your opponent from raising if he does have a straight. If he only has two pair, though, you will be called. By betting, therefore, you win a bet when he has two pair. If you had checked he may or may not have bet his two pair. You can't fold without risk if he does bet after you check.

Against some players, however, you can fold when they bet after you check in the dark. If you're sure that by checking dark, you'll prevent an opponent from betting unless he has three aces beat (the best hand you could possibly make), you never have to pay him off if he bets. This, then. is an example where it is correct to check in the dark even if there are cards you can catch which would cause you to bet had you looked.

So far all examples I've given suggest checking in the dark in order to make a fold easier. It can also be used to make a call easier.

Suppose you open with a pair of aces in draw poker. Someone calls and draws one card. You aren't sure whether he's on the come or has two pair, but you suspect the former. It's too dangerous to come out betting if you improve. If you check, he may bluff with a busted hand, or he may have had two pair and decide to bet them for value. That's a tough call to make if you don't help.

Fortunately, if you check in the dark, many players will show down two pair and still bet busted come hands. Against these players it is very profitable to call with just one pair, because they're more apt to be bluffing than to have made a completed hand.

Checking in the dark has made your call a lot easier. (The play will hurt you when you improve and your opponent shows down two pair, but this doesn't figure to cost you as much as when you don't improve and he bets two pair.)

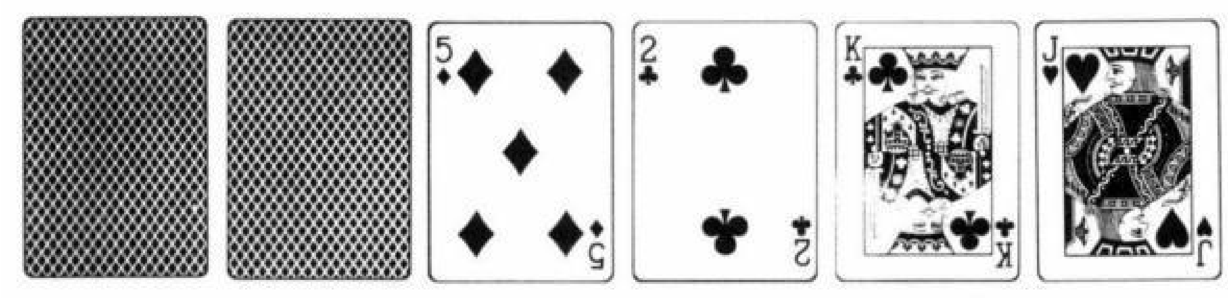
The general conditions to make the play correct would be that your opponent is probably "drawing," but may already have a decent hand. If checking dark will prevent him from betting a decent hand for value, you should consider doing so with the intention of calling if he bets. I leave it to the reader to think of an example from seven-card stud.

Now you've learned how to use checking in the dark as a sophisticated poker weapon. What about those opponents who check in the dark against you? There are two main reasons why you should normally like this:

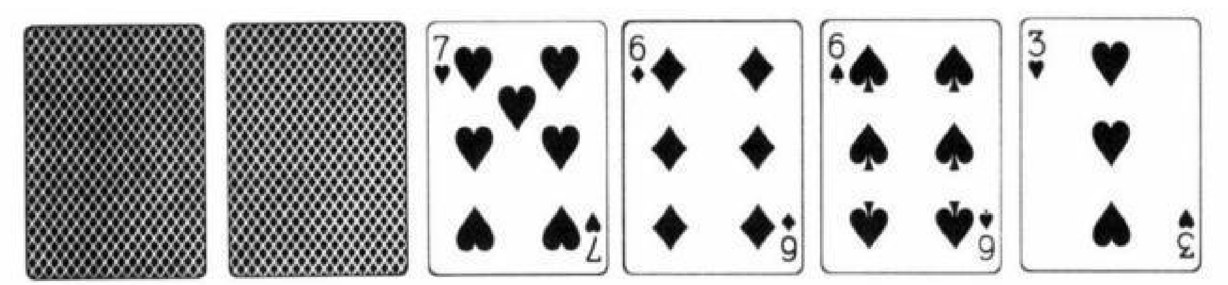
1.Your adversary is usually supplying you with information about the kind of hand he had - most often a mediocre hand he hopes you won't bet into.

2. Unwittingly he has made it easy for you to predict the chances that he has improved on the draw or, (as in the case of seven-card stud). on the last card. Since his check has not changed anything, you're dealing with his purely mathematical chances of improvement.

His play is most commonly to your advantage when you also have a hand that you are anxious to show down. A good example comes from razz (seven-card Your timid opponent shows



and checks in the dark, looking at your



If you have an eight or nine low, you can show it down.

Had your opponent been a good player, he would have looked first and then bet if he'd made his hand and occasionally when he hadn't. By doing so, a player costs you more money in the long run than if he had checked in the dark. Similar situations arise in all types of poker games.

Sometimes by checking in the dark, an opponent makes it easier for you to bet certain hands. As before, this happens when his nolook check gives you extra information about the type of hand he has.

For instance, suppose a player opens in draw poker and another player calls. You call with kings-up. You each draw one card. After the opener checks, the original caller checks in the dark. Now you can bet kings-up for value since by checking in the dark, that player has told you that he probably has two small pair. Since he thinks you're on the come, you should get a call. If he had checked after looking at his card, you probably should not bet (without improving), because it's too likely that you won't be called unless you're beat.

In seven-card stud, a player who shows an open pair and three suited cards might check without looking at his seventh card. Only a very, timid player would do this if he had a four-flush. Consequently, his check in the dark usually means he has no possibility of making a flush, and that makes it easier for you to bet a hand like aces-up.

The one time you would prefer that your opponent not check in the dark is when he gives up more information by looking and then checking. Many players cannot bring themselves to check if they're aware that they've made a good hand. Thus, if they look and then check, you can bet quite mediocre hands into them for value, whereas you would have to show down the same hands if they checked in the dark.

To illustrate this, let's take another look at stud. On seventh street, someone checks what figures to be a pair of kings and a straight draw. If you have a pair of aces hidden, you should bet if he checked after looking. You should usually show down your hand, however, if he checked in the dark, since he is about even money to have drawn out on vou.

This is one of the strange but occasional situations that come up in poker where an opponent's bad play has cost you money.

To reiterate, even though the subject of this chapter is not too important, it can do a lot of good if for no other reason than it shows how to think about this game.

A Poker Problem and a Paradox

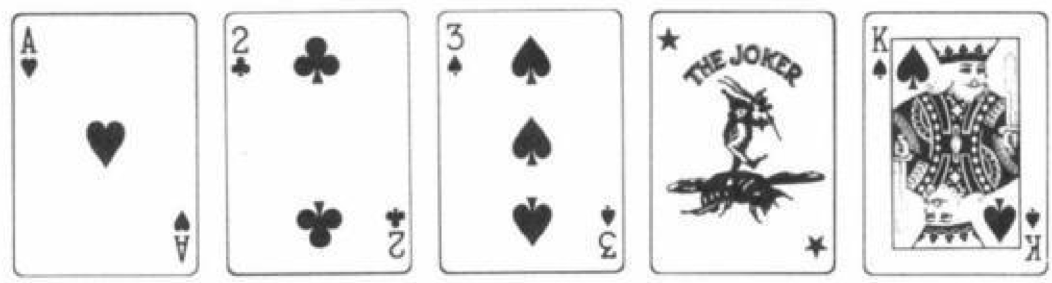
This is a reprint of a highly technical article I wrote for Gambling Times. Don't be upset if you do not fully understand it You won't if you are not somewhat conversant with the mathematical subject of "game theory" as explained in The Theory of Poker. However, even if you can't follow the math, take note of the moral behind this problem. It is that even the most obvious "common sense" play may not be the right one.

A few months ago an interesting, hypothetical poker situation came to mind. As an experiment, I asked about 20 top Vegas players what they would do in this situation. Only two got it right, yet the correct answer was not a matter of opinion. When the right answer was explained, the other players had to admit that they were wrong.

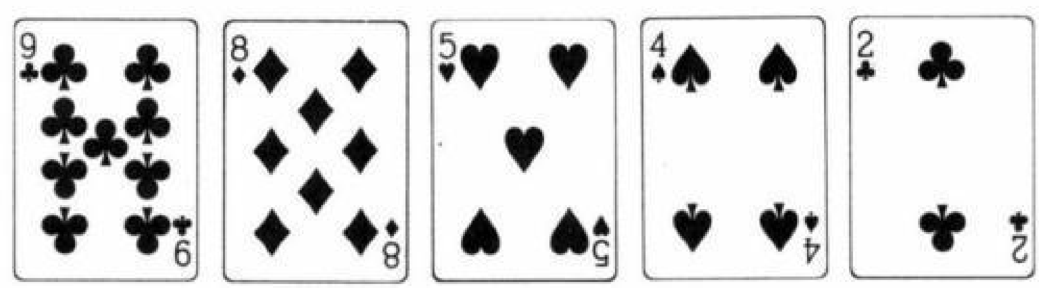
The problem does not involve some obscure principle of poker. On the contrary, it is a vivid example of a most important concept related to winning poker play.

I advise the reader to have some knowledge of poker game theory to correctly solve the problem. Those of you without this knowledge may correctly answer intuitively. However, I suggest you read the game theory chapter in my book The Theory of Poker.

Suppose that two poker experts are playing head-up, pot limit, lowball draw. The ante is $50 each. On one particular deal, some cards are accidentally exposed. Instead of calling a misdeal, they decide to play the whole hand face up before the draw, but the draw cards will be face down. One player picks up



(there is one joker in a lowball deck and it is wild, and this player has the best possible draw). The other player finds a pat



How should the hand be played before the draw (the 9 acts first)? Assume that both players have $10,000 in front of them. Also, assume that all bets and raises must be the size of the pot. That is, if one player bets $100 into the $100 pot, the other player must fold, call $100 or call the $100 and raise the $300 now in there. The reraise would be $900 then $2,700, etc.

First, calculate the chances that A♥2♣3♠joker will outdraw the pat 9. (The 9 does better by standing pat than by drawing one or two cards.) The bike-draw needs to catch a four, five, six, seven, eight, or nine. 4 of these cards are contained in the pat hand. The drawing hand has 20 of the cards remaining in the deck. Its chances of making a 9 or better are 20/43 or about 46½ percent. (The pat hand is a definite favorite to wind up with the best hand.)

If the game is no-limit instead of pot-limit, the pat-nine should simply bet all $10,000 (as an 1 I"/2-to-10 favorite). However, in this problem, the nine will have to worry about an after-the-draw bet. If he bets $100 before the draw and the bike-draw calls, he is subjecting himself to a $300 after-the-draw bet. Should he call the bet every time, the bike-draw can win $450

$$\$450 = \$50\,ante + \$100 + \$300$$

when it makes its hand, and lose only $150 when it doesn't. If the nine doesn't call every time after the draw, it risks being bluffed. The pat-hand. therefore, should not put more money in the pot, but simply check. In fact, most players got this part right.

What about the bike-draw? Should he bet? Before we can answer this questions, we must see how the hand will be played after the draw: this is where game theory applies. Basically, game theory tells a player how often to bluff in a particular situation. You must bluff randomly using the second hand on your watch, or the card that you caught, rather than your judgment. It is most effective against top players whose own judgment is equal or superior to yours.

Should the bike-draw bet after the draw? Let's assume the bikedraw actually told his opponent that he would come out betting if he made his hand, or if he caught any ten, any jack or the queen of spades. If the pat-nine was sure that the bike-draw was telling the truth, it wouldn't do much good. The bike-draw announced that, out of 43 cards he could catch, 14 cards would make him check and give up, 9 cards would make him bluff and 20 cards would make his hand.

Now, what would he do if the bike-draw bets the pot? Notice that in this situation he is getting 2-to-1 odds. However, since the odds are 20-to-9 that the bike-draw made a hand, it is not worth calling when it bets. Still, the pat-nine could fold the best hand 9 out of 29 times: and that's almost as bad.

We have a situation where the bike-draw will bet 29 out of 43 times after the draw and the pat-nine will fold. (Once again, the patnine must fold rather than take 2-to-1 odds on a 20-to-9 shot.) The bike-draw will min the pot 29/43, or 67'/2 percent of the time using this strategy. Therefore, the bike-draw should bet before the draw since it is the favorite to win the pot.

What should the pat-nine do? We have seen that the bike-draw wins the pot more than two-thirds of the time. The pat-nine should never call a bet after the draw, but wins a showdown 14 out of 43 times. Since it only gets 2-to-I odds on the original $100 pot-bet before the draw, it should simply fold.

That is your answer: The pat-hand should check, the bike-draw should bet and the pat-hand should fold before the draw! Such is the strength of the drawing hand that will occasionally and randomly bluff.

(Those readers who are familiar with game theory realize that the perfect "saddle-point'" strategy in this situation would have you bluffing with 10 cards rather than 9. 1 simply used 9 cards in order to make my point easier to see. If in fact the drawing hand was using perfect game theory. the following strategies would hold: If the drawing hands' chances are 4/9 or better, it should bet, and the pat-hand should fold. If the chances for the drawing-hand are between 3/9 and 4/9, it should bet and the pat-hand should call. If the drawing hand's chances are between 2/9 and 3/9, the pat-hand should bet. and the drawing hand should call. If the chances for the drawing hands are under 2/9, the pat-hand should bet and the drawing hand should fold.)

The importance of this example is to illustrate the strength of a drawing-hand, when there are bets to come and to exercise the mechanics of game theory. This is most true in a pot-limit game. However, the drawing hand can be the better hand, even though a slight underdog, even in a straight limit game, although the madehand should not actually fold. The made-hand, however, should not bet as a slight favorite into a big draw, a mistake many players make. (An example is betting 10-9 low into a smooth draw on sixth street in seven-card razz.)

Now, the poker paradox. Take the same problem, the same game, the same hands. This time both players have exactly $400 plus the ante. How should the hand be played? If the pat-hand checks and the bike-draw bets $100, as in the previous example, the pat-hand would be wrong to fold. The pat-hand should raise all in. There would now be no bet after the draw which would give it a nice edge, since the bike-draw can no longer bet. However, if the pat-hand bets itself, the bike-draw can simply call and command the position it held in the previous example. Therefore, they should both check!

We have a situation where two players see each other's hand. vet neither one should bet. It seems impossible that such a situation should arise, but here it is. This example proves that the intricacies and subtleties of this marvelous game of poker should never be underestimated.

Introduction to Hold ’em Poker

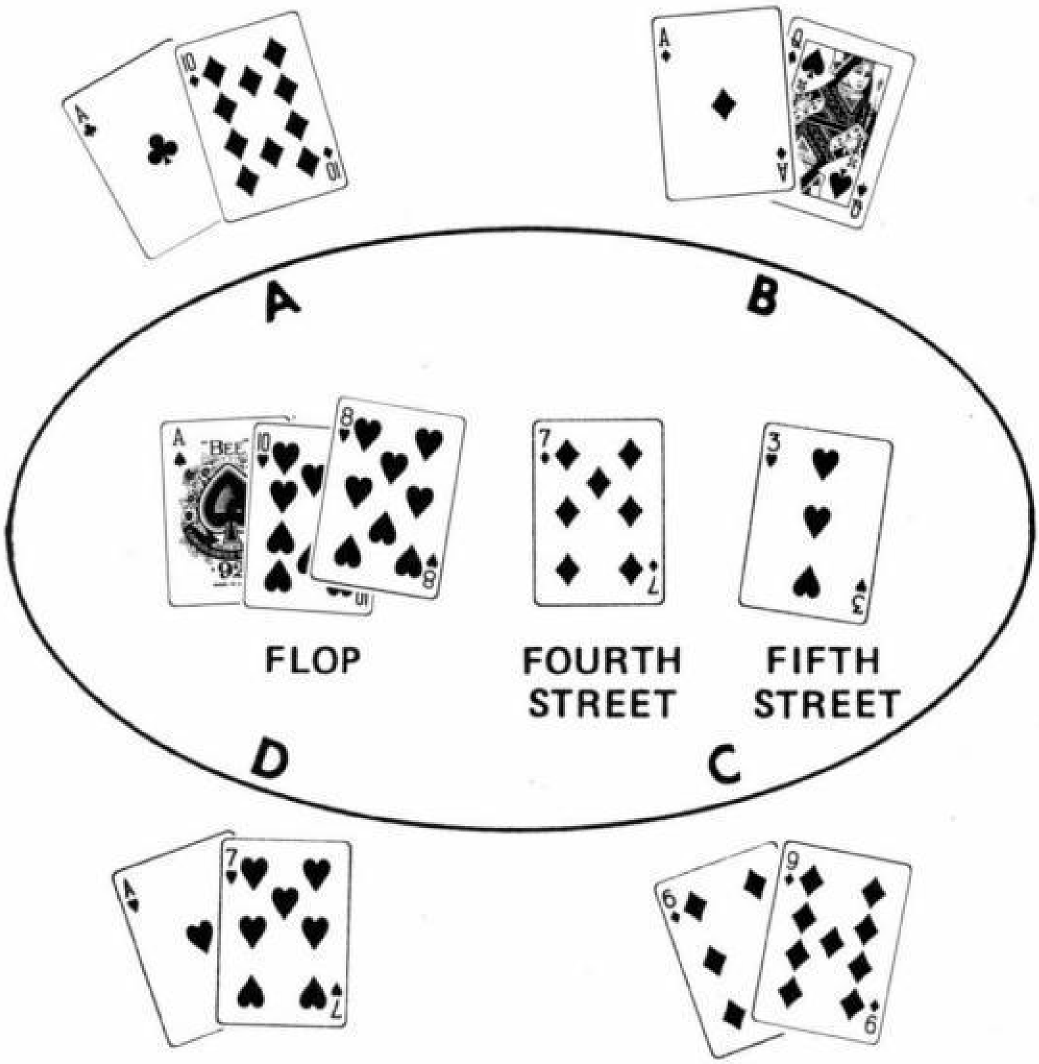
The following is simply a reprint of the first chapter of my book Hold 'ern Poker. Since I wrote this book, hold 'em has quadrupled in popularity. I expect that after you read this chapter you will rush out and buy the complete book. Also, once you have played for a while. you should pick up a copy of Hold 'em Poker. for Advanced Players which I co-authored with Mason Malmuth.

Introduction

Hold 'em poker is fast becoming one of the most popular poker games in America. However, there is virtually nothing in print at this writing that gives more than a cursory look at this game.' I intend to remedy that with this book. This is not simply an elementary introduction to the game. I believe that anyone who thoroughly understands everything in this book can become a top notch hold 'em player with a little experience. However, do not expect easy reading. Most people will find that they will have to read certain sections over two and three times before they can expect to fully understand the concepts contained in them. At the same time I will assume the reader has a prior knowledge of poker in general. Those readers who haven't played any kind of poker at all would be wise to pick up a book such as the Fundamentals of Poker by Mason Malmuth and Lynne Loomis before continuing with this book.

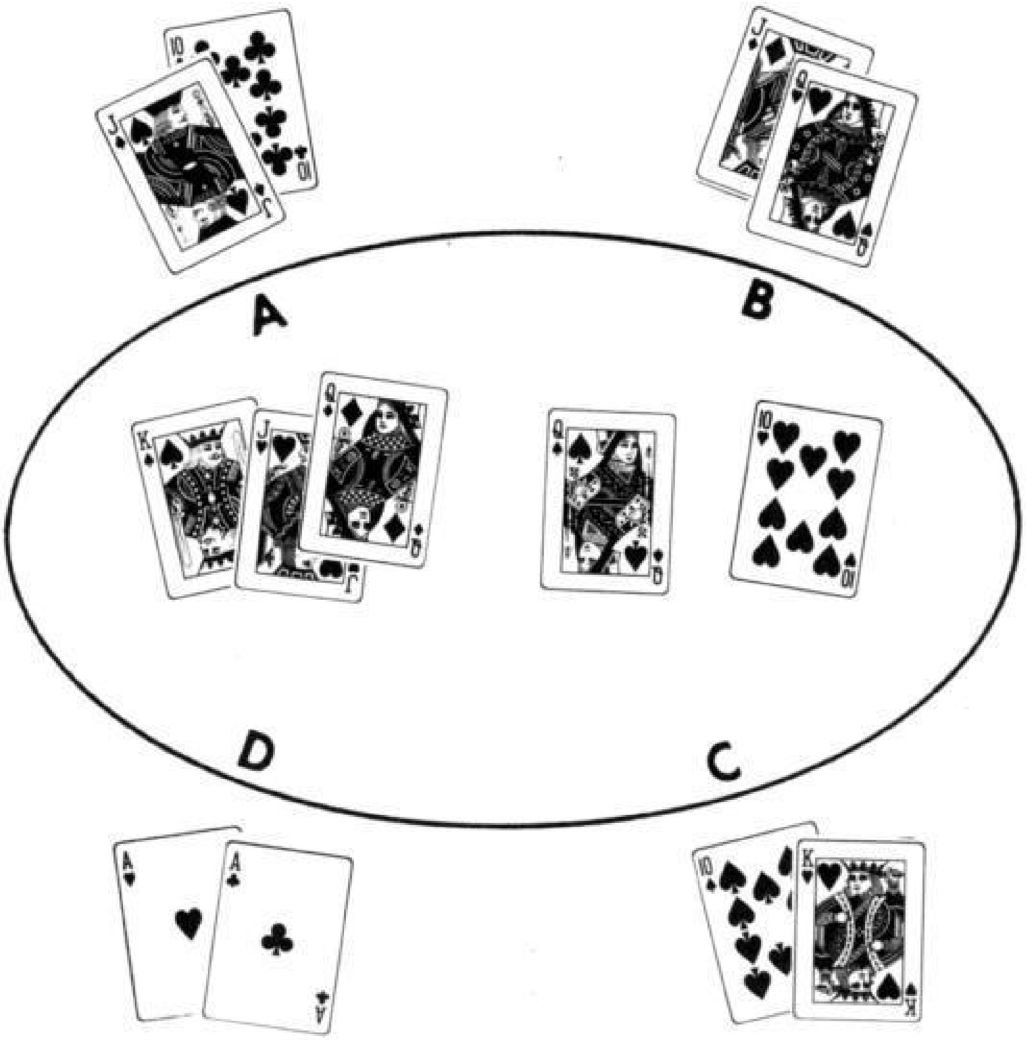
Dealing a Few Hands

At the beginning of each hand of hold 'em, the dealer starting at his left. deals around until each player has two cards face down. The deal moves to the left after each hand. (In casino games there is a house dealer so a rotating "button" is put in front of a player to signify that he is the dealer for that hand. and thus gets cards last and acts last in each betting round.) There is one round of betting on these first two cards. The dealer now deals three consecutive cards face up in the middle of the table. These cards are called the "flop." They are community cards used in common by every player still in the hand. There is then a second round of betting. After that, a fourth card is dealt face up and a third round of betting occurs. A fifth and Final card is then turned up and a last round of betting takes place. If there is more than one player left in the game at the end, the winner of the pot is the player with the best poker hand using the best five out of seven cards (the two in his hand and the five in the middle). Occasionally, there is a split pot. Before defining the rules of play more explicitly I will give some examples of possible hands:



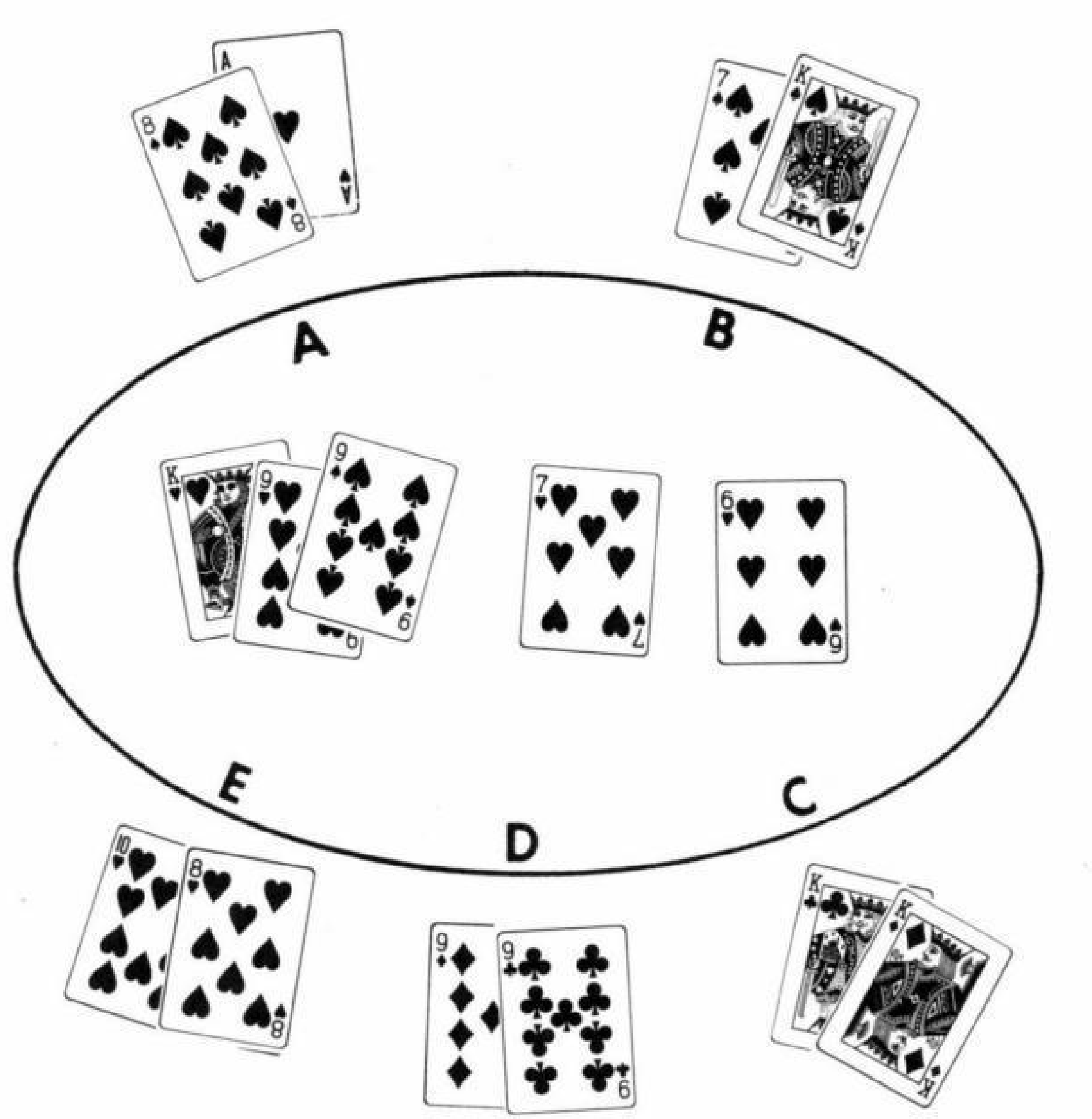
Player A's best five cards are A A 10 10 8
Player B's best five cards are A A Q 10 8
Player C's best five cards are 6 7 8 9 10
Player D's best five cards are A 10 8 7 3 All hearts

Therefore in this example, Player D has the best hand with an ace high flush. Notice he also has aces and sevens if he chooses to use a different combination of five cards. But, of course, he wouldn't do that and in this case he is lucky he made a flush on the last card since Player A has aces and tens and Player C has a straight. Here is another example:



Player A has Q Q J J K
Player B has Q Q Q J J
Player C has K K Q Q J
Player D has A K Q J 10 - straight

In this case player B wins with a full house. Notice that both hole cards are not always used to make your best hand. It is possible to have the best hand even if you are using only one of your hole cards and your opponent is using both of his. However, if neither card in your hand is used that is called "playing the board" and in this case the most you can hope for is a split pot (if your opponent also plays the board). One last example:



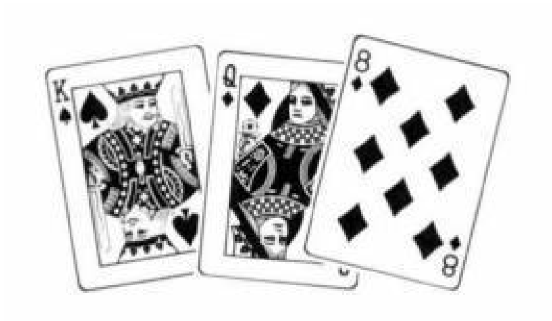
In this case Player B's two pair of kings and nines lose to Player A's ace-high heart flush. However, Player C beats that with kings full only to be beaten by Player D's four nines, and Player E's straight flush!

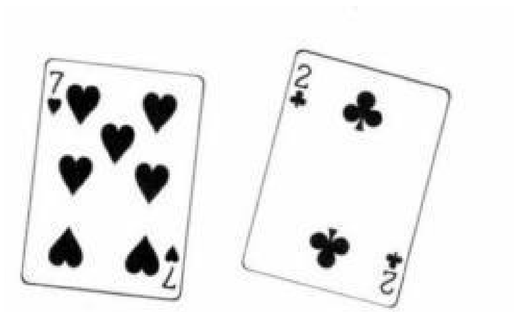
The reader should double check this last example. Figuring your best hand is not always as easy as it may appear. While I won't give any further examples, I urge the reader to deal out a few hundred hands himself until he can instantly recognize his best hand.

The Best Possible Hand

Another related concept is figuring the best possible hand that can be held, given the cards in the middle (also called the cards on board"). The second best hand, third best, etc. can also be figured.

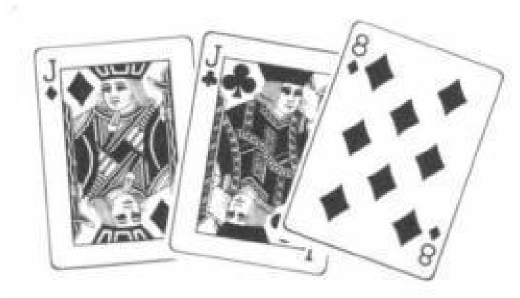
An Example:

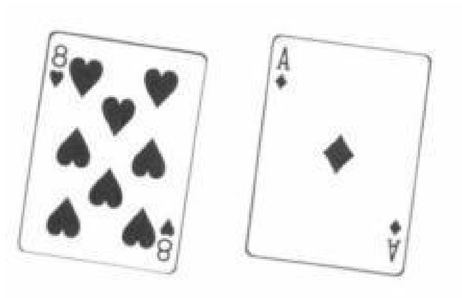




In this case the best possible hand anyone could hold would be three kings, followed by three queens, three eights, etc. The general rule is that there cannot be a full house or four-of-a-kind in the game without a pair on board. Also, there cannot be a flush or straight flush without at least three of a suit on board. Straights must be figured individually. In this example if the Q♦ was the J♦ instead, someone holding a T9 would have a straight.

A second example: The cards on board are:





In order, the best hands are:

Four' acks if you held JJ.

Four eights if y ou held 88.

Aces full if you held AA.

Jacks full of aces if you held AJ.

Jacks full of eights if you held J-anything.

Eights full of aces if you held A8.

Eights full of jacks if you held 8-anything.

A flush if you held two diamonds.

AAJJK if you held AK.

AAJJQ if you held AQ.

AAJJT if you held AT.

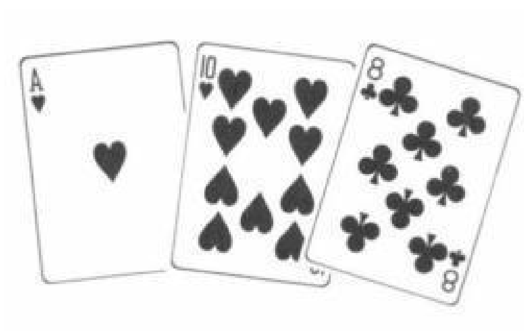
AAJJ9 if you held A9.

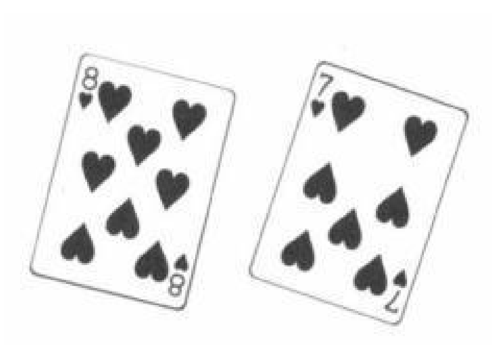
AAJJ8 if you held A7, A6, A5. A4, A3, or A2.

Kings over jacks if you held KK.

Let us notice something: One of the points of this exercise is to ascertain whether we, in fact, have the "nuts" (a lock). Sometimes, when it appears that a certain hand could conceivably beat us, one of our own cards might block this possibility. If the cards on board are KKJJ4 and we are holding KJ — nothing can beat us since we are blocking four-of-a-kind. If the board is 6♥8♥9♥K♦Q♦ we have the nuts with A♥7♥ since we block the straight flush.

A final example: The cards on board are:





You can't be beaten if you hold J♥9♥ or 9♥6♥. Your next best hole cards are 8♦8♠ followed by AA, TT, A8, T8, 87, 77, and then K♥-anything.

The reader should learn these concepts also by dealing out a few hundred hands and determining the nuts, the second nuts, etc. This is important because in hold 'em it is not that rare to get a hand that can't possibly lose and obviously, it is important to know when you have one.

Different Variations

Let us now look at the different variations in the betting rules of hold 'em poker. A well-known variation is "no-limit" (actually "table stakes"). This is the game that is played in the finals of the World Series of Poker every year at Binion's Horseshoe Casino in Las Vegas. However, no beginner should even consider playing it. It is not a friendly game. This book will concentrate on the "limit" variety of this game.

In all ensuing problems 1 will assume that the reader is playing in the $10-$20 hold 'em game at The Mirage in Las Vegas (unless otherwise noted). This will be done to simplify and standardize the examples to follow. These are the rules:

> There is no ante. However, the player to the left of the dealer button must put in $5 and the player to his left must put in $10. These bets are called the "small blind" and the "big blind." The next player must either fold, call the $10 or raise to $20. From then on raises before the flop are in increments of $10. There is usually a maximum of four raises allowed. If no one raises the player who put in the big blind he is given the opportunity to raise himself. This feature is called a "live blind." The second round of betting (after the flop) is in units of $10 starting from the small blind, if he is still in, proceeding to the left. The player may check at this point, or bet $10. Check and raise is allowed. The betting on "fourth street" and "fifth street" (fourth and fifth cards on board) once again starts to the left of the button, this time in units of $20.

Make sure you understand all these rules in order to best follow later examples.

Most limit games have the same general structure as The Mirage $10-$20 game. In a few places check-and-raise is prohibited. Some games do not have a blind. However, in every game I have seen without a blind, it was mandatory to bet or get out on the first two cards. Checking was not allowed on this round. In some games there is a choice of how much to bet. None of these variations significantly change playing strategy however.

There are two appealing characteristics of hold 'em that exist for any variation. One is that you can play with far more players than the usual number of seven or eight, even up to twenty can play since the cards do not run out! No one has to sit out of hands when ten people come to your poker party if you play hold 'em. Secondly, since a player's folding does not change the flop a player can see what hand he would have made even if he is out of the hand. He thus has the dubious opportunity to second guess himself.

Incidentally, some readers may have noticed that there appears to be a big advantage for the dealer (and disadvantage for those in early position) since he gets to act last on all four betting rounds. To those ofvou that did notice, congratulations, you are already on the way to becoming an expert hold 'em player.

World Class Poker Plays

The following two chapters were originally written as a change of pace from my usual Gambling Times columns. They are in the form of a quiz. The questions are pretty hard. Don't expect to be able to answer all of the questions simply because you read the previous chapters in this section. These chapters merely delve into a few selected topics in poker. For a complete treatment of this complex game I refer you to my book The Theory of Poker.

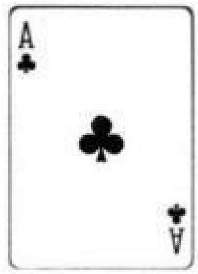
In his well-known book written many years ago, Oswald Jacoby suggests a play in draw poker which may raise some eyebrows. He said if you find your opponent drawing one card after you have raised him with two very small pair, you should throw one of your pair away and draw three! He is right, of course (assuming you are playing jacks or better without the joker). However, this is only one of many poker plays which may appear strange to the average player.

Most of these plays do not appear in any book, though the general concepts behind them are covered in my book.

The opportunity to make these unusual but ingenious plays comes up in even-type of poker, and I'd like to present a number of these advanced plays in somewhat of a quiz format. First, I'll lay out the situation, then I'll suggest the proper play. You can test yourself by seeing whether you can come up with the right play before reading my solution. I have confined myself to situations where the correct play is not really debatable. However, I do assume that your opponents are tough, rational players of the type normally encountered in Nevada casinos. Many of these plays would be wrong against poor opponents.

I should warn any expert players that some of these situations are very tricky. In a few cases, the amateur will suspect that one play is correct while you would choose another play. The amateur play, however, may be very well correct, albeit for different reasons than he might think. Try to avoid getting tripped up on these.

1. The Situation: The game is draw poker, jacks or better. Everyone has passed and you opened as the dealer with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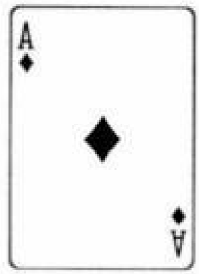
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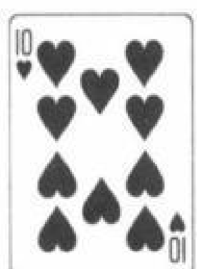
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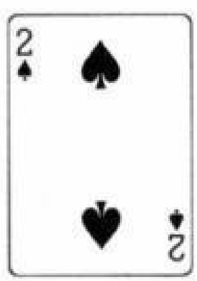
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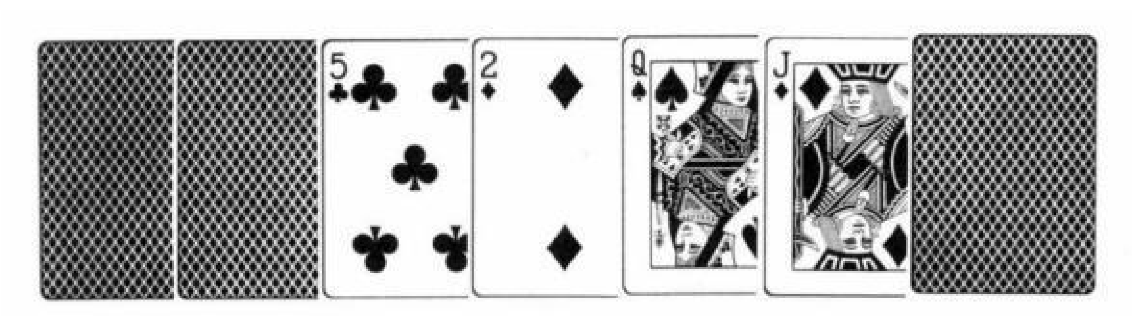
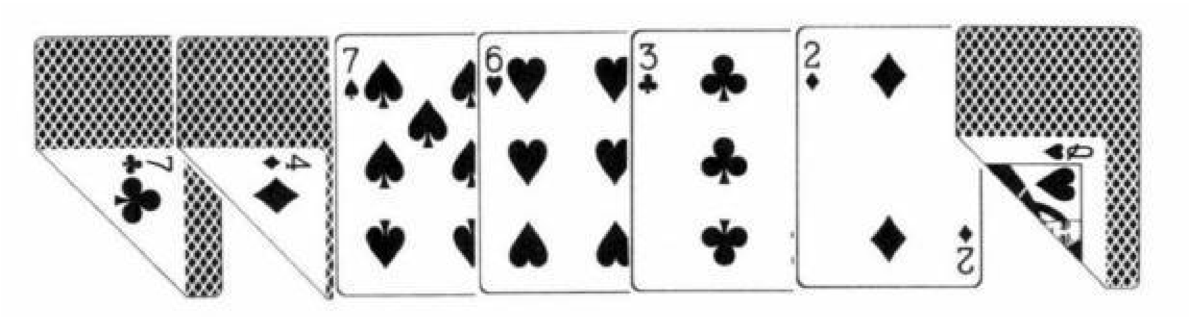
A player who had passed in late position has called and drawn one, obviously "on the come" for a straight or flush.

The Play: Stand pat! You have now stopped any chance of being bluffed out, as no player would dream that he could make you throw away a pat hand by bluffing after the draw. If he does bet, throw your hand away. You, of course, should not bet. You must be cautious about using this play too often because your opponents will wise up to it. Notice also that I said you have a pair of aces when you make this play. This is so that you do not have to worry about your opponent making a higher pair when he draws one card. Now the only risk is the approximate 500-to-I shot that your opponent makes his hand and your three-card draw would have beaten him. (I discuss this play in greater detail in The Theory of Poker.)

2. The Situation: The game is seven-card stud, hi-low split, with a simultaneous declare. Your opponent has



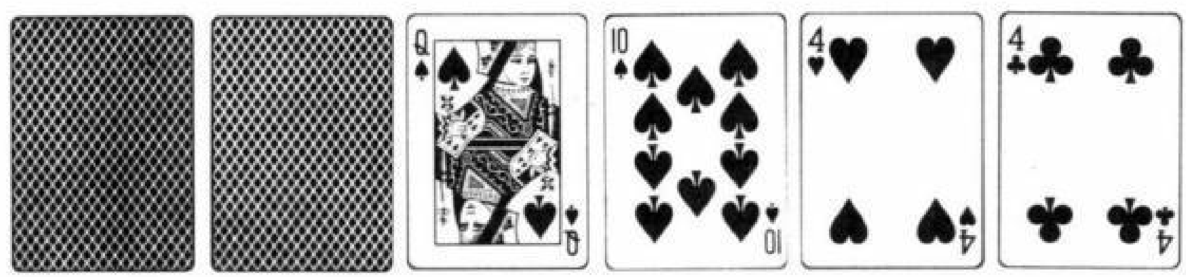
You have



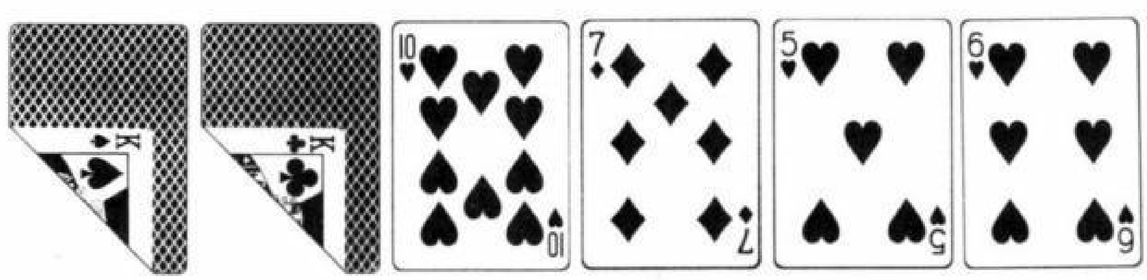
and have been betting all the way. How do you declare?

The Play: Declare "high" only. Your opponent is almost certainly drawing for a better low hand but will declare "low" only if he makes a very good low on the end. It does no good to declare both ways, since he will probably beat you if he declares low himself. Those of you who declare low only because you are afraid he might make a higher pair should not be gambling.

3. The Situation: Seven-card stud. Your opponent has been betting all the way. It is now the sixth card. He has



You have



You have been flat calling all along and your opponent now bets one more time when he pairs fours on board.

The Play: Raise! This is the right play, even if you are fairly sure he has two pair. As long as there is some chance that he has a four flush or four straight which he is planning to bluff with on the end if he misses. You are forced to call him on the end if you just call on sixth street, even if you don't improve. (Throwing your hand away on sixth street is, of course, the wrong play, even if you are sure he has two pair, because of your pot odds.) By raising, you have taken the initiative and have forced him to check to you on the last card. You can check right back if you don't improve and still lose only two bets if you don't have the best hand. This play is especially effective because you have a possible flush on board. This fact may very well induce your opponent to throw his hand away instantly when you raise on sixth street if he just has a straight draw. If he does, you have gained a lot. This play may also make you an extra bet when you improve your hand.

4. The Situation: The game is jacks-or-better draw poker with a joker (counts for aces, straights or flushes). A typical player opens in second position for $5 in a $5-$10 game. The ante is $1. You are in third position with two aces. What do you do? What would you do in this same position with nines and fours?

The Play: You should throw your hand away in both cases. Despite the fact that you are getting 13-to-5 odds, you must fold due to all the players behind you who have not yet acted, plus the fact than an early-position opener figures to be stronger than average. Failing to throw hands like these away in this spot is one of the most common mistakes made in the card rooms of California.

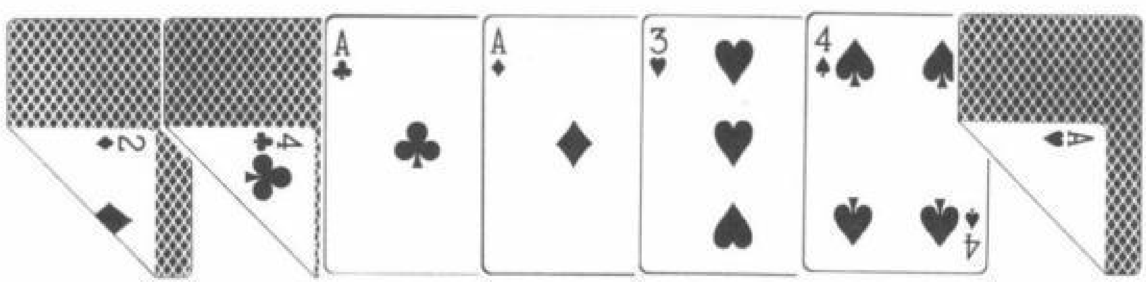
5. The Situation: The game is no-limit hold 'em. A player makes a very big bet before the flop; the bet is enough to put you all-in. You have played with this player before and know that he will only make this kind of bet with two aces, two kings or an ace, king in the hole. You have two queens.

The Play: Throw your hand away. This may seem obvious, since two out of three of his possible hands are better than yours. A somewhat more sophisticated reader may notice, however, that there is a flaw in this reasoning. The flaw comes from the fact that there are 16 ways to be dealt an ace-king and only 6 ways each to be dealt two aces or t o kings. Thus, the two queens are a 16-to-12 favorite to be the best hand. Unfortunately, this knowledge may incorrectly lead you to the conclusion that you should call. The reason why you should fold is based on the fact that you are almost a 5-to-1 underdog when he has an overpair but only about a 6'/z-to5 favorite when he has ace-king. Thus, in 28 typical hands you would only win about nine of the sixteen times he has ace-king plus about two of the twelve times he has an overpair. This makes him about a 17-to-11 favorite which is well over the 6-to-5 pot odds you are getting.

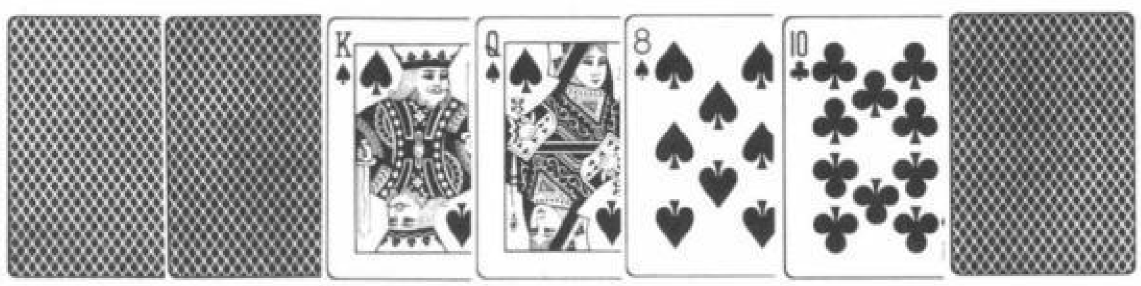
6. The Situation: The same as above against the same player except that this time you have two kings.

The Play: Believe it or not, it is only a very borderline call. You are not favored to win the hand. If you suspect that your opponent will not always raise with ace-king, you have a clear fold. The reason is because of the cards that are out. With two kings in your hand, there are now only eight ways that he can have an ace-king himself. However, there are still six ways he can have two aces in the hole. There is one way he can have two kings himself. So, in 15 typical hands you will be a little less than a 3-to-1 favorite 8 times, dead even once and almost a 5-to-1 underdog six times. You should therefore win 6 + 1/2 + I of those hands, only about half of them.

7. The Situation: The game is seven-card stud, hi-low split declare, with a bet after the declare. You have



(hidden aces-full).Your opponent has



The Play: After betting all the way through, you should declare both ways and check-raise after the declare. You have a pair for low but it makes no difference since you know that your opponent will declare high, so you are in no jeopardy by declaring both ways. Now, if your opponent has a flush, you should be able to make two bets on the end since he figures you for aces up, three aces or a small straight. If you declare high only, he may suspect that you have a full house and not fall for a check raise. Against a weak opponent it may be better to declare both ways and come out betting in hopes that he raises. Then you can reraise and thus make three bets on the end.

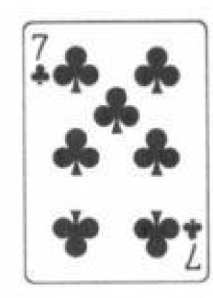
8. The Situation: You have opened with kings over queens in jacks-or-better draw poker (with one joker counting for aces, straights or flushes). A player who had initially passed now called and draws two cards to what you know is a small pair and an ace (or joker) kicker. From the standpoint of making the best hand as often as possible, how should you draw'?

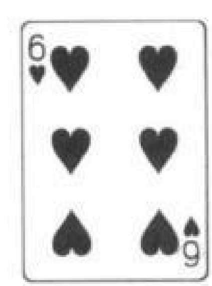
The Play: Draw one card to your two pair. This is another example where the common sense play made by the amateur is the right play, whereas the expert might very well play it wrong. This is because he knows that drawing three cards to his pair of kings gives him a better chance to beat his opponent if he makes aces-up or three-of-a-kind. The only drawback to drawing three cards is when the two-card draw catches a "cold pair," making it two small pair. Now it is better to draw one. Still, the expert knows that of the relevant times when the two-card draw improves, about 80 percent of those times will be trips or aces-up. Thus, drawing three is the correct play 4 out of 5 times when it makes a difference. All of this is very true, yet drawing three cards is a terrible play. The reason stems from the fact that when you draw one when he does make his trips or aces-up, you have only decreased your winning chances by about 3 percent (to 8 percent from 11 percent). However, if you draw three when you should have drawn one (because he made two small pair), you have decreased your winning chances by 72 percent (to 28 percent from 100 percent). Better to hurt yourself by 3 percent four times and help yourself by 72 percent once than the other way around.

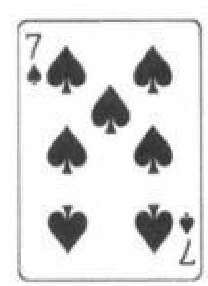
World Class Poker Plays
Part II

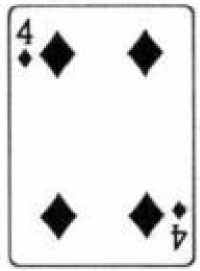
Enough interest was shown in my previous column on superexpert poker plays that I was prompted to write a sequel to it. Once again, I'll employ a "quiz" format where the reader can first consider the situation and see if he can come up with the right play before reading the solution. Remember that I am assuming you are playing against fairly tough, rational opponents.

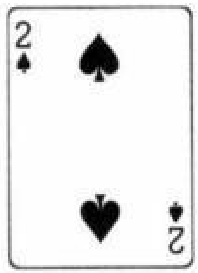
1. The Situation: The game is $20 straight lowball with a $5-$5 and $10 blind. You open for $20 in middle position with a









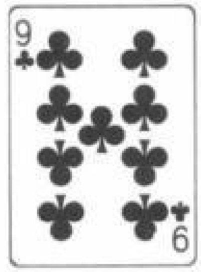


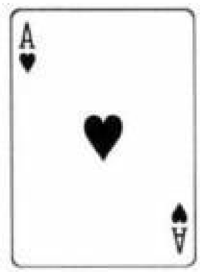
The player behind you calls and all the other players, including the "blinds," fold.

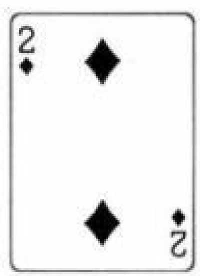
The Play: Stand pat and come out betting $20 into his one-card draw! While it is true that you originally opened with the intention of drawing one card yourself, the fact that you are against one opponent who is behind you with a one-card draw that is probably "smoother" than yours changes your strategy dramatically. By playing it this way your chances of winning the pot have gone from about 45 percent to over 65 percent, since he will usually only call with an eight-low or better. Even though you are risking an afterthe-draw bet as well as your opening bet, it works out better mathematically to play the hand this way than to draw one card yourself.

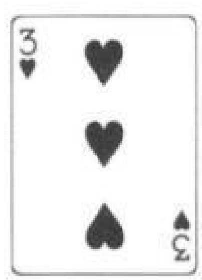
This is especially true when there is only a single bet, rather than a double bet, after the draw. This play is so powerful and so little known that it should never be shown if at all possible. If you are called, just throw your hand in and say that you were "snowing" with a full house. If the play does work, never show the bluff.

2. The Situation: The game is the same except that the after-the-draw bet is now $40. You open with a pat







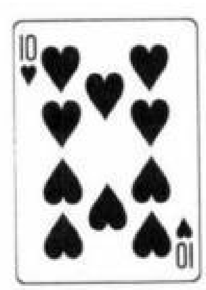


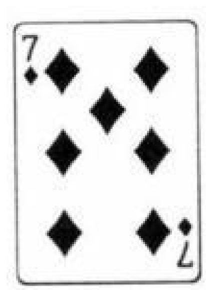
Only the big blind calls. He draws two cards.

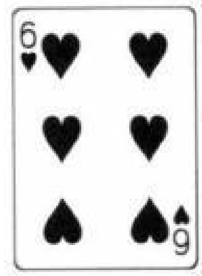
The Play: Throw the 9♣! If he checks, come out betting with a ten-low or better. While it is true that playing your hand this way reduces your winning chances from about 83 percent to about 70 percent, it is worth the risk because of the larger after-the-draw bet. Had you stayed pat, he would not have paid you off with a mediocre hand. Now that you have drawn a card, he will have a great tendency to call you with anything, hoping you are bluffing his two-card draw.

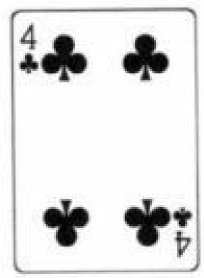
Another benefit from this play arises when his two-card draw makes him an eight or better and your one-card draw out draws him. Now your play has not only saved you the pot plus $40 but also makes you an additional $80 when you raise his bet. The only risk with this play is if he makes a fair hand and you bust out. Now you have cost yourself the $50 pot. However, the previously mentioned benefits out weigh the risks.

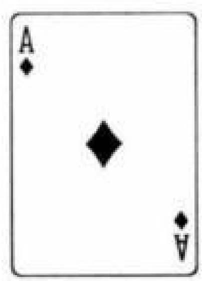
3. The Situation: It is the same game once again except that we are again back to a single bet after the draw. A player opens in early position, the next player raises and you make a marginal call behind him with a







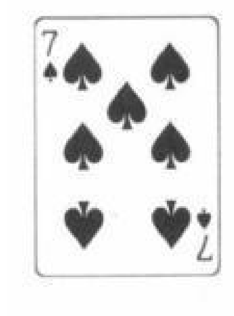


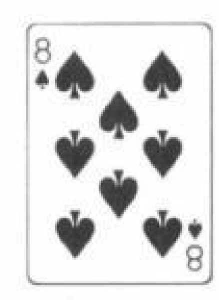


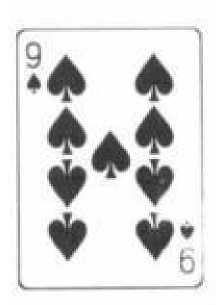
Everyone else now folds, including the original opener. The raiser draws one card.

The Play: Stand pat and call his after-the-draw bet unless you know he never bluffs. Playing your hand this way gives you just about a 50-50 chance to in the pot. He is almost certainly drawing to a better low and would thus be a favorite over you if you also drew a card. Of course. you had originally planned to draw a card yourself. You would have drawn if he had stood pat or if you were up against more than one player or if there was a double bet after the draw.

4. The Situation: The game is draw poker, jacks-or-better, Gardena style, where the one joker in the deck counts for aces, straights or flushes. A player opens in early position and you raise him with a



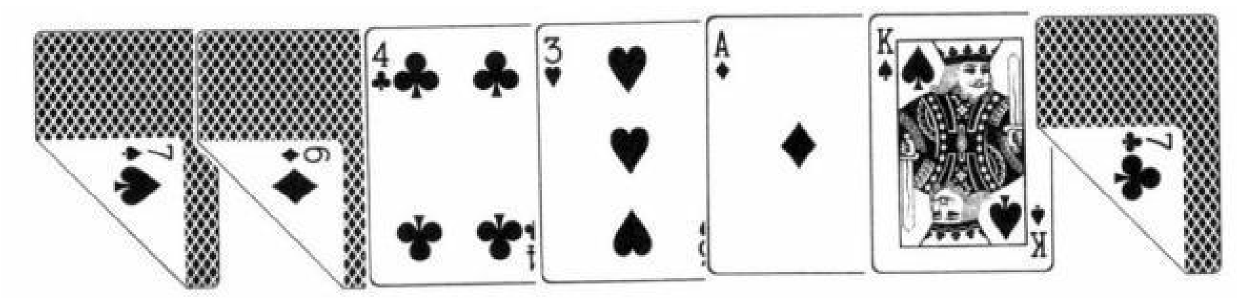




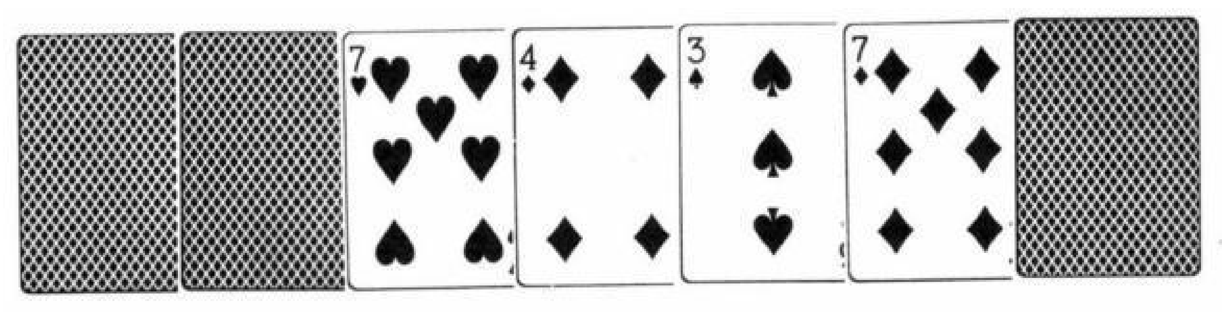
to draw to. He calls and proceeds to throw a queen away face-up while announcing that he is "splitting openers." You catch the jack of spades and raise with your straight flush. He reraises.

The Play: Throw your hand away! If he is a rational player he must have you beat. Once you raise him you are saying you have made a hand that beats the normal pat hand that a man who splits openers usually is drawing to. He now puts you on at least a full house. (I could have made this example absolutely airtight if I said that you raise and drew two cards with three aces and proceeded to make four). This being the case he will reraise only if he has a full house beat. Since a splitting hand cannot make either a full house or four-of-a-kind, he must have a straight flush himself, which must be at least queen high since he threw one of two queens away.

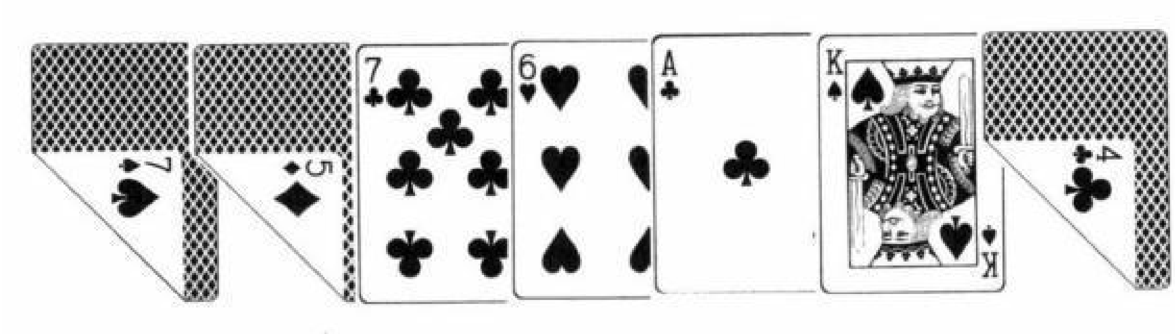
5. The Situation: The game is seven-card razz (lowball). You have been betting all the way and your opponent has been calling all the way. After the cards are out, you have



Your opponent has

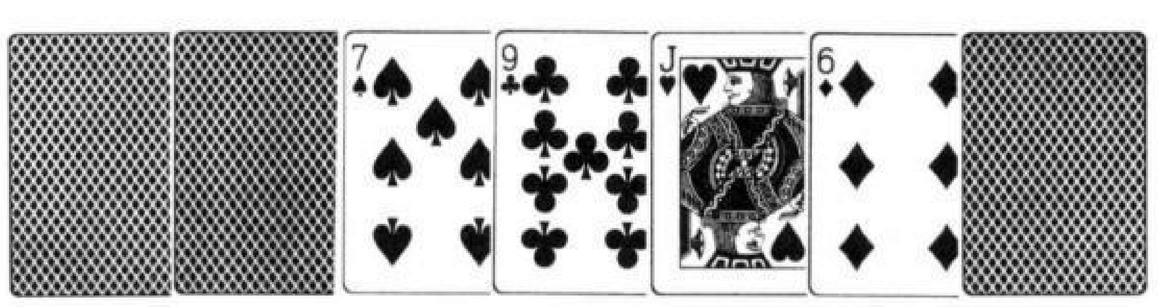


The Play: Bet right out. You may not have the best hand, but given your board, your opponent will not raise even if he has a better seven. By the same token, he will surely call with worse hands than yours (if he does raise, fold). If, in this same situation, you have

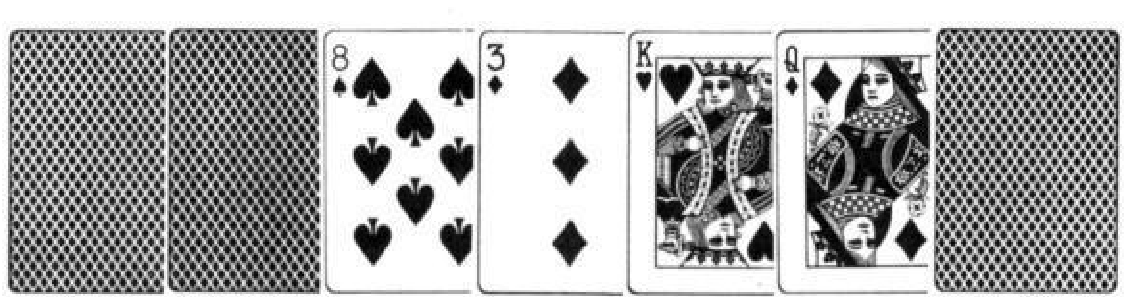


you should probably check (your seven-low.)

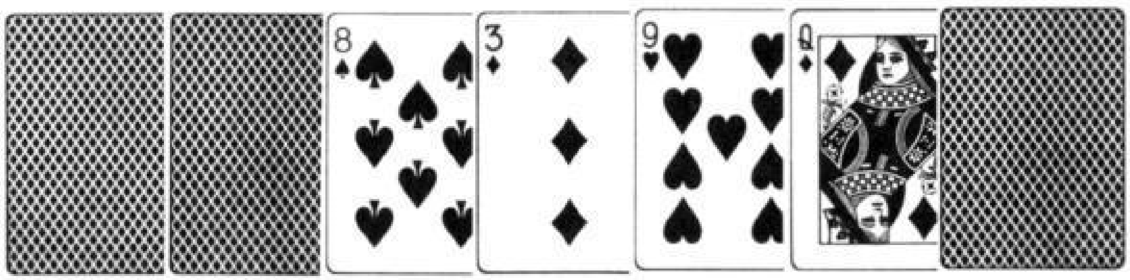
6. The Situation: It is razz again. You show



but have made a seven-low on the last card. Your opponent has

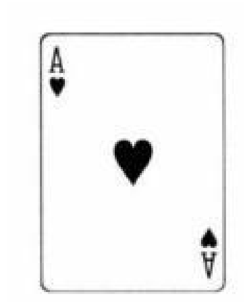


The Play: Try for a check raise. He will almost certainly bet an eight-low but might smell a rat if you bet. He won't call you anyway if he busts out. If. however. he has

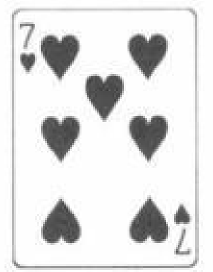


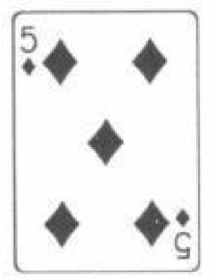
you should probably bet out to insure winning at least one bet from his nine-low. He may even raise you if he makes his eight.

7. The Situation: It is limit hold 'em and you have raised six players with



The flop comes



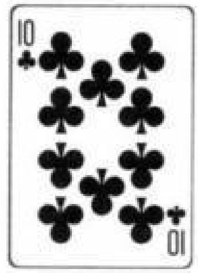


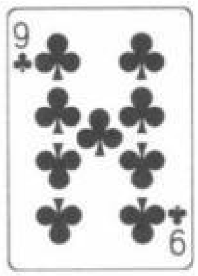
Everybody checks, you bet, and, surprisingly, get four calls. The next card is the 4♥. Everybody checks to you.

The Play: Bet again. You are a little worried that somebody is sandbagging, but its worth taking the chance. You may still have the best hand but even if you don't, you have a great draw. You want to get more money in the pot and don't really mind that much if you are raised.

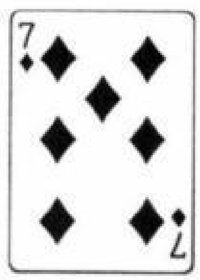
8. The Situation: Same as above except you are playing nolimit.

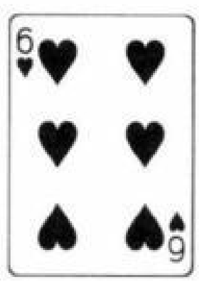
The Play: Check on fourth street after everybody else checks. You are drawing to the "nuts" but can't call a large raise if someone is sandbagging. By giving yourself a free card, you may win a big pot if a heart falls on the end and meanwhile will have kept yourself from being raised off a draw to a cinch hand. This is an important principle of no-limit hold'em, especially in situations where you are last to act. (A very good example of this principle would be where you check a

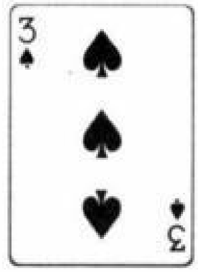




in last position when the flop comes

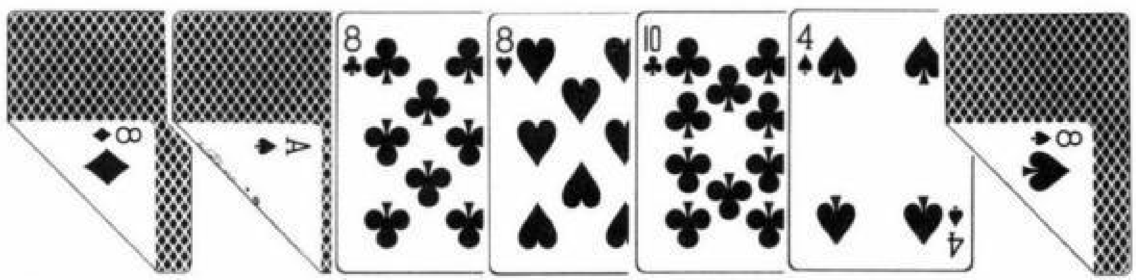




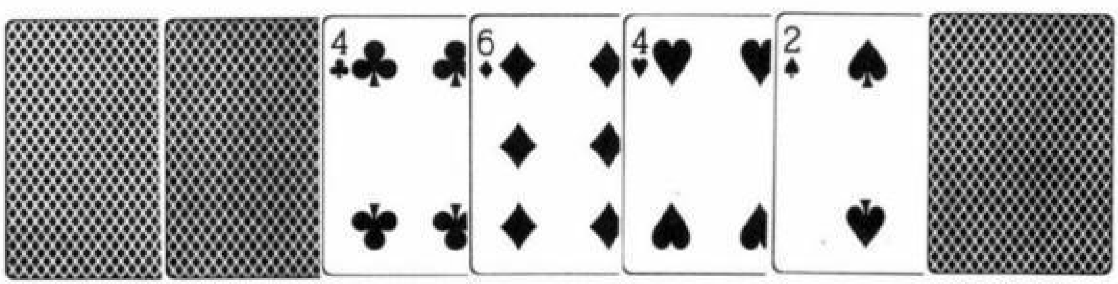


Give yourself a free card to catch an eight.)

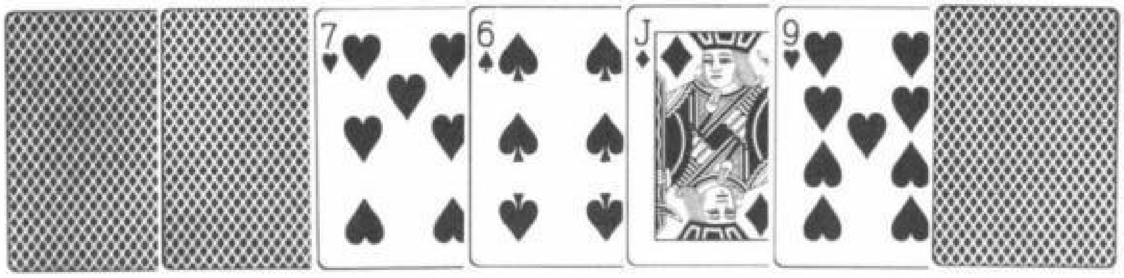
9. The Situation: The game is seven-card stud, high-low-split, declare. It is seventh street. You have



The man to your right has



The player to your left has



The man to your right bets. You are fairly sure that the player to your left will fold if you raise.

The Play: You should raise. This is the proper play even though you think the original bettor is going to declare low. Of course, if you knew he was declaring low, you should not raise in order to gain half a bet from the player's overcall. However, the small possibility that the bettor is also going high makes it worth trying to get the other player out, especially since he will likely declare low if you don't get him out. Now you have cost yourself half the pot. In other words, it is worth relinquishing a sure half bet if there is a decent chance you may gain half the pot.

Note: After reviewing these plays for the new edition, I realized that a few of these plays have now become somewhat debatable. This is especially true for questions number 2 and 3.

Poker, Bluffing, and Game Theory

"The Sklansky Bluffing System"

Clearly you cannot play poker by a "system," in the usual sense of that word, for as soon as he realizes you are playing by a strict set of rules a good player will be able to read your hand and will thereby have a tremendous advantage over you. To be a winning player, you must, mix up your play.

However, there is in fact a mathematical discipline that covers situations where you must alter your strategy. It even tells you how to alter it. Its name is Game Theory.

Game Theory is one of the ne\Ner branches of mathematics, which involves finding the best strategy for any game or game-like situation such as catching a thief or winning a war. It is most useful in analyzing games of imperfect information - that is, games in which the opponents' moves or holdings are not completely known. Card games are games of imperfect information, whereas chess and tic-tac-to~~ are games of perfect information.

Poker, a game of luck and imperfect information, is one of the best games to approach with the tools of Game Theory. Unfortunately, it is a monumental task to subject all the complexities of poker to Game Theory analysis. Strangely enough, though, bluffing is the one aspect of poker play that is fairly simple to analyze. This may seem ironic, since most players consider bluffing the least mathematical aspect of the game, but the invention of Game Theory has changed all that.

Before applying it to the bluffing in poker, I want to give a couple of examples of how Game Theory works. Consider the following game. I write the number one or two on a piece of paper as does my opponent. We cannot see what the other has written. We now compare numbers. If they match, I win $10. If they don't, he wins $10. On the surface this appears to be an even game.

However. if this game is repeated time after time, there may be an advantage to the player who can "outguess" his opponent and anticipate whether he will or will not change his previous number. The use of Game Theory will prevent an opponent with good judgment from outguessing you. With its help you would gladly play this game with anyone if they offered any kinds of odds. You simply play the Game Theory optimum strategy, which is to flip a coin to determine what number to write down. It now must be an even game.

This game illustrates two important concepts associated with Game Theory. One is the concept of a mixed strategy, which means a changing strategy, the use of two or more alternative strategies, according to some mathematically determined proportion. (In the previous example, there were two possible strategies: writing I or writing 2, and the proportion was 50-50.) However, this proportion must be implemented randomly in this case, by flipping a coin.

The second point to remember is that Game Theory is most effective only if your judgment could not do better. If, for instance, you knew you could predict your opponent's actions, it would be silly to bring it down to an even game by flipping a coin. This notion will come up again when we discuss bluffing.

Consider a second example. Once again my opponent and I write down a I or a 2 without showing what we have written to the other. But this time the payoffs are a little different. If we match, I win $20. If we don't match. I pay him $10 if I wrote I and he wrote 2. If I wrote 2 and he wrote 1, 1 pay him $30. Once again this appears to be an even game, and in fact it would be if I wrote as many I's as 2's. However, Game Theory tells me to use a different mixed strategy. It says to write I five-eighths of the time (randomly, of course). (To explain how I arrived at these figures would be beyond the scope of this article.) By playing this way, I assure myself of a profit in the long run. To illustrate, let's say my opponent wrote 8 ones while I wrote 5 ones and 3 twos. In this case, I would lose $30 3 times and win $20 5 times for a net profit of $ 10 or an average of $1.25 per decision. If instead he wrote all twos, I would win $20 three times and lose $10 5 times, which would also result in a net profit of $10 after 8 decisions or $1.25 per decision. Though it may not be immediately apparent, the fact that this strategy earns me $1.25 per decision when he uses one strategy or the other exclusively will assure me of an average of $1.25 per decision no matter how he mixes up his play. Game Theory only insures me a profit. not him. as the game is "stacked" in my favor. No other mixed strategy can do better.

In this example. as in the previous one, a random mixed strategy NNill assure that you get what's coming to you. Remember that it is important to be random. You do not write 5 ones for every 8 decisions except on average. While you can no longer flip a coin to determine your decision. you should use some kind of randomizing device such as the second hand of your wristwatch. Otherwise. your opponent can pick up a pattern. which is exactly what you can't let him do.

The Game Theory strategy is called the "optimum strategy." However. it is not always the best strategy. There may be a better strategy if your opponent is predictable. Thus, if in the second example you knew that your opponent would write mostly ones you could improve your edge (to. let's say. $1.75 per decision) by writing ones yourself more than five-eighths of the time. However, if your judgment were wrong, you might find yourself losing, while the optimum strategy guarantees you $1.25 per decision no matter \\hat your opponent does.

We are now almost ready to see how the foregoing concepts apply to bluffing in poker. When I speak of bluffing here, I am really speaking of only one kind of bluff, that is when you are up against one opponent, all the cards are out, and you have no chance of winning in a showdown. Whether you should bluff depends on three things: the size of the bet, the size of the pot, and the chances your opponent will fold if you bet.

A bluff is indicated if your chances of getting axvay with it compare favorably with your pot odds. Thus, if there is $ 100 in the pot and the bet is $20, you are getting 5-to-I odds on your bluff. You should try it if it will work better than I out of 6 times.

The problem, of course, is determining the chances that your opponent will fold. If you have very good judgment, your opponent is easy to read, and you know him well, your experience may help you to make a good prediction of his folding probability. There will be times when you know the chances are virtually nil and therefore should not bluff except to set up a play for the future. At other times it will be apparent that he will fold far more often than is necessary for you to show a profit, and so you must bluff.

Game Theory techniques are not usually indicated if there is an obviously correct judgment play. It should be used only if:

A. You have no idea what the chances are that your opponent will fold because you opponent is a stranger or an expert with as good or better judgment than yours: or

B. You assess that the chances he'll fold are right around the breakeven point, and thus, you can't accurately judge whether to bluff.

I should explain at this point that the Sklansky Bluffing System applies only to those situations where your opponent knows that he can only beat a bluff. In other words, if his hand is so strong that he might call you simply on its own value, the system is invalidated. To put it another way, you should not use these techniques if there is some chance that he will call because he thinks you might be betting a legitimate hand which is weaker than his. He must know that he can only beat a bluff and_ that if you're not bluffing you have him beat. A simple example of this situation arises in draw poker where he draws three cards and you draw one, obviously on the come for a straight or a flush. Neglecting the slight chance that he fills up, he can only beat a bluff if you bet.

If all the above criteria are fulfilled you should use the Sklansky Bluffing System. There are two parameters:

A. The pot odds your opponent is getting on your bet.

B. The chances that you have actually made your hand.

You now bluff in such a way that the ratio of your legitimate hands to your bluffs is the same as the odds he is getting. Thus, if you bet $20 into a $100 pot, your opponent is getting 6-to-I odds to call. If there is a 30 percent chance that you have made a hand, you should bluff 5 percent of the time. So of the 35 percent of the time you bet, it is 30-to-5 or 6-to-1 against your bluffing (the remaining 65 percent of the time you simply give up and check).

To give another example from draw poker, let's say there is $100 in the pot. The after-the-draw bet is $50 and you are drawing one card to a flush. He has drawn three cards and checked to you. (For the sake of simplicity we will neglect the slight chance that he has made a hand that will beat a flush.) We first notice that he is getting 3-to-1 odds if we bet ($150-to-$50). This means that our ratio of legitimate bets to bluff must also be 3-to-1. Since the chances of hitting the flush are 9/47 (9 cards to hit out of the remaining unseen 47), the chances we are bluffing should be 3/47. Thus, out of 47 similar situations, we will check 35 times, bet a flush 9 times, and bet a bust 3 times on average. Once again our decision is random.

A simple way to make this random decision is to use the card you caught on the end as a randomizing device. In this example, let's say you were drawing to four hearts. Just pick 3 other cards such as the ace, deuce, and trey of clubs to bluff with. You bet if you catch a heart or 1 of those 3 cards. You now can be sure of bluffing with an exact probability of 3/47. All you need to do is pick the correct number of "bluing cards" so that the ratio of cards that make your hand to bluffing cards is as close as possible to the odds your opponent is getting. (It won't always work out exactly using this method. If. for instance, he is getting 4-to-1 odds and you have 9 cards to hit, you would need 21/4 bluffing cards to keep this ratio. Just pick 2 bluffing cards in this situation as it is not essential to be perfectly accurate.)

To show why this Game Theory technique is your optimum strategy, let's go back to that four-flush example. You bet 12 times out of 47. 3 of those bets are bluffs. If your opponent folds every time you bet, you have made $1200 after 47 hands on average ($100 in each of the twelve situations).

$$\$1,200 = (12)(\$100)$$

If he calls every time, you win $150 9 times and lose $50, 3 times after 47 hands. This once again results in a net profit of or $25.53 per hand.

$$\$1,200 = (9)(\$150) - (3)(\$50)$$

$$\$25.53 = \frac{\$1,200}{47}$$

There is no sway he can prevent you from averaging this amount. However, if you tried to bluff too often, your opponent could cost you by always calling. On the other hand if you didn't bluff enough, he could save money by folding when you did bet. If you use the Sklansky Bluffing System, even a world champion couldn't outguess you because any strategy he uses results in the same gain to you.

Now I'll review the major points of the Sklanskv_ Bluffing System.

1. Only consider this technique when there are no more cards to come.

2. Your hand must have no chance to win in a showdoxNn. (When you do have a chance to win a showdoNNn, you still may use Game Theory bluffing techniques, but the strategy; would be different from that presented here.)

3. Use Game Theory to bluff only if you feel that your judgment will not give you a better pure strategy. Only use Game Theory if you think the situation requires a mixed strategy.

4. This technique is only accurate if it is clear that your opponent's hand is not so strong that he would call even if he didn't think you were bluffing. The Sklansky Bluffing System assumes that he can only beat a bluff and that if he does call you, it is only because he hopes you are bluffing, not because his hand may be better than yours.

5. Once you have decided that this technique is called for, you make note of the size of the pot and the size of your bet to determine the pot odds your opponent is getting if you do bet. Remember to count your bet as part of his odds.

6. Determine the chances of making your hand before you get that last card. These chances in comparison to the chances that you will bluff should be the same ratio as his pot odds. For example: there is $100 in the pot, you can bet $40, and there is a 21 percent chance that you'll make your hand. Since he is getting $140-to-$40 or 7-to2 when you bet, there should be a 6 percent chance that you are bluffing since 2 1 -to-6 equals 7-to-2.

7. The best way to bluff the proper percentage of the time in a random manner is to pick some "bluffing cards" in your mind before you look at the last card. You bet if you catch a card that makes your hand or if you catch one of your bluffing cards. The number of bluffing cards is chosen such that the ratio of good cards to bluffing cards is as near as possible to the pot odds your opponent will be getting if you bluff. In the previous example, if there were fourteen good cards for you, you would pick 4 bluffing cards since 14-to-4 equals 7-to-2.

And there you have the Sklansky Bluffing System.

Playing Tight

Almost all poker writers (including me) recommend tight play on the first betting round of a poker hand, yet many of the most successful players don't take this advice. So what really is the best strategy before the flop in hold 'em, before the draw in lowball, or on the first three cards of stud, razz, or hi-lo? Why is it that writers say to throw away 80 to 85 percent of your starting hands while many pros only throw away about 70 percent?

To answer this question, you must understand that there are quite a few conditions about a particular game which could make it correct to enter more pots than usual. I will list them and discuss them.

1. The ante is especially large compared to the future betting limits.

2. The initial bet is small compared to the ante.

3. The initial bet is small compared to future bets.

4. You play especially well.

5. Your opponents play especially poorly.

6. The game is short-handed.

7. The game has many rounds of betting.

8. "Second class" hands are only small underdogs to "first class" hands.

If even one of the factors exist, it may be right to loosen up your starting requirements. Obviously, if the ante is high, you cannot expect to play extremely tight and win. The money you make on the hands you play does not figure to make up for the antes you will have lost while waiting for those hands.

Not as obvious is the fact that you should also play more starting hands even in a smaller ante game if the opening bet is also small in comparison to the later bets. In Vegas, $1-$3 games typically have no ante and a 50-cent opening bet.

You must compare this 50 cents, not to what is in the pot now, but rather to what can be made in future bets. I coined the term "implied odds " many years ago to describe this concept. However. it is important that the not-so-good hands you pick to play have the possibility of turning into very good, or very profitable, hands. This also applies to games with an escalating limit and even more so to no-limit games. If you can get in cheaply, you can go ahead and play so-so hands if you think they might turn into a big money-maker. However, you should be careful with this strategy if you are not positive you have the skill to pull it off.

The more skillful you are compared to your opponents, the more you can get away ss ith "splashing around " at the beginning of a hand. It is for this reason, rather than the others, that many great players play a lot of hands. They are willing to give their opponents a little "head start" with the expectation of "catching up" on future betting rounds. I think that most of them overdo it. It won't work if later betting rounds are not much larger than the first round, and it wont work if the opposition is in the same class. However, these players play so well that they still are big winners playing too many hands. They don't realize that they would do even better if they tightened up a bit. One time even great players don't significantly loosen up from the "book's " opening strategy, is when playing any form of draw poker. One reason is that there are only two round of betting. Thus, there is not time to "catch up " with extra skill or to get highly implied odds. Conversely, some of the crazy home games with many rounds of betting and maybe a "replace " or two are set up so that the better players can see a card no matter what the starting hand.

Another reason why you can't get rambunctious when playing draw is that a lesser hand will draw out much less often on a better hand than in other types of poker. Thus, a pair of aces is about a 7to-2 favorite against a pair of kings in draw poker while in hold 'em ace-king offsuit is only about 8-to-5 against jack-ten suited. In Omaha, it is unlikely that one hand would be even this much of a favorite over another one. Thus great Omaha players are not that particular about starting hands when they can get in cheaply. The same would be true about great limit hold 'em players if it weren't for the fact of the large opening round bet in today's structure.

One other time to dispense with a tight first-round strategy is in short-handed games. The reasons are obvious. However, this is a treacherous game best left to the expert, shorthanded player. In fact, this whole subject of loosening up starting requirements is a dangerous subject that writers like to leave alone. Be aware that even the experts who correctly play, let's say twice the recommended number of starting hands, win only about 20 percent more than they would if they didn't play these hands. And their risk is at least four times as great.

Image

Let's settle this "image" question once and for all. There is no doubt that in most poker games, you can take advantage of the image you have been projecting up to that point in time. Only against inept unobservant opponents or experts who know you are apt to "change gears" does your image matter little.

However, that being said, it does not necessarily follow that a good strategy is to start a game off by projecting a wild image in the hopes that this will elicit calls later on in the game. This is because it may cost you too much money in the early going when you are "advertising," rather than playing your best game, in comparison to what you expect to gain in the future. Don't forget that there is no guarantee you will get the cards you need later on in the game to make the strategy pay off. It is true that this strategy, when done right, can be very profitable in no-limit or pot-limit games. After all, if you can set up a situation where you get called for a big bet just one extra time, when you have the nuts, that makes up for all the small loose-appearing bets you lost earlier. However, in a limit poker game it is debatable whether it is worth playing less than optimally early on in hopes of getting extra calls later. This is especially true because there are two very good alternatives.

One alternative is to project a tight, timid image and take advantage of that. Once again, developing your image costs youu some money, but you can get it all back in one fell swoop if this image allows you to steal a pot you otherwise could not have stolen. Your earlier tight, timid play may have cost you some bets, but now you figure to get everything back, plus interest, by stealing one pot!

This technique works especially well against fairly tough observant opponents who are capable of folding when they are sure they are beaten, even if the pot is large and it will only cost them one more bet. Against them, the play may be to bet out as a bluff, but a raise-bluff or even a check-raise bluff might work!

The main reason I tend to favor this type of image building is that when done right, it pays off so much better than the wild image strategy, especially in a limit game. Rather than hoping to get many extra calls if and when I do get a great hand. I simply need to steal one or two large pots any of those times I miss my hand. So this strategy is usually better in fairly tough games. (A word of caution however, your earlier extra tight, timid appearance should mainly be achieved by missing value bets where you are a slight favorite or by not playing marginal starting hands. Do not project this image by missing bets or raises where that error could cost you the pot. If you did this, you would be defeating your own purpose, which is to win a pot you otherwise couldn't.)

There are many games where it would be wrong playing anything but your best game. simply to take advantage of an image. However, this does not mean your image is not important. Rather. it means you should let the natural flow of cards and the game develop your early image and you take advantage of whatever that image happens to be.

For example, if you have been dealt very few good starting hands at first and thus have to fold an inordinate number of times, keep that in mind. You have projected a tight image whether you planned to or not and are now in a good spot to try a bluff.

Or the opposite situation may occur. In any case, the key is to constantly keep track of how you seem to be playing as far as your opponents' perception is concerned. Though you may simply be playing solid poker, it may look otherwise to them. So put yourself in their shoes before you decide on the best way to take their shirt.

Part Three
Blackjack

Blackjack Introduction

I have separated these five chapters from those on the other casino games for a very important reason. Blackjack is the one casino game (apart from poker) that can be consistently beaten. This is now a well-documented fact. There are now many good books that tell you how it can be done. I will not recommend any particular book except to say that the more recently written books tend to be slightly better. The serious gambler should read as many of them as he can.

All the good blackjack books are basically the same. They start you off with the basic strategy. which is how you should play your cards against any dealer up card if you are not keeping track of previously played cards (or if it is the first hand dealt and you have yet to see the other cards). Except for a few minor variations, all legitimate blackjack books and systems will give you the same basic strategy because there is only, one right one. If your book differs from the others on basic strategy it is simply wrong. (Unfortunately, there is one major work on gambling which has a completely incorrect blackjack chapter.)

The second step in beating blackjack is to learn how to keep track of the cards in some way. Each system does it somewhat differently but the basic principle is the same. Every counting method tries to determine whether there is a higher or lower proportion of big cards than normal. When there is a higher proportion of high cards the player usually has an advantage. The higher the proportion, the bigger the advantage.

Conversely. when the deck is proportionately light in high cards, the dealer has a greater-than-normal advantage. The main key in winning at blackjack is therefore to bet more when the remainder of the deck or decks is in your favor, and to bet less otherwise. A second aspect of winning play is to use your knowledge of the cards that are out to deviate your playing strategy from basic strategy on those occasional times that the `'count" indicates that you do so.

Development of Basic Strategy

The purpose of the following chapter is to give the reader a clearer understanding of why blackjack basic strategy as derived by computer. is what it is. What I have done is used the mathematical charts provided by Edward Thorp and Richard Epstein in their books to give specific examples of why one play is better than another against a full deck of cards. The reader should also be able to get a better idea of why certain plays may change based on the "count" while others do not.

Both players and casino management now agree that blackjack is the one casino game in which an expert player actually has an edge over the house. The fact has brought about a proliferation of blackjack systems. The good ones all work, though they may differ in their methods of "casing" the deck. There is one thing, however, that all these effective systems have in common, and that is the correct basic strategy.

As you probably know, basic strategy is the proper playing strategy "off the top of the deck," in other words, the right way to play your hand against the dealers up card, assuming you have seen no other cards. It is also the best possible strategy to play throughout the deck if you are not counting. (I might add at this point that basic strategy is not the same as the correct strategy at an "even" or "zero" count. The reason for this is that basic strategy does -count-the cards in your hand as well as the dealer's up card before dictating a decision. Thus, for example, while it is incorrect to double down on a total of 9 against the dealer's deuce when the count is zero. basic strategy calls for it because of the three low cards that are out (your's and the dealer's.)

How did the basic strategy originate? Contrary to popular opinion, Edward Thorp was not the first person to derive it. In 1956 four men, Roger Baldwin, Wilbert Contey, Herbet Maisel and James McDermott, working with old-fashioned hand calculators, came up with the correct basic strategy for one deck with only a few minor errors. Later, slight adjustments in basic strategy were developed to take into account multiple decks and rule variations such as whether the dealer hits a soft 17 and whether you can double down after splitting a pair. The work of Baldwin, Contey, Maisel, and McDermott has subsequently been verified by many different researchers working with high-speed computers.

Calculations and computers represent the two methods for determining correct basic strategy. One method is calculation and the other is simulation. Let us take the case of our total of 16 versus the dealer's ace showing. We are only concerned about those times he doesn't have blackjack. How should we play this hand? Either way looks bad. However, the question is simply which way will win more often. While a computer could use the method of calculation to answer this problem, it is far easier to have it simulated. In a simulation, the computer actually "deals out" a very large number of hands to itself by the use of a randomizing device. It plays the hand both ways, hitting and standing, and records the results. The right play is that which shows the greater profit or smaller loss.

Calculation is much more laborious. When figuring the problem mathematically, as those original four men did, one must take into account every possible combination of cards that could be dealt, calculate its probability and determine whether it is a win, loss, or draw. In the present case of 16 versus an ace, you might first determine the probability of winning by standing, which is the same as the probability that the dealer will bust with an ace showing, given he doesn't have a 10 in the hole. Assuming he stands on a soft 17, you are immediately dead if he has a six, seven, eight, or nine in the hole. Of the remaining approximately 5/9 of the time when he has an ace through five in the hole, we must figure all the possible combinations of cards he could catch that would result in his busting and then calculate the total probability. This is a monumental task, especially with an ace showing. In any case, your chances of winning by standing work out to about 17 percent of the time he doesn't have a blackjack.

To calculate how you would do if you hit your 16 is even more difficult. First notice that you will lose immediately about 8/13 of the time when you catch a 6 through king. (The probabilities mentioned are approximate because they depend to a small degree on the cards that make up your 16.) The remaining 5/13 of the time, you will not automatically yin. either. For instance, of the 1/13 of the time that you caught a 3, giving you a total of 19. we must calculate the chances that this 19 wins (because the dealer busted, or made. or had 17 or 18) and then multiply the probabilities. This must be done for each of the non-busting cards you could catch and then combined to give the total probability of winning (after adjusting for ties).

After doing all this, we would find that even though you would catch a non-busting card to your 16 about 381/2 percent of the time. you would only in about 25 percent of the hands (counting ties as half a win). Still, this is quite a bit better than the 17'/2 percent you would win by standing. Thus, hitting is clearly the right play. This play, incidentally, is one of those computed incorrectly by John Scame in his Complete Guide to Gambling. The error results from his incorrect handling of the dealer's possible naturals.

The main difficulty in calculating these problems is their extraordinary complexity given all the possible sequences of cards. To get around this, approximations are sometimes made. For instance, it is unlikely that Baldwin and his colleagues figured the chances of the dealer giving himself A. A. A. 3. 6, A. 2, 2 in that order. It is such a remote possibility that it can be neglected. Still, the chance exists, and omitting it keeps the results from being one hundred percent accurate. There is also the possibility of errors being made when using the calculation method.

Simulation therefore, has become the much-preferred method of solving blackjack problems. The reader may N\ onder about the accuracy of this technique, since it is subject to deviations from the true values due to the luck involved in dealing out hands. This is a good point. but you must remember that the computer deals out millions of hands for each situation. The theory of statistics tells us that it is almost impossible to be off more than one-tenth of one percent with that many trials.

We have seen that the basic strategy is devised by comparing plays in any situation, not by just analyzing one play. It is important to realize that reducing losses is just as important as maximizing wins. Standing on 16 against an ace wins about 8 percent less often than hitting the hand, and this 8 percent is the important figure to consider. It is a pretty big number. Let's look at a few other plays this way.

15 vs. 3	Standing wins 38% Hitting wins 29%	Standing gains 9%
12 vs.5	Standing wins 43% Hitting wins 41%	Standing gains 2%
16 vs. 6	Standing wins 42% Hitting wins 31%	Standing gains 11%
12 vs. 3	Standing wins 38% Hitting wins 38½%	Hitting gains ½%
A7 vs. 4	Standing wins 60% Hitting wins 57½%	Standing gains 2½%
A7 vs. 10	Standing wins 41% Hitting wins 43%	Hitting gains 2%
14 vs. 9	Standing wins 24% Hitting wins 29%	Hitting gains 5%
16 vs. 7	Standing wins 25½% Hitting wins 31%	Hitting gains 5½%
16 vs. 10	Standing wins 23½% Hitting wins 24½%	Hitting gains 1%

These figures should be accurate to about one half of one percent. They assume you are playing against one deck and that the dealer stands on a soft 17. Obviously, the correct basic strategy play in these cases is the one that will win more often. Thus, you would hit a 12 against a trey and stand with a 16 against a six.

As I already mentioned, the most important factor in these decisions is how much you gain when playing your hand the right way as compared to the wrong way. The less you gain, the closer the play. This is relevant because the closer the play, the more likely it is that you will deviate from basic strategy if you are counting. Counters find that they must often hit a 12 against a five or stand on 16 against a ten when their count so indicates. Very rarely, however, will you find them hitting a 16 against a six. These charts also explain why it is a much closer play to hit a 16 against a ten than to hit it against a 7 - something that confuses many players. It's not that you win more often when standing against a ten, but rather you win more often when you hit against the 7. When you hit 16 against a ten, it is much more likely that you will catch a small card yet still lose.

All normal hitting and standing decisions are derived this way. However, when analyzing double down or splitting decisions, there is a fly in the ointment. To catch this fly, we must turn to the concept of mathematical expectation, a subject I treat thoroughly earlier in this book.

Simply put, expectation is the amount a bet is worth. It is the amount you will average, winning or losing per bet in the long run. Thus, if I bet $1 (even money) in a situation where I will win 57 percent of the time, I have an expectation of 14 cents. (After 100 plays, I figure to win $57 and lose $43 for a net profit of $14 or 14 cents per play.)

$$\$14 = (\$1)(.57)(100) - (\$1)(.43)(100)$$

$$\$0.14 = \frac{\$14}{100}$$

A $2 bet in this situation would earn me 28 cents, a $ 10 bet $1.40. If I am a 40 percent shot, as in the case when standing with a 16 against a 4, I have an expectation of -20 cents. One way to analyze blackjack plays is by comparing expectations. We didn't use this method in the preceding chart, which simply stuck to percentages. If you want to change the percent gained in the first chart to pennies gained, simply double the percent gained. In other words hitting 16 against a seven gains 1 1 cents.

When evaluating doubling down or splitting plays, we must think in terms of expectation rather than simply the chances of winning. It is better to bet $2 on a 59 percent shot and earn 36 cents than to bet $1 on a 62 percent shot which earns only 24 cents. This is analogous to the situation that frequently occurs when analyzing doubling down plays. You will never win more often by doubling down. In fact, only in the case of 10, 11. or ace-six against a small card will we win just as often when we double down since in these cases. we would always take one and only one card anyway (as we must when \\e double down). In the case of let's say nine-deuce against a seven, we will not win as often when we double down since we have relinquished our option to take another card if the first card is a five or less. Any time we double down with a 9. we have hurt our chances when we catch a deuce. Nevertheless, the opportunity to double our bet frequently makes up for a decline in winning chances, as we still increase our expectation. Let's look at some examples:

11 vs. 9	Hitting wins 57% (+14¢) Doubling wins 55½% (+22¢)	Doubling gains 8¢ (for $1 bet)
11 vs.5	Hitting wins 68% (+36¢) Doubling wins 68% (+72¢)	Doubling gains 36¢
11 vs. A (no natural)	Hitting wins 58½% (+17¢) Doubling wins 55% (+20¢)	Doubling gains 3¢
10 vs. 10	Hitting wins 51¾% (+3½¢) Doubling wins 50¼% (+1¢)	Hitting gains 2½¢
10 vs. 9	Hitting wins 56% (+12¢) Doubling wins 54% (+16¢)	Doubling gains 4¢
10 vs. 2	Hitting wins 61% (+22¢) Doubling wins 60¾% (+43¢)	Doubling gains 21¢

(The reason we don't win quite as often when doubling with 10 against a deuce as we do when hitting is that we would hit again if we caught a deuce.)

9 vs. 7	Hitting wins 60% (+20¢) Doubling wins 54½% (+18¢)	Hitting gains 2¢
9 vs. 4	Hitting wins 59% (+18¢) Doubling wins 57% (+28¢)	Doubling gains 10¢
10 vs. 2	Hitting wins 54½% (+9¢) Doubling wins 52¾% (+11¢)	Doubling gains 2¢
8 vs. 3	Hitting wins 51% (+2¢) Doubling wins 48% (-8¢)	Hitting gains 10¢

Remember that the closer the play, the more likely the count would change it. It is often correct to double down on 10 vs. ten or 9 vs. seven. Similarly. it is frequently right not to double down on 11 vs. an ace. Rare. however. are the times you wouldn't double on 11 against five.

It is often correct to double on soft 13 through soft 18 when the dealer shows a little card even though this will usually decrease our winning chances. In the case of ace-deuce through ace-five, doubling has cost us the opportunity to hit again if we catch a baby card. With a soft 19 ww e win more often by standing. For instance, in the first chart ww e see that with ace-seven against a 4 we win 60 percent by standing and only 57'/2 percent by hitting. However, by doubling down we translate this latter figure to an expectation of 30 cents rather than the 20 cents we earn by hitting. Doubling down in this situation is clearly the right play. However, only in the case of ace-six does doubling down not decrease our winning chances. Here are some more examples:

A2 vs. 5	Hitting wins 58% (+16¢) Doubling wins 55¼% (+21¢)	Doubling gains 5¢
A3 vs. 3	Hitting wins 52% (+4¢) Doubling wins 50¼% (+1¢)	Hitting gains 3¢
A3 vs. 7	Hitting wins 53% (+6¢) Doubling wins 45½% (-18¢)	Hitting gains 24¢
A4 vs. 4	Hitting wins 53% (+6¢) Doubling wins 52% (+8¢)	Doubling gains 2¢
A5 vs. 6	Hitting wins 55¾% (11½¢) Doubling wins 55½% (+22¢)	Doubling gains 10½¢

(Here the only loss from doubling would be when we caught an ace and couldn't hit again.)

A6 vs. 5	Hitting wins 57% (+14¢) Doubling wins 57% (+28¢)	Doubling gains 14¢
A7 vs. 6	Standing wins 63% (+26¢) Doubling wins 59½% (+38¢)	Doubling gains 12¢
A7 vs. 7	Standing wins 70½% (+41¢) Doubling wins 56% (+24¢)	Standing gains 17¢
A8 vs. 6	Standing wins 74% (+48¢) Doubling wins 62% (+48¢)	No difference
A9 vs. 5	Standing wins 84% (+68¢) Doubling wins 63½% (+54¢)	Standing gains 14¢

Once again, I must stress that a play should be judged solely in comparison to its alternative. If someone offered to give you an aceseven every time against his seven showing, with the stipulation that you must double down, would you take the deal? Since you know that doubling on ace-seven versus a seven is a bad play, you might decline his offer. However, you would be making a terrible mistake if you did.

Doubling on ace-seven against a seven earns you 24 cents for each original dollar bet (12 cents for each of the $2 you ultimately risk on this play). It is in fact quite a profitable play. The only reason it is thew rong play under normal conditions is that standing is even more profitable (41 cents per hand).

The preceding concepts must also be used in evaluating pair splitting. Unlike the case of doubling down, where it is only profitable to do so when it increases an already positive expectation, splitting may be correct because it decreases a negative expectation or even changes a negative expectation to a positive one. When we split, we are not only doubling our bet but also changing our hands. Thus. two 8s are always split, not so much because starting with 8 is that good, but rather because starting with a total of 16 is that bad. In fact, splitting two eights against a seven is the most important play in blackjack.

To put it another way. those players who hit (or even worse, stand) with two eights against a seven are making the worst play you are ever likely to see at a blackjack table. It is worse than hitting hard 18 against a ten! In the following examples, I will include only your expectation for each play and dispense with the percentages. Of course. one can be derived from the other. I assume that you can double down if you catch a good card after splitting. Without this rule, splitting becomes somewhat less profitable.

22 vs. 4	Hitting loses 3½¢ Splitting wins (2)(5½¢) = 11¢	Splitting gains 14½¢
22 vs. 7	Hitting loses 9¢ Splitting wins (2)(0¢) = 0¢	Splitting gains 9¢
33 vs. 9	Hitting loses 31¢ Splitting loses (2)(20¢) = 40¢	Hitting gains 9¢

33 vs. 7	Hitting loses 16½¢ Splitting loses (2)(3½¢) = 7¢	Splitting gains 9½¢
44 vs. 6	Hitting wins 17¢ Doubling wins 19¢ Splitting wins (2)(9¢) = 18¢	Splitting gains 1¢ over hitting, but is 1¢ worse than doubling if allowed
66 vs.4	Standing loses 15¢ Splitting wins (2)(2¢) = 4¢	Splitting gains 19¢
77 vs.7	Hitting loses 39¢ Splitting loses (2)(3¢) = 6¢	Splitting gains 33¢
88 vs.5	Standing loses 13¢ Splitting wins (2)(6½¢) = 13¢	Splitting gains 26¢
88 vs 7	Hitting loses 37¢ Splitting wins (2)(13¢) = 26¢	Splitting gains 63¢
88 vs. 10	Hitting loses 51¢ Splitting loses (2)(22¢) = 44¢	Splitting gains 7¢
99 vs. 7	Standing wins 40¢ Splitting wins (2)(18¢) = 36¢	Standing gains 4¢

99 vs. 8	Standing wins 6½¢ Splitting wins (2)(10¢) = 20¢	Splitting gains 13½¢
99 vs. 9	Standing loses 19½¢ Splitting loses (2)(4½¢) = 9¢	Splitting gains 10½¢
1010 vs.5	Standing wins 67¢ Splitting wins (2)(26¢) = 52¢	Standing gains 15¢
AA vs. 8	Hitting wins 9¢ Splitting wins (2)(20¢) = 40¢	Splitting gains 31¢

Thus we have some good examples of basic strategy and how it is derived. Even without counting, playing pure basic strategy and therefore maximizing your typical expectation for any play gives you at least an even game against one deck with Vegas rules. While the foregoing concepts may be somewhat difficult to master, a full understanding of them will make it that much easier to forge ahead to become a world-class blackjack player.

Why Count?

Once you have mastered blackjack basic strategy, you are playing very close to an even game. The next step is to learn to play in a way that \tiill actually give you an edge over the house! In order to do this you must keep track of the cards in some manner. There are many ways of keeping track and I will be going into most of them in the next chapter. In this chapter I will explain why "counting" the cards can give you the advantage.

There are two basic reasons to count. One reason is so that you know how to play your hand. Secondly. so that you know how much to bet. It is usually the second reason that is more important. Even if you stick to basic strategy in all situations, you can still have an edge over the house by altering the size of your bet.

There are many types of counting systems that are presently in use. However, they almost all have one thing in common. Evenlegitimate counting system's basic idea is to determine the proportion of high cards left in the deck as compared to the low cards.

Not all counts consider the same cards as "high" or `'low." However, tens are always considered high and fours, fives, and (usually) sixes are considered low in all counts. Certain cards (usually deuces, sevens, eights, or nines) are sometimes considered neutral. Aces are frequently counted as high cards but they are in a special category as will be explained shortly.

Let us first see how the count is used to determine whether the player has an edge (for the upcoming hand). It works out that when there is a higher-than - normal proportion of high cards left in the deck, the player probably has an edge. By high cards I mean in this case, aces, tens. and much less importantly, nines. Why is this so? There are a few reasons.

Probably the most important reason why a higher proportion of big cards helps the player is that blackjacks become more common. Of course the dealer will also receive more blackjacks in this situation but he only wins even money while you get paid 3-to-2 odds for your "naturals." Also you will sometimes not lose to the dealer's blackjack as you will have many opportunities to make a profitable insurance bet when the dealer has an ace showing. Notice that your edge from extra blackjacks can best be taken advantage of if your count includes aces as high cards (or a "side count" of aces are kept) and does not include nines. Unfortunately, the second reason why a high proportion of high cards helps the player can best be taken advantage of if nines are counted and aces are ignored.

When there is a preponderance of tens (and to a lesser extent, nines) and a scarcity of small cards left in the deck this helps the player for reasons other than the higher possibility of being dealt a blackjack. It helps him because of the fixed rules that the dealer must follow. The player on the other hand may play as he likes.

The player. for instance, may split a pair or double down on his first two cards. options the dealer does not have. The most important option. however, that the player has (especially when the deck is rich in high cards) is the option to stand on a "stiff' (12 through 16). This helps the player dramatically.

For instance, if the player is dealt 13 against the dealer's five showing he normally only has about a 42 percent chance of winning when he makes the basic strategy play of standing. However, when the deck becomes "rich" he may very well be over 50 percent to win the hand with a 13 vs. five since the dealer may actually be favored to bust! Thus, the second major reason why decks with a high proportion of high cards helps the player is because his option to stand on a stiff becomes more valuable. (Notice that the fact that the player will be dealt more 19s and 20s than normal does not add to his edge, as the dealer will also be dealt more of these good hands. Notice also why extra aces in the deck do not help the player as far as this second reason is concerned.)

There is a third, less important, reason why a rich deck helps the player. This is, splits and double downs become more profitable especially against small dealer upcards. However, this is somewhat balanced out by the fact that they occur slightly less often with less low cards in the deck. As far as this third reason is concerned, extra aces are usually helpful. However, there is a major exception. You don't want extra aces in the deck when you are doubling down on 11.

The second advantage of keeping track of the cards is so that you know how to play your hands. The basic strategy play is correct about 80 percent of the time. If you are using a good count, you will usually know those times when You should deviate from basic strategy. This should add about another one half of one percent to your edge. Examples of when to deviate from basic strategy would be to hit 13 vs. a trey or refrain from doubling on 10 vs. eight when the deck is "poor." Other examples would be to stand on 16 vs. ten, double on 9 vs. seven, or by insuring against an ace, when the deck is "rich."

Knowing when to deviate from basic strategy is more important when playing against single decks rather than multiple decks. It is especially important if you are forced to use a small "bet spread." This means that your maximum bet is not much larger than your minimum bet. Betting this way does not yield the highest percentage advantage. However, professional blackjack players frequently have to play this way, especially against single decks, in order to avoid detection. When they do play this way, a good part of their edge comes from their strategy changes rather than their betting changes.

Unfortunately, there is a problem. It works out that the counting methods that are most accurate in detecting betting advantages are not as good for strategy purposes. Conversely, those counts which are best for strategy, lack something when it comes to spotting edges. (One of the main reasons for this is the dual nature of aces. They should be counted as a high card for betting, but not for playing.) Thus it is important to know how and when you will be playing and under what conditions. You should then choose a counting method that will be most appropriate. How to make this choice will be covered in the next chapter.

Which Count?

Once you have made the decision to become a counter, the next step is to decide which count to use. There are three main factors that are important in making this decision. They are:

1. How accurate is the count far as betting decisions are concerned`?

2. How accurate is the count as far as strategy decisions are concerned"

3. How simple is the counting system to apply?

Unfortunately, there is no system that maximizes all three factors. Systems that are best for betting accuracy are not as good for strategy and vice versa. Likewise, the simpler counts are usually not quite as accurate as the more complicated ones. Thus your decision will be based on the type of game you will be playing against as well as your ability to use a difficult count (which in turn may depend on the number of hours you plan to play per day and also against how many decks). Expert counters will sometimes learn more than one count in order to use the best one for each particular situation.

Nowadays almost all professional blackjack players use some type of "point count." However, before describing and evaluating point counts, I should first mention the original count used in the first legitimate blackjack book (Edward Thorp's Beat the Dealer). I speak of the "Ten-Count." The Ten-Count involved the forming of a ratio of the non-tens left in the deck compared to the number of tens left in the deck. A single deck starts out with 36 non-tens and 16 tens which gives us a starting ratio of 36-to-16 or 2.25-to-1. This ratio changes as cards are played.

For instance, if the first 10 cards played were two tens and eight non-tens, the new ratio for the remaining cards would be 28-to-14 or 2.00-to-1. It is this non-tens-to-tens ratio that is used to determine when you have an edge and also how to play your hand. For instance, you have a small edge when the ratio is 2.00. Your edge increases as the ratio decreases. At 1.50 your edge is a few percent. Conversely, at ratios above 2.25 you figure to have a disadvantage. Thus, you tend to make relatively small bets at ratios above 2.00 and to increase your bets as the ratio falls below this point. (The ratio will almost always be between 1.0 and 5.0).

The Ten-Count ratio is also used for strategy decisions. For example at ratios above 2.5 you would hit 12 against a four, five, or six, hit 13 against a deuce, stop doubling down on 10 against a nine or 11 against an ace, as well as making other deviations from basic strategy.

The Ten-Count has recently fallen into disrepute. It is claimed that it is too difficult to keep track of the cards this way and that it is inaccurate. However, the fact of the matter is that the strategy accuracy of the Ten-Count is greater than that of the widely used Hi-Lo Count. It is 'Weaker for betting purposes but against a single deck it can still be very useful if the dealer will deal far down into the deck or if you don't vary your bets very much.

It is also not true that the count is all that difficult to use. At least 10 percent of the population has the arithmetical skills to learn it. All in all, I think the Ten-Count has been given a bad rap. In fact, I still usually use it myself when I play the single deck game up in Reno.

The effort to make counting simpler resulted in the development of "point counts." When a player uses a point count he assigns certain values to each of the cards. For instance, in the previously mentioned Hi-Lo Count, aces and tens count as minus one and deuces. treys. fours, fives, and sixes count as plus one, as they come out of the deck. Sevens, eights, and nines have no value. The deck starts out with a count of 0.

If the first five cards out of the deck are a four, two fives, an eight, and an ace, the count would now be "+2." The player usually has an edge at a count of +1 or higher. He usually is at a disadvantage at a negative or even count. The higher the count the greater the edge (another factor, however, is the number of cards left in the deck. A count of "+4" is much stronger for the player if there are 20 cards left than if there are 40. Experts adjust for this by using what is called the "true count" but I won't go into it here).

A second simple count that is sometimes called the Hi-Opt I. counts treys, fours, fives, and sixes as and tens as -1. Aces and deuces are now counted as 0 along with sevens, eights, and nines. Once again a positive count is usually an edge for the player. Notice that the cards out of the deck in the previous example that results in a "+2" count using the Hi-Lo results in a count of "+3" using the Hi-Opt 1. This is because the Hi-Opt I doesn't count the ace. (In this case the Hi-Opt would somewhat overestimate the edge.)

Many point counts use other numbers besides plus and minus one. The following two, summarized in chart form are sometimes called advanced point counts.

A	2	3	4	5	6	7	8	9	10
-4	+2	+3	+3	+4	+3	+2	0	-1	-3
0	+2	+2	+3	+4	+2	+1	0	-2	-3

The first count is more accurate for betting purposes while the second one is better for strategy decisions. The reason once again is due to how the ace is handled. It was once thought that the extra difficulty associated with these counts was worth the accuracy it gave. However. this now appears not to be true.

Recent research has shown that these two card counts are only very slightly better than the Hi-Lo and Hi-Opt I and thus should only be considered by those rare individuals who can handle these large numbers with ease. In fact, it appears that the second Advanced Point Count, where the ace is counted as zero, should never be used. This is because there are simpler counts involving only zeros, one and twos that are equal or better in every respect. For instance, the count in the following chart is just as good and easier to do.

A	2	3	4	5	6	7	8	9	10
0	+1	+1	+2	+2	+1	+1	0	0	-2

This count is just about the best one you can use against a single deck especially if you keep a side track of aces for betting purposes. (Of course, it is important that you obtain an accurate strategy chart so you can know at which count you start deviating from basic strategy in different situations.) However, there is nothing wrong with using the Hi-Lo or Hi-Opt I even against single decks if the extra ease allows you to play longer with less errors. You should also use as simple a count as possible if you are keeping a side count of specific cards such as sevens and eights for advanced strategy decisions (see next chapter).

If you are plying against a shoe where strategy decisions are not nearly as important as betting decisions. (since the casinos are more Milling to let you jump your bets more drastically against a shoe) it is important to use a point count where the aces are counted.

The first Advanced Point Count is almost perfectly accurate for betting purposes. However, most successful pros that I know are content to use the slightly less accurate Hi-Lo Count against a shoe because it is so much simpler to use for hours on end.

To summarize:

A. If your advantage against the casino will come mainly from bet variations use a count that counts aces. This is almost certainly your best option when playing against a shoe.

B. If you are playing against a single deck where you are using little or no bet variation use a count that doesn't count aces, possibly even the Ten-Count.

C. Unless you have a computer mind or are playing very short sessions it is almost certainly better to use the simple Hi-Lo or HiOpt I count (or one of a few other good counts that involves only zeroes and ones). They are only slightly less accurate than more advanced counts and the relative ease in learning and applying them make them the best choice for most players.

Key Card Concept

The following chapter was originally written in 1977. It can be extremely important to the serious blackjack player when he is playing against a single deck. Unfortunately, it was initially misinterpreted by some people when it first came out. It is not a complete system. Rather it is an idea upon which a more detailed system can be based.

The major principle is that there are times when the proportion of one or two denominations of cards have far more importance on your playing decisions than the "count" has. Obviously, a card that will make you 21 is important. Frequently however, there are other cards that may be even more important. These I call key cards. The following is an introduction to how they can be used to increase your profits in single deck blackjack.

This chapter will serve as an introduction to some new and very powerful techniques to beat one-deck casino blackjack. They have been independently developed by a handful of mathematicians and computer experts. I have integrated them into one system as well as adding a few of my oven discoveries, notably my Key Card Concept. While these techniques are very difficult to master, the rewards will justify the effort as they generate profits at a rate at least twice that of normal counting procedures. Not only that, these methods require little if any fluctuation in your betting and will therefore not bring as much heat down on you. Finally, since your edge doubles, the bankroll you need to avoid ruin is reduced drastically. Even if you cannot master these techniques thoroughly, some basic knowledge of them cannot fail to increase your edge.

For my non-counting readers, I must first digress into history before describing these advanced concepts. The original system to beat blackjack was based on keeping track of the cards in such a way as to determine the proportion of tens remaining in the deck, especially as related to the number of fours, fives, and sixes remaining. It was also important to determine the proportion of aces left.

If there were a higher-than-normal proportion of aces and tens left in the deck, the player who kept a count and knew how to play his hand actually had an advantage over the house. There were two main reasons for this:

1. The player would be dealt more blackjacks than usual. (The dealer would also get more blackjacks, but he wasn't getting paid 3-to-2 as the player was.) In addition, the player could advantageously take insurance when the dealer showed an ace.

2. The dealer would bust much more often when he had a small card showing, thereby enabling the player to take greater advantage of the fact that the dealer had to hit 16 or less, while he could stand on these "stiffs." The player's double-downs and splits become more advantageous too.

When these situations occur, the deck is said to be "rich" and the counter will thus increase his bets. Though the deck is not rich the majority of the time, counters still have the edge over the casino since they bet high only when they have an advantage. A good counter using standard techniques should have an edge of about I percent if he varies his bets from one to four units and is playing against usual rules and conditions.

Besides using the count to vary his bets, the expert will also use this same count to decide how to play his hand. Of course, before he reaches this level of expertise, he must first know the basic strategy cold. This is the computer-derived strategy for playing any two cards against any up card off the top of the deck. It is usually the correct strategy whether a player is counting or not and can be found in any good blackjack book.

Playing pure basic strategy against one deck gives you about an even game. Therefore, if you count and vary your bets, you will have a small edge even if you don't vary your play. In fact, until recently your bet variation was the main reason for your edge. Still, it made sense to increase this edge somewhat when whatever count you were using indicated that you should. For instance, if the deck were ten poor, it would now be correct to hit a 13 against a trey even though basic strategy says to stand. To give another example, it would become correct to double down on a total of 9 against the dealer's seven if the deck was rich. In general. it is correct to double down more and stand more than usual against a rich deck, and conversely one should hit more and double less against a poor deck. The exact point at which you deviate from basic strategy depends on your hand. the dealer's up card. and the count.

The reason that these strategy variations do not add much to your edge lies in the fact that the count does not always give accurate enough information for all strategy decisions. For instance, in the 13 vs. trey situation the number of sevens and eights left is of more importance than the proportion of tens, but these cards are rarely counted. Advanced techniques have now changed this situation. Previously a good player who changed his strategy but not his bet size would not do as well as the player who stuck to basic strategy but altered his bets. These new techniques I am about to describe have changed all that.

Strategy changes can now account for about two-thirds of the edge in the hands of an expert. Even without changing his bet, he will have an edge well over half the time. Besides winning more than twice as fast, the expert now needs a much smaller starting bankroll in order to make the same initial bet. Finally, the expert now faces little danger of being barred from a casino since bet variations are the main source of heat. The only drawbacks are that you must play against one deck, which rules out most places not in Nevada, and that these advanced techniques are difficult to learn and apply. However, even a cursory knowledge of them should add to your edge.

Modern blackjack theory has shown that for many strategy decisions the count is not as good an indicator of the right play as the proportion of certain specific cards. Which cards are important depends on what you hold and what the dealer shows. Here are a few examples:

1. When all the sevens are gone, it is correct to stand on 14 vs. a ten.

2. When all the eights are gone, it is correct to double down on ace-trey vs. a deuce.

3. When all treys are gone, it is correct to double down on 9 vs. an eight.

4. When all the nines are gone, it is correct to stand on 12 vs. a deuce.

5. When all the deuces are gone, it is correct to stand on ace-seven vs. a ten.

There are many other situations in which apparently strange plays are. in fact, the proper play when all cards of a certain value are missing from the deck and everything else is equal. These plays were in fact computed quite a while ago by mathematicians, for they are simply basic strategy for depleted decks.

What was not known was how to change one's strategy when certain cards were scarce but not completely gone or when there were a higher than normal proportion of them.

Here are some examples of new strategy changes:

1. Don't double with a 10 vs. a nine if there are many deuces left.

2. Stand on 15 v's. a nine if there are many sevens left.

3. Doubling on ace-deuce vs. a six is no longer close if there are many sevens and eights left; it is now profitable even with a poor deck.

4. Split fours vs. a six with extra sevens left.

5. Double down on 9 vs. a seven with extra nines left.

At this point the reader is probably asking himself three questions: How can I possibly remember how many of each card are left in the deck? Even if I could remember every card, how do I use this information to adjust my strategy? How do I know which are the important cards to remember for a particular strategic situation?

It turns out that it is not all so impossibly difficult as it seems. While you need a good memory, it is not necessary to be perfectly accurate. You only need a general idea of how many of a particular card or group of cards are left. If you are not sure, just play your normal count strategy. What is important is that you have a full understanding of expectation and blackjack theory as well as some common sense.

In actual play what you would be doing is this. You would be playing your normal count preferably a simple one since you have a lot else to think about: any of the simple plus-minus counts will suffice. At the same time, you would be getting a general feel of the various cards that are out. Just counting the sevens, eights, and nines individually or even as a group will get most of the benefit of these techniques since these cards seem to crop up most often as the relevant cards. In fact, an exact count of sevens, eights, and nines along with the plus-minus count is probably better than a general idea of every card.

When you are faced with a strategy decision, you would then decide how "close" this decision is, which requires knowing some blackjack theory. In general, the closer your count is to the number in your strategy charts, the closer the decision is. If the decision is clear-cut, you simply play according to your count. If, however, the decision is close, you should now look at the situation and decide which particular card or cards you should know about. If these cards are sufficiently out of line from their usual proportion, it might swing the play away from the one indicated by the simple count. For example, if you have 16 vs. a ten, you should stand with all or most of the fives gone even when the deck is slightly negative and especially if the sixes are still in there.

Let us now see how to identify the relevant cards in any situation. First of all, the cards that will make you 20 or 21 are very important. When they are gone, you will now stand in situations where you might normally hit. If there is a higher-than-normal proportion of these cards left, you will now hit if the play is close. Seeing few sevens and eights gone halfway through the deck will frequently make it correct to hit 13 vs. a deuce or trey. It is also important to consider how a card will affect the dealer's hand. If there are many cards left that will give the dealer a 15 or 16, this may indicate standing in close situations. Similarly, his 17 or 18 cards may have some small effect on your decision.

We now come to key cards. which are generally the most important cards to consider. To determine whether a card is a key card, ask yourself two questions.

1. How would I play my hand if this card was on the top of the deck and would thus be my card if I took it?

2. How would I play my hand if I knew the dealer had this particular card in the hole'?

If the answer is the same to both questions, especially if neither answer is close, then the card is a key card. For example, suppose you are dealt a 15 against the dealer's nine. Which is the key card'? There can be more than one key card or no key cards, but in this case there is exactly one. While it is true that fives and sixes are important because they make you 20 and 21. they are not key cards. If you knew a five or six were coming off the top, you would hit your 15. However, if you knew the dealer had a five or six in the hole, you would stand. Thus, these are not key cards. Let's check eights. If one was coming off, you would stand. If the dealer had an 8 in the hole you would hit. No good. By now sharp readers will have picked out sevens as the key card since this is the only card that will result in the same answer (in this case stand) to both questions. Thus, a preponderance of sevens in this situation might sway a close decision toward standing, while a lack of sevens would sway it the other way.

When a key card is also your 20 or 21 card. I call it a super key card. Eight is a super key card when you have a 13 vs. a ten, and it can actually be correct to stand in this situation when all the 8s are gone. Super key cards have the greatest impact of all on your strategy decisions. If the cards involved make the dealer 16 or 17, they are also more important than the typical key card.

Let's look at some more examples:

1) vs. ten. Here six, not five. is the key card. Five is important as your 21 card. but in fact the rule "stand on 16 vs. ten when there are more sixes left than fives - otherwise hit" is a much better indicator of the correct play than the count. Of course, the expert knows how to weigh all the evidence and would hit even if he broke the aforementioned rule when the deck was very negative.

2) 14 vs. ten. Besides being a key card. the seven is also your 21 card and his 17 card! You could call it a superduper key card. Basic strategy says to stand on two sevens against a ten: the two sevens gone from the deck have that much effect.

3) 12 vs. ten. Sevens. eights and nines are all key cards. If they were gone. you would be crazy to hit. The dealer either has a 20 or a stiff. You wouldn't hit against a stiff and you have almost no chance to beat him if he does have 20.

4) Ace-live vs. Four. Sixes and sevens are key cards. If they are gone. doubling down is a great play. If they're not, watch out. The key card concept is extremely important for soft doubling situations. It is a much better indicator for the correct play than the count, which handles these situations very poorly.

The Worst Play in Blackjack

This short essay while addressing one particular bad play is also a good summary of some of the principles in the other chapters in this section.

All experienced blackjack players have witnessed many incredibly dumb plays by other players at the tables. Hitting 16 against a five showing: standing on 12 against a ten; splitting tens; standing on soft 17: these are just a few examples.

Now, the fact of the matter is that other bad players at your table do not significantly affect your results in the long run. These bad players are just as likely to help you as hurt you and that can be proved quite simply; however that is not the subject of this chapter. Rather, it is to identify the worst common play at a blackjack table. Before reading further the reader should try to figure it out for himself.

Picking the worst play is not just a trivial exercise. This is because in order to reach the right answer it is necessary that one understand the underlying concepts used in determining the correct basic strategy. These concepts really all come down to one thing: mathematical expectation.

Mathematical expectation is the amount one should expect to average winning or losing per play if the situation were repeated many times. For instance, if a player bet $1, was dealt a 14 and stood against the dealer's four showing, his mathematical expectation would be about minus 19 cents.

If you were in this situation 1,000 times you would figure to be down about $190 when it was over. In other words, you figure to have won about 405 and lost about 595 of these decisions. Had you been dealt a 19 against the same four, however, your expectation would be positive, 41 cents.

Another way to look at mathematical expectation is the fair amount you would sell your bet for. Thus, if you had that 19 against a four showing and someone offered to "buy" your hand, you should refuse unless he offered you 41 cents profit or more for, every dollar bet. In the case of your hand having a negative mathematical expectation. you should pay for someone to take it off your hands.

Notice that Ntihen you "surrender' you are paying 50 cents per dollar bet: however, this is only correct in those rare cases where your expectation is less than negative 50 cents.

NoNN it is important to understand that a play does not have to have a positive expectation in order to be correct. Rather, the criterion is that it have a better expectation than any of the alternatives. Standing with a 14 against a four showing has an expectation of minus 19 cents, but that's better than minus 34 cents, which is your expectation by hitting. Thus, standing gains you 15 cents per dollar bet in the long run. Hitting would be a 15 cent mistake.

It can also happen that your choice is between two alternatives that both have positive expectations. For instance, if you are dealt a soft 18 against an eight showing, hitting the 18 results in a positive expectation of five cents: however, the correct play of standing is 7 cents better at a positive 12 cents. (Doubling down, by the way. loses 1 cent.) So, hitting ace-seven against an eight is a 7-cent mistake.

We now have our criterion for determining the worst common play in blackjack. It is the amount of expectation lost by the play. Standing on 12 against a ten showing is about a 17-cent mistake. Hitting 16 against a five is a 27-cent mistake. Splitting tens against a deuce is a 30-cent mistake. Not doubling down with I I against a five is about a 38-cent mistake. But none of these plays is the worst. Try one more before I tell you.

Okay, the answer is standing on two eights against a seven rather than splitting them. If you split. each split is worth about 13 cents (assuming you are allowed to double after splitting). Standing loses about 50 cents. It is a 76-cent mistake for every dollar bet! You go from a 26-cent winner to a 50-cent loser when you stand. which is a 76-cent mistake. Hitting two eights is slightly better but it is still a 63-cent mistake to hit rather than split.

This is definitely the worst mistake you will normally see in blackjack play. It is even worse than some very abnormal plays (such as hitting a hard 18 against a nine, which is only a 40-cent mistake).

Part Four

Other Casino Games

Other Casino Games Introduction

Unlike poker and blackjack, most casino games are of such a nature that there is normally no way to get the best of them. This is because the odds are against you on each individual bet and because the outcome of the preceding bet has no effect on the present bet. Mathematicians say that they are independent events.

While there is no routine wav to beat these other casino games, there are three possible ways (short of cheating) to occasionally have the odds in your favor.3 One way is during casino promotions. Sometimes a casino. in order to attract business, will in one way or another pay higher odds than normal on some of their games. It has happened that these higher odds are enough to give you the best of it.

A second possibility is to use physical techniques to somehow help predict the outcome of an event. At present, this seems most feasible for the game of roulette. Finally, the addition of progressive jackpot slot and video poker machines, has opened up the possibility of having an edge over the "one-armed bandits" once their jackpots have reached a certain level.

In the following chapters, I explore each of these three ways of beating the other casino games. I also include a chapter which gives a brief introduction and explanation of all the games. In addition, I discuss baccarat, crapless craps, and keno.

An Analysis of Gambling Games

To start off the section on casino games, the first chapter is an article I wrote summarizing each of them (as well as horse and sports betting). It indicates which games have the potential to be beaten and which don't. It is must reading for any aspiring gambler.

Any gambling game can be fun and exciting, but only a few can be "beaten." By beaten, I mean played consistently with the percentages in your favor. If you are gambling,you may decide to have an evening of entertainment with the full realization that you are taking the worst of it. However, there is an irrefutable mathematical law which states that any game that has a negative mathematical expectation for each individual bet must have a negative total expectation. Thus, no betting system can change an unfavorable game into a favorable one. Even the occasional gambler should be aware of this if he intends to gamble seriously. Why not be a winner rather than a loser! Let's look at the various standard gambling games.

Poker

This is your best bet. In a private poker game, where no one cuts the pots, the best players will win over the long run. This is the game of most professional gamblers. You are not playing against the "house" but against other players. Play better than them and you must eventually win. It isn't even necessary to be the best player if there are a few weak players ("live ones") in the game. When I was just starting out, I would occasionally sit down in a game where all but one of the players played as well or better than I did. That one player would be throwing his money away and we all figured to get our share.

At this point I must offer a few words of caution: All poker players think they are champions. This is why it is such a good game to `'hustle." No one thinks they need a "spot" as they would in games like chess or pool. This means that you should be certain that you are really as good a player as you think you are. If you think you are better than your opponents you should be able to say why. For example, in a lowball draw game, you might notice that other players are consistently drawing two cards which you know is a bad play. If you simply have some kind of instinctive feeling that you can beat the game, you may be in trouble. While there is not space here to go into the best strategy in poker, a general rule is that a good player plays quite tight until he gets into a pot and then plays quite aggressively. If you don't play this way, you are probably not as good as you think you are.

Another problem with poker comes up in casino games. Besides frequently having to buck a bunch of pros, (many a hometown champ has been cut down to size by these fellows) - you also have to contend with the "rake" (pot cutting) or "collection" (time rental). In the "collection" case you pay a fixed amount per hour to play. Even in the pot cutting case you can figure an average hourly rate you are paying. If this amount is greater than the amount you feel you should average winning per hour, your expectation has now gone from positive to negative. Even when this is not the case, the rake may bring down your earning rate to a negligible amount.

Poker, all in all, is the best game if you are the favorite, but is certainly the worst game if you are not. By the way, for those players who strive to reach the expert level in this game, I would like to immodestly recommend my books Hold 'Em Poker, The Theory of Poker, Sklansky on Poker, and Poker, Gaming. and Life. In addition, I would like to recommend Hold 'em Poker For Advanced Players. which I co-authored with Mason Malmuth, Seven Card Stud For Advanced Planers, which I co-authored with Mason Malmuth and Ray Zee, and Gambling For a Living, which I co-authored with Mason Malmuth.

Roulette

This game is generally unbeatable. On American wheels you have at least a 5 5/19 percent disadvantage on any bet. No betting system can change it. Even if you are not out to "beat" a game but are just hoping to get lucky, you should stay away from roulette. Craps or baccarat have much better odds, and lucky streaks in these games are more likely.

There are two techniques which could theoretically beat the game. One is to find a "biased" wheel where, due to mechanical imperfections, one or more numbers is coming up better than 1-in36 times, as opposed to the expected frequency of I-in-38 times. Only a very long trial - tens of thousands of spins, can accurately find these numbers (when they exist). The modem wheels with their safeguards make your chances of finding a biased one remote. The second technique is to mechanically predict where the ball will fall based on its speed and the speed of the wheel rotating under it. This is theoretically possible since the ball will always leave the rim at the same escape velocity. It is not necessary to predict the ball's final resting place; simply knowing the general vicinity gives you a big edge. I have heard that there are small computers that can do this in the few seconds you have available to you. It may even be possible with experience to estimate the portion of the wheel where the ball will fall by sight alone. If you can do this, just bet on any number in that portion. Eventually, the percentages in your favor will catch up with you. Of course, this technique assumes you can still bet while the ball is in motion. The casinos will be quick to eliminate this option if they think it will hurt them.

Casino Blackjack

This is the second most popular game among professional gamblers. It most definitely can be beaten. Anybody who devotes some time to learning this game thoroughly can have at least a 1'/2 percent edge over the house (especially against one-deck games). There are many good published blackjack systems which are similar. They involve keeping track of the cards already played and changing your bets and your strategy based on the remaining unplayed cards. When there is a higher proportion of aces and tens in the deck and/or a lower proportion of fours, fives, and sixes than usual, you have the best of it. There are two reasons for this. For one, blackjacks will appear more often than the normal once in 21 times; the dealer's blackjacks will appear also but he doesn't get paid 3-to-2, plus you can buy insurance when he has the ace up. Secondly, the dealer will bust often when he has a "stiff' (12 through 16) but you can stand on your stiffs. You thus increase your bets in favorable situations and keep your bets smaller otherwise.

Recently more advanced techniques have appeared. They involved keeping track of the cards in a manner designed to indicate the best playing strategy rather than betting strategy. The older methods did this too but were inaccurate. By playing the best possible strategy at all times you are rarely at a disadvantage. It is no longer that important to change your bet. If you play this way you will never be "barred" from a casino since they use fluctuating bets as the tip-off that you are a counter. I wish I knew these techniques fifteen years ago. Most of these techniques came out only a few years ago at a mathematical gambling conference in Lake Tahoe. One aspect of them, the Key Card Concept, was developed by me. Whether you want to be a counter or not, any blackjack player must learn the basic strategy. This is the correct way to play your two cards against any dealer upcard if you are not counting. If you stick to this strategy you are playing an even game! - against one deck and Las Vegas strip rules. Also, if you simply notice when all the fives have been played and bet more in this situation you would now have an advantage over the house! Any player who will not do at least this should be ashamed of himself.

Craps

Unfortunately, the casino variety of this game is unbeatable. As with roulette, any system player is lying if he claims that he can actually make a living at this game using his system. Most of these systems tend to produce winners in the short run but the occasional losses tend to be catastrophic and these losses must accumulate to a greater amount than the combined winnings. As long as each outcome is independent of the previous outcome as it is in craps, (but not in blackjack) you can't get around the negative expectation. Anybody who thinks he has a craps system can bet me instead of the house. I'll even give him some of his losses back.

Nonetheless. craps is the best game to play if you are gambling simply for the excitement and the chance to make a score. If you stick to pass, don't pass, come or don't come bets and lay or take the odds. you are playing at a disadvantage of less than 1 percent. Please confine yourself to these bets. Most of the other bets on the table have a disadvantage of between 3 and 16 percent. The greater the edge against you the luckier you have to be to overcome it. It is impossible to win taking 10 percent the worst of it over even a moderately long haul. Playing craps correctly, however, is your best casino bet unless you are an expert poker or blackjack player. For those of you who play in private crap games, you can have the best of it by constantly fading the shooter since double sixes are not barred as they are in casino play. Your advantage is 1.4 percent. Even better are those games where players will take even money on six or eight versus a seven.

Fading them produces a 9.1 percent advantage.

Keno

Every bet has at least a 25 percent disadvantage! `Nuf said.

Baccarat

This game is not composed of totally independent results since cards are dealt out. In fact, the player has an edge when there is an extremely high proportion of deuces and treys remaining, the bank has an edge if there is an extremely high proportion of eights and sevens left. Baccarat aficionados should be able to see why. Other extreme deck configurations may also give an edge to one side or the other. Unfortunately these situations come up much too infrequently to beat the game. Normally you are playing with a little more than 1 percent disadvantage. Those of you who think you can win by detecting patterns in the previous runs of wins and losses are sadly mistaken.

Horse Racing

This game can theoretically be beaten. Since it is only necessary to bet overlays and the public sets the odds, there is nothing intrinsic about this game that makes it unbeatable. I know two people who do beat it.

By an overlay I mean a horse who is going off at higher odds than it truly deserves. If you correctly assess a horse as a 4-to-1 shot and it goes off at 5-to-I you have a 20 percent edge when you bet it. Simple as it seems there are at least three reasons why this is one of the most difficult games to beat:

1. You can never be sure of the closing odds. The aforementioned horse may drop to 7-to-2 on the last flash, after you bet him.

2. It is very, very difficult appraising a horse's true chances. If your opinion differs greatly from the tote board you had better have a good reason for it since the public is very good at assessing a horse's chances, on the average. A friend, who bets the harness races is at the track every day. By watching every race he occasionally sees something that is not shown in next week's program such as a horse's driving finish but then getting boxed in, or a horse forced to go wide

around a turn. Coupled with his vast knowledge of the game he finds about two overlays a night. If he had to stick with information from the program he would be hard pressed to win. At the flats it may be a little easier to win just using the Racing Form because the horses are not always running against each other and the distance of the races vary, unlike harness racing. Once again you must know something the public does not. A good example was related to me a few years ago by another friend who beats the flats. When there was a mile and an eighth race at Aqueduct my friend would look for a sprinter with good early speed who was quitting at six furlongs, especially one situated near the rail. The public naturally assumed that horse which couldn't go the distance at a sprint certainly couldn't make a mile and an eighth. Therefore, it went off at a nice price. What my friend realized was that the very short run to the first turn tired the slower horses on the outside, while it was a breeze for our sprinter. He had plenty left for the stretch while the others didn't. This principle has produced many great bets. (Don't forget however that no principle can produce a good bet if everyone knows it!)

3. The worst thing about trying to beat the horses is the percentage of the total amount bet which the state takes before distributing the remaining money to the winners, including "breakage" (a horse that should pay $8.79 will pay $8.60). The total figure is almost 20 percent. This means the public's assessment of a horse's chances have to be off quite a bit in order for you to find a bet. If the money bet would normally make a horse 3-to-1, the tote board will show 2-to-1. Even money becomes 3-to-5. 10-to-1 becomes 8-to-1. For this reason even real professionals are hard pressed to find more than one good bet a day.4

Sports

There is no question that some people make a living by betting sports. Betting at random, results in about a 5 percent disadvantage, not nearly as bad as horse racing where the average bet has about a 20 percent disadvantage. The "line" does not have to be that incorrect to swing the edge to your favor. Of course in this case you are not bucking public opinion but the rather expert opinion of the line makers. However, bookies are more interested in putting up a line that will get equal action than one that is "correct." Thus to a large degree you are betting against the public. By the way, if a bookie does put up a line that will obviously not get equal action you would be well advised to go against the obvious since the bookie now desperately needs this side to win, a situation he chose to be in. Almost every bet I've seen like this has won.

Normally, one's ability at assessing intangibles is the key to sports betting. Everybody can read the statistics. Much has been written on this so I won't add anything. I do have one tip which has worked well for sports betting pros I know: Concentrate on small underdogs. They always need to win the game. The favorites don't need to cover. (For more on this subject, see the chapter on Sports

Casinos and Their Mistakes

This book is about getting the best of it. I have little to say about those games where the odds cannot be turned in your favor. Those games that can be turned in your favor are usually poker, blackjack, and occasionally progressive slot machines. That is why these games comprise the bulk of this book. Games like craps, roulette, keno, and the big wheel, where the odds are fixed against you are almost totally ignored. However, there are times when even these games can be beaten. It can happen when casinos put on promotions.

Every once in a while, a casino or race and sports book, in order to attract business, starts a promotion whereby they offer better odds or add an extra prize to a normally unfair game. Sometimes they even change the game itself (as in the case of the joker in the Fremont's blackjack game). Very often these promotions swing the odds to the player's favor. The casinos may or may not be aware of it.

In any case, whenever casinos put on these promotions the professional gambler should know how to evaluate them in order to decide whether he has the best of it and by how much. In order to do this he must be very conversant with the mathematics of gambling as detailed in the first section of this book.

Between 1970 and 1980, there were at least 20 promotions in Nevada betting places that were very profitable to the player. Most pros missed them, usually because they didn't have the mathematical knowledge to see the profit in them. I, however, was there for most of them. (Hopefully you will be there for the ones in the future.) The following chapter describes just a few of them.

Over the last few years, I have tried in my various articles to give my readers both theoretical and practical advice. The practical advice has usually been along the lines of explaining certain plays or concepts in poker, blackjack, sports betting, etc. These articles have always been well received.

Along with these articles, I have also written on how to figure parlays, permutations and combinations, Bayes' Theorem, and most importantly, (from a bettor's point of view), mathematical expectation and percentage edge. In fact, mathematical expectation and percentage edge was the subject of my first column in Gambling Times 'inaugural issue.

While it is certainly true that the theoretical articles are usually not as popular, this shouldn't be the case. Most gamblers are lazy by definition. However, they should not let their laziness interfere with their learning of probability. If they do, it will cost them money, because most casino personnel are usually just as lazy, as well as less intelligent. This attribute shows up most often when casinos put on various promotions.

At least 20 times in my gambling career, a casino or sports book has promoted money giveaways, but didn't know it. I don't mean that the promotion just cut down their edge. They actually had the worst of it. In all cases, the man in charge of the promotion didn't even have enough understanding of simple probability to see what he had done. This isn't the 1940s. Casino executives had better know their math or hire someone who does.

Since that probably will not happen, it is mandatory that the serious gambler learn how to mathematically evaluate a proposition, whatever it may be. Let me describe just a few of the many mistakes casinos have made in the past. Mistakes that have lined the pockets of myself and a few other sharp gamblers.

It was about 1973; I had just come to Vegas and didn't have much money. Then one day while walking down Fremont Street, I looked up and saw, on the promotional sign for the Union Plaza, something like ""$1 Big Wheel Bets - 75¢." I couldn't believes my eyes. It only took me a second to realize that if the sign meant what it said, a player had the "nuts."

I quickly walked into the casino to check for details. Lo and behold, the promotion was exactly as it seemed! The cashier was selling books of 20 coupons for $15 each. One coupon was equivalent to a dollar bill as long as it was bet on the wheel. If you lost, they took your coupon. If you won, you won the same as if you had bet one dollar, plus youu kept the coupon.

For those of you who are unfamiliar with the Big Wheel, it is an upright, carnival-type wheel with 54 positions. There is a bill of some denomination in 52 of the positions, and the remaining two contain special symbols, usually a joker and a flag.

Altogether, the wheel has:

24 - one dollar bills — pays even money
15 - two dollar bills — pays 2-to-1
7 - five dollar bills — pays 5-to-1
4 - ten dollar bills — pays 10-to-1
2 - twenty dollar bills — pays 20-to-1
1 - flag — pays 40-to-1
1 - joker — pays 40-to-1

Thus, there are seven different bets you can make. Normally, they are all terrible, as anyone can see.

Let us now examine how you would do if you had the special coupons. After 54 spins, you'd average hitting each of the denominations as many times as they appear on the wheel. Thus, if you were betting one coupon on the one dollar bill, you would win $1, 24 times and lose 75o, 30 times for a net profit of $1.50.

$$\$1.50 = (24)(\$1) - (30)(\$0.75)$$

Betting on the $2 bill would win $2, 15 times and lose 75¢, 39 times, netting you 75¢ after 54 spins.

$$\$0.75 = (15)(\$2) - (39)(\$0.75)$$

Incredibly, a bet on the $5 bill still shows a loss, even with the coupon (+$35, -$35.25). The $10 bet wins $2.50 after 54 spins and is the best bet from the point of view of mathematical expectation.

$$\$2.50 = (4)(\$10) - (50)(\$0.75)$$

The edge on this bet is 6.2%. Bets on the $20 and joker and flag show profits of $1, 25¢ and 25E respectively after 54 spins.

The Union Plaza was giving money away and the only problem was figuring the best way to take it. While betting on the $10 bill would show the quickest profit, there were two compelling reasons to consider a bet on the $1 bill.

1. There would be much smaller fluctuation while I was grinding away if I bet on the $1 bill, since it would be much more difficult to run into a prolonged losing streak with 24 winning slots on the wheel.

2. Since you kept the coupons after a winning bet, it was much more likely that you could reuse a coupon if you bet on the $1 which was less likely to lose. This meant that a $15 book of coupons was actually worth about the same amount whether you bet on the $1 or the $10 because of the increased likelihood of reuse when betting on the $1.

In fact, each $15 coupon book was worth $16. If some limit were imposed on the number of coupons you could buy, betting on the $1 would win just as much as betting on the $10. It would just take a little longer.

I finally decided to bet 20 coupons on the $1 and 10 coupons on the $10 on every spin of the wheel. This would show an average profit of almost exactly $1 per spin. To make a long story short, I took $300 and bought 20 books of coupons. After exhausting them, I went back and bought 20 more. Each $300 I spent became an average of about $315. After about fifteen hours of this, they ran out of coupons.

I won $2,000, just as expected. They told me to come back the next day. When I did, the promotion was over. The Union Plaza may claim they knew all along they had the worst of it, but were just promoting the wheel. If that were true, why did they only have it one day?

A couple of years later, I looked up and saw another sign I couldn't believe. This time it was at a small casino called (ironically enough) the Big Wheel. Only this time the promotion they were advertising had nothing to do with any wheel. The sign read they were paying 2-to-1 odds on a blackjack. Normally, the payoff is 3to-2.

This may not seem like a big deal, but I knew different. In Nevada a blackjack is considered to be any 10 value card with an ace. With one deck, your chances of being dealt blackjack are

$$\left(\frac{16}{52}\right)\left(\frac{4}{51}\right)[ten\,followed\,by\,an\,ace] +$$

$$\left(\frac{4}{52}\right)\left(\frac{16}{51}\right)[ace\,followed\,by\,a\,ten]$$

or approximately one time in 20. Each time you get one, you win a half bet more than usual if the payoff is 2-to-1, (except for the rare times that the dealer ties your blackjack). You make 2'/2 extra bets per 100 hands. Since basic strategy is an even game in Las Vegas, you started off with a 2 'h percent edge if you weren't counting. A good counter had a 4 percent edge.

A few other players and I were there every day - until they went out of business in about two weeks. Meanwhile, where were all the other gamblers in town? They didn't know their math and their instinctive ability wasn't enough in this spot. Most weren't at the Gamblers' Hall of Fame about a year later when that establishment, in a desperate attempt to attract business, offered 2to-I on blackjack just as the Big Wheel did. They got business all right, but not the kind of business they were looking for - namely the best players in Nevada. (Of course, even the bad players were breaking about even with the extra 2%s percent gift.)

To illustrate their ignorance, one of the pit bosses recorded the frequency of blackjacks to determine how much the promotion cost. Not surprisingly, the blackjack frequency was I in 20, but they wasted all that time recording blackjacks instead of doing some sixth grade arithmetic. They closed the casino about 10 days later.

By the way, some of you may know that Binion's Horseshoe Club in Vegas pays 2-to-I on blackjack for $5 bets or less during Christmas week. Unlike other casinos I mentioned, the Horseshoe is well aware that they are taking the worst of it, but they do this as a Christmas present to their customers during this typically slow time of the year. Don't play anywhere else if you find yourself in Vegas during this week.

The Fremont Hotel in downtown Las Vegas was the next place to show that they didn't understand the game of blackjack. Someone there came up with the bright idea of putting a joker in the double decks. It was wild for the player, but not for the dealer. (The dealer just discarded it and took another card.)

It is easy to see that any time a player had the joker, he would always make an automatic 21 if he knew what he was doing. Since the average hand contains about three cards, a player was almost a sure winner about once in 35 hands. This is even better than getting 2-to-1 on blackjack since you gain about three bets per hundred hands.

A counter now had about 4'/z percent edge. Bad players figured to break about even. The promotion lasted four days and the Fremont lost a ton. Once again, a lot of otherwise smart gamblers weren't there because they didn't know their math.

Before mentioning other promotions where the casino had the worst of it, I would like to mention one where they don't. This is double exposure 21 which was originally conceived by Bob Stupak for his Vegas World Hotel and since adopted by many clubs throughout Vegas.

In this game, the dealer exposes his hole card before you play your hand. However, you get paid only even money on blackjack and lose ties. Stupak realized that these two restrictions were just enough to keep even the great players from having the best of it. Perfect play would yield a 2%2 percent advantage if the blackjack payoff were the normal 3-to-2.

Just as a 2-to-1 payoffin the previous example added 2'/2 percent for the player, an even money payoff subtracted 2'/2 percent, giving the expert players only an even game.

A promotion that I'm sorry I did not take advantage of, occurred around 1975 at the Sahara Tahoe. This time the game was keno. The hotel had obtained a number of gold South African Krugerrands, worth about $200 at the time. They decided to give them away as a bonus to those players who hit 7 or more on a 70cent, I0-spot. They were sure they still had the best of it and they were right. In fact, they only gave away a few Krugerrands and decided to make one small change. They would give away the coin to those who hit 6 spots or more.

I wasn't there at the time, but my friends were. They realized that is was about nine times easier to hit 6 out of 10 than 7 out of 10, but keno bosses at the Sahara didn't know it. It was a relatively simple problem of permutations and combinations. Each 70-cent ticket was now worth over a dollar.

There are many other promotions I could tell you about, but I'll close this chapter by telling you about a somewhat different one that came up in 1981 at the Bingo Palace in Las Vegas.

This time there was no gambling involved and their mistake was not one involving probability. Instead, they underestimated the discipline of mathematics and science in general when it came to finding flaws in the idea. The Bingo Palace had decided to give away money; they just didn't know how much they would be destined to give away.

The promotion was based on $20 bills. Every $20 bill has a serial number, eight digits plus a letter on either side. Every four hours, the Bingo Palace drew two letters and eight numbers at random in a specific order. Any patron who had a $20 bill that matched enough of the randomly drawn numbers and letters in their same position could simply turn it in during that four hour period and collect their prize. If I remember correctly, the prize structure was:

Match 10 spots	$250,000	Match 7 spots	$1,000
Match 9 spots	$50,000	Match 6 spots	$100
Match 8 spots	$10,000	Match 5 spots	$10

Even a $10 winner figured to be very rare. It seemed almost impossible that they would be hit for $1,000 or more. After all, even a gambler with thousands of twenties at his disposal would need 10 to 20 seconds to peruse a bill. His expectation would be less than $10 per drawing. Their mathematics were correct in a sense, yet they found that they were giving away much more than they expected.

They changed the promotion to a drawing every two hours but with only a half an hour to collect. They still got annihilated. Why? Because they forgot that it was the beginning of the era of the home computer. The sharpies simply got hold of a bankroll in twenties, put them in some kind of filing order, and entered the serial numbers in a Radio Shack or similar computer. When the number was drawn, they simply told the computer and in a few minutes, the winning bills were identified and located.

Players were camping outside the casino in vans, getting up every two hours to collect an average of a hundred dollars or so. The Bingo Palace threw in the towel before they had planned to. I still don't know if they ever knew what hit them.

Alas, I was also out of town for this one. When I heard about it, I knew it was another example of how underestimating the power of technical science and math can cost you money. Don't let this happen to you.

The Ultimate Play

Previously I have written about bad mistakes made by casinos that didn't know how to figure out their own promotions. I've gone on to tell how I have made money from these promotions. But I left the best one out. It happened twenty years ago, but to this day it is still probably my greatest triumph and one of my biggest scores. And there was no way I could lose.

I have kept this story a secret for all this time, mainly because there was someone else involved who wanted it that way. He has now left the gambling scene. A second reason was that I kept a slim hope that it might come up again, but it is now obvious that the casinos have learned their lesson. So here's the story.

It was football season. There was lots of action going on. Casinos had just recently been allowed to open up their own sports books to compete with the older established individual legal books around Las Vegas. Thus, started the parlay card wars.

A parlay card has traditionally been the method by which a small bettor can bet football games. All the college and pro football games for that weekend are listed on the card along with their point spreads. The bettor circles the teams he likes and writes down the amount he wants to bet. However, he cannot bet only one team. Usually he must bet at least three teams and all his teams must win (versus the point spread) in order for him to collect. That is why it is called a parlay card.

The odds you get if you win depend on how many teams you pick. Before the "war" these odds were terribly against the bettor. A three-team parlay paid 5-to-1 (true odds are 7-to-1). A four teamer paid 9-to-1 (true odds: 15-to-1). Hit a nine team parlay and get 49-to- l (instead of true odds of 511-to-1)! The more teams you bet the worse it was, with the exception of the 10 team bet. (The maximum number of teams you were usually allowed to bet on the card.)

You got 99-toI on a ten teamer if you won all the games, but you also got a 9-to- I consolation prize if you won 9 out of 10. Thus the ten-team bet was somewhat better than the eight or nine teamer, but it was still quite horrible in that the bookie figured to hold about 80 percent of the amount bet.

As I said, parlay cards were traditionally meant for the $5 and under bettor. That was why the bookies could get away with such disadvantageous odds. A serious bettor risking $10 or more could get much better odds, even on parlays, if he didn't use a card. There was only one advantage to using a parlay card rather than a regular bet. This advantage occasionally occurred because the cards were printed four or five days before the start of the game. By game time, the point spread on a particular game would sometimes have moved two or three points. especially if there was a late key injury. Thus the parlay card bettor might be able to get a couple of points the best of it on one or two games. but this would rarely make up for the terrible odds he was getting.

All this changed during the parlay card wars. Each week, in order to lure customers, a different sports book would issue the most liberal card vet, without actually giving the player the best of it. This ryas easy to do, since the chances of hitting a parlay for any number of games are quite simple to calculate.

Here they are in a nut shell.

Go 3 for 3 1/8 chance; true payoff 7-to-1 or 8-for-1.
Go 4 for 4 1/16 chance; true payoff 15-to-1 or 16-for-1
Go 5 for 5 1/32 chance; true payoff 31-to-1 or 32-for-1
Go 6 for 6 1/64 chance; true payoff 63-to-1 of 64-for-1
Go 7 for 7 1/128 chance; true payoff 127-to-1 or 128-for-1
Go 8 for 8 1/256 chance; true payoff 255-to-1 or 256-for-1
Go 9 for 9 1/512 chance; true payoff 511-to-1 or 512-for-1
Go 10 for 10 1/1,024 chance; true payoff 1,023-to-1 or 1,024-for-1

As long as the payoffs didn't exceed these odds, the bookies should still have an edge (as long as the point spreads were right). Because the correct payoffs for a straight parlay were so easy to figure. I had little hope that even a war would push the prices to a point that I could bet them. The card paybacks were creeping up to 80 and even 90 percent of the correct amount, rather than the 20 percent of the past, but still that wasn't good enough for me.

In the case of the ten-team bet with its typical bonus for 9 out of 10 winners, the math was also quite simple. Betting every possible outcome for 10 games would take 1,024 cards. Only 1 of these would win. However, 10 cards would produce exactly 9 winners. One card would have everything right except game one, another all but game two, another all but game three, and so on. The early cards paid 100-for-l for 10 out of 10 and 10-for-I for 9 out of 10. This meant that if you bet every combination at $1 each, it would cost you $1,024 and you would wind up with I ticket worth $100 and 10 tickets worth $10 each. Your $1,024 would return $200, which was more than an 80 percent disadvantage.

Anyway, it appeared that all the bookies understood that when figuring the ten-team payoff, you would take the 9 out of 10 bonus, multiply it by 10, add this to the payoff for 10 out of 10, and make sure this figure was less than 1,024.

By the war's end, this just-mentioned sum was up to 800 or even 900 and was achieved in a variety of ways. Some cards paid 500for-l on 10 out of 10 and a 30-for-I bonus for 9 out of 10 (800 total). Others paid 300-for-1 on 10 out of 10 and a 50-for-1 bonus for 9 out of 10 (also an 800 total). Many different combinations were used, and they were all much better for the bettor than in the past. However, the bookie still had a nice edge. It never really occurred to me that things could be otherwise.

Then one day while glancing through the newspaper, I saw, an ad that literally made me jump out of my seat. In the ad was a reproduction of the parlay card that the Stardust Hotel was to use that week. After listing all the games, along with their point spreads that were to be played that Saturday, Sunday and Monday, they went on to list their payoffs.

They looked something like this:

3 out of 3 pays 6-for-1
4 Out of 4 pays 12-for-1

5 Out of 5 pays 20-for-1
6 out of 6 pays 35-for-1
7 out of 7 pays 50-for-1
8 out of 8 pays 100-for-1
9 out of 9 pays 150-for-1
10 out of 10 pays 160-for-1
    Bonus 9 out of 10 pays 60-for-1
    Bonus 8 out of 10 pays 20-for-1

I am not sure that I remember all the payoffs accurately - except for the 10 team payoffs, I remember them like it was yesterday. I couldn't believe my eyes. After five seconds of calculation, I realized that of the 1,024 ten-teamer combinations, there would not only be the usual one card with all 10 winners and 10 cards with 9 winners but also 45 cards with 8 winners, each paying 20-for-1 ! Bet $1 on every combination and you would get back

$$\$160 + (10)(\$60) + (45)(\$20)$$

for a grand total of $1,660 on your $1,024 investment! Remember that you would get this return no matter who won the games. You got a 62 percent return on your money in three days with absolutely no risk.

No wonder I jumped out of my seat. I couldn't believe the Stardust could make such a monumental error. Even if they had, wouldn't they correct it almost immediately? And if they didn't correct it, was it conceivable that they wouldn't pay? It was this last concern (which I realize now was probably silly) that prompted me to get someone else to put up the money even though there was theoretically no risk. Another problem was the fact that to make the most out of this situation would take more money than I had at the time. So I got me a backer.

Now the parlay cards had a $100 limit. Theoretically we could bet 1,024 different combinations at $100 each and make an automatic profit of $63,600 regardless of the outcome. There were, however, two reasons not to do this. One problem was that we were sure that betting 1,024 cards at $100 each would "wake up the dead" and thus alert the Stardust that something was amiss. It would have been neat to hand in all those cards along with $102,400 and then say "OK gentlemen, pay us $166,600," but we weren't willing to risk the Stardust's canceling the whole deal.

Another problem involved the fact that most of the point spreads were whole numbers rather than "half points," and on parlay cards unless otherwise specified ties lose. Only seven games had half point lines that couldn't tie. Betting both sides of other games meant taking the risk of the game falling on the number exactly, which would mean a loss for both sides. We didn't want to take that chance.

Instead, I devised a scheme in which we bet every combination on the seven half-point games but bet only one side on three "key" games. This required putting in 128 different cards rather than 1,024 cards, but it also meant that the cards weren't guaranteed to win. We bet $100 on 128 different cards, but every card had the same side of the three key games. If all three games lost, every card would be worthless, as none would have even eight winners. If only one of the three games won, we would have one measly card with eight winners worth $2,000 and thus a total loss of $10,800. However, if two of the three games won, we would have one card with 9 winners worth $6,000 plus seven cards with 8 winners worth $2,000 each, for a total of $20,000. And if all three key games won, one card would be worth $16,000, seven cards would be worth $6,000 each, and 21 cards would be worth $2,000 each, for a grand total of $100,000!

Now we could simply bet these 128 cards and keep our fingers crossed. Mathematically, we had way the best of it. We were going to lose a little, win a little or win a lot. But I saw a way to guarantee a profit plus give us a chance for a big score. What we had to do was bet the opposite way from the way we bet on the card, on the key games. Thus, for the sake of argument, if the three key games on the card had us picking Penn State, the Chicago Bears, and the Miami Dolphins, we would turn around and bet against.these teams with a bookie on the day of the game. Using a technique that I invented, I came up with the following scheme.

A. Start off by betting $13,200 to win $12,000 against Penn State.

B. If Penn State won the game, we would lose our bet but hit our first team on the card. Now bet $24,200 to win $22,000 against Chicago. If instead Penn State lost, bet $5,500-to-$5,000 against Chicago.

C. If the first two bets win (thus lose on the card), bet $1,100-to$1,000 against Miami. If both bets lose against the bookie, bet $41,800-to-$38,000 against Miami. If the first two bets split, bet $10,450-to-$9,500 against Miami.

By following this procedure we guaranteed ourselves an automatic profit of around five or six thousand dollars no matter what happened. (The numbers are slightly off in this article in the interest of simplicity.) For instance, if all the games on the card won, we would win $87,200 from the Stardust but lose about $80,000 to the bookie.

To use this scheme, two other factors had to be considered when selecting our 3 key games. For one they had to be played at different times, since we wouldn't know how much to bet the "other way" until the previous results were in. Secondly, we had to take care that we couldn't "middle" ourselves. This meant finding three games that not only were at different times, but also had a different point spread from the one printed on the card. Well, we found three such games. Again I'll call them Penn State, which played Saturday, the Chicago Bears, which played Sunday, and Miami on Monday. This is what happened.

On Saturday we bet $13,200 to win $12,000 against Penn State, getting 5 points. We had given 3 points on the card. Penn State won easily, so our first key game won, but we were down $13,200 with the bookie.

On Sunday we bet $24,200 to win $22,000 against the Chicago Bears, getting nine points. We laid seven points on the Bears on the card. Guess what happened? The Bears won by eight points! We won both ways! We got our first two key games in on the card, but instead of being down $37,400 to the bookies, we were up $8,800.

Since Monday's game was still the difference between $20,000 and $100,000 as far as the card was concerned, we still bet $41,800 to win $38,000 against Miami and we once again had a two or three point middle going for us. Well, it was not to be. Miami lost badly which meant we hit only two out of three of our key games. The Stardust paid us $20,000, so we beat them for $7,200. Not too bad for no risk. But what about the poor bookie that we were making our hedge bets with? He was the innocent victim of the Stardust's stupidity. Miami lost, remember. This cost him $38,000. He was down $8,800 already. We beat him for $46,800. Add this to the $7,200 from the Stardust and you have a $54,000 weekend.

Of course, the Stardust wised up and changed their card the next week. The fun appeared to be over. But it wasn't. The Stardust card was gone. but both the Churchill Downs and the Golden Nugget came out with the same payoff schedule as the Stardust had the previous week. So there was in fact one more week of fun before the party was over.

A Couple of Ways to Beat the Wheel

Craps, roulette and keno, can unfortunately never be "beaten" by a betting system or technique. (Unless of course a promotion is under way where the casino is paying higher than the correct odds.) This is not my opinion. It is an irrefutable fact that can be absolutely proven mathematically. Short of cheating, your only hope of beating these games in the long run would be to use some kind of physical or mechanical technique. I can think of no physical technique that can be used for craps or keno.

However, there are two possibilities for roulette. Both of the techniques mentioned in the following chapter have been used in casinos. Biased wheels were first written about by Allen Wilson. The mini-computer was just an idea in my head when this article first came out. I have learned that it has actually been implemented since that time with various degrees of success. The techniques described are easier said than done. Still they should be reported. Hopefully my conscientious readers will be able to win some money with them.

My position (and the truth) on roulette and other games of that ilk should be well known by now. To put it simply, if every individual bet has the "worst of it," there is no way you can get the "best of it" simply by altering the size of your bets. There are, however, two techniques, physical rather than mathematical, that can possibly beat the roulette wheel. One of these methods has already been used with success in the past. Unfortunately, greater precision in the manufacturing of roulette wheels and paraphernalia casts doubts on the future effectiveness of this technique.

The second method has never, to my knowledge, been used in a casino. It requires the development of a specialized minicomputer. If there is any one who has the technical expertise to develop such a computer, he should be able to win a fortune in a very short time. Add to this the profits that would be made from selling the device, and you have a big incentive to any computer expert to start working on such a computer, which I shall describe shortly.

The method which has been used already is finding and betting against a biased wheel. A full treatment of this subject can be found in The Casino Gambler's Guide by Dr. Allen Wilson. Dr. Wilson, a physicist from Cal Tech, was, in fact, one of those people who used this method to win money from some Reno casinos. A biased wheel is one that for some physical reason produces certain numbers more often than others. The two most common physical imperfections that can cause this result are an imbalance in the wheel or lack of uniformity in the slats between numbers. In extreme cases, these imperfections can be spotted by the naked eye. Most of the time this is not true.

In any case, the only way we can find out if the wheel is "off" enough to make for profitable betting situations is by clocking the wheel. This involves recording many thousands of consecutive spins of the wheel while betting nothing or the absolute minimum. Fortunately, the management would think nothing of your sitting there with a pencil and paper. They would assume you were just another sucker trying to pick up patterns. It is important to have one or two partners before embarking on such an effort. You will have to work in shifts and there should always be someone at the wheel to make sure it has not been changed. It has also been suggested that separate data be kept for both roulette balls as well as for both directions of spin, as there is the possibility that certain numbers will occur more often than is normal only under certain conditions. But before jumping to any hasty conclusions, it is far better if your combined results still show such a "hot" number.

What kind of data do we need before we can be reasonably sure that we have come up with a profitable bet? Remember that there are 38 numbers on most American wheels, but the payoff is only 35-to-1 instead of the fair odds of 37-to-1. Any number we find must be hot enough to overcome a 5 percent disadvantage. To obtain a reasonable edge, we would like our number to be no more than a true 34-to-I shot (1 out of 35). This gives us an edge of slightly less than 3 percent. Those of you who may clock a singlezero wheel still need to find a 34-to-I shot and should still follow the guidelines that I will give you. The only difference between single-zero and double-zero wheels is that the former do not have to be as biased in order to show beatable results. They should therefore be checked out first.

Even after we have obtained data that seems to point to a hot number, it is difficult to evaluate. Thus, even if we obtain results that will occur only once in a thousand times by chance alone, it may not be totally conclusive. If, for instance, we knew that the chance of finding a biased wheel was one in a million, it would indicate that our experimental data was still only a lucky streak rather than the result of a one-in-a-million shot. This is an example of Bayes' Theorem, a subject I covered more fully in a previous chapter. Unfortunately, other writers on biased wheels fail to take this factor into account. Not knowing the a priori chances of finding a biased wheel, I can only give a subjective judgment of what would be good evidence of a sufficiently hot number on a biased wheel. For those mathematically inclined readers, the following guidelines are about five standard deviations away from a result of one "hit" in 35.

In 1,000 spins a number would have to show at least 54 times (1 out of 18'/s) to give good evidence of sufficient bias.

In 5,000 spins a number would have to show at least 200 times (1 out of 25) to give good evidence of sufficient bias.

In 10,000 spins a number would have to show at least 366 times (I out of 271/3) to give good evidence of sufficient bias.

In 25,000 spins a number would have to show at least 841 times (1 out of 293/4) to give good evidence of sufficient bias.

In 100,000 spins a number would have to show at least 3,110 times (1 out of 32 1/6) to give good evidence of sufficient bias.

The reader can interpolate from these guidelines to get a good estimate for a number of spins other than those mentioned. It is unlikely that one can reach any trustworthy conclusions with fewer than 10,000 readings. At two spins a minute, this would take about 3'/2 days. For those who want to gamble a bit, my guidelines can be loosened somewhat (to 345 out of 10,000, for instance). If there is some obvious physical evidence of bias in the wheel, Bayes' Theorem tells us that we can loosen our requirements still further. Under no conditions, however, should we be satisfied with a result worse than one hit in 32 or 33, assuming a reasonable number of trials.

The other method for beating the wheel is much more foolproof The necessary technology now exists, and it only remains for someone to build the previously mentioned mini-computer. The effectiveness of this system or technique is based on three facts. One is that the casino will allow you to place bets well after the ball has been spun around the outer rim, right up to the moment it drops onto the wheel. The second fact is that the ball will fall off the wheel at a consistent, predetermined speed. This is an immutable law of physics, with no regard for the initial velocity imparted on the ball. Third, the speed of the ball will always decrease at a predetermined rate. This is another law of physics. As an example, let's say the ball's "escape velocity," i.e. the speed at which centrifugal force no longer holds it on the rim, is 20 mph. Suppose further that we know it slows at the rate of 5 mph per second. If we knew that it was now going 40 mph, we could predict that it would leave the wheel in exactly four seconds. With this information, along with the speed and circumference of the wheel itself, a computer could tell us exactly over what point on the wheel the ball would drop! The initial speed the croupier imparts to the ball has no effect on the outcome of this experiment.

Two questions immediately arise. How can these calculations possibly be made unnoticed in a casino setting? Second, even if this could be done, the ball bounces around so much after it hits the wheel, how could you predict where it would land?

Taking the second question first, you must remember that you don't need to win every bet. A 9 percent edge will eventually get you all the money you'll ever need, and a 9 percent edge can be obtained by eliminating only five numbers! In other words, it is only necessary to find a section of the wheel that the ball figures to fall into. (This section is not necessarily directly under the point where the ball falls onto the wheel due to the spinning of the wheel itself. You should be able to adjust for that fact with experimentation.) For those of you who ask how one could bet on all these numbers in the few available seconds, shame on you! Just bet one number in that section and you will show just as great a profit if the total amount bet is the same. In other words, it is just as good to bet $10 on one number in that section as to bet $1 on 10 numbers (though you will have larger fluctuations betting on one

The other question is harder to deal with. How can this method be used in a casino? Richard Epstein, in his book The Theory of Gambling and Statistical Logic, suggests that someone program a small hand-held computer in such a way that you need only press a button as the ball passes a specific number, say double zero, and then press it again as it passes the second time and again the third time. With this information the computer could instantly calculate the ball's speed and its rate of deceleration. (If the deceleration rate is already known, the button would only have to be pushed twice.) Since the computer will already "know" the escape velocity as well as the position of all the numbers on the wheel, it would be simple for it to read out the number above which the ball will fall. If you want to read out a number a few compartments away (because of momentum), that would be no problem.

Unfortunately, this method does not take into account the wheel, which is spinning in the opposite direction of the ball. Various wheel speeds would cause different apparent speeds of the ball as it passed a double zero. It is true that the nearly frictionless wheel slows very little during the spin of the ball, but it does change quite a bit from spin to spin, especially after the croupier gives it an occasional push. The computer must therefore be further refined in such a way that would allow us to push buttons during the wheel's path in order to determine its speed along with the real speed of the ball. It would then have to make the necessary adjustments to determine over what number the ball will fall. This still seems to be within the realm of present-day technology.

With a machine like this, anyone would become wealthy before the casinos knew what hit them. Their only defense would be to forbid any bets by anyone after the ball has been spun, and every casino I know is much too greedy to give up their chance at winning all those last-minute sucker bets.

Beating The Progressive Slots

The information contained in the following chapters is becoming increasingly important to the professional and occasional gambler alike. In recent years the progressive jackpot slot machine craze has swept the state of Nevada. Progressive slot machines have now become the instrument of serious gambling. Progressive jackpots are jackpots that constantly increase as more money is put into the machine until they are hit. The figures are staggering. A million dollar hit is almost commonplace.

What we are concerned with, of course, is whether we can get the best of it on these machines. In other words, can the jackpot reach a point where the machine is actually worth playing? The answer is a definite yes, although it is much more likely that machines with relatively smaller jackpots fit the bill.

Many jackpots do increase to the point where the player has a definite edge. However, it is not that easy to ascertain what that point is. That is what the next two chapters will show you how to do.

Anyone who follows gambling cannot help but notice the gigantic slot machine jackpots that have been paid in the past several years. Quite a few have paid over a million dollars!

Big payoffs have come about largely because of a relatively new innovation in slot machines. I am speaking of progressive jackpots. There are many different types of progressive machines, but they all have one thing in common: Their jackpots steadily increase until they are hit. (Some casinos post an upper limit on the progressive jackpot, not allowing it to increase past this point. Avoid these casinos.)

From a serious gambler's point of view, the beauty of a progressive slot is its jackpot. It will sometimes increase to a point in which the player has the best of it. This seems hard to believe, but it is true.

The "one-armed bandits" may not be bandits at all. Some slots give the player a very high positive expectation. The trick is in finding those machines.

To determine whether any progressive slot machine gives you an edge, you must evaluate quite a few factors.

Among them are:

1. The size of the jackpot.

2. The size of the smaller jackpot. (Many machines alternate between two different progressive jackpots. They are called double progressive machines. Their attraction lies in the fact that there still figures to be a large jackpot available, even if one of them is hit).

3. The number of reels.

4. The number of symbols on each reel.

5. The number and denomination of coins that must be played in order to be eligible for the progressive jackpot.

6. The sequence whereby the machines alternates from one jackpot to the other on double progressive machines.

7. The number and size of smaller payoffs.

8. The speed at which the jackpot increases.

All eight factors can be important. They are all related to the mathematical expectation of the machine. Let us examine them one by one.

First, we must figure the probability of hitting the jackpot. This can be instantly calculated if we know factor numbers 3 and 4. In other words, we must know the number of reels and the number of symbols on each reel.

Since there is invariably only one super' ackpot symbol on each reel, the probability of hitting the super jackpot is the number of symbols on each reel multiplied by itself as many times as there are reels. (See warning at the end of the next chapter.)

If there are three reels with 22 symbols on each reel, the chances of hitting the jackpot are I out of

$$(22)(22)(22) = 10{,}648$$

The chances of hitting the super jackpot on a four-reel machine, with 20 symbols on each reel, are 1 out of

$$(20)(20)(20)(20) = 160{,}000$$

Notice that if there were 30 symbols on each of the four reels the chances would dwindle down to 1 out of 810,000! (In algebraic language, if there are r reels and n symbols on each reel, the probability of hitting the jackpot when there is only one jackpot symbol on each reel is

$$\left(\frac{1}{n}\right)^{r}$$

Determining the number of symbols on each reel may not be that simple. Possibly, you can ask someone in the slot department. They may or may not tell you, and you can't be sure they are giving you the correct information.

A second method is to wait until that machine, or a similar machine, is opened for repairs. When it is being serviced, you can count the symbols.

The third way is to "clock" the machine as you play it. This is the best way. By clocking the machine, you eventually discover how many of each symbol is on every reel.

Just knowing the chances of hitting the jackpot does not by itself give you enough information to determine whether a slot is worth playing. Knowing the chances, however, are a vital part of the equation. Take a moment to look at the following table.

Number of Symbols (Stops) on Reel	Chances of Jackpot on 3 Reel Slot	Chances of Jackpot on 4 Reel Slot
20	1 in 8,000	1 in 160,000
21	1 in 9,261	1 in 194,481
22	1 in 10,648	1 in 234,256
23	1 in 12,167	1 in 279,841
24	1 in 13,824	1 in 331,776
25	1 in 15,625	1 in 390,625
26	1 in 17,576	1 in 456,976
27	1 in 19,683	1 in 531,441
28	1 in 21,952	1 in 614,656
29	1 in 24,389	1 in 707,281
30	1 in 27,000	1 in 810,000

Notice how much tougher it is to hit the jackpot on a four-reel machine than on a three-reel slot. Even more striking is the vast jump in the odds against you as the number of symbols on the reel increase from 20 stops to 30 stops. (Most slots in Nevada have between 20 and 30 stops.) (No longer! See warning at the end of the next chapter.)

After determining the chances of hitting the jackpot, the next step is to note the amount of money you must put into the machine to gain eligibility for the super jackpot. Most machines require that you put in three coins or five coins although some machines only require one coin.

(One advantage to multiple coin machines is that many casual players do not play the maximum number of coins all the time. Since the progressive jackpot keeps increasing until it is hit, this player will be feeding in coins to increase the jackpot with no chance to hit it himself. He is not putting in the required number of coins, and he will receive a much smaller payoff if he hits the jackpot.

In other words, he can line up all the jackpot symbols, but he won't win the progressive jackpot. Thus, it will keep increasing until it is ready for you to move in.)

Your next step is to compare the size of the progressive jackpot (Factor No. 1) to the amount of money you must insert per pull (Factor No. 5). Then form a ratio.

For example, if the jackpot was $300,000 for $3, the ratio would be 100,000-to-1. If the jackpot was $2,000 for 5 nickels, the ratio would be 8,000-to-I ($2,000-for-$0.25). If the jackpot was $60,000 for three quarters, the ratio would be 80,000-to-1.

If the machine had no smaller payoffs and simply one progressive jackpot, it would be easier to decide whether the machine gives you the best of it. You would simply compare the aforementioned ratio to the chances of hitting the jackpot. Let us look at some examples: A 3 reel nickel machine, that requires 5 nickels to be eligible for the jackpot, is paying $3,000. There are 22 stops on each reel - $3,000-to-$.025 = 12,000-to-1. Since the chances of hitting the jackpot are 1 in 10,648, this slot is worth playing even if there were no smaller payoffs.

Let''s say a four-reel quarter machine, that requires three quarters, is paying $75,000. There are 24 stops on the machine. The jackpot is paying 100,000-to-1 odds ($75,000-to-$0.75). The odds against hitting it are 331,776-to-1. It may not be worth it. It certainly wouldn't be worth it if there were no smaller payoffs. It is possible, however, that these minor payoffs can swing the odds in your favor.

In order to determine whether the smaller payoffs are enough to swing the progressive jackpot to your favor, you must know how much the machine "holds." I will show you exactly how to do this in the next chapter. For now, suffice it to say that almost all progressive slots hold between 10 and 40 percent of your money. (This is not counting the big jackpots) This leads us to the next step in determining whether a machine is worth playing.

Since virtually all progressive slots have smaller payoffs, as well as the big jackpot, you should not simply compare the size of the jackpot to the amount you must insert per pull. What you must do is first multiply your cost times the percentage hold of the machine. Then compare this figure to the size of the jackpot.

As an example, take the same four reel quarter machine that has 24 stops and requires three quarters. Suppose this machine has a hold of 20 percent. Taking 20 percent of 75 cents gives you 15 cents for your average "loss per pull."

Now, the ratio for $75,000-to-$0.15 cents is 500,000-to-1. Thus, if the hold is 20 percent this slot would give you the best of it. If the hold is 40 percent, the ratio would be $75,000-to-$0.30 cents or 250,000-to-1. This is less than the required 331,776-to-1 which would make the machine not worth playing.

There is another fly in the ointment. Most progressive machines have two different jackpots. One is often much larger than the other. The adjustment in your figuring is easier when the machine takes one coin per pull. When a double progressive slot takes just one coin, you simply take the average of the two super jackpots. Then compare this average jackpot to the average loss per pull (money times hold).

To give you another example, let's say the two jackpots, on a Harold's Club three-reel dollar machine (with 26 stops), are $6,200 and $2,400. Say the hold is 25 percent. Is this machine worth playing'?

The average of $6,200 and $2,400 is $4,300.

$$\$4{,}300 = \frac{\$6{,}200 + \$2{,}400}{2}$$

The net hold is $1 times 25 percent, equaling 25 cents. $0.25 is your average loss per pull. The odds are $4,300-to-$0.25 cents or 17,200-to-1. Unfortunately, your chances of hitting the super jackpot on a 26 stop machine are one in 17,576.

Notice that any one of the three factors could be slightly changed to give you a positive expectation:

1. The jackpots could be slightly larger,

2. The hold could be slightly smaller;

3 There could be 25 stops rather than 26 stops on the machine.

Any of these changes would make the machine worth playing. Notice once again the large swing in the odds caused by a slight decrease in the number of stops.

When a double progressive slot requires more than one coin, it is not always necessary to take the average of the two super jackpots when computing whether or not you have an edge. If one super jackpot is much larger than the other, you may be able to insert coins in such a way that would enable you to shoot for the bigger jackpot with the majority of your coins.

For instance, in certain five-coin machines, you can put in 5 coins twice, followed by 2 coins, and then repeat this pattern. By playing in this way, you would always be shooting for the larger jackpot when you put in five coins.

Look for the best way to do something similar when only one of the two jackpots is worth playing. How to do it will vary from slot to slot. This method is usually better than blindly putting in the maximum number of coins, which would result in your shooting for the average of the two progressive jackpots.

Factor No. 8 is important if you intend to seriously invest a lot of money in a machine. In this case, it may be worth playing with a very small negative expectation to start with since your own playing will increase the jackpot to a profitable amount.

Important: Please see the warning at the end of the following chapter.

Beating The Progressive Slots Part II

Last chapter, we learned that progressive slot machines may actually have jackpots which have grown to the point where the player has the best of it. We learned the various criteria to use when determining whether the super jackpot (or jackpots) has reached this point. However. two of these criteria weren't fully examined.

They were:

1. The number of reels and the number of symbols on each reel.

2. The percentage of your total money bet which is paid back in smaller (non-super jackpot) payouts. (Loss per pull.)

It is only after we have the answers to the above that we can be sure a particular machine is really worth playing.

What follows is based on a very important assumption. That is that all honest slot machines are set up so that each position or stop" on every reel will show just as often as any other positions; and it will show in a totally random manner. (No longer true! See warning at end of chapter.)

All this boils down to one thing: in order to accurately compute the relevant percentages for any slot machine, you need to know the exact make-up of each reel. By this I mean the number of cherries, oranges, plums, bars, etc. It is not necessary to know the order of the symbols on each reel.

I can think of three methods to ascertain this desired information:

1. Get a slot mechanic or pit boss to open the machine for you so that you can record the data.

2. Take slow motion film of the reels as they are turning and get your data from the film.

3. Play the machine and gradually recreate the reel from the symbols you see in the window.

For those of you who are unable to use either of the first two methods, I will go into greater detail about the third. Meanwhile, don't pass these first two possibilities off lightly. If you plan to play slots seriously, it will save you a lot of time. It will also save you a little bit of money (the money you figure to lose while you play to recreate the reels) if you can find a way to use one of the shortcuts.

When using raw data to recreate the reels, it is important to try to record as many of the symbols in sequence on each reel as you can possibly see after each pull. You can see at least five symbols on each reel on most machines.

Often, you can spot six if you crane your neck. In other words, besides seeing the three symbols directly displayed in the window for each reel, you should be able to see the symbols just below them. You should also see one or two symbols above those three symbols.

(Unfortunately, this method has major drawbacks on video machines. These slots display only three symbols at a time on each reel).

For non-video machines, start recording data for each reel after every pull. For example. your data might be similar to the information in Figure 1. When recording the data, it is not necessary to specifi' which symbols were directly in the window. I noted it in the sample for the sake of clarity. Anytime you come up with a duplicate sequence on any reel from a previous pull you need not record it again.

Basically, you can be fairly sure that you have recreated a whole reel if you had about 100 consecutive pulls without coming across a new sequence on any reel. However, don't forget that there are the same number of symbols on each reel. For example, suppose you came up with 26 different sequences on the first and third reels but only 25 different sequences on the second reel. You can be almost certain that there is still some unrecorded sequence on the second reel.

The only other possible explanation is that the same sequence appears twice on two different sections of the reel. This is extremely unlikely, if you are recording sequences in groups of five or six. (The fact that duplicates are not that unlikely in groups of three is why video slots are hard to recreate.) To guard against this possibility, it is advisable to record any nicks or smudges next to the symbols that have them, in the eventuality that there are duplicate sequences.

There's another piece of information that can be used to save time. That is, it's almost a certainty that there's only one seven (or the equivalent super jackpot symbol) on each reel. For example, if the only missing sequence was the one where the seven was the second from the top (i.e. ?, seven, ?, ?, ?, ?), you could deduce this sequence from the other sequences involving the seven. After you are sure that you have recorded or deducted all the distinct sequences on each reel, you can start making calculations.

First, how many distinct sequences are on each reel`? The answer better be the same for all of them or you made an error somewhere. This number is the number of symbols on each reel.

The next step is to count the number of each different symbol on a reel. For instance, to count the number of oranges on a particular reel, count the total number of oranges that show in all the distinct sequences. Then, divide this number by 6 (if 6 is the number of symbols on your recorded sequence).

Thus, if you counted 36 oranges altogether on the various sequences for the first reel, there would actually be 6 oranges on this reel. This is because your recorded data showed each particular orange in 6 different positions on each reel.

Another method would be to count the number of times an orange shows in, let's say, the top position. You should, with this same data, count 6 times. This will give you the number of each kind of symbol on each reel. Let us look at the hypothetical example in Figure 2.

Most progressive slots have more than 20 symbols on each reel. Many have four or more reels. (A machine with five or more reels is almost impossible to hit and not worth playing.) I use the example in Figure 2 to make things simpler.

In order to calculate the payback on any machine, first you need to note the payoff schedule. Let's suppose that on this machine the schedule turned out to be what is listed in Figure 3.

We must now compute the number of possible ways of hitting each of the various winning combinations (permutations). For instance, let's take the case of hitting three oranges. All we need to do is multiply the number of oranges on each reel. This would be (6)(1)(8) = 48. Next, we multiply 48 times the payoff, which in this case is 10, and obtain the figure 480.

Orange, orange, bar can be made in (6)(1)(1) = 6 different ways. This is also multiplied by 10 resulting in a figure of 60.

Using similar techniques, we calculate that bar, bar, bar gives us a figure of 1,500 {(3)(5)(1)(100)}; plum, plum, plum results in 252 {(6)(1)(3)(14)}; plum, plum, bar yields 84 {(6)(1)(1)(14)}; bell, bell, bell yields 540 bell, bell, bar yields 90 {(1)(5)(1)(18)}; cherry, cherry, cherry gives us a figure of 210 {(3)(7)(1)(10).

The case of cherry, cherry, - is a little different. What we are really speaking of is cherry, cherry, non-cherry. Thus, we multiply {(3)(7)(19), and then multiplying this by 5. The result is 1,995.

Similarly, cherry, -, - is actually cherry, non-cherry, anything. Here, we multiply {(3)(13)(20), and then multiply by the payoff of 2. This gives us 1,560.

The next step is to add up all these individual results:

$$480 + 60 + 1{,}500 + 252 + 84 + 540 + 90 + 210 + 1{,}995 + 1{,}560$$

$$= 6{,}771$$

This means that 6,771 coins will be paid back by the slot machine (assuming it is a one coin machine) after each of the equally likely combinations has come up exactly once. This does not count the super jackpot payoff. The total number of combinations for this particular machine is simply

$$(20)(20)(20) = 8{,}000$$

On average, our hypothetical machine would pay back 6,771 coins in similar payoffs for every 8,000 coins it took in. Thus, it pays back 84.6 percent and holds 15.4 percent.

$$84.6\% = \left(\frac{6{,}771}{8{,}000}\right)(100)$$

$$15.4\% = 100\% - 84.6\%$$

Use this "hold" percentage to figure your loss per pull. In this case, it would be 15.4 percent of the total amount that you must put into the machine per pull.

A dollar machine requiring I coin would result in a cost per pull of 15.4 cents. A nickel machine requiring 5 nickels costs you 3.9 cents per pull.

With this information, we now have everything we need to know; combine this chapter with the previous and you have what it takes to decide whether a machine actually gives you the best of it. (In this particular example the super jackpot would have to be over 1,229 coins (8,000 - 6.77 1) to give us an edge.)

Figure 1

	Reel 1	**Reel 2**	**Reel 3**
Pull 1	orange	bar	cherry
	cherry	bell	seven
directly	orange	plum	bell
in	bar	cherry	plum
window	plum	orange	bell
	orange	cherry	plum

Figure 1 (con't)

Pull 2	Reel 1	Reel 2	Reel 3
	bar	seven	orange
directly	cherry	bell	bell
in	plum	cherry	plum
window	orange	plum	cherry
	bar	orange	seven
	orange	bar	bell

Figure 2

(20 symbols on each reel)

Reel 1	Reel 2	Reel 3
1 seven	1 seven	1 seven
3 bars	5 bars	1 bar
3 cherries	7 cherries	1 cherry
6 oranges	1 orange	8 oranges
6 plumbs	1 plum	3 plums
1 bell	5 bells	6 bells

Figure 3

cherry	—	—	2 coins
cherry	cherry	—	5 coins
cherry	cherry	cherry	10 coins
orange	orange	orange	10 coins
orange	orange	bar	10 coins
plum	plum	plum	14 coins
plum	plum	bar	14 coins
bell	bell	bell	18 coins
bell	bell	bar	18 coins
bar	bar	bar	100 coins
seven	seven	seven	super jackpot

Important warning: I regret to say that since the publication of the original edition of Getting The Best Of It, things have changed. Many slot machines are now computerized in such a way that the probability of a particular symbol coming up has no relation to the number of symbols on the reel. Thus, the point at which the progressive jackpot reaches a profitable point, can only be determined by statistical analysis for these machines.

Card Counting and Baccarat

Of all the high rolling gambling that goes on in Nevada casinos, none surpasses the game of baccarat. Now that many hotels have giant limits for baccarat, it is the one game that occasionally produces a multimillion-dollar winner - or loser. It is also a reasonably good game for a gambler, because he is bucking a disadvantage of only slightly over 1 percent. The question is whether keeping track of the cards can reduce the disadvantage or even swing the advantage to the player. The answer is theoretically yes.

The card counting technique that I have discovered has never before appeared in print as far as I know. (Edward Thorp developed a counting method that would beat the proposition bet on a "natural" 8 or 9. This bet, however, has since been removed from the layout. My method, on the other hand, is used for finding good bets on the "player" or the "bank."

To start with, the "player" and the "bank" are both dealt two cards. Either one or both may then draw a third card, according to fixed rules. The object is to get as close to a total of 9 as possible.

Tens and picture cards count as zero. Also, only the last digit of the total is counted. Thus, a 23 or a 13 counts as 3. Should either the player or the bank get a two-card total of 8 or 9, the other side cannot draw and thus loses unless it also has a two-card total of 8 or 9. Ties are possible.

If neither side is dealt a natural 8 or 9, it is up to the player to draw first. He must draw a third card if his two-card total is 5 or less. He must stand on 6 or 7. The rules for the bank are not that simple and are designed to give him a small edge over the player.

The bank may stand on as low a total as 3, or may draw on as high a total as 6. It all depends on the value of the player's third card. For instance, if the player's third card is specifically an eight, the bank must stand on 3 or higher. Notice that the player's total after catching an eight must be either 8, 9, 0,1, 2, or 3, as this eight was drawn to an original two-card total of 0 through 5.

The bank, therefore, has a better chance by standing on 3 than by hitting. Similar considerations are used for determining the other rules for the bank.

These other rules are:

1. The bank stands on 4 or higher if the player's third card is an ace, eight, nine, or ten. Otherwise, it hits:

2. The bank stands on 5 or higher if the player's third card is an ace, deuce, trey, eight, nine, or ten;

3. The bank stands on 6, unless the player's third card is specifically a six or seven, in which case it takes a third card, and

4. The bank always stands on 7; if the player stands pat, the bank hits up to 5 and stands on 6.

All these rules are designed to give the bank an edge. This edge, however, is small. The player will still win about 49.3 percent of the time (not including ties). This is about a 1.4 percent disadvantage. A bet on the bank also has the worst of it because a 5 percent commission is charged on winning bets.

To see how this works out, let's assume you make 1,000 typical $10 bets on the bank. You figure to lose 493 of the bets and win 507. Your losses will equal $4,930 and your winnings will total $5,070. However, you must pay 5% of this, or $253.50 in commissions, for a net loss of about $114 per $10,000 bet.

$$\begin{aligned} -\$113.50 &= -(493)(\$10) + (507)(\$10) - (.05)(507)(\$10) \\ &= -\$4{,}930 + \$5{,}070 - 253.50 \end{aligned}$$

This is 1.14 percent. (The exact figures are close to -1.37 percent against the player and -1.16 percent against the bank after commissions.)

As soon as I learned these rules, I started to think about whether certain types of decks can favor the player or the bank. Could the asymmetrical nature of these rules be occasionally taken advantage of? I found that they could be. Favorable situations can exist for both the player and the bank.

Let us first take the case of finding good bets on "player." If you look back at the rules, you will see that the bank will stand on a total of 5 or better if the player's third card is a deuce or trey (as well as an ace, eight, nine, or ten). This can be the bank's downfall.

Suppose, for instance, there were nothing but deuces or threes left in the deck. Both the player and the bank will always be dealt a total of 4. 5, or 6. Both the player and the bank will be hitting their totals of four and standing on their totals of six. However, the player will be hitting when he has a total of 5 (and thus always improving to 7 or 8), whereas the bank will stand on 5 unless the player stands pat.

In other words, the bank is forced by the rules to stand pat on 5 (because the player's third card is a deuce or trey) even though this play will guarantee a loss since the player has made a total of at least 6. Thus, in this case, the bank's normally favorable rules have worked against it. To show how strong this can be, let us examine the extreme case where there are only 6 cards left in the deck - 3 deuces, and 3 treys.

If the player is dealt deuce-deuce and the bank is dealt trey-trey, the player will win or tie.

If the player is dealt deuce-deuce and the bank is dealt treydeuce, the player will always win.

If the player is dealt trey-deuce and the bank is dealt deucedeuce, the player will always win.

If the player is dealt trey-deuce and the bank is dealt trey-deuce, the player will always win.

If the player is dealt trey-deuce and the bank is dealt trey-trey, the player will always win.

If the player is dealt trey-trey and the bank is dealt deuce-deuce, the player will tie or lose.

If the player is dealt trey-trey and the bank is dealt trey-deuce, the player will always lose.

The preceding argument should make it clear how much of an edge the player has with this particular subset of a deck. Out of the seven possible cases, only 1'/2 of them results in a loss for the player. Another two halves result in a tie. The other four and one half cases are a win for the player.

However, it is even stronger than it appears. This is because treydeuce vs trey-deuce comes up 40 percent of the time, while all the other cases only occur 10 percent of the time each.

When this is taken into account, we find that the player only loses 15 percent of the time and ties 10 percent of the time. A bet on the player wins a whopping 75 percent of the time. Ignoring ties, this makes him a 5-to-1 favorite!

This, of course, is the absolute extreme case. A larger deck of half deuces and half treys is still a great edge for the player, though not quite as much as in the preceding example. (Notice that it is necessary that there be both deuces and treys remaining for this concept to work. If the deck contains nothing but deuces or nothing but treys, every hand would be a tie.)

It is not necessary that there be only deuces and treys for the player to have an edge - just a high proportion of them. Likewise, if the deck contains little or no deuces and treys, the bank probably has a big enough edge to overcome the commissions. There is, however, a particular subset of the deck that will result in a much larger edge for the bank. Can you think what it is? It once again takes advantage of the rules of the game.

The answer is a deck that contains one-half eights and one-half sevens. If there were only 6 cards left in the deck, 3 eights and 3 sevens, we would have the results below:

Player's Hand	Bank's Hand	Prob. of Occurrence	Results
7,7	8,8	10%	Bank Wins
7,7	8,7	10%	Bank Wins
8,7	8,8	10%	Bank Wins
8,7	8,7	40%	Bank Wins
8,7	7,7	10%	Bank Wins
8,8	8,7	10%	Player Wins
8,8	7,7	10%	Player Wins

The bank wins 80 percent of the time and loses 20 percent. This is almost as strong as the tree/deuce situation, except this time it is in the bank's favor. Once again. It is only necessary that there be a high proportion of eights and sevens (and approximately half of each) in order for a bet on the bank to have an edge. Likewise, most decks that contain little or no eights and sevens should be in the player's favor.

What do we have here? We see that deuces and treys are in the player's favor, while sevens and eights favor the bank. (I suspect that aces and fours also favor the player and nines and tens favor the bank. but the effect of these cards does not appear to be nearly as significant as the deuce, trey, seven, and eight.)

So. how do we make the best use of this information? I'm afraid I must leave this to the computer experts who can experiment on various counts that make use of my general idea. However. I do have some observations and suggestions.

For one thing, it is clear to me that a favorable bet on the player or the bank comes up quite infrequently. especially against the big shoes. It may even occur so rarely that this chapter will turn out to be only of theoretical interest. However. you must remember that there is not yet any paranoia among the casinos regarding baccarat counting.

Thus, players can bet anywhere from $5 to $50.000 at any time (or even not bet at all, counting on the sidelines, until the deck is favorable for one side or the other). Thus, even if there is a good bet only I in 100 deals, there is money to be made.

As far as what count is --best," an obvious choice would be to use a type of "point count" where the sevens and eights are counted against the deuces and treys. High plus counts would favor one side, while high minus counts would favor the other. If this technique works well, that's fine. However, it is not necessary that the counting technique be that simple. This is because you can use a pencil and paper!

It is common to see gamblers recording wins and losses in baccarat in order to "pick up patterns." This is hogwash. However, while all the suckers are writing down patterns, you can be recording the exact number of critical cards that are out. For this reason, computer specialists can consider developing counting techniques that would be too difficult to apply mentally, but which would be amenable to pencil and paper or even a calculator at the table! The casinos may not know what hit them.

Note: Since this article was published, Peter Griffin, Edward Thorp, and Joel Friedman, among others have done computer work on baccarat. which shows that the game, while theoretically beatable by counting, is. in fact, not beatable as a practical matter. This is because profitable bets occur too rarely.

Crapless Craps
12 Times Odds

This chapter is now a little outdated since 20 times odds is easy to find and Vegas World is no more. But the unique way to figure the house edge still deserves your attention.

There's a game in Las Vegas that deserves analysis. It also deserves a rather strong recommendation for those gamblers who are looking to make a big score from a small investment while bucking only a tiny house advantage. The game is crapless craps with 12 times odds, available at this writing only at Bob Stupak's Vegas World.

Contrary to what you might think, I am not unequivocally opposed to playing a game where the house has a small edge. As a professional gambler, I won't normally play these games. However, those gamblers who are not gambling for a living are actually "getting the best of it" in a sense when they play a game with a small casino advantage where there is a chance to make a score.

There are two reasons for this. First of all, the entertainment value for the player while playing the game. When you are facing less than half a percent disadvantage (as you are in this game) you could put $10,000 in action and still have the evening's entertainment cost you an average of $50, which is less than you would be spending somewhere else.

The second reason involves something that I think economists call "reverse marginal utility." It works like this: The player decides to risk $100 he can easily afford and expects to lose. However, if he doesn't lose it he won't quit until he wins $5,000. On the surface, it appears that he is getting 50-to-1 odds where the true odds of his making the score are maybe 55-to-1.

However, to the casual gambler out for some fun, this really isn't the case because the $100 is almost valueless to him. He is ready to lose it and doesn't really care if he does. Thus, his "psychological odds" are closer to 50-to-O. If you only play occasionally and for the fun of it, this is the better way of looking at it.

Still, even the occasional player should not play games with a large disadvantage. The chances of winning over any length of time are too small. It is especially dumb to play in these games when there are much better games available. The occasional player should also shy aN\ay from games involving only even money bets. as it is harder to make a big win. Of course. the preceding discussion is all meaningless to those players for whom gambling is a business, where having fun is irrelevant and a long-term win is a must. Their games are probably poker or blackjack, and never craps.

However, even those players who would never take the worst of it ought to read the rest of this chapter as the techniques I will use to analyze crapless craps involve some methods that are different from the standard methods but are actually much simpler and can be used in analyzing other games as well.

Whose Advantage?

First of all, for those of you who don't know the rules of crapless craps, they are just as the name implies. You can't crap out on the first roll. If you roll a two, three, or twelve, that is simply your point. You no longer lose automatically. However, if you roll an eleven, this is also a point rather than an automatic win. Of course, a seven on the first roll is still an instant winner. (I am assuming the reader knows the rules of regular craps. However. to refresh your memory, once you establish a point, you win by rolling it again before rolling a seven.)

Before going on with the discussion, I am presenting the accompanying chart. If you don't already know it, you will need to refer back to it at times as my analysis proceeds.

You should immediately be able to see by looking at the chart on the next page, that out of the 36 possible dice throws, the rules of crapless craps are to your advantage on 4 of the come-out throws (two, three, and twelve) and disadvantage to you on only two throws (the two ways of throwing eleven). Thus, it seems that the rules of crapless craps are better for the player when compared to regular craps.

This however. is wrong. See if you can see why before I tell you. The basic reason is that you are not helped that much when your come out roll is a two, three, or twelve, but you are hurt quite a bit when it's an eleven. In regular craps an eleven will win 100 percent of the time. In crapless craps, it wins 25 percent, for a reduction of 75 percent. A three goes from 0 percent in regular craps to 25 percent in crapless craps for an increase of 25 percent. Two and twelve move from 0 percent to 14.3 percent.

Total on Dice	Number of Ways	Odds Against Making Point
2	1	6-to-1 or 14.3%
3	2	3-to-1 or 25.0%
4	3	2-to-1 or 33.3%
5	4	3-to-2 or 40.0%
6	5	6-to-5 or 44.4%
7	6	—
8	5	6-to-5 or 44.4%
9	4	3-to-2 or 40.0%
10	3	2-to-1 or 33.3%
11	2	3-to-1 or 25.0%
12	1	6-to-1 or 14.3%

So what is the disadvantage in crapless craps'? The long way to do this is to go through a weighted average of all the combinations. This is how most mathematicians would do it. Instead I want to show you a better technique that can be used for many different gambling situations. Rather than go through all the possibilities, let's just see how to adjust the percentages from normal craps after taking into account the changes in the rules. To do this we do have to know that regular craps has a disadvantage of about 1.4 percent which means that the player wins about 49.3 percent of the time.

First notice that this 49.3 percent includes the 2 of 36 rolls that are automatic winners on 11. This happens 5.56 percent of the time. In crapless craps only one-fourth of these I 1 s are winners. Almost 4.2 percent of this 5.56 percent now lose. This brings the 49.3 percent down to 45.1 percent. Three is rolled 5.56 percent of the time, also. It is no longer a sure loser, but now wins one-fourth of the time. So we can add about 1.4 percent back onto that figure and get 46.5

percent. Two and 12 combined are also rolled 5.56 percent of the time. These sure losers in regular craps now win one-seventh of the time which adds another .8 percent, bringing the final total up to 47.3 percent.

$$47.3 = 49.3 - 4.2 + 1.4 + .8$$

This is your answer. Without taking odds, crapless craps has not quite a 5.4 percent disadvantage!

It seems wrong, but it's true. Bob Stupak has been known to delight in fooling the "smarts" and this was a good example. (Of course, if they had been truly smart they would have known how to figure it out themselves.)

Taking Odds

Stupak, however, has now completely eliminated this extra edge and then some in this game by offering 12 times odds. The opportunity to take odds in crapless craps helps the player even more than it does in regular craps.

One reason is that this opportunity arises after 30 out of 36 come out rolls in crapless craps rather than 24 out of 36 come out rolls in regular craps. Also, crapless craps affords the opportunity to make a really big win by taking the 3-to-I odds on three or eleven and even more so when taking 6-toI true odds against making a point of two or twelve.

To see this clearly, imagine that you bet $1, roll a 12, take full odds and then make your point. Your Si pass line bet has turned into a $73 profit. Come bets also allow 12 times odds. Thus if you are constantly making come bets and taking full odds, a "hot hand" without a seven will wreak complete havoc on the game. (By the way, for those of you who don't see why the true odds of making a point - let's say twelve - are 6-to-1, even though there is only 1 way to roll a twelve, it is because there are only 6 ways of rolling a seven. The other numbers don't matter, which leaves us with a ratio of 6-to-1.)

How do we figure the percent for crapless craps with 12 times odds? Once again, I'll show you an easy way. Assume 36 come out rolls betting $1 each time. Six times on the average the bet will stay $1 when the roll is a seven, but the other 30 times we will get a total of $13 in action after taking the odds.

Altogether that is $396. But we still average losing 5.4 percent of $36 or about $1.94 (since the odds bets have no house edge). That $1.94 is less than 0.49 percent of $396. The house edge is less than one-half of 1 percent.

$$0.49\% = \left(\frac{(.054)(36)}{6 + (30)(13)} \right) (100)$$

Let's use the same technique to calculate double odds in regular craps. In 36 come outs at $1 each, you have 12 come outs where only $1 is bet. The other 24 will have a total of $3 put into action. Altogether, $84 is bet and an average of 50.4 cents is lost (1.4 percent of 36); 50.4 is about.6 percent of 84.

$$0.60\% = \left(\frac{(.014)(36)}{12 + (24)(3)} \right) (100)$$

Double odds craps is thus not as good as 12 times crapless craps, even from a purely mathematical standpoint.

Complete Keno Odds

I have never before written an article about keno. The reason is simple: what is there to write about a game with a 25-30 percent disadvantage? The only way to beat the game is if the casino is putting on some kind of promotion where the player has the best of it (as happened at the Sahara Tahoe a few years ago when they were giving away South African Krugerrands to anyone who could hit 6 or more on a ten-spot).

This will be my one and only chapter on keno. It contains eyen1hing anyone needs to know about the game in the form of a chart. Furthermore. this chart is set up in such a way as to make it as simple as possible to evaluate or design any payoff schedule. It can be used by a casino to design a schedule with any desired hold. Or it can be used by a player to evaluate the advantage or disadvantage for any pay-off schedule a casino may be using.

The figures in this chart are accurate to at least 1/10 of a percent.

For example. let's say a $1 eight spot is paying off as follows: Hit 4 numbers, pays $1: hit 5 numbers, pays $8: hit 6 numbers, pays $80: hit 7 numbers, pays $1,600: hit 8 numbers, pays $20,000.

Using the following chart we see that in 230.116 trials we should Win:

Individual Payoff	Chances out of 230,116	Total Payoff
$20,000	1	$20,000
$1,600	37	$59,200
$80	545	$43,600
$8	4,212	$33,696
$1	18,755	$18,755
Total	23,550	$175,251

Thus. the typical payoff schedule only pays back an average of $175,251 for every $230,116 it takes in - which is 76 percent.

If you are devising keno payoffs you simply work backwards. For instance. ifvou want a six spot to hold 20 percent, you want to hold $1.550 of the $7.752 you take in. All you do is manipulate the payoffs in any way that will pay back about $6,202 for every $7,752 taken in. It's easy to do using this chart.

One Spot

In four trials you will average hitting:

1 number	-	1 time
0 numbers	-	3 times

Two Spot

In 16⅔ trials you will average hitting:

2 numbers	-	1 time
1 number	-	6⅓ times
0 numbers	-	9⅓ times

Three Spot

In 72 trials you will average hitting:

3 numbers	-	1 time
2 numbers	-	10 times
1 number	-	31 times
0 numbers	-	30 times

Four Spot

In 326 trials you will average hitting

4 numbers	-	1 time
3 numbers	-	14 times
2 numbers	-	69 times
1 number	-	141 times
0 numbers	-	101 times

Five Spot

In 1,551 trials you will average hitting

5 numbers	-	1 time
4 numbers	-	18¾ times
3 numbers	-	130¼ times
2 numbers	-	419½ times
1 number	-	629 times
0 numbers	-	352½ times

Six Spot

In 7,752 trials you will average hitting:

6 numbers	-	1 time
5 numbers	-	24 times
4 numbers	-	221 times
3 numbers	-	1,006 times
2 numbers	-	2,390 times
1 number	-	2,818 times
0 numbers	-	1,292 times

Seven Spot

In 40,980 trials you will average hitting:

7 numbers	-	1 time
6 numbers	-	30 times
5 numbers	-	354 times
4 numbers	-	2,139 times
3 numbers	-	7,172 times
2 numbers	-	13,386 times
1 number	-	12,916 times
0 numbers	-	4,982 times

Eight Spot

In 230,116 trials you will average hitting:

8 numbers	-	1 time
7 numbers	-	37 times
6 numbers	-	545 times
5 numbers	-	4,212 times
4 numbers	-	18,755 times
3 numbers	-	49,425 times
2 numbers	-	75,512 times
1 number	-	61,317 times
0 numbers	-	20,312 times

Nine Spot

In 1,380,700 trials you will average hitting:

9 numbers	-	1 time
8 numbers	-	45 times
7 numbers	-	817 times
6 numbers	-	7,897 times
5 numbers	-	45,013 times
4 numbers	-	157,460 times
3 numbers	-	339,800 times
2 numbers	-	436,900 times
1 number	-	304,700 times
0 numbers	-	88,020 times

Ten Spot

In 8,912,000 trials you will average hitting:

10 numbers	-	1 time
9 numbers	-	55 times
8 numbers	-	1,207 times
7 numbers	-	14,359 times
6 numbers	-	102,310 times
5 numbers	-	458,340 times
4 numbers	-	1,312,900 times
3 numbers	-	2,383,100 times
2 numbers	-	2,631,400 times
1 number	-	1,600,300 times
0 numbers	-	408,100 times

Eleven Spot

In 62,382,000 trials you will average hitting:

11 numbers	-	1 time
10 numbers	-	66 times
9 numbers	-	1,770 times
8 numbers	-	25,665 times
7 numbers	-	225,062 times
6 numbers	-	1,260,350 times
5 numbers	-	4,621,300 times
4 numbers	-	11,140,600 times
3 numbers	-	17,366,200 times
2 numbers	-	16,723,000 times
1 number	-	8,977,700 times
0 numbers	-	2,040,400 times

Twelve Spot

In 478,264,000 trials you will average hitting:

12 numbers	-	1 time
11 numbers	-	80 times
10 numbers	-	2,596 times
9 numbers	-	45,627 times
8 numbers	-	487,640 times
7 numbers	-	3,361,000 times
6 numbers	-	15,404,000 times
5 numbers	-	47,533,000 times
4 numbers	-	98,410,000 times
3 numbers	-	133,784,000 times
2 numbers	-	113,717,000 times
1 number	-	54,410,000 times
0 numbers	-	11,108,700 times

Thirteen Spot

In 4,065,200,000 trials you will average hitting:

13 numbers	- 1 time
12 numbers	- 97½ times
11 numbers	- 3,835 times
10 numbers	- 81,558 times
9 numbers	- 1,056,540 times
8 numbers	- 8,875,000 times
7 numbers	- 50,064,000 times
6 numbers	- 193,103,000 times
5 numbers	- 511,724,000 times
4 numbers	- 923,946,000 times
3 numbers	- 1,108,736,000 times
2 numbers	- 839,951,000 times
1 number	- 361,032,000 times
0 numbers	- 66,652,000 times

Fourteen Spot

In 38,910,000,000 trials you will average hitting:

14 numbers	- 1 time
13 numbers	- 120 times
12 numbers	- 5,752½ times
11 numbers	- 148,287 times
10 numbers	- 2,324,400 times
9 numbers	- 23,667,000 times
8 numbers	- 162,708,000 times
7 numbers	- 772,414,000 times
6 numbers	- 2,558,620,000 times
5 numbers	- 5,913,260,000 times
4 numbers	- 9,424,250,000 times
3 numbers	- 10,079,420.000 times
2 numbers	- 6,859,600,000 times
1 number	- 2,666,100,000 times
0 numbers	- 447,521,000 times

Fifteen Spot

In 428,010,000,000 trials you will average hitting:

15 numbers	- 1 time
14 numbers	- 150 times
13 numbers	- 8,850 times
12 numbers	- 278,038 times
11 numbers	- 5,282,712 times
10 numbers	- 65,083,000 times
9 numbers	- 542,358,000 times
8 numbers	- 3,137,931,000 times
7 numbers	- 12,793,100,000 times
6 numbers	- 36,957,840,000 times
5 numbers	- 75,394,000,000 times
4 numbers	- 107,003,400,000 times
3 numbers	- 102,894,000,000 times
2 numbers	- 63,319,400,000 times
1 number	- 22,376,000,000 times
0 numbers	- 3,430,990,000 times

Part Five
Sports and Horse Betting

Sports and Horse Betting

Introduction

Does the subject of sports betting legitimately belong in a book on getting the hest of it? Some professional gamblers think not. The bookies' `'vigorish" must put you at a disadvantage, they contend. You are never getting your fair odds.

These pros would be right if they were speaking of cold inanimate objects such as cards or dice. However, the fact that you are betting on sporting events allows for the possibility that the bookie's odds for a particular event are wrong and that you can take advantage of it. My experience has been that the bookie does occasionally make a big enough error to produce a decent edge for the player. Thus, I include sports betting in this book.

However. I must caution you. It is very hard to beat sports. That 4-5 percent "vig" that a bookie typically has is enough to take care of small errors in his line. As I explain in two chapters in this section, if there is not a large enough error in the bookie's line, there is no bet. This unfortunately is what happens most often.

Anyone who wants to in at sports betting must understand two things. He must first know the various types of bets that are available to him and which ones are bucking the smallest percentage. Secondly. he must know which teams to bet on.

The following chapters should give him a good start in his understanding of both aspects of sports betting. This new edition now includes a chapter on horse race betting as well.

Mathematics of Sports Betting

I consider the following chapter must-reading for anyone who bets sports at all. This goes double for any readers who may be bookies.

Both bookie and player alike should understand how to evaluate the bookie's vigorish. How much is it and where does it come from. The greater the bookie's mathematical edge, the harder it is to overcome it by handicapping. (See the following chapters on beating sports.) This chapter analyzes every aspect of the bookie's edge; the point spread and how it is derived, the money line, over and under bets, hockey lines, parlays and future bets. It also discusses going for "middles."

Can sports be beaten? Cynics say there is no way you can win since you are bucking a disadvantage usually at least twice as great as the one you fight in craps. Others say, the vigorish makes no difference since you are betting on human beings, not little white cubes. As for me, I'm sure sports can be beaten. I have two friends who are beating it. However, beating sports takes a unique talent and the patience to bet very few games. Otherwise, even knowledgeable sports bettors will eventually be eaten up by the vigorish, which is the bookie's edge. In this article I will explore the various ways the bookie obtains this edge.

The Point Spread

The money line and the point spread are the two main tools with which the bookie works. The point spread, sometimes called the "line" is the main factor in football and basketball betting. To illustrate how it works, let's take an example. Suppose the Oakland Raiders are playing the Miami Dolphins at Miami. The bookie might make Oakland a 4 point favorite. This is the point spread. It means that if you bet Oakland, they must win by more than 4 points in order for you to collect. If you bet Miami, you win not only if they win the game, but also if they lose the game by less than 4 points. If Oakland wins by exactly 4, it is a tie and everybody gets their money back (with the exception of those rare and unfair cases where the bookie specifies he wins on a tie). If the line were 4%2 (or 6V2. 10V2 and so on). there could be no tie.

How is the point spread determined'? There are two possible methods. One is to try to determine a number such that the final scores will fall above and below that number exactly equally. If the line maker can do this accurately, no one can have an edge betting the game, for in order to have a decent edge betting the game the "true line" must be at least two points different from the bookies' line. as I'll explain shortly. If the line maker is determining his point spread by, this first method, our only hope is that he is making a mistake and that we are smart enough to recognize this mistake. This won't happen very often.

The other method of determining a point spread is to find a number that will split public opinion. In this case, the line maker is not so interested in finding the true line but rather a line that figures to be bet into equally on both sides. With equal money bet on both sides the bookie makes a sure profit. It is this method that is more frequently used, and because of this the smart bettor has a chance since he is no longer fighting an expert handicapper, but rather public opinion. Even if the line maker thinks that Oakland is a good bet giving 4 points as in the previous example, he will still usually make the line 4 if he knows that number will split the betting.

(Many experts feel, however, that the line maker will sometimes get stubborn and thus force the bookies to gamble. Using the same example, let's say the line maker thought that it was an even game or possibly that Miami should even be a slight favorite. He knows that Oakland (-4) should split the betting. In spite of this, he comes with a line of Oakland (-2). Now the majority will bet on Oakland, and the bookie will need Miami to win. Of course, if Miami does win or "cover" as the line maker suspects, the books will show a windfall profit. Only in clear-cut situations will they take this chance. This is why experts say to bet against the public when the line looks strange and the public is betting mostly one way. Now you are betting on the bookie, who didn't have to gamble on this game.) The bookie's edge. when using point spreads, comes from the fact that he makes you lay I I -to-10 odds when making a bet. If you want to win $50 you must risk $55. The average player will be down $5 after even, two games for a negative expectation of $2.50. Since he will lose $5 per $110 bet his disadvantage is 5/1 10 or 4.54 percent. The bookie has a positive expectation of $2.50 but an advantage of 5 percent since he figures to in $5 per $100 he risks. Some bookies have the nerve to make you lay 12-to10 which puts you at an 91/3 percent disadvantage.

To break exactly even laying 1 1-to- i 0. you must win 1 1 out of every 21 games or 52.38 percent of your bets. Since one point in football is worth about 21/2 percent on average, we see that the true line must be off at least two points from the bookie's before wye can have an kind of an edge. With two extra points going for us, we should in about 55 percent of the time. It should now be obvious Nyhy the bookie almost al\t ays x\ Ins in the long run.

The Money Line

Most baseball games are too even to put up a point spread involving runs. Still the games are not so close that the bookies could in simply by asking for I I-to-10 on either side. Usually, one team is more of a favorite than that. To solve this problem, bookies use what is called a money line. For example. let's say the Yankees are playing the Twins at New York. After the line maker does his homes\ork using one of the two methods discussed in the previous sections. the bookie will have something like this on the board:

Twins +1.30
Yanks -1.50

This means that if you want to bet the Twins. you must bet $1 to yin $1.30; if you want to bet the Yankees, you must bet $1.50 to in $1.00. In other words, if you want the Twins, you get 6'/z-to-5 odds. If you want the Yankees, you must lay 71/2-to-5 odds. You may, at times. see this written as "Yanks 71/2-to-6`/`.- It should be obvious why the bookie has an edge here. If the line maker is right. the true line should be about 7-to-5. The Yanks should therefore win 7 out of 12 games, and a player laying $15-to-$10 would find himself down $5 after 12 bets ($180) for a disadvantage of 2.78 percent

$$-\$5 = (7)(\$10) - (5)(\$15)$$

$$2.78\% = \left(\frac{-5}{(12)(15)} \right)(100)$$

Those players taking $13-to-$ 10 on the Twins will also be down $5 after 12 bets, but in their case the disadvantage is 4.17 percent, since their total amount wagered is only $120.

$$-\$5 = (5)(\$13) - (7)(\$10)$$

$$4.17\% = \left(\frac{-5}{(12)(10)}\right)(100)$$

We see that the percent disadvantage is less when we bet the favorite, though the negative expectation remains the same.

Both the percentage disadvantage and negative expectation decrease as the line moves further from even money. The reader should verify for himself that a line like

Favorite -1.80
Underdog +1.60

results in a smaller disadvantage to the bettor than a line like

Favorite -1.30
Underdog +1.10

Notice that in the examples of money lines given thus far, there has been a 20-cent difference between the price on the favorite and that on the underdog. When this is the case it is called the twentycent line, which is the line most baseball bookies deal. Notice that football and basketball are simply special cases of this twenty-cent line where both teams are -1.10. Interestingly, it is at this point that the bettor's disadvantage is at its peak.

Another special case of the twenty-cent line is

Favorite -1.20
Underdog Even

It also could be written

Favorite -1.20
Underdog +1.00

In both these cases the difference in prices is 20 cents.

The Ten-Cent and Other Lines

Some bookies, eager for business, are willing to operate on a very small profit margin and thus deal the I 0-cent line. An example would be

Favorite -1.40
Underdog +1.30

When betting the ten-cent line your disadvantage is cut approximately in half to a figure usually less than 2 percent.

it is more likely that you will see a line higher than 20 cents such as the forty-cent line. This is especially apt to occur as the line strays further from even money since the bookie's edge decreases on any particular line the further it strays. It is therefore likely that a true 2to-I shot \yill be quoted as

Favorite -2.20
Underdog +1.80

rather than

Favorite -2.10
Underdog +1.90

so that the bookie can get his edge back to where he thinks it should be.

As a game becomes more and more one-sided or as an event such as a boxing match becomes harder to handicap, the difference in the line increases. For example. a bookie who was dealing money lines on football games had it

Redskins -6.10
Packers +5.40

or a seventy-cent line on a Monday night game. The point spread on this game was 14. Fights are usually even worse:

Foreman -4.00
Young +3.00

Here you took 3-to-1 or laid 4-to-1.

Over & Under

A bet that has become increasingly popular. especially in Nevada. is the over and under bet. Here you are betting on the total number of points or runs scored by both teams. The bookie puts up a number and you can bet that the total will be above or below that number. A tie is usually no bet. Typical numbers range from 30-48 in football. 180-210 in pro basketball. and 7-12 in baseball. The player usually has to lay I I -to-10 on these bets but some bookies charge 6-to-5. It is also possible that you will see (especially in baseball), a line like this:

Over 7½ -1.20
Under 7½ Even

In this case. you would have to lay 6-to-5 odds ifyou like the over but could get the under at even money.

Hockey Lines

Because of the likelihood of ties in pro hockey, those bookies who book hockey action have found it very profitable to deal another kind of line. An example might be:

Rangers +1½ goals
Montreal-2 goals

Thus. if you bet the Rangers you get I"/2 goals. but if you bet Montreal you must lay 2 goals. Where is the bookie's edge? Well, what happens if Montreal wins by exactly two goals? He collects from the Ranger bettors but ties with the Montreal bettors. This should happen in the neighborhood of 20 percent of the time if not more often. The bookie therefore N\ins an average of half the money bet about 20 percent of the time for a 10 percent edge. When the game doesn't fall on the number, the bookie will break even on average.

Here's another way of looking at it. If the line is right, Montreal will win by more than two goals 35 percent of the time. Montreal will win by exactly two goals 20 percent of the time and Montreal will either in by less than two goals, tie or lose 45 percent of the time.

Thus. Ranger bettors will win 45 percent of the time and lose 55 percent. When there is a decision, Montreal bettors will win about 44 percent and lose about 56 percent. Both get even money where they should be getting about 6-to-5 odds, assuming conservatively that there are ties 20 percent of the time. This is approximately equivalent to a forty-cent line!

When a baseball game is a mismatch, bookies frequently put up a runs line rather than a money Iine. Thus, when the Tigers play the Yankees in New York, the Yankees may be a 2 or 2'h run favorite. Just as in football or basketball, you would pick a side and lay 11 to-10. However, greedy bookmakers will put up the more profitable hockey line in this case. You \\ill see something like

Tigers +2 runs
Yanks -2½ runs

One casino had the unmitigated gall to do both at the same time! They were actually not ashamed to have on their board something like

Tigers +2 runs -1.10
Yanks -2½ runs -1.10

You had to lay 1 1-to-10 for the privilege of making a bad bet at even money! I can't say much for the people N\ ho bet into this line.

Parlays

When you parlay two or more bets. they must all win (or tie) for you to collect. Your odds should therefore be correspondingly higher. Theoretically, the odds on a parlay should be the same as if you took your original bet plus all your winnings, put them on the second and succeeding games yourself and «on all the bets right up to the last one. If you made a $110 parlay on two football games yourself, your $210 bankroll after one winning bet will win you about $191 on the second game for a total profit of $291 on your original $110 bet. This works out to about 13.2-to-5 odds. Most bookies give you 13-to-5 odds, which isn't bad. Some are not so generous and give only 5-to-2 odds or even 7-to-3 odds on a twoteam parlay. Stay away from them. It is best to parlay two games yourself if you can - something which is obviously impossible when the games are simultaneous. (Strangely enough liberal bookies lay 6-to-1 on a three-team parlay, which is slightly better than if you parlayed the games

Most bookies will give you the correct parlay price when you parlay baseball Notice how the effect of the vigorish is compounded when you bet a parlay. Since the correct price on a tvvo-team parlay when both games are even is 3-to-1, you have a 10 percent disadvantage taking 13-to-5 instead of your normal 4.54 percent disadvantage. Taking 6-to-I on a three-teamer, rather than the correct 7-to-1, gives you 12.5 percent the worst of it.

Teasers

Many bookies give you the opportunity to bet what are called teasers. They apply mainly to football and basketball. When you bet a teaser, you get extra points in return for lower odds. If, for instance, you bet what is normally a 10-point favorite in a six-point teaser, you only lay 4 points. If you bet the underdog, you would get 16 points. Since each point occurs about 2'/z percent of the time, 6 extra points should make your bet about a 65 percent shot. The bookie might make you lay something over 2-to-I odds. However, what is usually done to disguise the high price you are laying is to make you bet two or more games (and win them all) in the teaser. A sign on the wall of the bookie shop will look something like this:

SIX POINT TEASER (Ties lose)
Bet 2 Teams: Even Money
Bet 3 Teams: 9-to-5
Bet 4 Teams: 5-to-2

Now the chances of two 65 percent shots both winning is 42'/4 percent. The bookie is only paying even money. Your disadvantage is over 15 percent and that is not including the fact that many bookies count ties as losses. Teasers with three or more teams are even worse.

Six-point teasers are only one of many different types of teaser. Basketball has four-point teasers and football frequently has seven, ten, and fourteen-point teasers with appropriate shifting of the odds. Interestingly enough, some teasers -

especially 14 pointers, where the bettor is usually required to win 4 out of 4 and lay 6-to-5 odds - have been unprofitable for the bookie in recent years. The reason is that teasers are the one kind of bet NN here bettors on both sides of a game can win. All that is required is that the final scores fall within 6 points (or 10 points or 14 points) of the point spread. Since this has been happening quite a lot lately. the bookies have found themselves losing to the random dummy since any teaser bet has been winning. With football results running so true to form, many bookies have been forced either to abandon teaser bets (especially the fourteen-point variety) or reduce their odds.

Middles

Hitting a middle for big money is the dream of every sports bettor. The opportunity arises when two bookies have a major difference in their lines on a particular game. It can also be done on over-under prices. This will occasionally happen because one bookie will move his line to encourage action on one particular side in order to even up his books. It is more likely to happen when the two bookies are in different parts of the country.

Suppose a team is an 8 point favorite with one book and a 5'/z point favorite with another. Let's say I bet $1,100 to win $1,000 on the underdog with the first book and $1,100 to win $1,000 on the favorite with the second book. If the favorite wins by 6 or 7 points I've won both bets or $2,000. If it wins by eight points I tie one bet and win the other for $1,000 profit With any other result I win one bet and lose the other for a net loss of $100. Thus, I am risking $100 to win as much as $2,000. Using the rule of thumb that each point occurs about 2%2 percent of the time (of course, some numbers such as 7 occur more often than other numbers such as 8), I find that after 200 such, bets I Figure to win $2,000 10 times, win $1,000 five times and lose $100 185 times for an overall profit of $6,500 or $32.50 per bet.

$$\$32.50 = (\$2{,}000)(10) + (\$1{,}000)(5) - (\$100)(185)$$

Since I am really only risking $100 rather than $2,200. I have an edge of 32'/2 percent on money risked. Any discrepancy of two points or more is worth a try at a middle. Even a 1'/2 point difference is okay if it involves a frequently hit number such as 3, 4, or 7.

A middle in baseball is a different animal. Here, instead of risking a little for the small chance of winning a lot, you bet a lot for the sure chance of winning a little. Thus, in the rare case where you may be able to lay 1.30 on the favorite and take 1.40 on the underdog, you might bet $1,300 to win $1,000 one way and bet $1,000 to win $1,400 the other way. If the underdog wins you win $100. If the favorite wins you break even. If instead you bet $960 to win $1,344 on the underdog. you would make a sure profit of either $44 or $40. Some experts try to anticipate middles by betting early in the week on a game where they think the line will move. If they are right, they bet the other way at the end of the week. However, if the line doesn't move, they either have to sweat out a bet or pay the vigorish to "off' the bet. If. God forbid, the line moves in the wrong direction, offing the bet will result in the bookie having a middle on them! For this reason only those players who are real pros at predicting line movements should attempt to anticipate middles.

Pennant Prices

As a season is about to start, many bookies will put up odds on each of the teams to in the pennant or division or conference, as the case may be. A typical example would be:

To Win AFL		
Los Angeles	3-to-2	40%
Pittsburgh	3-to-1	25%
Baltimore	4-to-1	20%
New England	4-to-1	20%
Cincinnati	6-to-1	14%
Denver	8-to-1	11%
Miami	10-to-1	9%
Cleveland	10-to-1	9%
Houston	12-to-1	8%
San Diego	20-to-1	5%
New York	50-to-1	2%
Kansas City	100-to-1	1%
Buffalo	100-to-1	1%

To figure out the bookie's edge we must change all these odds to approximate percentages using the technique I explained in Chapter Two. We then add them up to obtain a total. In this example it is 165 percent. The larger the total the greater the edge for the bookie. 165 percent is typical for Las Vegas, although I have seen totals over 200 percent. Your exact percentage disadvantage can be obtained by taking the excess in this total above 100 and dividing it by the total itself. In this case it would be

$$\frac{65}{165} \textit{ or } 39.4\%$$

Beating Sports: Part I

Although the following two chapters appeared in Gambling Times, they were originally written especially for this book. I felt they had to be written to make this book complete.

You can have the best of it when betting on sports in spite of the bookie's vigorish against you. However, most people approach sports betting the wrong way. Read on and learn the right way.

Over the years my readers have, I hope, come to expect one thing from my writings, if nothing else. That is when I write something it is correct (save for a couple of minor errors.) I won't write something because I think it's true. I have to know it's true. Players have come to trust that when they ask me any questions about gambling, my answer can be counted on. If I'm not sure, I'll tell them.

It's because of this reputation that I am reluctant to write this chapter. You see, there is a 2 percent chance that some of the things in this chapter are wrong!

This would normally be unacceptable to me. But since we aren't dealing with purely mathematical considerations, I must risk writing something incorrect, if I want to communicate to my readers what I know about sports betting.

I have never mentioned this in print before, but I used to bet sports for a living. I had no other income. I still bet sporting events occasionally and earn a hefty profit. (I don't do it full time, simply because the time it would take away from poker isn't worth it to me.)

There is no doubt in my mind that sports can be beaten. This article will show you how. (I assume that the reader is conversant with sports betting, as far as, understanding the meaning of point spreads, money lines, etc. Those of you who aren't, should read the previous chapter "The Mathematics of Sports Betting.")

There are four or five methods that will beat sports, but they all have one thing in common: In order for any of them to find a good bet, the bookie's line must be wrong. This is an extremely important concept which is frequently misunderstood even by so called experts.

To understand this concept, let's say you know the greatest football handicapper who ever lived. You might even say he is a `'perfect" handicapper. You ask him who you should bet on the Monday night football game. Suppose the Redskins are a 3'/z point favorite over the Giants. If your friend is honest, he may tell you that there is no bet.

This is not the same thing as saying that he has no opinion. He may have a very strong opinion and still have no bet. This is because there is some luck in all sporting events. The same two teams playing each other many times under exactly the same circumstances will not have exactly_ the same result every time. Your friend may forsee that if this game were to be played 100 times under the same conditions, the Redskins would win by 4 points or more 52 times. and the Giants would cover the spread 48 times. (A lesser handicapper might see the Giants covering 58 out of 100: he will make a bad bet on the Giants.)

Unfortunately. our perfect handicapper cannot profit from his perfect knowledge of the possible outcome of the game. Neither side xyi11 in the approximate 52.5 percent of the time necessary to break even after laying 11-to-10. All the handicapper can do even «-ith his expertise is lay off the game (or possibly have a small edge by betting the Redskins man-to-man without paying the juice.)

To put this another \tiay. when the bookie's line is right (meaning the game figures to fall on either side of the point spread about half the time), there is no good bet - no matter how good a handicapper you are. Consider also that the bookie's vigorish gives him some margin for error. His line may be wrong by as much as a point in football and basketball. and as much as 10 cents in baseball. yet you would still have no real edge on these games in most cases.

This is the reason that the results of the famous Castaways Football Handicapping contest resembled the results of 200 people flipping coins. When the line is right, as it usually is, it is a coin toss. A much truer test of handicapping ability would result if contestants were required to pick. let's say. only their three or four best bets a week.

If the bookies' lines were always correct or even close, there would be no way to beat sports. This is an undeniable fact. Furthermore. it is almost certainly true that the line on most games is so close to being correct that the game is unbettable. But a significant minority of bookies' lines have been moderately to severely "off' and this can be counted on to continue. There are three reasons for this:

1. The line maker may not have properly evaluated the factors involved in a particular game.

2. The line maker does properly evaluate the game but knows that the betting public is biased toward one team. (He might then put up what he knows is an "incorrect" line in order to even up the betting on both sides.)

3. The line maker puts up a correct line, but the betting action moves the line to a point where it is no longer correct. (In this case, if you did bet. you would bet against the public money.)

The foregoing is all absolutely true. There is no 2 percent chance it is wrong. It is only when we try to find these incorrect lines that no method is foolproof. However, the following methods have worked for me in the past and I'm quite sure they will still work in the future.

The easiest way to find an incorrect line is simply to find two bookies with two significantly different lines. For instance, if one bookie has the Lakers a 3 point favorite and another bookie has them a 6 point favorite, at least one of them must be wrong by the 1'/~ points necessary to have a pretty good bet.

One way to take advantage of this situation is simply to bet both ways - lay the 3 and take the 6. This is called going for a middle. With a 3 point discrepancy in either football or basketball, a middle must certainly show a profit in the long run. Even if you are quite sure that the 6 point line is the right one. it still might be better to play a middle rather than laving the 3 points. This is because you can afford to make a much bigger bet with a limited bankroll than if you are betting one side only.

For instance, you might only be able to afford a $220 bet on the favorite. You can bet $1,100 each way, however, and really only risk losing $100 when the result doesn't fall in the middle. You have a smaller percentage edge this way, but you have put 10 times as much money in action thereby providing a higher mathematical expectation.

If you strongly believe that laying the points is a far better bet but you have a limited bankroll, you can get the best of both worlds by, let's say, betting $1,100 on the favorite and $770 on the underdog. You still give yourself a chance for a middle, but you now also show a profit if the favorite wins by more than 3 points.

Unfortunately, it is very rare to find a 3 point discrepancy between two lines. An analogous discrepancy in baseball lines would give the player the opportunity for an automatic profit.

For instance, the Yankees may be 1.60-1.40 with one bookie and 1.30-1.10 with another. With this much of a discrepancy, you can guarantee a small profit by laying the 1.30 and taking the 1.40.

Once again, this is an extreme case. It's much more likely that you will find 1.60-1.40 in one place and 1.50-1.30 somewhere else. This is no longer automatically profitable as the true line may be about 1.45. In this case your handicapping ability must still come into play.

It is virtually unheard of to find this much line discrepancy between two Vegas or Reno bookies at the same time. The line may move much during the week, but you can't take advantage of this unless you can anticipate the move. A big middle may, however, occasionally occur between two illegal bookies, especially in two different states. Just don't complain to me if playing the middle lands you in a state or federal penitentiary.

When there is a smaller discrepancy, say I'/2 points, between two lines, it is still possible to take advantage of it. If either of the lines is correct, the other line is therefore wrong by the 1'/2 points necessary to have a decent bet. But it may be that the true line falls between the two bookies' lines. In this case, neither bookie is off by more than a point and there is no good bet.

That's why in this situation handicapping ability comes strongly into play. But, you can probably show a profit in this situation even without handicapping if one bookie's line is 1'/2 points or more away from all the other Las Vegas bookies. If they have the Knicks a 5 point favorite and one bookie has them 6'/z points, you can usually assume the Vegas bookies are right and take the 6'/z. If, however. in this same case the Vegas bookies have them 5 points and you find a 4'/z point and a 6, you have found a 1'/z point difference, but neither line is off by more than a point. Now there is no decent bet.

Most of the professional sports bettors in Nevada use a combination of handicapping and finding the best line. They bet a lot of games and the better ones win in the neighborhood of 55 percent of their football and basketball bets. I think these pros' successes come more from the access to all the lines than their handicapping ability. Still their opinion does have some merit. If it didn't, they would not even do as well as they do just from simply finding the best line.

There is nothing wrong with doing some legwork to bet into the best line. On the contrary, even when using the deeper handicapping techniques that I will be explaining in the next chapter, you should always try to find the best line. But unlike the Vegas pros, you should win with these techniques even when you take a point or two the worst of it.

Beating Sports: Part II

There are times, especially in football and basketball, when the bookie's line is way off The line may be the same all over the country. but this is simply because the public has missed a key point that makes the correct line on a particular game far different from what common sense dictates. That key point usually involves either matchups or the psychological situation. especially the latter. If you know your teams well, you can frequently find a good bet involving favorable matchups. The reason for this involves the way the bookies and the general public decide on a line. It involves "power ratings." If team A is rated at 98, Team B at 94. and Team C at 91, Team A is typically a 4 point favorite over B and a 7 point favorite over C. Team B would be a 3 point favorite over C. An adjustment is then made for home-team advantage and other factors.

About the only sport where this method would be accurate would be weight lifting. In games involving defense, this technique for determining the line may be quite wrong. It could well be that A deserves to be 4 points over B yet deserves to be only 2 points over C. This can occur if for some reason C matches up well with A.

The reason may not always be discernible. However, even when it is not, this situation may be spotted from the results of the two teams playing each other in the past. If one team has been doing consistently better than expected against another team, there may well be a reason involving matchups rather than just luck. If the line appears to be a "trap" (to be discussed later) this could confirm your suspicion.

A good matchup situation can frequently be predicted the first time it arises if you know your team and the players. For instance, take an NBA team whose major strength is a great defensive center. They now play a team with a low-scoring, relatively weak center. What should you do? The possibly surprising answer is that you bet against the team with the good center! Its power rating has been derived on the assumption that the center will prevent a lot of points. But in this game, the center's value will be wasted to quite a degree since the other teams center doesn't figure to score much anyway.

More specifically. suppose this particular defensive center usually holds the man he is defending down to 60 percent of his average. His contribution might normally then be about 7 points saved. However, if his opponent averages 8 points a game, he might only save 3 points of this and thus be over-valued for this game. Of course if this team happened to be playing a team with a center that averages 30 points a game, the converse holds true and the team with a good defensive center should now be bet.

An example from football might arise if a team's major strength is its pass rush. If it now plays a good running team, you should consider betting against it. Its normal strength is overvalued.

I could give you many more examples of how matchup situations may be exploited, but space does not permit. I think I've given you the idea. though. Just realize that this type of betting requires a thorough knowledge of the teams and players involved. Since I no longer have time to do this research, I personally no longer make bets of this nature.

In sports betting there are opportunities to find good bets, including the best kind of'het, without knowing the teams well. This occurs at times when the most important factor is not which teams are involved as much as the situation. Evaluating the situation is probably the most important thing in sports betting.

Before going into situation evaluation in detail, I should point out a couple of technical factors. Almost all my football and basketball bets are on small underdogs.

Besides the fact that the psychological edge is usually with the underdog (to be explained shortly), you have some other things going for you. For one, your team is always on your side. I mean, they are always trying to win the game which automatically means they are trying to cover the spread.

Such is not the case with favorites. You might properly evaluate a team as being easily capable of covering a 6 point spread, yet it won't, simply because it is trying to ensure a win, rather than score points. A team may be up by 1 l points with two minutes to go and now permit its opposition to march down the field slowly in order to prevent the "bomb." They now wind up winning by only 4. Hasn't this happened to you in the past'?

A second technical reason to lean toward small underdogs is that the public tends to over bet the favorite, thereby driving the line up. They reason, incorrectly, that if a team figures to win the game. what's the big deal about laying 3'/z points? Well, it is a big deal! Even if you think a team will win the game 60 percent of the time, it isn't worth laying 3'/z. Your team figures to win by exactly 1, 2 or 3 points over 10 percent of the time.

I don't contend that you show a profit laying 11-to-10 on every small underdog. However. there is a bias in your favor for the foregoing reasons. Should you bet a small favorite, you are going against this bias.

Very large favorites, especially in college football, are another story. Many bettors will automatically take the points when they see a number like 27. For this reason, these numbers tend to be too small. I suggest that you can automatically show a small profit by laying the points in this situation.

Let's look at some of the other situations I've found to be very profitable. One of them is based on the fact that whether you win is far more important than your margin of victory. Some teams like to run up the score. Others with more class tend to win almost all their games by a small but comfortable margin. When these two teams meet. the high-scoring team is typically a small to moderate favorite. The underdog in this case is usually an excellent bet. They may well be the true favorite because the other team is accustomed to having its own way and may fold under the stiffer competition. This situation comes up repeatedly in post-season bowl games. The underdog covers the majority of games for exactly the reasons mentioned. You can't lay points against a winner.

Another automatic bet occurs when a tough winning team incurs a key injury, especially when they are playing another good team. Bet on the team with the injury. The bookies overadjust the line for the injury. In fact, the team may very well play better knowing it needs to make up for the missing star. However, do not make this bet if the depleted team is a weak one. The injury now tends to be too demoralizing.

Let me now examine how psychology can affect a game, especially in football and basketball. The fact is, there is really only a very small difference in strength, speed and skill from one team to another. This small difference usually determines the winner. But if one team is really "up" for the game and the other isn't, the extra adrenaline and intensity may well make the lesser team the better one on this day only. The situation is especially profitable if the normally better team is down or "looking past" the game. Betting an "up" team against a "down" team while getting a few points is one of the best bets in sports. It is my bread-and-butter bet. The problem is finding the situations.

One of the more automatic examples of the foregoing occurs during the regular college basketball season. When a team that is ranked in the top 10 by the wire services finds itself a small favorite visiting a good, but unrated, team, you should bet against them. It is just another game for the ranked team, but one of the most important games of the year for their opponents who are at home. The underdog has been waiting for this game for weeks. A win here could make their season. Take the points.

A second automatic bet occurs during the NBA play-offs. In order for it to come about, the play-off must be set up so that the first two games are at the same location. When the home team wins these first two games, taking a 2-0 lead in a best-of-seven series, you should bet against them in the third game. The other team is coming home with their backs against the wall. The team with the 2-0 lead figures to have a let-down. This situation has historically been a great bet. The only draw back is that you usually have to lay a few points. Plus, the bookies are starting to catch on to the situation and have been adjusting the line (but not enough).

Most situational bets are not quite as automatic as the foregoing examples. It usually takes some insight to come up with them. This insight involves trying to see and feel the attitudes of the teams involved as if you were in their shoes. The best bets come about when one team is really up for the game and the other team is taking the game for granted. It is normally better if the team you're betting is getting a few points and, preferably, is at home. Another important criterion is that the game not be a mismatch. I have found that psychological factors don't seem to help a team that is significantly inferior to its opposition.

An example of this type of bet occurs in pro football, usually early in the season.

A mediocre team has surprisingly won its first two or three games. It also seems to be playing pretty well. It now hosts a Super Bowl contender from the previous year. This contender has also been winning, though not playing exceptionally well. Last year's contender noxti finds itself a 4'/2 point favorite on the road against this mediocre upstart. lake the points. This is the game of the year for the underdogs. It's just another game on the schedule for the favorite. It is a classic bet that has won consistently over the years.

(I must point out at this point that the principles I have been speaking of do not apply as strongly to baseball betting. Baseball can be beaten but it's much more necessary to know your teams and players and other factors before you try it.)

I have given you just about all the general principles I use in beating sports. Find a good situation. Find a good matchup. Bet into the best line. Obviously, if you do all three at once it is even better.

Before ending this chapter I would like to briefly discuss what I call the bookie's trap. Simply stated, the player should beware whenever the bookie puts up a line that does not look like it will get equal action. Since bookies usually hate to gamble, the only justification for this line would be if they have a strong suspicion that the majority of the money will be bet on the side that they think will lose. If they make Oakland a 2 point favorite, knowing that most people will lay the points, betting the other way is probably a good bet.

In other words, whenever you see a strange line, if you bet at all, bet against what seems to be common sense. Now you are betting the same way as the bookie. Of course, after reading this chapter you should see that some of these strange lines are really quite understandable. The public, for instance, may think that the Super Bowl contender in the previous example should be a lot more than a q'/z point favorite. You and the bookie know why they're not.

Hedging and Middling

With the end of the regular football season not too far off, I thought it a good time to discuss vet another widely misunderstood concept among gamblers: that of hedging bets. This is a technique that is used most commonly in sports betting. but it can come up in other gambling events. In general, hedging is usually the wrong play. There are times, however, when it isn't. Simple logical analysis can tell us which times are which.

When someone hedges a bet they are. in essence, betting against themselves. They are making a second bet after an original bet where, if they N\ in the second bet. they probably lost the first one and vice versa. It is frequently done to ensure a profit.

For example. if a player has blackjack and buy's insurance against a dealer's ace showing. he is really making a second bet (that the dealer has a ten in the hole) that is a hedge against his first bet. By doing so he ensures a payoff of even money. rather than either getting a push or a payoff of 3-to-2. Other examples of hedges are:

1. Betting a football game one x\ ay and then, after jumping off to an early lead, betting the other «ay at halftime,

2. Betting the daily double and then betting other horses in the second race after the first horse has won:

3. Betting a five-team football parlay and then betting the opposite team in the fifth game after your first four teams have wont

4. Betting on a baseball team to win the pennant before the season starts and then betting against them in the divisional playoffs, and:

5. Betting one side of a game or sporting event and then betting the other side after the odds have shifted.

Take the case of a half-time hedge in football. Suppose you originally bet the underdog getting 10'/2 points. Your team now jumps off to a 7 point lead at the half. The bookmaker installs the other team as a 6V2 point'favorite in the second half. (In Nevada it is quite common for bookmakers to put up a point spread on the second half of a televised game.)

You reason that by laying the 6'/2 points, you ensure winning at least one of your bets and may very well win both. (If the original favorite outscores the original underdog by between 7 and 17 points in the second half you do, in fact, win both bets.) If not, you win either your original bet or your half-time bet. This seems like a golden opportunity to go for a 10-point "middle." In spite of the foregoing reasoning, however. making this second bet at the half is probably a bad play.

The basic principle is this: If the second bet has the worst of it, You will cost yourself money in the long run by making it. This second hedge bet is really a separate bet. Previous bets should not usually be an excuse to make a bad bet now.

Let's look at a clear-cut example of this principle from the game of craps. Suppose you bet $7 on don't pass and the shooter comes out with a point of four. You now get the idea that you can hedge your bet by placing the four for $5, getting 9-to-5 odds. This ensures a $2 profit. If the four comes up before a seven you lose your $7 don't pass bet but collect $9 on your place bet.

If the seven shows before the four, you lose your $5 place bet, but win your $7 don't pass bet. Either way you show a $2 profit. The flaw in this idea is that your average profit is higher if you don't hedge. If you just let your $7 don't pass bet ride, you will win 2 out of 3 of them (when the point is four). Thus, you figure to be $7 ahead after 3 such decisions. This works out to about $2.67 per decision. Hedging your bet costs you an average of 67 cents.

Of course, hedging is correct if the second bet is a good bet by itself. Say, for example, you have won the first half of the daily double and now notice that a horse you don't have in the second half is going off at much higher odds than expected. You should certainly bet it if it has become a good one. It's only a coincidence that it appears to be a hedge bet: you would have bet it even if you weren't alive with another horse in the double.

One time it is acceptable to make a hedge bet purely as a hedge is when this bet is dead even. In this case you are only making the second bet because of the existence of the first bet, but you aren't costing yourself anything by doing it and are reducing your fluctuations.

There was a situation recently in Reno that almost forced you to hedge if you bet. Harold's Club had a football parlay card that was paying something like 20-for-t on 5 out of 5 winners and 50-for-1 on 6 out of 6. The true odds are, or course, 32-for-I and 64-for-1, respectively.

However, let's say that because of line movements during the week you find five very good bets on the card. You still shouldn't bet a five teamer. Rather than bet a $10 five-team parlay card you should bet two $5 six-team cards. You do this by picking any game for your sixth game and going both ways on it while betting your five key games on both cards. To show you why this is better, let's say your five teams win. If you only bet one $10 card, it would be worth $200. Had you bet two $5 cards, one of the cards would lose but the other would be worth $250. Yet in spite of this, I observed many people betting five-team parlay cards though it couldn't possibly be right even if they knew the winners.

There are some situations where a good case can be made for taking the worst of it on hedge bets. They occur when you are on a limited bankroll and the knowledge of a future hedge opportunity allows you to make a bigger wager than you could otherwise afford on what you know to be a very good bet.

For instance, suppose you thought that the Cincinnati Bengals had a very good chance to win the Central Division of the AFC. However. the only bet available to you was on their winning the Super Bowl (you could get 100-to-1 in some sport books). Well, even though you really only liked them to win the division, you should still take the 100-to-1 on the Super Bowl and then perhaps start betting against them in the playoffs (taking slightly the worst of it on these bets) in order to ensure a profit. This is still better than making a much smaller bet to start with because of your limited bankroll and then not hedging.

To make this principle crystal clear, take this hypothetical example from baseball. The Yankees are playing the Dodgers. One bookie has the Yankees a 1.80-1_60 favorite. A second bookie has them 1.40-1.20. You are sure the game is even money. You have a $10,000 bankroll. You can middle this game by laying 1.40 and taking 1.60 and thus make a sure profit, but this seems silly since you know laying 1.40 is a bad bet.

I've been saying in this essay that you shouldn't make bad bets just to ensure a profit - but this is an exception. If you were to only bet the Dodgers. you could really only afford to bet about $1.000 (getting $1,600-to-$1.000 odds). If the game is truly even money. this should give you an average profit of $300. However. if you bet both ways you can put all $10,000 in action, since you can t lose.

By betting about $6,025 on the Yankees and $3,975 on the Dodgers, you must come out with a profit of about $330. It is important to understand that if the bookies only had a $1,000 limit, or even a $3,000 limit, it would have been better to bet the $1,000 on the Dodgers rather than to go for the sure profit. The middle is worth it (when one side is a bad bet) only if it allows you to put far more money into action.

In the example just discussed, had you originally bet $1.000 on the Dodgers (getting 1.60) and then discovered you could lay 1.40 on the Yankees. you should not do it just to ensure a profit unless you think- you can go back and bet far more on the original bet and a similar amount on the hedge bet.

To put it another way you should not try to middle a game if you had originally bet it one way just because the line has moved. That is. unless this neN\ line means that your second bet has also become a good bet or you have the opportunity to put much more money into action. It is especially important to understand the foregoing concepts when betting football. Suppose one bookie has the Colts a 4'/z point favorite and another one has them a 7'/2 point favorite. Almost all bettors would go for a middle by laying the 4'/2 and taking 7'/2. Suppose ho\\ever, you're sure that the right line is 2. Shouldn't you just take the 7'h? Don't you do better in the long run? The answer is yes, if you can afford to bet the limit.

If the bookie's limit is $2,200, but you can only afford to bet $550 taking 7 %2 points, you're better off betting $2,200 both ways. You are really only risking $200 now and stand to win $4,000 if the Colts win by 5. 6, or 7. (There is another option if you are sure it's a great bet taking the 7'/2. Bet both ways, but bet more on the underdog. If you bet $2,200 on the dog and $1,650 on the favorite you still have a nice middle going for you but still show a profit if your side

In practice, if there is a 3 point line discrepancy, it is likely that the true line is somewhere in the middle. The discrepancy has to give you the best of it - so go for it.

Beating The Racetrack

In this chapter. I will discuss some of the general principles that you must understand if you are to have any hope of beating the racetrack. I. myself, have never made a living betting horses, though I have made some money at the "trotters." It is, however, at least theoretically possible to beat the races in spite of the large takeout against you.

On average, every dollar that is bet on a horse returns about 82 cents. This is an 18 percent disadvantage. But unlike craps or roulette where the chances for each outcome can be calculated precisely, the chances for a horse can only be determined using skill and judgment. If your skill is better than the general public's, you will not lose 18 percent. The question is whether your expert judgment is enough to actually give you an edge rather than just reducing your disadvantage. After all, even an expert poker player could not beat a game with an 18 percent rake. Why should a horse player be able to?

The reason that the poker analogy isn't correct is that the horse player doesn't ante. He may pass as many races as he likes and lose nothing. A poker player could also beat an 18 percent rake game if he didn't ante. He would merely wait for a hand with a big enough advantage to overcome the rake. Having said that the track can theoretically be beaten (even for a living), I must caution that it appears to be extremely difficult in practice, with very few people actually doing it. It requires more than just handicapping ability.

The first thing to understand is that you are not necessarily looking to bet the horse that you think has the best chance of winning. Rather, you are looking to bet a horse (or exacta, or quinella, etc.) that will win often enough to show a profit at the odds at which it is going off. This might be the favorite. It might be the second choice that you think should be the favorite. Or it might be your fourth choice to win the race. All that really matters is that the price the horse will pay is higher than you think is the correct price. For instance, if you as an expert think that a particular horse has a 10 percent chance of winning, and it is going off at 15-to-1, then it is a good bet.

Note that it is important to understand that your evaluation of the horses chances include not only handicapping data, but also the fact that the horse is going off at 15-to-1. If, for instance, you expected 8-to-1, these surprisingly high odds should make you revise your estimate of the horse's chances downward. This factor will tend to eliminate spur-of-the-moment "overlay" bets. Rather, most of your bets will be anticipated well before you see the odds. More on this later.

There is no getting around the basic fact that the idea is to find a good bet, not to find winners. You must compare your opinion of a horse with its odds (keeping in mind that the odds themselves may help shape your opinion) before deciding whether to bet. There is nothing wrong with betting a 4-to-5 shot if that horse figures to have a 60 percent chance of winning. Conversely, there is nothing wrong with betting the worst horse in the race if it has a three percent chance but is going off at 80-to-1.

How do we find these elusive good bets? There are three main reasons why they are not so easy to find. For one, the public sets the odds and the public is usually quite accurate. This is because it is not really the average fan who sets the odds, but the big bettors who tend to be more knowledgeable. You, the pro, must be better than they are. Unfortunately, the second problem is that you must be much better than they are - at least as far as the races you bet are concerned because of that monstrous takeout.

For instance, if the public has 20 percent of the money bet on a particular horse and you know that the horse has a 25 percent chance of winning, you still won't get the 4-to-1 odds on your true 3-to-1 shot that you would get if there was no take. Rather, you would only get 3-to-1 and thus have no bet. A horse going off at, let's say, 4-to-1, has less than one-sixth of the total money bet on it. What all this means is that your opinion of a horse's chances must be significantly higher than the public's opinion for you to have a good bet.

The third reason it is difficult to find good bets is that you have no inside information. You don't know which horses were slightly lame but are now better. But some people do know. And they bet. Fortunately. some of this disadvantage can be negated by including the tote board odds themselves as one of your parameters as I will discuss shortly.

Even with all these negative factors, good bets can be found. They will occur one or two times a day when the public has greatly underestimated a horse's chances and you have assessed its chances correctly. (Most races, unfortunately, will have to be passed.) If you think about it you will see that the only way this can happen is if there are factors that the public does not know how to correctly take into account.

Since I'm not a good handicapper, especially at the flats, I don't know what these factors might be. But I do know that they can't be obvious. If they were they would be reflected in the odds. It also might be that this hidden factor might not show on the program but instead can only be found by watching the horse's last race. This is especially true for harness racing where a horse may have been boxed in, or four wide around a turn, and it doesn't show on the program. (In fact, I'm quite sure you can't beat the trotters just off the program. You must watch the races to find upcoming good bets.)

Thus, if there is not something special about a horse or a race that only an expert can understand, then there is no chance to find a good bet. You still may not find one unless this special factor makes a particular horse much better than it appears or makes the short-priced favorite much worse than it appears. (In this second case, eliminating the favorite almost has to mean that there is at least one good bet in the race.)

A bet based on special factors should usually be foreseen before you ever see the odds. Don't forget that your bet doesn't have to be on the horse you think has the best chance of winning, but rather on a horse that you know will he underrated or running against a favorite that you know will be overrated. There is one special factor, however, that can't be foreseen until you see the odds. That is the factor of odds themselves.

The concept here is almost the opposite of what we had been discussing before. The idea is that if a horse is going off quite a bit shorter than what is expected. it is a good bet, and if it is much higher than expected it is a bad bet. A horse you think will be 4-to-I that goes off 5-to-2 should be bet, but it should be passed at 6-to1. The underlying reason behind this is, of course, that there is some inside information not available to the public that is affecting the odds.

You might say that this betting method is exactly the opposite of what I have been saying all along, but it is not. To use this method the expert does not assess what he thinks the true odds are on a horse, but rather he predicts what he thinks the odds will be on a horse. If they are much shorter he might have a bet. If they are longer, he passes. (I must caution you that this method only works at some tracks where the "smart money" is truly smart.) This is why a last minute unexpected rise in price shouldn't really entice you in many cases.

To prove that the two methods are not really opposite, I will give an example where both methods pick the same horse. Suppose you see some veny unique factors about a horse in a particular race that you know will be misinterpreted by the public. You make the horse a 2-toI shot but you expect to get 15-to-1. It goes off at 8-to-l. Either method picks the horse. It is well above the true price of 2to-1 and well below the expected price. (In this case the lower odds might not be due to inside information, but the existence of some other pro handicappers on the grounds.)

Bets like these are probably the best kind. To make certain that you understand the aforementioned ideas, I will give some examples in the table on the following page.

We now see the general principle behind beating the races with win betting. There are other methods such as finding place or show pools out of line as well as exacta picks. You may have situations where you know that if one horse wins, another horse will likely come in second, thereby indicating an exacta overlay. If you are going to play the horses for a living you should certainly know how best to take advantage of these opportunities when they arise. Space does not allow a detailed discussion of them in this essay.

I must admit that little in this chapter will help you if you are not a great handicapper with the ability to assess a horse's chances as well as an ability to predict the odds the public will make a horse. But I do know that there are some fine handicappers out there who were not aware of all the other important principles that I've explained and without them they couldn't quit their jobs. Now I hope they can.

True Odds	Expected Odds	Actual Odds	Action Taken
3	3	5	Unexpected overlay; Risky bet
3	3	2	Smart money; Bet at some tracks
3	5	5	Expected overlay; definite bet
3	5	7	Odds worry you, but still worth a bet
3	5	4	Everything says bet
3	2	2	Typical underlay; Horrible bet
3	5	3	Smart money and fair odds indicate bet
5	4	2	Smart money, bad odds; risky bet
8	1	1	Should be some other good bet in race
1	1	2	Odds indicate another horse might be worth it

Part Six
General Gambling Concepts

General Gambling Concepts

Introduction

The following section contains articles about gambling concepts that apply to many different gambling situations and games. A full understanding of the logic and mathematics behind them is necessary if you NN ant to achieve what I call a "full conception" of gambling.

The first two chapters in this section are quite similar and are extremely important. They both describe how to make a decision when you have a choice of two or more options in a gambling situation. I show in some detail why the best decision is not always that obvious. These decisions not only involve making the best play in a particular game, but also choosing a game in the first place as well as choosing what aspect of a game needs the most study. Even pros frequently make the wrong decision on all three counts. These first two articles will show you why.

The third chapter in this section explores the most overrated and misunderstood concept in gambling: money management. People think that it is very important to know when to quit a game. The better you are at this, the better "manager" you are. Well, this is almost complete hogwash. If you have the best of it, are not tired and can afford to play, there is no reason to quit regardless of whether you are winning or losing. The chapter on "Knowing When to Quit" explores this subject in more detail.

Another overrated concept that is explored in this section is the Law of Averages. While the law is true, it is the second most misunderstood concept in gambling. It does not say that a gambling outcome. eventual ly will even out to its true odds in the long run. It only says that it will approach the true percentage in the long run. This distinction is very important if you want to avoid the mans' fallacious "systems" around. I explore this further in the fourth chapter of this section.

The fifth chapter analyzes a technical fact that can almost be called a paradox. It shows why the "best horse" or the "best hand" in a poker game may actually not be the best when up against many opponents. This concept has many implications for expert poker players and can also be applied to horse betting.

The sixth and seventh chapters are new for this edition. "What the Pro Knows" gives some of the insights and secrets of various games that are largely unknown to amateurs. Some Words to the Other Side gives my philosophy about what I think is fair and ethical methods of operation for casinos - an important subject to me now that I am a consultant.

Finally. I close with a short but difficult quiz that originally appeared in Gambling Times.

Making The Best Decision

This chapter and the next one both discuss a subtle point that is quite critical to gambling or any type of decision-making for that matter. In this first chapter I show that it is not necessarily true that when you have a choice of to or more alternatives you should choose the one \\ ith the highest probability of being the best one.

Most of the time you will choose this alternative. However, before making this decision. you must also evaluate the possible loss if your choice backfires. Similarly, you must evaluate the possible windfall that may occur to you if the "lesser" alternative is chosen. Only after evaluating all these factors can you make the best decision. This chapter will show you how.

This chapter \\ill discuss a very important concept that comes up in virtually all gambling or game-playing situations where luck is involved. For that matter, the concept can even be applied to investment or business decisions.

Suppose the serious gambler, game-player or businessman is forced to make a decision about how to `'play' in a particular situation when given a number of alternatives. Typically, what he does is evaluate these alternatives and pick the one that is most likely to be the best. This seems reasonable enough and this method will frequently give you the right answer. But not always.

A second alternative may be better even if it is less likely to be the best one! This can happen either because it will occasionally be much better than the first alternative or because the first alternative can occasionally be much worse than the second one.

Both of these situations would result in the average value of the second alternative being better than the first, even though it figures to fare worse a majority of the time. Let's look at some examples.

Suppose you are playing blackjack in a casino. You are dealt an ace and a five against the dealer's five showing. Should you double down? Let's say you hadn't read any of the recent blackjack books, but you did know that the dealer will bust with a five showing 42 percent of the time. You therefore try to reason out the best play. You realize that the double down will hurt you if you catch a six, seven, eight, nine, ten, jack, queen, or king (since you have in these instances doubled your bet as a 42 percent shot), while you will profit only if you catch a deuce, trey, four, or five. (Catching an ace is about neutral.) Thus, the chances are about one-third that you would double down if you knew what card was coming off. There is a two-thirds chance you wouldn't.

Yet, as you blackjack students know, doubling down is the correct plan. How can this be true if doubling down appears to be wrong in this situation more often than it is right? The answer stems from the fact that the play is very profitable when it is right, but only mildly unprofitable when it is wrong.

Thankfully. the decisions in blackjack have already been computed by experts. Unfortunately, such is not the case for other games or endeavors. Normally the participant is forced to make the decisions on the spur of the moment without any help. It is at these times that the error we have been discussing is the most apt to occur. Even the fact that there is time to evaluate a decision does not guarantee the correct choice since this choice frequently violates "common sense."

How can a play that is favored to be wrong, be right? To reiterate, it can happen if the potential gain of the play is much more than the potential loss. It can also happen that a play is right if the other play has a much greater potential for disaster.

This error in decision-making is seen very often in the game of pro football. I don't claim to be able to do a better job of coaching than a pro coach, but that does not change the fact that they almost all make the same strategical errors of the type we have been discussing.

One of the most flagrant of these is the failure to go for a first down on fourth and shortyardage on the opposition side of the field (but out of field-goal range). The proper way to make the decision in these cases is to compare two factors. They are: (1) the chance of success multiplied by the gain and (2) the chance of failure multiplied by the loss.

By this criteria, it is usually right to go for a first down on fourth and 3 on the opponent's forty-five yard line even as an underdog to make it. The gain that is achieved when it works, namely possession of the ball, is far greater than the average loss of about 30 yards in field position when it fails. Coaches seem to be gambling in this situation more nowadays than they used to, but still not nearly enough.

Another time when coaches miss the opportunity to make a "good bet" is when they fail to go for on-side-kicks in routine situations early in the game. Once again, the gamble may be slightly more likely to fail, but the potential gain (ball possession), is far greater than the average loss in field position (about 25 yards.)

It is possible that some coaches understand these concepts, but are afraid to make a play that is apt to backfire, since the owner, who pays his salary. may not understand the concept himself.

An example from football can also be used to show when an alternative that is usually right should be avoided because of its potential for disaster. It is my contention that a fair catch of a punt should never be made inside the 15-yard line. If you can't run it back, let it drop. Catching it will probably save you a few yards, but there is always a chance that the ball will bounce backward or end up in the end zone for a touchback, giving you the ball on the twenty-yard line.

More importantly, you have avoided the possibility of a fumble or a muffed catch Which can mean an easy touchdoxNn for the opposition. Even if this possibility is only two or three percent it is usually better not to risk it.

Most coaches tell players to let the ball drop inside the ten-yard line. I say this should be extended at least 5 yards. There are many other mathematical errors of this type that are made in both football and baseball. Any coach who would like to discuss this should feel free to contact me.

Let us now get to poker. I give a contrived example of this error in The Theory of'Poker. You are playing draw poker and are dealt two kings, two queens and a deuce. After the opening round of betting, your opponent draws two cards. You somehow see his hand and notice that he is drawing to a pair of jacks with an ace kicker. How should you draw?

While it may appear automatic to draw one card, the expert may very well throw his two queens away and draw 3 cards. He reasons that he has a better chance of making three-of-a-kind by drawing 3. He will need three-of-a-kind if the jacks make aces-up or three jacks and will still win with one pair if the jacks don't improve..

The only way the play can hurt him is if the' catch a "cold pair," making two pair. This happens less often than the jacks improving to three jacks or aces-up. Thus drawing 3 will help you much more often than it will hurt you. In spite of this, drawing 3 is the wrong play. This is because you only gain slightly (about 3 percent extra chance of drawing out) when he does make aces-up or three jacks, but are hurt very badly those occasional times he makes two smaller pair.

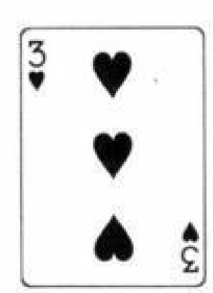
When he does make two pair, you have hurt yourself about 72 percent of the time since you threw away a sure winner and only have a 28 percent chance to improve your one pair. This is another example of a play that will be better more often than its alternative, yet it still isn't right. The correct play is to draw one.

A more practical example of errors of this type comes up all the time in poker. I myself am guilty of it more often than I should be. The basic principle is that it is worth making a play that will probably cost you one or two bets if not making the play will occasionally cost you the pot. This typically occurs in a multi-way pot where you have a chance to raise out the other players who might out draw you, but you think the initial bettor has you beat.

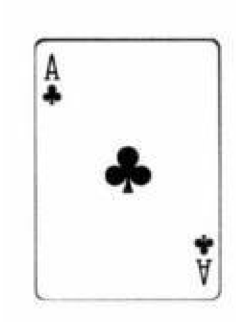
Raising is still correct as long as there is a reasonable chance you have the best hand (or will make the best hand).

Recently, while playing hold 'em, I had

in the hole in a big raised pot. The flop came



Another player bet with what I knew to be a minimum of ace-queen in the hole. This was the only hand I could beat. He could have also had three jacks, three queens or two aces. I was probably beat. Thus, I called his bet and let other players in behind me. When the pot was over, one of these players out drew my two kings with his



It will be quite a while before I make that mistake again.

I am no backgammon expert, but I'm quite sure similar errors are made by good players in this game. For instance, in reading backgammon books, I find that the "right play" is frequently found by counting the number of "good rolls" that are left for your opponent.

For instance, it may be that a certain move will leave your opponent 14 good rolls and 22 bad rolls while its alternative will provide him with 12 good rolls and 24 bad rolls. The second play is therefore supposed to be better. It may not be.

It's possible that, if you choose the second move, many of his 12 good rolls would be especially bad for you while many of his 14 good rolls, if you choose the first move, don't help him as much comparatively.

On the other hand. it may be that some of the 22 bad rolls you leave him, using the first play may be disastrous for him if he rolls one of them whereas the 24 bad rolls the second play leaves him don't hurt him as much. In either case, the first play would in fact be the better alternative. Backgammon pros should mull this over and see if they can isolate mistakes of this type that they are making.

All of the errors mentioned are related to misunderstanding the concept of mathematical expectation. In order to be sure you are making the right decision about anything, whether it be gambling, business or war, you must not simply pick the one that is most likely to be best. You must evaluate both the gain when it is right and the loss when it is wrong. Sometimes, under this criteria, a different decision may actually be the better one.

Knowing What's Important

Basically, this chapter further expands on the principles in the previous chapter. This time I discuss which factors to concentrate on in a gambling or business endeavor if you can't concentrate on everything. If there are many factors involved it may not be right to simply "maximize" as many of them as possible. It is important to consider the relative importance of each factor. You may very well do better by concentrating on only a few factors rather than on a great number of factors with lesser importance.

Many serious gamblers suffer from a very common problem. The problem is not easily seen, but frequently keeps them from being as successful as they should be.

This problem affects players in all types of gambling: poker, blackjack, backgammon, sports betting. Ironically, it frequently affects the conscientious student of a game more than the intuitive player. I used to be a victim myself.

However, as with everything else in gambling, a thorough logical analysis Finds and cures the problem. The intuitive player must eventually wind up second best.

The problem I'm speaking of is the propensity of some players to learn as much as possible about a certain game without attaching a relative importance to each of the concepts. Basically, every gambling game involves knowing and using the right play for each given situation.

Often the game is too complex for one person to learn everything. It stands to reason, then, that the goal should be to learn as many of the right plays as possible. The more often you are right, the more you yyill yyin.

Unfortunately, this is not quite correct. It is possible that you will know a higher percentage of the proper plays without playing as profitably as someone who knows less.

The reason this can happen stems from the fact that not all plays are equally important. Even though you know the right decision for more of the situations that may arise, an opponent may be correct more often when it comes to an important decision. Thus, he wins more money.

What do I mean by an important decision'? One of two things: First, a decision may be important because there's a big difference between playing it right and playing it wrong. More on this in a moment. Second, a decision is important if it comes up often (even if there isn't a great difference between the right and wrong plays). Obviously a decision that fits both of these categories is extremely important.

To be a really successful gambler, you must do the right thing as often as possible when confronted with important decisions. You need not worn, as much about less important decisions.

Let's look at some examples.

It's sad but true that most of the world class poker players are not thoroughly knowledgeable regarding poker odds. (Though their intuitive estimates come pretty close any way.) Conversely, many lesser players know the odds exactly.

They can still be weaker than the intuitive players because knowing the right odds is not that important! The reason is twofold. First, it rarely happens that the odds the pot offers are near the break-even point. When, for instance, a good player is a 6-to-1 underdog to make his hand and is only getting 3-to-1 pot odds, he would know to fold just by his fairly accurate intuition. Likewise, he would know to take 8-to-1 odds. He is only likely to make a mistake those rare times when he is faced with a close decision.

In other words, your ability to always make the correct decision in this spot will only occasionally make a difference those few times when it's close. Moreover, making the correct decision is no big deal when it's close.

For instance, suppose you're an 8-to-1 underdog and the pot offers 7-to1. (I disregard future-round bets for the purpose of this discussion.) Say someone bet $10 into a $60 pot. You would now fold and save $10. But you really didn't save the full $10. The intuitive player might call in this spot and cost himself $10 7 out of 8 times. However, he gains $70 the 1 out of 8 times that he makes his hand. His bad play only costs an average of $1.25, which isn't much.

Now let's take a somewhat different situation. Once again, a player bets $10 into a $60 pot, except this time it's the last betting round. Your hand can only win if he's bluffing, and you're almost sure he isn't. Thus, you fold in order to save $10.

The intuitive player, confronted with this same situation, might realize that there is a I-in-3 chance that his opponent is bluffing. So he calls, thereby winning $70 1 out of 3 times while losing $10 2 out of 3.

In this case, his ability to assess his opponent made him an average of $16.67 more than you. Furthermore, even if he agreed with you that the guy was almost certainly not bluffing, this intuitive player might realize that a $10 raise could win the pot, let's say, 30 percent of the time. If this assessment were accurate, $20 risked would win $70 3 out of 10 times for an average gain of $7. That's a lot better than folding as you would have.

In the previous example, we saw how a poker decision could be classified as important, because there was a large difference between making the wrong play and making the right play. This usually occurs late in a hand when the better player may be able to save or win a pot that a poor player loses.

The other kind of important decision (namely one that occurs frequently) usually arises early in a poker hand. I'm talking about the decision of whether or not to play a hand. Many players are too loose with their starting hands, committing themselves with any pair in seven-card stud or any ace in hold 'em.

While these are only moderately bad plays the problem is that they come up often. Eventually, they will cost you a lot of money.

We have now seen examples of both types of important decisions as they apply to poker. What about the super important decisions? Is there a poker situation that comes up frequently where the right play is vastly superior to the wrong play?

Yes. It is game selection. Whenever you have a choice, your ability and willingness to find and play in the game that is most profitable is the most important aspect of your earning power. No matter how well you play any game, it is the abilities of your opponents that is the main component to your expected win.

This is why I think that a good player should learn to play all games. Then, if he finds an exceptional game, he can play even if it isn't his specialty.

It is better to be the third-best player in a good hold 'em game than the best razz player in a tight game. (Obviously game selection only comes up if you're in a large public casino or cardroom.) However, there are also important decisions applying to private games. One would be to keep the game going on a steady basis.

Another important factor would be to deal the type of poker (if it's dealer's choice) that gives you the biggest edge even if it isn't your best game! As far as keeping a game going is concerned, this involves things far removed from poker strategy and is another reason that some of the most successful players are not always the best players.

Let's see how this principle applies to other games. Blackjack gives the clearest example. In almost all cases, the best players are not the most successful. In most cases (one exception to be noted later), the ability to use advanced counts and the knowledge of obscure strategy variations is almost irrelevant. Not only is the opportunity to use this knowledge very infrequent, but the play is usually so close that making the wrong choice is not that bad.

The important thing in blackjack, like poker, is getting a good game. By this I mean a game in which they are dealing a large portion of the deck or decks and are allowing you to vary the size of your bets significantly.

The most advanced counter, betting 1 or 2 units and being dealt 20 cards from one deck, will not win as fast as someone using a simple plus-minus count while betting from 1 to 4 units and being dealt 30 cards.

Thus, the most important thing in blackjack is finding and keeping a good game. What this boils down to, in this day and age, is disguising the fact that you're counting. The count can be simple, but you must know it cold in order to appear that you're not counting.

(The exception I mentioned earlier sometimes arises in casinos that deal single decks rather than a multiple-deck shoe. These casinos have a tendency to be veny paranoid about winners who alter their bets -no matter how well those winners disguise their efforts to count.

These casinos, though, will usually deal a good game to anyone who doesn't change his bet. Thus, the best way to beat a one-deck game is to use an advanced count that will show a nice profit on a flat bet. A simple count won't win enough. This is why most professional blackjack players now play mainly against shoes.)

What about backgammon'? Once you become a decent player, there is only one important decision (besides finding a good game, of course). It can be said in one word. The cube. Knowing how to play the cube is almost always more important than any individual move might be. Yet, otherwise great backgammon players are frequently inept in this aspect of the game.

It is also important in backgammon to distinguish between a very bad move and a clearly bad move. I know some hustlers who get good games and sometimes even a "spot" by occasionally making a move that is obviously incorrect but doesn't hurt them much.

A good example of this occurs when you have a closed board with your opponent on the bar. If you must bring some men into your home board, you should do it in such a way that double sixes will not leave a shot. Failure to take this precaution, however, will cost you the game much less than 1 in 100 times.

Still, a good opponent who sees you make this elementary mistake may very likely be persuaded to over spot you. A similar wrong move that's no big deal is making the two point on a 6-4 opening roll. If this causes an opponent to lay 11-to-10 odds on each future game, it is well worthwhile.

Sports betting is another field to which there are important and non-important aspects. I have found that it is usually not that important (believe it or not) to know how good the teams are! It is usually far more important to know

1. How to evaluate a particular situation;

2. How to evaluate individual match-ups; and

3. How to get the best line.

For now. I hope I've helped both the experienced and the aspiring gambler to decide what's really important while studying his or her particular game.

Knowing When To Quit

Ideally this chapter should not be considered very important. Unfortunately, however, many gamblers consider knowing when to quit a major factor in gambling success. (They call this "money management.") Well, they are wrong.

If you have the best of it, it doesn't much matter when you quit a gambling situation as long as you can afford to play. More importantly if you have the worst of it, no method of determining whether to quit or play on can show a profit in the long run. This is an absolutely irrefutable .fact, not just my opinion. Yet, literally billions of dollars are lost by people who refuse to acknowledge this fact.

There are, however, some legitimate criteria that should be used in deciding whether to quit or play on in those situations where you appear to have the best of it. The criteria is not usually how much you are winning or losing at the time. The correct criteria is explained in this chapter.

Money management is one of the most misunderstood concepts in gambling. In fact, it is not really a concept at all. Many misinformed gamblers are under the delusion that an important key to success is to be a good "money manager." What they mean by a good money manager is someone who knows "when to quit." However, the criterion money managers often use in making this decision is simply how well they are doing, i.e., are they winning or losing and how much. Typically, the "good" money manager quits after small losses but "let's his winnings ride." Unfortunately, this is not usually a sound principle to follow for maximizing winnings. When it is accurate, it is usually for reasons other than those normally considered. More on this later.

The intelligent gambler knows that the time to quit is simply when he is tired and not functioning well or when the game has become "bad" - meaning unprofitable. Determining whether the game is good or bad is the single most important consideration to the professional gambler. He will frequently play tired if the game is extremely profitable. (I do). Even more important is this: if the game has become bad you must quit even i f you are a big winner (or losing badly).' More than anything, the ability to follow this rule separates the winning gamblers from the losers. Even many pros go wrong here. (An exception is the Las Vegas pro who stays in a tough game in order to "hold it together" in hopes that a "live one" will sit down.)

At this point let me stress that these principles hold for any game, not just poker. When playing blackjack, you should quit if the dealer doesn't deal down far enough, if you are subject to scrutiny by the pit bosses, or if you are tired. Once again, the question of whether you are winning or losing should have little to do with your decision. Furthermore, if you are playing keno, casino craps or roulette, I have no advice for you on when to quit except the sooner the better. (If you are gambling for fun, fine - but this book is directed at those who want to win.) It is for games like this that money management techniques are most often applied, yet they are totally useless. The fallacious concept applied to these games is that luck somehow runs in predictable streaks rather than totally randomly. (See the following chapter on the Law of Averages.)

Proper techniques can be applied to other games of skill such as backgammon, gin rummy, golf or even a private craps game where you are betting against the shooter. How you are doing as an indicator of "how your luck is running" should have no bearing in deciding when to quit. (One exception would be where you have reason to believe you might be getting cheated. In this case your losses will reinforce your suspicions and you would be wise to quit. A bad loss is also an indication that you have incorrectly evaluated the goodness of the game. For this reason it is usually correct to set some limit on your losses unless you're absolutely sure you have the best of it and have just been temporarily unlucky.) Predicting luck has no place in the successful gambler's repertoire. In general, all he does is make every effort to play when he has the best of it and no effort when he doesn't. He pays little attention to how he is doing.

There are times, however, when how you are doing should have some bearing in what you do. As previously mentioned, a large loss may indicate you are being cheated or do not really have the best of it. A large win may also be the signal to quit in order not to "kill the goose that lays the golden egg." If you have a gin sucker it may be wise to even lose a little back to him so that he will come back tomorrow for more. The same goes for a private poker game where you are in danger of not being invited back. This would not apply to casino poker. If you are playing blackjack, in order to avoid being "barred." you should never take an extra large win in any one casino. (I am naturally assuming that you are a counter.) Other similar exceptions can be made.

At this point, I will introduce the only correct interpretation of good money management. It concerns betting within the limits of your bankroll. It is yet another place where many gamblers - even good ones - go astray. In fact, good gamblers - more often than amateurs - have a tendency to bet more than their bankroll would indicate they should (when they have the best of it). They blindly follow the practice outlined earlier in this article of betting whenever there is an edge. Unfortunately, they fail to follow one last precept which should override all others:

If you are a winning gambler, it is usually wrong to risk going broke, to risk putting yourself out of action or even to risk diminishing your bankroll to the point where you must now gamble on a lower level with a smaller win rate.

To illustrate that this is true mathematically, as well as intuitively. I give you the following very hypothetical example: Suppose your total gambling bankroll is $1,000 (but you have more money coming in next week). Let's say someone now offers you 6to-5 on the flip of a coin. If you take $1,200-to-$1,000 your mathematical expectation is plus $100. Normally, you should accept this bet. However, suppose you know of another maniac who is willing to lay you 2-to-I on the flip of a coin tomorrow (with a $2,000-to-$ 1,000 maximum). Is it now correct to pass up the first bet? Let's see: If you only bet tomorrow, you will wind up with $3,000 half the time and nothing the other half, for an average of $1,500 and an expectation of plus $500.

$$\$1,500 = \frac{3,000}{2} + \frac{\$0}{2}$$

$$\$500 = \$1,500 - \$1,000$$

If you make the first bet, you will wind up with nothing half the time (when you lose the first bet and are then forced to quit because you are broke), $4,200 one-quarter of the time (when you win both bets) and $1,200 one-quarter of the time (when you win, then lose). This averages out to $1,350 for an expectation of plus $350.

$$\$1{,}350 = \frac{\$0}{2} + \frac{4{,}200}{4} + \frac{\$1{,}200}{4}$$

$$\$350 = \$1{,}350 - \$1{,}000$$

Thus, on purely mathematical considerations, it is correct to pass up the first bet even though it is giving you the best of it.

Let's look at a second example of this principle that actually happened to a friend of mine. He was playing a head-up $5,000 freeze-out no-limit seven-card stud with a complete sucker. He estimated that he was a 5-1 favorite to win the match. Early in the game a hand came up where he had two pair on fourth street and was up against an obvious four flush. However, the four flush bet the whole $5,000 into a $100 pot! At this point my friend was at least a 3-to-2 favorite (with three cards to come), yet he folded, correctly figuring that his chance of losing his $5,000 was 40 percent if he called, but much less if he played on. Once again, it was correct to pass up a bet with the best of it.

I have another friend - one of the best no-limit hold 'em players in the world - who I have seen assess this situation incorrectly. I have seen him call very big bets before the flop with ace-king suited or two jacks (I should point out that he's usually calling a reraise). However, while it is frequently correct for the average player to call in this situation, it is correct for my friend to fold. This is especially true if losing this pot will leave him short or encourage his opponent to quit. Why give the other players the chance to get lucky and maybe leave when he is a cinch to grind them down if he takes his time. This same fellow has a tendency to bet with both hands when he thinks he has the best of it on a sporting event. Once again he is frequently wrong to do this even from purely theoretical considerations. This one flaw is the only thing that has kept him from attaining a top spot in the Vegas gambling hierarchy.

The Law of Averages and Other Fallacies

This chapter was written to explain the seeming contradiction between the so-called "Law of Averages" and the fact that the outcome of a particular event is usually independent of the previous results. If things tend to even out, then why can you not say that a certain event is "due" if it has failed to happen for quite a while? Well, you can say what you want, but you had better not bet on it, as this chapter shows.

Possibly the most misunderstood concept in the realm of gambling is the so-called law of averages. In the title of this chapter, I imply that it is a fallacy. Actually it is not, but it is seriously misunderstood by most gamblers. Under the guise of following the law of averages, more useless systems have been invented than could be listed on all the pages of this book. To some extent, however, the inventors of these systems should not be faulted. They fell into a trap - one that is very easy to fall into. With this chapter, I hope to prevent the reader from ever falling into that trap again.

Before discussing the law of averages, I must bring in the concept of independent events. It is extremely important. In fact, when a gambler places a bet, he is usually betting on independent events, whether he knows it or not. The definition of this term is almost self-explanatory. Each outcome depends on no other previous outcome. Where the roulette ball lands does not depend in any way on where it landed before. The chances of rolling snake eyes remains at 1 out of 36 regardless of all previous rolls. More to the point; when you are betting on independent events, the fact that you won or lost your last bet has nothing to do with whether you will win the next

Sometimes there might be a tenuous dependence between two events that can usually be ignored. Whether a baseball team has won its last game, for instance, may have some small impact on their chances of winning the next game. If you have just won your previous hand of blackjack, there is a tiny change (for the worse) in your chan ces for the next hand, as the deck figures to have become "poorer" in aces and tens. Events such as these, however, are close enough to being independent that any system focusing only on previous wins and losses is doomed to failure. In fact, even events that are quite dependent on each other may act like independent events when previous wins and losses alone are considered.

Card games provide the best example of dependent events in gambling. As more cards are exposed, your odds are constantly changing. If you are betting according to these exposed cards, you may be playing a legitimate system. However, betting according to previous results can never be right except in those rare cases where previous results might correlate with some other factor that changes the odds, as in the blackjack example. In other words, the dice don't know whether you won your last bet.

There are two main techniques used in most fallacious systems. The first of them has to do with riding streaks. It involves betting more after you have won a few bets and cutting back after a series of losses. The theory is that luck runs in streaks - hot and cold. As it happens, this is true, but only after the fact. Five successive wins does not mean a sixth win is more likely than normal. If you are flipping coins all year, it is true that you will get quite a few runs of many heads. However, the fact is that the number of runs of 8 straight heads should be just about twice the number of runs of 9 heads and half the number of runs of 7 heads. Many gamblers have difficulty believing that streaks will end* ust as easily after they have been going for a while as any other time. But this is only because we tend to remember the long streaks and forget the interrupted ones.

Riding streaks is almost always a totally useless method of betting. There are, however, some situations that might be called exceptions. Betting on sports, as I already mentioned, is one. Athletes are not little plastic cubes. They seem to have some tendency to keep winning when they are winning and keep losing when they are losing. A team that has won eight games in a row probably has a better chance of beating a particular team in their 9th game than they would if they played that team at some other time. Similarly, a sports bettor who has been running well might have a legitimate reason to increase his bets, as his previous success could well be an indication that he is handicapping well rather than just getting lucky. If he thought he was just getting lucky, this would not be a good reason to press his bets.

Along these same lines, even if you are gambling on purely independent events such as dice or roulette, there could be some validity to jumping on a hot number because the hot streak could be evidence of a defect and therefore a bias in the wheel or the dice. More likely, of course, the streak was simply luck, especially in a legal casino, but jumping on a streak certainly can't hurt and just might be a profitable play. Those who would get off a hot number because the streak can't continue never have a legitimate basis for their action. At least those players who ride streaks may be playing correctly, although for different reasons than they imagine. A good example of this is playing "rushes in poker," i.e., playing more loosely and aggressively after winning a couple of pots in a row. Those who think that this rush has somehow increased their chances of catching better cards for the next few hands are dead wrong. However, the fact of the matter is that if all their opponents also believe in rushes and thus fold good hands against them during this rush, an otherwise absurd idea has become a self-fulfilling prophecy. (Some readers may see an analogy to certain stock market

We have just examined the first of two techniques which are used in an attempt to buck the mathematical law of independent events. It was based on the idea that winning bets tend to keep on winning and losing bets tend to keep on losing. While theoretically ridiculous, there are rare times when this method will show results for reasons other than those put forth by its proponents.

The second technique doesn't even have this shaky evidence going for it. This technique is, as noted earlier, to bet against the streak, using the rationale that things must even out according to the law of averages. Many people use the law of averages as a weapon against the law of independent events. The law of averages states that as we perform more and more trials of an experiment, the ratio of successes to failures tends to get closer and closer to the true probabilities. For even money situations the results should get closer and closer to 50-50. This law is correct. Unfortunately, there is a key word that many people miss or at least don't see the significance of.

It is easy to see how gamblers might think doubling up to catch up (the Martingale System) and similar systems might work. Since wins and losses must eventually even out according to the law of averages, it would seem that a tail should be more likely to come up on the 100th flip of a fair coin if the first 99 flips have resulted in 69 heads and only 30 tails. If tails remained an even-money shot there would be no reason for them to overcome their 39 flip deficit but the Law of Averages says they eventually will.

But does it really say this? That is the question. They key word I spoke of is ratio. Only the ratio gets closer and closer to what it should be. This will still happen if the losing side simply behaves normally. It does not have to catch up. Suppose I flip a coin 100 times and get 70 heads and 30 tails. I keep on flipping and make a note of the results after a total of 500, 1,000, and 10,000 flips. If the coins are fair and the law of independent events holds, the expected results can be summarized in the table below:

Number of flips	**Expected No. of Heads**	**Expected No. of Tails**	**% of Heads**
100	70*	30*	70%
500 (1st 100+400)	70+200=270	30+200=230	54%
1,000 (1st 100+900)	70+450=520	30+450=480	52%
10,000 (1st 100+9900)	70+4950=5020	30+4950=4980	50.2%

*70 and 30 are not the expected number of heads and tails but the assumption is that this is how they started out.

Similar results could be shown for non-even-money situations. The point is that the ratios do in fact get closer and closer to the expected value without the losing side ever catching up. In this case, heads remain 40 ahead.

In the past, both the law of independent events and the law of averages have been much maligned. Ironically, detractors of one law have invariably used the other law as the logical basis for their argument. With this chapter, 1 hope once and for all to explain and thus eliminate the apparent contradictions between them and to put them both in their proper places as irrefutable laws of mathematics and gambling. Together they prove conclusively that no system can work if every individual bet has the worst of it.

Another Gambling Paradox

It was a close decision as to whether to put this article into the Poker or General Gambling Section. Its main application is to poker, but the general concept involved can also be applied to horse racing and possibly other games.

The general concept is that it is frequently to the disadvantage of the best competitor (or the best poker hand) to have many opponents even if he is getting proportionally higher money odds. In some situations you do better, even in the long run, if you take a surer smaller win rather than gamble on a larger one. In other words, you may not necessarily welcome one extra opponent even if his hand is inferior to your hand and is putting up just as much money as you. This chapter will show you those times this is (and is not) the case.

Imagine a race between two horses. One horse is the model of consistency. It is a good, though not great, race horse and you can be virtually certain that it will run a good, though not great, time.

The second horse is erratic. It will sometimes run an excellent race, but will more often make a poor showing. Let's give this hypothetical erratic horse a 40 percent chance of running a faster time than the steady horse in a match race. Thus it is a 3-to-2 underdog.

Now add a second erratic horse that also has a 40 percent chance of beating the consistent horse. In this three-way contest, the steady horse has less than a 50 percent chance of winning. However, it's still your first choice to win the race, and it would be a good bet if you could get 2-toI odds.

The consistent horse remains the one with the best chance to win. (Its chances are 36 percent. The answer is found simply by multiplying the probability that it beats erratic horse A by the probability that it beats erratic horse B.)

$$36\% = (60\%)(60\%)$$

It seems like common sense that the consistent horse would still have the best chance to win the race even if you added a third erratic horse to this race. Although adding more irregularly performing horses will reduce the best horse's prospects, it would still remain the horse most likely to win.

That may be common sense, but it's wrong! Adding just one more similarly erratic horse to the race makes the best horse the least likely to win. You're better off picking any one of the other three horses.

Best Horse, Worst Pick

The chances of the consistent horse beating all three of the other horses is

$$(60\%)(60\%)(60\%) = 21.6\%$$

The remaining 78.4 percent of the time some other horse will win. Since each of the other horses are equal. their chances are 1/3 of 78.4 percent or 26.13 percent - which is quite a bit better than the "best" horse's prospects.

If you add more erratic horses to this race, each with a 40 percent chance of beating the steady horse, it becomes even a poorer choice. With five opponents, this horse would be far worse than a 5-to- l underdog. In fact, it would be almost 12-to-1 !

Strangely enough. this horse does have the best chance to finish specifically second against 2 to 7 opponents. With 5 or less opponents, this horse is still the most likely to finish in the top 2.

This anomaly applies to things other than horse racing. In a moment, we'll see how to use it to play better poker. First let's see how it can be applied to horse betting.

Most experts believe that the only way to play the horses, if you're to have any shot at all, is by betting to win. Although I agree with this approach in general, the previous example shows that there exist situations where the best bet is to place.

If the place price figures to be about half the win price, it may be correct to bet the favorite to place rather than win. In order for this to be true:

1. The favorite must be a very consistent performer that is clearly the best horse day in and day out, but is unlikely to run a really outstanding race;

2. Many of the other horses, while not as good, do have occasional super races.

Under these circumstances, it's quite likely that one of the inconsistent horses will run a faster race than the favorite, but not very likely that two of them will. Thus, a good handicapper would do well to consider a place bet (where you win as long as your horse finishes first or second). Another possibly even better alternative is to bet the steady horse in exactas with other contenders but only in the second slot!

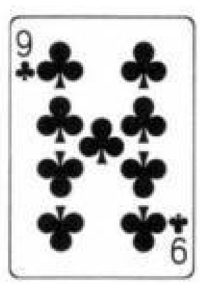
Poker: No Second Prizes

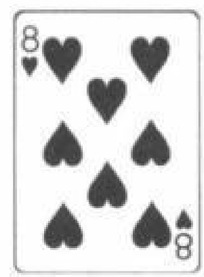
Unlike horse racing, poker does not give you the opportunity to make money when coming in second. On the contrary, the second place man usually loses more than anybody else.

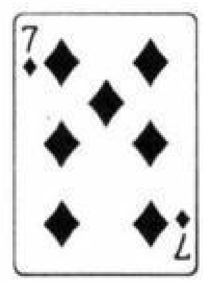
For this reason it's especially important for players to recognize poker situations analogous to the previous horse racing example.

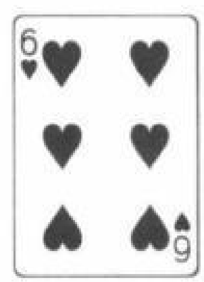
Any time you have the probable best hand, but there are several players drawing to beat it, you are in a similar situation. This is especially true if your own hand has little or no chance for improvement itself.

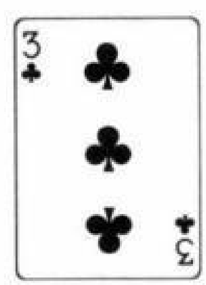
A clear example of this phenomenon arises in lowball draw. Suppose you're dealt a pat











Your hand will stand up against a single one-card draw about 55 percent of the time. However, against 2 such one-card draws, you can expect to win only 30 percent of the time (depending on the exact cards that are in each hand).

In other words, against I player you should bet all you can because you're the favorite. Against 2 players, you're getting the worst of it even though you're receiving 2-to-I odds. In order to have an advantage against 2 opponents, your chances of beating each individually must be almost 58 percent. Keep in mind that this assumes that your hand can't improve enough to "redraw" on a hand that out draws you.

Against 3 opponents, it's necessary that you be about 63 percent against each one individually in order to have the best of it. In this case your chances of winning are

$$(63\%)(63\%)(63\%)$$

which is just over 25 percent - or 3-to-I against you.

Following is a chart showing how much of a favorite you must be (with a non-improvable hand) against each single opponent to avoid having the worst of it against varying numbers of opponents.

Number of Opponents	**Minimum Chances Against Each Opponent in Order to Have Best of it Overall**
1	50.0%
2	57.7%
3	63.0%
4	66.8%
5	69.9%
6	72.3%
7	74.3%
8	76.0%
9	77.5%

As an example, against seven opponents with similar hands all drawing against yours, you need to have more than a 74.3 percent chance against each one individually in order to be less than a 7-to-1 underdog against the field.

What this means is that you must be more cautious than you might have thought when you hold a decent hand with little improvement chances in an unavoidably multi-way pot.

This extra degree of caution should not be exercised if you have a chance to thin out the field by raising.

Don't forget that in all gambling situations the goal is to maximize the mathematical expectation. By this I mean your average profit in the long run if the circumstances were repeated infinitely.

For example. if you had a poker hand that will win 60 percent of the time against a single opponent, you would show a profit as long as you had no more than two such opponents - as the previous chart illustrates. However. your average profit would be greater against just one opponent. (In these problems I neglect the effect of the antes which usually makes it even more desirable to thin out the field.)

Betting $100 as a 60 percent shot against I opponent yields an average profit of $20 per bet. Against 2 opponents the same $100per-player conflict yields just $8 per decision since you will only win 36 percent {(60%)(60%)} of the hands. In fact, you would need to be about 67 percent against each of these hands individually to prefer to play against 2 hands rather than 1.

Your hand would have to be about 76 percent against each opponent before you do better when 3 opponents are drawing to beat you. And this is not taking into consideration the antes and future betting rounds.

How to Make Money

In a real game it is even more important than these statistics indicate, to narrow the field with a good-but-not-great hand. You may sometimes be in a situation where you should actually throw a decent hand away if you would otherwise be forced to play against many opponents.

The classic example of this arises in five-card draw when you hold two small pair. Another comes up in lowball draw when you have a rough pat hand, such as the 9-8 discussed earlier. Don't play these hands if a lot of players are drawing against you.

Here's another way to explain the principle under discussion: Let's say the last player to call a bet has raised your pot odds from 5-to-1 to 6-to1. Frequently, this increase in pot odds is more than offset by your decreased chances of winning the pot.

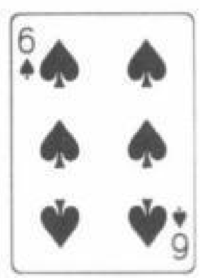
Only certain kinds of poker hands do better against many opponents, namely those that may improve to very big hands. Usually, you want a lot of players in the pot only when you have a big draw. Even then you have an optimum situation only if none of your opponents has an even bigger draw.

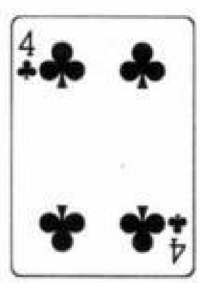
When you have an already-made hand, you normally prefer as few opponents as possible drawing to beat it.

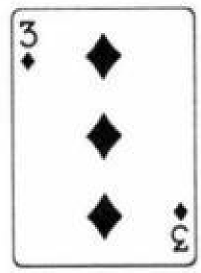
This chapter could well end here, but I feel an obligation to my readers who are serious poker players to point out a few of the tricky ramifications of the mathematical principle just described.

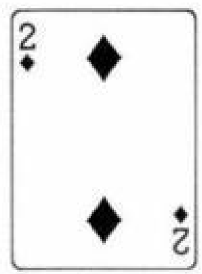
Advanced Advice

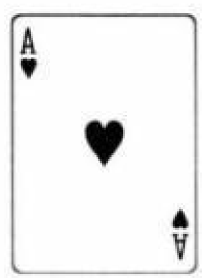
If you have a very strong hand that's unlikely to get out drawn, you have no desire to thin out the field. A pat





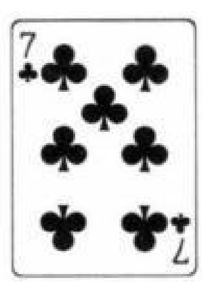


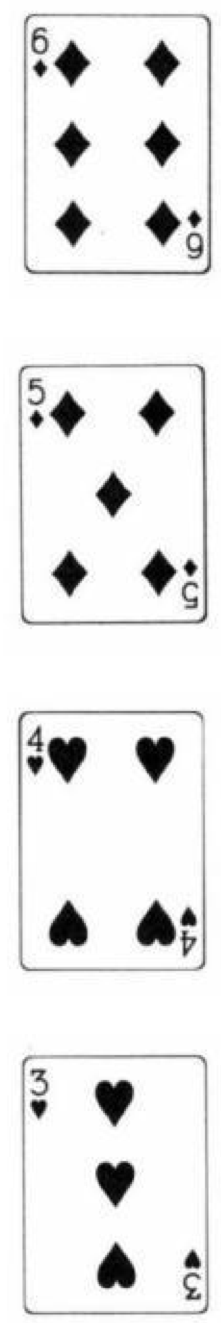




in lowball draw is an example. If you raise with this hand before the draw, it is to get more money in the pot, not to knock players out. It's worth taking the small chance that someone will make a twocard bicycle in order to get more action.

However, don't go overboard with this principle. If you have a pat





you'd rather not have 6 players drawing to beat you.

Another situation that should be fully understood is the one where you have a mediocre hand against many opponents. As I said, this should often be thrown away. Nevertheless, sometimes the antes and early money in the pot make your hand worth a call but not a raise.

For instance, you may have what appears to be the best hand, but there are 5 players drawing at it. It looks like your hand is about an 8-to- I underdog to stand up. A call is still proper if the antes and the earlier bets, along with the present round of bets, provide you with odds of close to 9-to-1 or more.

A bet or raise is incorrect, however (if it will make no one fold), since you would be getting only 5-to-I on this bet or raise.

An example of this is when you have a pair of aces on fifth street in seven-card stud against what appear to be 3 open end straight draws. Another is when you have a 7-6 low made on fifth.street in seven-card lowball and are against 3 bike draws. In both cases, you should usually call a bet but not raise or initiate the betting.

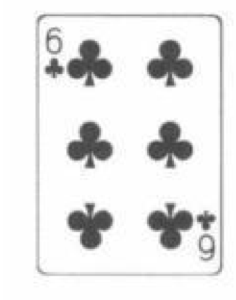
Come One, Come All

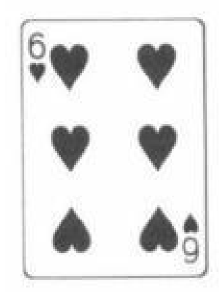
Sometimes you may find yourself with what is not only the best hand but is also the best draw. Take the case where you open as the dealer in a jacks-or-better draw game (53-card deck which includes a joker) with a pair of aces and a one-card draw to an ace joker flush. Even if you miss the flush, you'll still retain the pair of aces, since the joker can be used as an ace.

Assuming that every caller is "on the come" to a straight or smaller flush, the more calls the merrier. Normally two aces would prefer no more than 2 or 3 callers, but here the possibility that you may "redraw"- making a flush - changes things.

This situation also arises when you have three-of-a-kind on fourth street in seven-card stud. The more four-flushes out against you the better. The chances of making a full house - which leaves them "drawing dead," together with your increased pot odds from the calls, will compensate for your decreased chances of winning the pot. (The only exception would be if a lot of money had been put in the pot on third street.)

Occasionally, you'll prefer either very few or very many opponents. This happens when your hand is mediocre but has a small chance of becoming an excellent hand. For instance,





in hold 'em will probably show a profit against less than 2 or more than 5 opponents.

This anomaly arises from the fact that two sixes should stand up against I player, but otherwise probably needs to make three sixes to win. Similarly, two small pair on fourth or fifth Street in sevencard stud may fare better against 1 or many opponents rather than 2 or 3 opponents.

I won't bore you with the math, but rest assured that the concept is valid. I now leave it to you to turn these mathematical principles into expert poker strategy.

What The Pro Knows

What is it that the pros know that the merely good players don't? Just a few things here and there, but these few things allow them to gamble for a living while the merely good player has to keep his job. I'm not going to give away all the secrets, but I'm going to give you some samples.

The pro knows the importance of getting an extra half point when betting .basketball or football and will go to some lengths to find the best line. The fact is that a measly extra half point just about makes up for the vigorish; an extra full point gives a monkey the best of it.

The pro knows that if everybody checks on the flop in a limit hold 'em game, he has a good bluffing opportunity on fourth street, especially from early position and especially when a small card falls.

The pro knows that the most important thing in blackjack counting is finding a good game rather then memorizing extremely intricate counts and strategies. Better to play quite well under good conditions than brilliantly in a poor game. (A good game is one with advantageous rules and/or deep deck

The pro knows that in order to beat the races you must watch the past performances, not just read them off the program. When you watch a race you will see things that aren't shown on the charts.

The pro knows that it's still usually better to start with low cards in eight-or-better high-low stud as well as regular high-low. Big pairs are especially dangerous against an ace showing or in multi-way pots.

The pro knows to play by the month rather than by the day. Should he find himself a big loser one day, he may continue to play, but only because he is hoping to get just part of it back. He doesn't change his game or play wildly simply so he can get even for the day.

The pro knows that in large ante seven-card stud games it is rarely right to "limp" in for the minimum bet on third street unless other players have already come in. Even hands like three flushes are usually best raised with, if there is a chance that that raise will steal the large ante. If it doesn't, that's okay, too.

The pro knows that good poker players are aggressive poker players, usually betting or folding - but merely good poker players know this, too. What pros also know, however, is the right time to back off and be a passive player. Sometimes this is the better way to win the money.

The pro knows how to play all poker games reasonably well, not just his specialty. If there is a "live one" playing a game they don't usually play, he is able to get in the game and take advantage of the situation.

Along these same lines, most pros are looking for any gambling opportunity with the best of it, even if it isn't in their "chosen field." Thus, if there is a special promotion involving parlay cards, keno, blackjack, or whatever, the pro takes advantage of it if he has the best of it.

The pro throws away a pair of aces without the joker or two pair up to jacks-up when he is in middle position and it is opened in front of him (except large ante situations). However, he will raise with these same hands if the opener is in late position.

No-limit hold 'em pros know not only whether to bet, but how much to bet. This alone could be the subject of a whole book.

Pros know that it is absolutely impossible to make a living playing games like craps or roulette where you are betting on inanimate objects and independent events, unless they are cheating or taking advantage of a promotion that puts the odds in their favor.

Pros always consider the size of the pot as well as their hands and their opponents' hands before making a decision on whether to bet, fold, or raise. They also consider what is in the minds of the opponents.

Pros are always thinking about whether to bluff, but their decision is not based on whether or not they think it will work. Rather, their criteria is whether it has a good enough chance of working to show a long-range profit.

Pros put a lot of importance on "extra outs" when they have an otherwise marginal hand. For instance, a three-flush with two cards to come in seven-card stud or hold 'em is of little value by itself but is significant if you have something else with it.

This chapter could have gone on indefinitely, but I only wanted to give a few examples to make you start thinking about some things you might not have thought about before. You may have noticed that I included nothing about lowball draw. This is because at this writing,6 it is presently the main game at which I make my living. Which leads me to one last comment:

Pros don't go around teaching others the really important things for nothing - just to show how smart they are.

Some Words to the "Other Side"

Contrary to what most people might think, I am not opposed to gambling with the "worst of it," that is, where the odds are not in my favor. I only object to those people who contend that you can make a living, or be sure of a long term profit when you gamble this way. But, if you are out to have fun, or out to try to make a "score," I see nothing wrong with playing at a slight disadvantage. The best place to do this is usually in a casino.

First off, I believe that it is certainly fair for casinos to only offer games where they have the best of it. Though I regularly beat casinos by playing blackjack, or by taking advantage of some of their ill-conceived promotions, I don't believe that I actually have a "right" to beat casinos simply because they have an advantage over the vast majority of players. On the contrary, I believe that casinos have a night to a small advantage over every player because of what they offer in return. The casino offers the gambler two things for which it is entitled to win more than it loses. For one, it gives the gambler the choice of which bets to make, when to bet, how much to bet: or whether to bet at all. Normally, for a very small "fee" or vigorish, it agrees to always take the other side of your bet. Of course, from a mathematical standpoint, this doesn't really hurt the casino in most cases, as long as each outcome is random. Nevertheless, most players are willing to give up something in return for being able to make all the choices.

The second advantage to playing in a casino is more important. The casino is allowing you the possibility of winning huge sums of money, even if you started with very little. It is not all that rare for a player to run a hundred dollar bill into tens of thousands. And, there is no stigma attached to quitting with this money. You have no moral obligation to give the casino a chance to win its money back, as you might in a private poker game.

The fact is, if you are gambling just for fun, or with the intention to try to run a little into a lot, it doesn't matter very much if the percentages are against you. Your chances of winning after an evening of fun gambling are almost the same if you have one percent the worst of it, or if you had one percent the best of it. This is especially true if you are trying to turn a $20 bill into $10,000. Having an advantage is almost meaningless if you only try this once. In fact, you might be better off with the worst of it.

To illustrate, suppose I gave you the choice of taking $100,000to-$100, on a 1050-to-1 shot, or $200 million-to-$100, on a 1,800,000-to-I shot. Unless you were already quite wealthy you would be silly to take the second bet though you have a slight mathematical edge. The first bet, with a small mathematical disadvantage, gives you a far better chance of winning serious money.

So, you see, it is only if you are a professional gambler, or trying to grind out steady profits that it is clearly better to have the advantage. (Actually it is imperative that you do.) Thus a fair, intelligently operated casino has my wholehearted support to offer no games that can be "beaten." With the exception of poker, of course, where the better players beat the weaker players, and the house makes its money from the "rake." Nevertheless, I don't feel guilty about taking lots of money from casinos. You see at the moment, the number of casinos that are both fair and intelligently operated by my standards is one (though some are a lot better than others).

There are three reasons why I still love to see smart players get the best of it. First, almost all casinos offer bets to the unwary that give themselves far too much the best of it. This is true for both table games and slot machines. Secondly, almost all casinos offer at least one game (blackjack) with rules such that a skillful player can beat them. But they then treat these skillful players almost like criminals even though the publicity surrounding the "beatability" of blackjack has vastly increased their business. The casinos should either change the rules so the game cannot be beaten or allow the counters to play with a slight edge. When they do neither, they deserve everything they get.

Finally, many casinos have no one on their staff who is mathematically competent. And, they don't seem to care. They even seem to have contempt for mathematics, not realizing its great importance in a casino operation. No wonder they frequently come up with promotions where they don't realize they have the worst of it. Well, I believe that people who deride mathematics deserve to lose.

Now, let me expand on the foregoing. Casinos do quite well, on an edge of less than 2 percent. especially on their even money type bets. This is where they make most of their money. Craps is 1.4 percent (much less when odds are taken). Baccarat is 1.2 percent. Blackjack is about 2 percent, depending on the player's skill. So why have the Big 6 or Big 8 on the craps table, a bet with 11 percent the worst of it? To win a few extra dollars from beginners or drunks? Why not pay 4'/2-to-1 instead of 4-to-1 on any seven or eliminate the bet altogether?

Another disgrace is the big wheel. It only attracts the neophyte gambler with a few bucks to blow. Why put a black eye on the whole casino with this ripoff game (with the present odds)? The vast majority of a casino's profits already comes in a very acceptable way, so why not let all the profits come that way? Give up the double zero on the roulette wheel, casino owners. Five and a quarter percent is too much of an edge to have in this game. Cut it in half with one zero.

Interestingly enough, I don't think that this would cut your profits in half, but rather increase them with greater and longer play. It is important to realize that if a smaller edge entices a player to play longer, the casino figures to make just as much, if not more, from an individual player. This is even more true of slot machines. Keno also should be more liberal for the player, especially for those who bet more than the minimum. In the case of bets that offer very high odds it is reasonable that the casino have more than a 2 percent edge, maybe 5, or 8 percent. After all, if you tried to parlay your money at even money bets with a 1 percent disadvantage you would be paying well over five percent vigorish on your initial bet. But the 30 percent disadvantage casinos have in keno is just way too much.

Slot machines are another peeve of mine. There is no reason why all machines should not pay back at least 95 percent. Once again, this shouldn't decrease profits at all. Increased play should more than make up for the smaller edge. That has been the experience with the new 97.4 percent payback dollar slots. But even if it wasn't, I oppose anything less than 95 percent payback. Give the customer a good chance to win.

There are a couple of other problems with slot machines that I consider very serious. For one, I think the player should know what the odds of hitting the biggest jackpot are. Some casinos do post the payback on their machines, but none post the jackpot odds. Of course, if they did this, they would probably have to eliminate those machines where the odds are outrageously high, like 10 million-, or more, to-1. And that is a good idea.

Another problem I have with slot machines are the new "virtual reels." These machines may have only 20 symbols on each reel, but the jackpot symbol is set to come up much less often than the other symbols.

In a nutshell, all I'm saying is that the casinos should eliminate the sucker bets. They add little or nothing to their profits, and only serve to cast the casino in a bad light.

What about blackjack? I see nothing wrong with offering games where skill matters. But skill should only allow the player to approach a break even game. In the case of blackjack this could be achieved in a number of ways. Experto Twenty-one' which I invented for Vegas World is one example. But if the casino insists on offering a game that can be legitimately beaten, they shouldn't get mad when it does get beaten.

Finally, I believe that all casinos should have a mathematical consultant. His job would be to see that the aforementioned ideas be implemented both in regular games and in special promotions. Management should not consider the players suckers who need to be fooled. Both the players and the casino should know exactly what the cost of an evening's entertainment, or the chance of a score is. They should know that it is almost as easy to win as to lose. This would truly legitimize the casino industry.

A Quiz

The following quiz should not be considered a general test of the concepts explained in this book. In fact, a few of these questions are almost impossible if you haven't read The Theory of Poker, Hold 'em Poker, Hold 'em Poker For Advanced Players or Seven-Card Stud For Advanced Players. Still, anyone who has thoroughly read and understood everything in this book up to this point, should be able to answer most of the questions in this short, but difficult quiz.

This quiz deals with all aspects of gambling. Some of the questions are hard, but none is exceedingly so. Still, anyone who gets them all correct can be very proud of himself. That's because few gamblers have "full conception" of all games.

I have been careful to include only questions which are not a matter of opinion. If you disagree with the answer, you are simply wrong.

Questions

1. In the game of keno, which is a better bet: 5 different $2 nine-spots or 1 $10 nine spot?

2. Which is the worse play in blackjack: Standing with 16 against a dealer's nine showing or standing with 16 against the dealer's seven showing?

3. If a certain play in backgammon will increase your chances of winning a gammon by 20 percent, you should make it as long as the play doesn't increase your chances of losing a "single" game by more than X percent. What is X? (Disregard any effect of the doubling cube.)

4. Stud poker. There is $60 in the pot. The bet size is $20, and two rounds of betting remain. You are considering making a bluff. You are sure that your opponent will call on this round, but he may fold on the last round. How often must your bluff be successful to justify your betting twice?

5. Similar to the last question. (There is $60 in the pot and the bet is $20. Two

rounds of betting remain.) This time your opponent bets and you can only beat a bluff. You know he is going to bet twice. You should call twice if your chances of winning are greater than what?

6. Three-team parlays pay 6-to-I in Las Vegas. If the three games are at different times, should you take the 6-to-I parlay or parlay the games yourself, laying 11-to-10 on each game'?

7. More blackjack. If you double down with a total of 10 against a five showing, you will win twice as much in the long run than if you don't. That's a fact. But the same is not true if you double with a 10 total against a deuce. Why?

8. Hold 'em poker. Which hand has a better chance against a set of trips on the flop: A flush draw or an open-end straight draw'?

9. A three-reel slot machine has a progressive jackpot of $3,000. The smaller payoffs are set to return 75 percent of what is taken in. There are 20 symbols on each reel. If it costs $1 per pull, do you have the best of it playing this machine? What must the progressive jackpot be in order for you to have an even bet?

10. You're playing the pass line in craps. If you start with $100 and plan to play until you go broke or have $1,000, what is your best strategy? (Assume you cannot "take the odds" and must restrict yourself to pass-line bets.)

Answers

1. It is worse to bet 1 $10 nine-spot because the maximum payoff is usually $100,000. Hitting 9 out of 9 on a $2 nine-spot will usually pay $100,000 - or at least $50,000. 9 out of 9 for $5 should pay 5 times as much, but it doesn't. It only pays the $100,000 limit, and you would get shortchanged quite a bit if you hit the big one.

2. Standing against the seven is much worse. The reason is not because the dealer has less chance of breaking with a seven showing. Your chance of winning by standing is about 25 percent against both a nine and a seven. However, you have a much better chance when hitting against the 7 because you're in better shape when you don't bust. For instance, if you catch a deuce you are still an underdog against a nine, but a favorite against a seven.

More precisely, hitting a 16 vs. a seven increases your winning chances by about 5 percent. Hitting a 16 vs. a nine helps by only 2 percent (as opposed to standing.)

3. X = 10 percent. The general rule is that it is only worth taking the risk of losing a -single- game if your chances of scoring a gammon increase by more than twice that risk.

As an example: Suppose you have a 60 percent chance of winning a single game, a 20 percent chance of losing and a 20 percent chance of scoring a gammon. If a risky move can increase your gammon chances to 30 percent (and a safe move will keep things the same), it's worth it if your chances of losing has not gone above 25 percent. This is because you only gain one unit when your gamble works, but lose double when it fails and you lose a "single" game. (When you lose a unit that you would have won, that's a twounit swing.)

4. One-third of the time. When the bluff fails, you lose $40. When it wins, you gain $80 - the original $60 in the pot plus the $20 that your opponent added when he called the first half of your two-bet bluff. Thus, your effective odds are $80-to-$40 or 2-to-1. If you succeed once out of three tries, you'll break even.

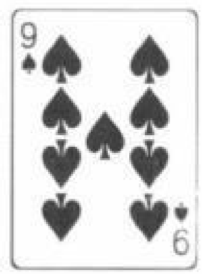
5. Two out of seven. This time your effective odds are $100-to-$40 or 5-to-2. You stand to lose $40 or gain $100 - the original $60 plus your opponent's two $20 bets.

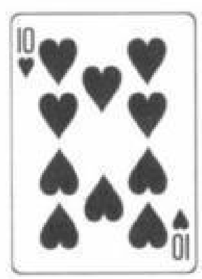
6. It's slightly better to take the 6-to-1 odds. The do-it-yourself parlay, laying 11-to-10 each time, returns a $5.95 profit for each $1 wagered rather than the $6.00 profit from a parlay.

7. Since it is correct to hit 12 against the dealer's deuce showing, you will occasionally "wish" you could take another card when you double down with a 10 against a deuce (namely when you catch a deuce.)

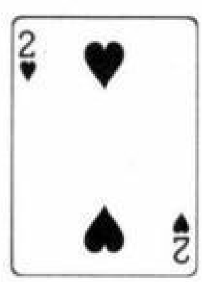
However, no matter what you caught to your 10 total, you would stand against a five. Thus, doubling down is exactly twice as profitable as hitting when the dealer shows a 5, but not quite twice as profitable against a deuce.

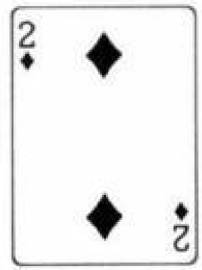
8. A straight draw. Let's say the flop was





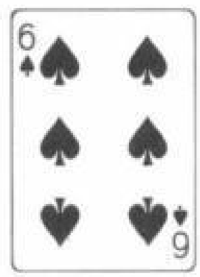
Your opponent has





You would rather have

than



The straight draw starts off with 8 "wins" (4 eights and 4 kings). If you hit the straight on fourth street, your opponent (with three deuces) has 10 cards to make a full house or four-of-a-kind. If the fourth street card is a deuce, nine, or ten (7 of these cards remain), then the Q♦J♦ is "drawing dead." If fourth street helps neither hand, your straight draw still has 8 wins on fifth Street.

Now let's look at the A♠6♠. It also starts off with 8 wins (any spade except the 10). Similarly, if it hits a spade on fourth street, the trips have 10 cards to "redraw." Also, the same 7 cards immediately change the three-deuces into a full house or four-of-a-kind which means the four-flush is drawing dead.

The difference between the two hands (Q♦J♦ and A♠6♠) shows up when the fourth street card is one of the 30 that help neither hand. When this occurs, the flush draw is down to only seven wins on fifth street, because one of the spades can pair the card that turned on fourth street.

For example, say the fourth card is the 3♥. The flush draw now only has seven winning cards, since it can't win with the 10♠ or 3♠. Thus, the normally superior flush draw is not as good as an open-end straight draw against a "set."

9. You have the best of it as long as the jackpot is over $2,000. With 20 symbols on each reel, you figure to hit the jackpot once out of 8,000 pulls.

$$8{,}000 = (20)(20)(20)$$

During these 8,000 attempts, you should take in about $6,000 from the smaller payoffs

$$\$6,000 = (\$8,000)(.75)$$

Therefore, a jackpot of more than $2.000 gives you the edge. (Unfortunately, not all machines are this simple. See the chapters on beating the progressive slots. Also beware of virtual reels.)

10. Bet everything you have unless winning the bet puts you over $1,000. In that case, bet only the amount you would need to win to reach $1,000. To put it another way, bet it all if you have $500 or less, otherwise bet the difference between $1,000 and your bankroll.

This is called the "maximum boldness strategy" and is almost always the best way to bet in order to maximize your chances of attaining a goal when each bet has the worst of it.

The mathematical proof is too difficult to explain here, but the concept is common sense. The smaller your bets, the more likely it is that the house edge will grind you down. Alas, if you must take the worst of it, don't be timid. Bold betting is your only real chance to make a score.

Sklansky on Poker by David Sklansky is a combination of **Sklansky on Razz** and **Essays on Poker** with much new material added. This book not only contains the definitive work on razz but also contains chapters discussing such concepts as choosing your game, middle round strategy, the protected pot, and many others. Finally, there is a section devoted to tournament play.

Gambling Theory and Other Topics by Mason Malmuth is must reading for all serious gamblers. The theme of the book is the dynamic concept of non-self weighting gambling strategies, and the author shows how these apply not only in the gambling world but in real life as well. Also included is accurate advice on fluctuations, bankroll requirements, poker tournament strategies, the new games of Pai Gow Poker and Super Pan Nine, and much more.

Blackjack Essays by Mason Malmuth is designed to help today's modern blackjack player succeed in today's modern casino environment. The text assumes the reader already knows how to count cards and introduces techniques and concepts that should be useful to the successful player well into the 1990s and beyond.

Poker Essays by Mason Malmuth is similar in format to **Blackjack Essays**. The book contains many of the author's current ideas on poker and related subjects. Topics covered include general concepts, technical ideas, structure, strategic ideas, image, tournament notes, in the cardrooms, and poker quizzes.

Poker Essays, Volume II by Mason Malmuth contains those essays that the author wrote on poker from 1992 through early 1996. Topics covered include general concepts, technical ideas, structure, strategic ideas, cardroom theory, and erroneous concepts.

Winning Concepts in Draw and Lowball by Mason Malmuth is the ultimate book for anyone trying to master these forms of poker. This book is for both the typical player and the professional player. It is partitioned into sections that are designed to help all players grow and improve their games.

'Technically the chances of losing the 200 bets honestly is not (1/100) but rather

'This was in 1977. Now of course there is Hold 'em Poker For Advanced Planers by myself and Mason Malmuth.

There is actually a fourth way, namely by playing in casino tournaments. That subject is not covered in this book but is covered in Gambling For a Living by myself and Mason Malmuth.

'This of course is no longer true now that you can bet and watch races from many tracks at once.

;For a fuller treatment on the proper method to compute parlay prices, see the chapter on parlays.

61985

'One deck dealt to the bottom - blackjack pays even money.